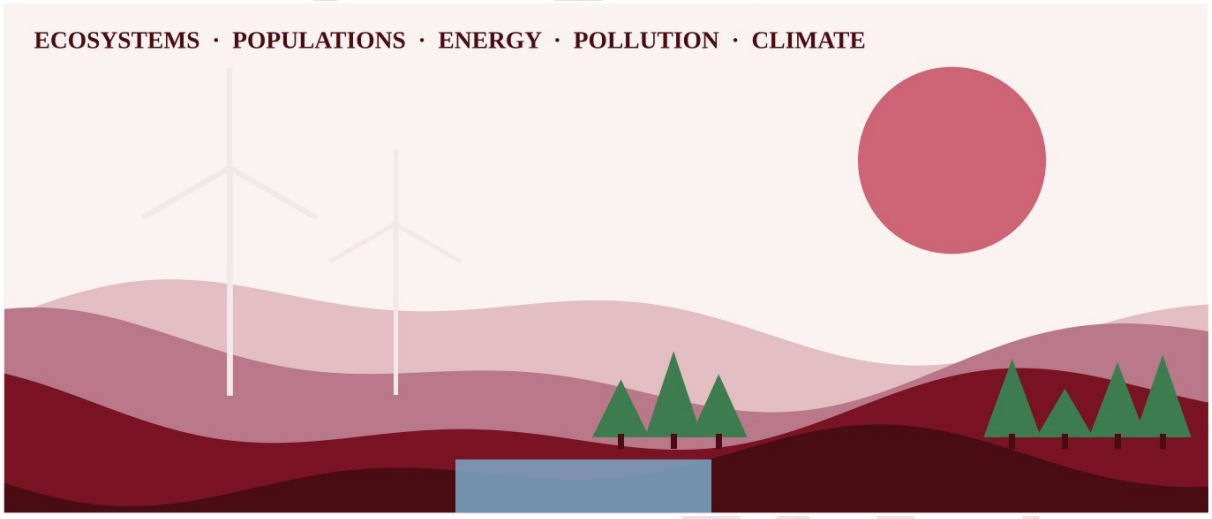
Success4Sure Academy

BOOK THREE

AP ENVIRONMENTAL SCIENCE

COMPLETE SOLUTIONS



Every classroom example, every classified question and every predicted-exam question — fully solved.

Prepared by Mr. Waseem Allabadi

Published by Success4Sure Academy

2026–2027 Edition | Aligned to the 2026 Course and Exam Description

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Prepared by Mr. Waseem Allabadi. Published by Success4Sure Academy. First edition, 2026–2027.

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How to Use This Book

Book Three contains the complete solutions for Book One and Book Two of this collection. It is organized in three parts that match the question books exactly.

LEARNING OBJECTIVES

- **Part I** — worked solutions to every classroom example in Book One, organized by unit.
- **Part II** — answers and explanations for every classified question in Book Two: MCQ first, then FRQ, for each unit.
- **Part III** — complete solutions to the four predicted examinations.

Solutions to the College Board released questions in Book Four are printed at the end of Book Four, not here.

Getting the most from a solution

- Attempt the question first. Look only at the answer line to check your result, then read the explanation to confirm your reasoning.
- For calculations, check the formula, the substitution, the units, and the significant figures, not only the final number.
- For free-response questions, compare each part with the scoring idea shown in italics. A point is earned for the specific idea, so be explicit in your own answers.
- Each solution lists its question ID. IDs beginning **SG** are Book One classroom examples, **QB** are classified Book Two questions, and **PE** are predicted-exam questions.

Part I — Book One Classroom Example Solutions

Worked solutions to every unsolved classroom example in Book One, organized by unit.

Unit 1 — The Living World: Ecosystems

SG-U1-E001

Answer: (a) resource partitioning; (b) it separates the species' use of the seeds so they rarely compete; (c) competition increases.

(a) The four species feed at different plant heights (0–10, 10–30, 30–60 and 60–100 cm), so they share the seed supply by **resource partitioning** (here, spatial partitioning).

(b) Each species takes seeds from a different layer, so a seed eaten by one species is not one that another species needed. Overlap in resource use is small, so interspecific competition is weak and each species keeps enough food to survive and reproduce.

(c) **Competition increases.** Half as many seeds means each bird must find food from a smaller supply, so individuals within each species (intraspecific) and, where feeding zones touch, between species compete harder for what is left. A common wrong answer says competition falls because "there is less food"; scarcity makes interactions stronger, not weaker.

SG-U1-E002

Answer: (a) A = tropical rainforest, B = tundra, C = desert; (b) decomposition is fast in A and very slow in B.

(a) Station A (25 °C, 300 cm/yr) is hot and very wet: **tropical rainforest**. Station B (−9 °C, 22 cm/yr) is very cold with low precipitation: **tundra**. Station C (18 °C, 12 cm/yr) has precipitation under about 30 cm/yr: **desert**, even though it is not especially hot.

(b) At Station A, warmth and moisture make decomposers very active, so dead matter is broken down quickly and nutrients are returned and taken up by plants almost at once. Little is stored in the soil, which is thin and nutrient-poor, and turnover is fast. At Station B, the cold (and permafrost) slows decomposers, so dead matter decays slowly, organic matter accumulates and nutrients are released and recycled slowly.

SG-U1-E003

Answer: (a) Site X = coral reef; Site Y = estuary; (b) clear water lets light reach the producers; Site Y offers abundant nutrients and shelter.

(a) Site X (salinity 34 ppt, warm, 8 m deep, clear, corals) is a **coral reef**. Site Y (12 ppt, where a river meets a bay, sediment-laden, nutrient-rich, young fish and oysters) is an **estuary**, with salinity between fresh and marine.

(b) Low turbidity at Site X lets sunlight penetrate to the algae living in the corals and to other producers, so photosynthesis can occur throughout the shallow water and primary productivity stays high. Site Y supports many young fish because it is shallow and sheltered, and its high nutrient supply sustains abundant producers and food; the cloudy water also helps hide the young from predators. Either reason (nutrients/food or shelter) earns the point.

SG-U1-E004

Answer: (a) combustion → photosynthesis → feeding → cellular respiration; (b) source; the coal carbon had been buried for millions of years.

(a) In order:

1. **Combustion** of the coal: carbon in coal (fossil organic carbon) becomes CO_2 in the atmosphere.
2. **Photosynthesis** in the wheat: CO_2 becomes organic carbon (glucose and other compounds) in the wheat plant.
3. **Feeding** (the cow eats the wheat): carbon is now organic compounds in the cow's body.
4. **Cellular respiration** in the cow: organic carbon is oxidized and leaves as CO_2 in the exhaled breath.

(b) The coal seam acted as a **source**, because it released carbon to the atmosphere. Before the plant was built that carbon was locked in a long-term reservoir (buried, compressed remains of ancient organisms) for millions of years, so it was not cycling among the atmosphere, organisms and surface ocean. Burning it added "new" carbon to the fast cycle.

SG-U1-E005

Answer: (a) denitrification, with nitrogen gas (N_2) leaving; (b) nitrate is the main nitrogen form roots absorb; (c) ammonification.

(a) Waterlogged soil is low in oxygen, which favors **denitrification**: bacteria convert nitrate (NO_3^-) to **nitrogen gas (N_2)**, which escapes to the atmosphere. (Leaching of dissolved nitrate also contributes to loss, but the process asked for is denitrification.)

(b) Nitrate is the main form of nitrogen plant roots absorb, and nitrogen is needed for proteins, DNA and chlorophyll. With less nitrate, the corn makes less chlorophyll (yellowing) and less protein (stunted growth), so nitrogen limits production.

(c) **Ammonification** (decay): decomposers convert the nitrogen in the dead leaves to ammonium (NH_4^+), which plants can take up or which nitrifying bacteria can convert to nitrate.

SG-U1-E006

Answer: (a) phosphorus; (b) 280% increase; (c) phosphorus has no atmospheric source and is released only by slow rock weathering.

(a) **Phosphorus** is the limiting nutrient. Adding phosphorus alone raised algae from 10 to 38 µg/L, a large response, while nitrogen alone barely changed it (11 µg/L). Adding both gave only a little more (40 µg/L) than phosphorus alone, so nitrogen is not the main limit; once phosphorus is supplied, growth rises sharply.

(b) Percent increase = (new − original) ÷ original × 100 = (38 − 10) µg/L ÷ 10 µg/L × 100 = 28 ÷ 10 × 100 = **280%**.

(c) Phosphorus has no significant atmospheric component, so it cannot be added from the air as nitrogen can be fixed from N_2. It enters ecosystems mainly by the very slow weathering of phosphorus-bearing rock, and much of it settles out into sediments, so little is available in the water.

SG-U1-E007

Answer: (a) 8.0×10^6 m³/yr; (b) evapotranspiration, with plant transpiration as a biological contributor.

(a) Precipitation depth = 80 cm = 0.80 m. Area = 25 km² × 10^6 m²/km² = 2.5×10^7 m². Precipitation volume = 0.80 m × 2.5×10^7 m² = 2.0×10^7 m³/yr. Stream flow = 40% (100% − 60%) of that: 0.40 × 2.0×10^7 m³/yr = **8.0×10^6 m³/yr**.

(b) **Evapotranspiration** carries away the largest portion (60%). Biological contributor: **transpiration**, the release of water vapor from plant leaves (evaporation from soil and surfaces is the non-biological part).

SG-U1-E008

Answer: (a) NPP of A = 1,800 and B = 1,800 kcal/m²/yr; (b) 30% and 75%; (c) the claim is not supported.

(a) NPP = GPP − R. Ecosystem A: 6,000 − 4,200 = **1,800 kcal/m²/yr**. Ecosystem B: 2,400 − 600 = **1,800 kcal/m²/yr**.

(b) NPP ÷ GPP × 100. A: 1,800 ÷ 6,000 × 100 = **30%**. B: 1,800 ÷ 2,400 × 100 = **75%**.

(c) The claim is **not supported**. The energy available to consumers is NPP, not GPP, and both ecosystems have the same NPP (1,800 kcal/m²/yr). Ecosystem A's higher GPP is mostly spent on the producers' own respiration (70% of GPP), so it supports no more consumers than Ecosystem B, other things being equal.

SG-U1-E009

Answer: (a) 25 kcal/m²/yr; (b) 40,000 kcal/m²/yr; (c) each transfer loses energy as heat.

(a) Three transfers (phytoplankton → zooplankton → small fish → tuna), each × 0.10: phytoplankton 25,000 → zooplankton 2,500 → small fish 250 → tuna 25 kcal/m²/yr. Check: 25,000 × $(0.10)^3$ = **25 kcal/m²/yr**.

(b) Go down two steps from small fish to phytoplankton, dividing by 0.10 each time: 400 ÷ 0.10 = 4,000 kcal/m²/yr of zooplankton; 4,000 ÷ 0.10 = **40,000 kcal/m²/yr** of phytoplankton.

(c) By the second law of thermodynamics, every energy conversion is less than 100% efficient and some energy becomes low-quality heat (mostly through the respiration the small fish use to move, grow and stay alive). Only the energy stored in the small fish's tissues, about 10%, can be eaten by the tuna, so the tuna receive less than the small fish received.

SG-U1-E010

Answer: (a) grass → grasshoppers → lizards → hawks; the hawk is a tertiary consumer; (b) grasshoppers increase; (c) negative feedback.

(a) The longest chain is **grass → grasshoppers → lizards → hawks** (four levels). Grass is the producer (level 1), grasshoppers are primary consumers (level 2), lizards are secondary consumers (level 3) and the hawk is a **tertiary consumer** (level 4) in this chain. (In the shorter chain grass → rabbits → hawks the hawk is a secondary consumer, so it occupies more than one level in the web.)

(b) The grasshopper population **increases**. Lizards are their only predator, so removing the lizards removes the main source of mortality; grasshoppers survive and reproduce more, limited only by food or other factors. (Hawks would then rely more on rabbits, which can affect the rabbits as well.)

(c) This is **negative (balancing) feedback**. A rise in rabbits increases the foxes' food and fox numbers, and more foxes then eat more rabbits, which reverses the original rise and pushes rabbits back down, stabilizing the system.

SG-U1-E011

Answer: (a) M = savanna, N = temperate grassland; (b) the dry season selects for drought-tolerant, fire-adapted plants; (c) N would shift toward shrubland or desert.

(a) **Station M is a savanna.** Evidence: (1) temperature is warm all year (24–28 °C, almost no seasonal change); (2) rainfall is strongly seasonal, with a wet season (150–230 mm/month, November–April) and a long dry season (under 15 mm/month, May–October). A rainforest would have rain in every month. **Station N is a temperate grassland.** Evidence: (1) a cold winter (−6 °C in January) and a warm summer (20 °C in July), a large seasonal temperature range; (2) low-to-moderate precipitation (15–25 mm in winter), with rainfall peaking in summer (60–90 mm, May–August). Too little rain for forest, too much for desert.

(b) Plants in the savanna must survive six months with almost no rain. Grasses die back or go dormant in the dry season and regrow when rains return, trees are scattered and often drop their leaves or have deep roots, and dry grass fuels fires, so the plants that dominate are drought-tolerant and fire-adapted; the dry season also limits tree density so grasses dominate.

(c) If the region warmed and became drier, evaporation would rise and soil moisture would fall, so tall grasses would be replaced by more drought-tolerant shrubs and short grasses. The biome would move toward **shrubland or, if precipitation fell low enough (under about 30 cm/yr), desert**, because biomes are determined by temperature and precipitation. Station N's biome boundary would shift accordingly.

SG-U1-E012

Answer: (a) 13,200 Mg CO_2; (b) draining lets oxygen in and speeds aerobic decomposition; (c) a source.

(a) Carbon stored = 50 ha × 600 Mg C/ha = 30,000 Mg C. Carbon oxidized = 0.12 × 30,000 Mg C = 3,600 Mg C. Mass of CO_2 = mass of C × 44 ÷ 12 = 3,600 Mg × 3.67 = **13,200 Mg CO_2** (about 1.3×10^4 Mg; exactly 3,600 × 44 ÷ 12 = 13,200).

(b) Waterlogged peat has little oxygen, so decomposition is very slow and organic matter accumulates. Draining (and plowing) lets air into the soil, so aerobic decomposers become active and oxidize the stored organic carbon to CO_2.

(c) The peatland is now a **source** of carbon: it releases CO_2 to the atmosphere instead of accumulating carbon.

SG-U1-E013

Answer: (a) 250 years; (b) −3 km³/yr, shrinking; (c) pavement reduces infiltration and increases runoff.

(a) Residence time = volume stored ÷ rate of inflow = 3.0×10^3 km³ ÷ 12 km³/yr = **250 years**.

(b) Change in storage = inflow − outflow = 9 km³/yr − 12 km³/yr = **−3 km³/yr**. The aquifer is **shrinking** by 3 km³ each year.

(c) Pavement is impermeable, so rain cannot soak into the soil (infiltration and percolation fall). More of the precipitation leaves quickly as surface runoff, less water reaches the aquifer, and recharge drops.

Unit 2 — The Living World: Biodiversity

SG-U2-E001

Answer: (a) genetic diversity, low; (b) habitat diversity, high; (c) species diversity, high.

(a) The problem describes variation in the genes of individuals within one species, so this is genetic diversity. It is low: the population passed through a bottleneck, and the survivors carried only a small sample of the original alleles.

(b) Four distinct physical environments (pine forest, lake, peat bog, grassland) in 40 hectares is variety of habitats, so this is habitat diversity. It is high, because several different habitats exist in a small area.

(c) A count of the different kinds of fish is species richness, a measure of species diversity. 63 species in 100 m^2 is high richness for such a small plot.

A common wrong answer calls (a) "species diversity" because a species is mentioned; the clue is that the variation is among individuals of the same species.

SG-U2-E002

Answer: (i) regulating; (ii) provisioning; (iii) supporting; (iv) cultural.

(i) Soaking up floodwater moderates a natural hazard, so it is a regulating service. (ii) Wild nuts are a material good, so this is provisioning. (iii) Decomposition returning nitrogen to the soil is nutrient cycling, which underlies all the other services, so it is supporting. (iv) Canoe tours are recreation and tourism, so this is a cultural service.

Consequences of clearing the forest for farmland:

Ecological: the forest no longer absorbs floodwater or holds soil, so spring floods are higher and faster and eroded soil and sediment wash into the river (also acceptable: loss of habitat and species, less nutrient cycling).

Economic: the downstream town faces flood damage and repair costs, dredging or water-treatment costs from sediment, and lost income from the canoe business (also acceptable: loss of nut harvest).

SG-U2-E003

Answer: (a) 50% decrease from P to Q; (b) richness rises with area, from 9 to 24 species (+167%); (c) island Q.

(a) P and Q have the same area (4 km^2) but differ in distance (10 km versus 120 km), so any difference is due to distance. Richness falls as distance from the mainland increases.

Percent change = (new − old) ÷ old × 100 = (9 − 18) ÷ 18 × 100 = −50%, a **50% decrease**. (The other matched pair gives (24 − 41) ÷ 41 × 100 = −41.5%, a 41% to 42% decrease, which is also correct.)

(b) At 120 km, Q (4 km^2) has 9 species and S (90 km^2) has 24 species. Richness increases with island area: 15 more species, a (24 − 9) ÷ 9 × 100 ≈ 167% increase.

(c) **Island Q** is most likely to lose a specialist by chance extinction. It is small, so its populations are small and more likely to die out, and it is far from the mainland, so a lost species is unlikely to be replaced by new colonists. It already has the lowest richness (9), which fits this pattern.

SG-U2-E004

Answer: (a) Species A 14 to 26 °C; Species B 10 to 26 °C; (b) Species B; (c) Species B is affected more.

(a) Reading only the tested temperatures, survival is above zero for Species A from 14 °C (lowest) to 26 °C (highest), and for Species B from 10 °C (lowest) to 26 °C (highest). The true limits lie somewhere between the tested values (A between 10 and 14 °C and between 26 and 30 °C; B between 6 and 10 °C and between 26 and 30 °C).

(b) **Species B** has the wider tolerance. It survived across at least 16 °C of tested range (10 to 26 °C), compared with 12 °C (14 to 26 °C) for Species A. B's range reaches colder water, so it is the better generalist for temperature. Note that A is more tolerant at the warm end, which matters for part (c).

(c) The plant will raise the stream from 20 °C to 28 °C. At 26 °C survival is already down to 70% for A but only 40% for B, and both are at 0% at 30 °C. 28 °C lies in the steeply falling section between those two points, so both species will be stressed. Species B is affected more because it loses more of its population at the warm

end (40% versus 70% at 26 °C). A common wrong answer says B is safe because it has the wider range overall; the wide range is toward the cold side and does not help at the new, warmer temperature.

SG-U2-E005

Answer: (a) 75% decrease; (b) 120 g/m²/yr each year; (c) episodic; (d) see below.

(a) Percent change = (new − old) ÷ old × 100 = (300 − 1,200) ÷ 1,200 × 100 = −75%, a **75% decrease**.

(b) Average rate = (value at end − value at start) ÷ time = (780 − 300) g/m²/yr ÷ (5 − 1) yr = 480 ÷ 4 = **120 g/m²/yr per year**.

(c) **Episodic.** Hurricanes come in occasional bursts with some pattern (active and quiet years) but not on a fixed schedule. They are not periodic (regular, clock-like) and not random (no pattern at all).

(d) Any one of: animals flee or migrate away from the damaged area and return later; they shelter in burrows, caves, or tree cavities; or they change their diet or behavior to use new food sources such as fallen fruit and insects. These are individual responses, not genetic adaptations.

SG-U2-E006

Answer: (a) population bottleneck; genetic diversity decreased; (b) fewer alleles means less chance of resistance; (c) behavior is individual, adaptation is a population change.

(a) The collapse from 2,000 to 40 mice was a **population bottleneck**. Only the survivors passed on genes, and rare alleles were lost by chance. Five alleles fell to two, a loss of (2 − 5) ÷ 5 × 100 = 60% of the alleles. Genetic diversity is therefore **lower** even though the population size is back to 2,000; regrowing in number does not restore lost genes.

(b) With only two immune-gene alleles, the chance that any mouse carries a form that resists the new virulent disease is much smaller than in the original population, which had five. If none carries resistance, nearly all could die. The original population had more variation, so more individuals were likely to survive and rebuild it.

(c) A mouse that changes its behavior (for example, avoiding sick mice) is making an individual, within-lifetime response that is not inherited. Adaptation is a change in the population across generations: mice with heritable traits that improve survival leave more offspring, so those traits (such as a resistant allele) become more common. Behavior counts as adaptation only if it is inherited.

SG-U2-E007

Answer: (a) X is primary, Y is secondary; (b) ≈210% increase; (c) rises then falls; (d) lichens build soil.

(a) **Plot X: primary succession.** It began on bare scoured rock with no soil, and the data agree: 0 species in Year 1, only 3 by Year 10. **Plot Y: secondary succession.** The abandoned field kept its soil and seed bank, shown by 12 species and measurable biomass already in Year 1.

(b) Percent change = (new − old) ÷ old × 100 = (6.5 − 2.1) ÷ 2.1 × 100 = 4.4 ÷ 2.1 × 100 ≈ **+210%** (about 209.5%).

(c) Plot Y richness **increases** from 12 (Year 1) to 26 (Year 10) and 31 (Year 30), then **decreases** to 24 by Year 80. After Year 30, the canopy closes and shade-tolerant dominant species (and large long-lived trees) crowd out sun-loving pioneers, so richness falls even as biomass keeps rising.

(d) Lichens are pioneers. They slowly break down rock and, when they die, add organic matter. This produces a thin soil that holds water and nutrients, so mosses, grasses, and later plants can establish. In this way the pioneers change the habitat and make it suitable for later colonists.

SG-U2-E008

Answer: (a) see below; (b) 93% decrease; (c) remove the rats; (d) bottleneck and low genetic diversity.

(a) Any two of the following, tied to the concepts:

- Island biogeography: the island is small (6 km²), so species G has a small population (about 300) that is prone to chance extinction, and it is 80 km from the mainland, so few colonists arrive to replace lost birds (no "rescue" by immigration).
- Specialist versus generalist: G is a specialist that eats only the seeds of one native grass, so it cannot switch foods if the grass declines. The invasive rat is a generalist that eats many foods, reproduces quickly, and preys on or outcompetes the specialist. Ground-nesting birds that evolved without mammalian predators have no defenses against rats.

(b) Percent change = (new − old) ÷ old × 100 = (20 − 300) ÷ 300 × 100 = −93.3%, about a **93% decrease**.

(c) Remove or control the rats (trapping, poison baiting, or fencing off the nesting area). Justification: the rats are the direct cause of the decline, and G cannot be rescued by immigration from the distant mainland. Other acceptable answers: captive breeding and release, or protecting the native grass that G depends on, each with a justification.

(d) A drop from 300 to about 20 is a population bottleneck. If numbers stay near 20 for several generations, alleles are lost by chance and relatives breed together, so genetic diversity is low. With little variation, there may be no individuals able to survive a new disease, climate change, or another invader, and natural selection has little variation to act on.

SG-U2-E009

Answer: (a) $86,400 per year; (b) plant $16,500,000, marsh $216,000, difference $16,284,000; (c) regulating and provisioning; (d) see below.

(a) Value = area × rate × price = 120 ha × 90 kg/ha/yr × $8/kg = 10,800 kg/yr × $8/kg = **$86,400 per year**.

(b) Plant: cost over t years = one-time cost + (annual cost × t) = $9,000,000 + ($500,000 × 15) = $9,000,000 + $7,500,000 = **$16,500,000**.

Marsh: annual cost = 120 ha × $120/ha/yr = $14,400/yr; over 15 years = $14,400 × 15 = **$216,000**.

Difference = $16,500,000 − $216,000 = **$16,284,000** less to protect the marsh (about $16.3 million). Counting the shellfish harvest as well (86,400 × 15 = $1,296,000) would make the marsh an even better choice.

(c) Filtering nutrients from the bay is a **regulating** service (it controls water quality). Shellfish are food, so they are a **provisioning** service.

(d) Any one of: the marsh can be damaged or overwhelmed by a very large nutrient load, storms, or sea-level rise, so it may not give the guaranteed performance of an engineered plant; protecting the marsh means the land cannot be developed for housing or industry, so its owners give up that income; or a marsh treats only as much water as its size allows, so a growing city may still need a plant.

SG-U2-E010

Answer: a model design is below. Other reasonable designs are acceptable if every element is present.

(a) **Hypothesis:** If the pH of the water is lowered below a certain level, then the percent survival of salamander larvae will decrease, because the acidity exceeds the larvae's range of tolerance.

(b) **Independent variable:** pH of the water. **Dependent variable:** percent survival of the larvae after a set time (growth or deformities could be added).

(c) **Treatments:** use a series of graded pH levels, for example 7.0, 6.5, 6.0, 5.5, 5.0, 4.5, and 4.0, made by adding dilute acid to pond water. **Control:** the natural pH of the pond (about 7.0), with no acid added. **Replication:** at least 3 to 5 identical containers at each pH, each with the same number of larvae (for example, 20), so that chance differences between individuals average out.

(d) **Constants** (any two): water temperature, water volume, food type and amount, light and day length, age and size of larvae, source (same pond or clutch) of larvae, duration of the experiment, container type.

(e) After the set time (for example, 14 days) count survivors in each container and calculate percent survival for each pH, using the mean of the replicates. Graph survival (y-axis) against pH (x-axis). The tolerance limit is the pH where survival drops sharply toward zero; the range of tolerance is the span between the upper and lower limits. Survival that is reduced but above zero marks the zone of stress.

SG-U2-E011

Answer: (a) resistant share rose from 3% to 93%; (b) natural selection; (c) rotate herbicides or leave untreated refuges.

(a) The percentage of resistant plants **increased every season**, from 3% to 93%, a gain of 90 percentage points. The increase was slow at first (+6 points from season 1 to 2), fastest in the middle seasons (+32 points from season 3 to 4), and slowed as it approached 100% (+12 points from season 5 to 6).

(b) **Variation:** weeds in the population differed in a heritable trait, and a few plants (3%) already carried genes for resistance. **Selective agent:** the herbicide, which killed susceptible plants. **Outcome:** resistant plants

survived and reproduced, passing resistance genes to their offspring, so each generation had a larger share of resistant plants. No individual plant became resistant by being sprayed; the population changed across generations.

(c) Rotate to herbicides with a different mode of action (or leave untreated refuge strips). Justification: a plant resistant to one herbicide is likely to be killed by a different one, so selection pressure on any single trait is weaker. Untreated refuges let susceptible plants keep breeding with resistant survivors, which dilutes the resistance genes. Spraying only when needed is also acceptable. Higher doses or more frequent spraying would strengthen selection and speed up resistance.

Unit 3 — Populations

SG-U3-E001

Answer: (a) Beetle A is a specialist and Beetle B is a generalist; (b) A crashes and may disappear locally while B persists or grows; (c) A would be favored in a completely stable forest.

(a) Beetle A has a narrow niche (one tree species for eggs and food), so it is a **specialist**. Beetle B eats more than twenty tree species and lives in several habitat types, so it is a **generalist**.

(b) Beetle A's only resource has been removed, so it has nowhere to switch. Its population should fall sharply over the five years and could go locally extinct. Beetle B simply feeds on the many other tree species that survive, so its population should hold steady or even grow (it may gain if A's loss reduces competition). A common wrong answer says both decline because "the forest was damaged"; that ignores niche width.

(c) **Beetle A** (the specialist). In an unchanging habitat a specialist is fine-tuned to its one resource and uses it more efficiently than a generalist, so it out-competes the generalist for that resource. With no disturbance, the specialist's weakness never shows.

SG-U3-E002

Answer: (a) P is r-selected and Q is K-selected; (b) Q would be harmed more; (c) R has a mix of traits and sits between the extremes.

(a) **P is r-selected**: it is tiny (2 g) and produces 400 eggs per season, matures at only 3 months, and lives about 1 year (any two of these). **Q is K-selected**: it is huge (4,500 kg) and has one calf every 3 years, matures at 12 years and lives 60 years (any two).

(b) **Species Q.** It reproduces so slowly (one calf per 3 years, first breeding at 12 years) that it cannot replace individuals lost to a predator. Species P can produce hundreds of offspring and mature within months, so its population rebounds quickly. A common wrong answer says P is harmed more because it is smaller; body size does not determine recovery, reproductive speed does.

(c) R has traits from both strategies. Its moderate mass (25 kg), moderate maturity age (2 years) and moderate life span (12 years) are intermediate. Two pups per year is a modest number (far fewer than P's 400 eggs, but repeated every year rather than once every three years like Q). r and K are ends of a spectrum, and most species fall between them, so R is best described as having a tendency toward one side rather than as a pure r or pure K species.

SG-U3-E003

Answer: (a) 20%; (b) 80 individuals; (c) Type III, and the species is more likely r-selected.

(a) Percent surviving = (number alive at age ÷ starting cohort) × 100 = (160 ÷ 800) × 100 = **20%**.

(b) Deaths in the interval = survivors at start − survivors at end = 160 − 80 = **80 frogs**.

(c) **Type III.** In the first year 800 − 160 = 640 individuals (80% of the cohort) died, which is the steepest loss. After that the loss is much slower: 80 deaths in year 2, 40 deaths over the two years 2–4 (about 20 per year) and 20 over the four years 4–8 (about 5 per year). A very high early death rate followed by slow loss is Type III. This fits an **r-selected** species: it produces very many eggs with little parental care, so most offspring die young but a few survive.

SG-U3-E004

Answer: (a) K ≈ 500 deer, 20 deer/km²; (b) years 4–6, an increase of 140 deer; (c) resources became limiting.

(a) The population levels off at 500 (years 14 and 16), so **K ≈ 500 deer**. Density = 500 deer ÷ 25 km² = **20 deer per km²**.

(b) Increases per two-year interval: 0–2: 50; 2–4: 110; 4–6: 140; 6–8: 100; 8–10: 50; 10–12: 15; 12–14: 5; 14–16: 0. The greatest increase is **years 4–6**: 330 − 190 = **140 deer** (70 deer per year). This is the steepest part of the S-curve, which occurs about halfway to K (the population passes 250 in this interval).

(c) By year 8 the herd (430) was close to K (500). Food, water and space per deer were becoming scarce, so competition increased, per-capita births fell and deaths rose, and the number added each interval shrank. The curve flattened as the population approached its carrying capacity.

SG-U3-E005

Answer: (a) about 87% over capacity; (b) dieback; (c) overgrazing and soil erosion; reduced forage lowers K.

(a) Percent over capacity = (N − K) ÷ K × 100 = (560 − 300) ÷ 300 × 100 = 260 ÷ 300 × 100 ≈ **86.7%, or about 87%**.

(b) The sharp decline from 560 to 140 is called **dieback** (a population crash). It followed overshoot, with the drought speeding it up.

(c) Two impacts of overshoot (any two): goats consumed forage faster than it regrew, so vegetation was depleted (overgrazing); bare soil eroded in rain and wind; water sources and nutrient cycling were degraded; plant species may have been lost. A new K below 300 is possible because the damaged habitat produces less forage: topsoil lost to erosion and fewer plants regrowing mean less food is supplied each year, and K is set by the food supply.

SG-U3-E006

Answer: (a) 150, 200 and 38 fish per year; (b) fastest at N = 1,000; (c) −312 fish per year, a decline.

(a) dN/dt = r × N × (K − N) ÷ K, with r = 0.40 per year and K = 2,000 fish.

- N = 500: 0.40 × 500 × (1,500 ÷ 2,000) = 200 × 0.75 = **150 fish per year**.
- N = 1,000: 0.40 × 1,000 × (1,000 ÷ 2,000) = 400 × 0.5 = **200 fish per year**.
- N = 1,900: 0.40 × 1,900 × (100 ÷ 2,000) = 760 × 0.05 = **38 fish per year**.

(b) Growth is fastest at **N = 1,000**, which is K/2 = 2,000 ÷ 2. Growth is slowest at N = 1,900 because the population is almost at K: only 5% of the capacity is still open, so the factor (K − N) ÷ K is tiny. Resources are nearly used up, so the population adds few individuals.

(c) dN/dt = 0.40 × 2,600 × (2,000 − 2,600) ÷ 2,000 = 1,040 × (−0.30) = **−312 fish per year**. The negative sign means the population is above K (overshoot) and is shrinking by about 312 fish per year.

SG-U3-E007

Answer: (a) A is growing fastest, C is most likely declining; (b) continued growth after fertility falls, strongest in A; (c) a shrinking workforce supporting many older people.

(a) **Country A** is growing fastest: 41% of its people are under 15, the highest share of young people, who will soon become parents. **Country C** is most likely declining: only 13% are under 15 while 51% are 45 or older, so too few children are being born to replace the large older groups. B (25% young, 34% old) is in between.

(b) **Population momentum** is continued population growth after fertility drops to replacement level, because a large group of young people is entering their reproductive years and the number of births stays high even with fewer children per woman. It is strongest in **Country A**, which has the largest share of people under 15.

(c) Country C will have fewer workers for each older person, so there will be strain on pension systems and health care (and the tax base). Other acceptable answers: a labor shortage, closing schools, rising costs of elder care.

SG-U3-E008

Answer: (a) TFR = 1.59 children per woman; (b) below replacement, so the population is likely to decline in the long term; (c) see below.

(a) TFR = (sum of age-specific fertility rates) × (width of age group) ÷ 1,000. Sum = 10 + 70 + 105 + 85 + 40 + 8 + 0 = 318 births per 1,000 women per year. TFR = 318 × 5 ÷ 1,000 = 1,590 ÷ 1,000 = **1.59, about 1.6 children per woman**.

(b) 1.59 is below the replacement level of 2.1, so, ignoring migration, each generation is smaller than the last and the population is likely to **decline** in the long term (after population momentum has run its course).

(c) Examples (any two): greater access to education for girls and women would lower TFR; wider access to family planning would lower TFR; a rise in the age at first child would lower TFR; a pro-natalist policy (child allowances, paid parental leave) would raise TFR; a cost-of-living change that makes children cheaper, or an anti-natalist penalty being removed, would raise it.

SG-U3-E009

Answer: (a) 1.4% per year; (b) 50 years; (c) net migration (200 people) contributed more than natural increase (150 people).

(a) Growth rate = [(births + immigrants) − (deaths + emigrants)] ÷ starting population × 100 = [(400 + 350) − (250 + 150)] ÷ 25,000 × 100 = 350 ÷ 25,000 × 100 = **1.4% per year**.

(b) Doubling time ≈ 70 ÷ growth rate (%) = 70 ÷ 1.4 = **50 years**.

(c) Natural increase = births − deaths = 400 − 250 = **150 people**. Net migration = immigrants − emigrants = 350 − 150 = **200 people**. Net migration is larger, so most of the growth this year (200 of 350, about 57%) came from **net migration**.

SG-U3-E010

Answer: (a) 2.0%; (b) 1.85%; (c) about 38 years and 8 times larger; (d) earthquake density-independent, disease density-dependent.

(a) RNI = (CBR − CDR) ÷ 10 = (28 − 8) ÷ 10 = **2.0%**.

(b) Growth rate = (CBR − CDR + immigration − emigration) ÷ 10 = (28 − 8 + 0 − 1.5) ÷ 10 = 18.5 ÷ 10 = **1.85% per year** (about 1.9%).

(c) Doubling time ≈ 70 ÷ 1.85 ≈ **38 years**. After three doubling times the population is 2^3 = **8 times** larger.

(d) An earthquake is **density-independent**: it harms a place regardless of how crowded it is. An increase in disease transmission is **density-dependent**: spread is easier when people are crowded together.

SG-U3-E011

Answer: (a) 0.3%, 1.3%, 2.4%, 1.5%, 0.2%; (b) Stages 1, 2, 2, 3, 4; (c) 1960; (d) near-vertical sides, stable or aging.

(a) RNI = (birth rate − death rate) ÷ 10:

- 1900: (44 − 41) ÷ 10 = **0.3%**
- 1930: (43 − 30) ÷ 10 = **1.3%**
- 1960: (38 − 14) ÷ 10 = **2.4%**
- 1990: (24 − 9) ÷ 10 = **1.5%**
- 2020: (12 − 10) ÷ 10 = **0.2%**

(b) 1900: **Stage 1** (both rates high and nearly equal). 1930: **Stage 2** (death rate falling, birth rate still high). 1960: **Stage 2** (death rate low and falling, birth rate still high). 1990: **Stage 3** (birth rate falling clearly, death rate low, growth slowing). 2020: **Stage 4** (both rates low and close together).

(c) **1960**, with an RNI of 2.4%. The gap between birth and death rates (24 per 1,000) was largest because the death rate had dropped sharply (better food supply, clean water, medicine and sanitation) while families were still large and the birth rate had barely fallen.

(d) With a birth rate of 12 and a death rate of 10, the diagram would have a **narrow base and nearly vertical sides**, widening a bit in the middle and upper ages: a stable or slowly aging population with a low proportion of young people.

SG-U3-E012

Answer: (a) 25 years; 144 million; (b) Stage 2; (c) see below; (d) population momentum.

(a) Doubling time ≈ 70 ÷ 2.8 = **25 years**. In 50 years there are 50 ÷ 25 = 2 doublings, so the population = 36 million × 2^2 = 36 × 4 = **144 million**.

(b) **Stage 2** (transitional). Supporting values: a growth rate of 2.8% per year (rapid growth); a TFR of 4.8 (high births, large families); an infant mortality rate of 62 per 1,000 (still high but falling in this stage); only 28% of girls finish secondary school (low female education, characteristic of an early stage). Any two values with reasons earn credit.

(c) Examples (any two): *Expand girls' education* – educated girls have more job options, marry and have children later, so fewer children per woman. *Expand access to family planning* – couples can choose the number and timing of children, lowering births. *Improve maternal and infant health care and nutrition* – when more infants survive, families no longer need many births to ensure survivors, so fertility falls after a lag.

(d) **Population momentum.** The age structure has a very large share of young people who are about to reach reproductive age. Even if each woman had only 2.1 children, the number of births would still exceed the number of deaths for decades because there are so many parents, so the population keeps growing.

SG-U3-E013

Answer: (a) K = 6 voles/ha; (b) +0.20 per vole per year, growing; (c) about 5.5 voles/ha, a decrease.

(a) Carrying capacity is the density at which births equal deaths. At **6 voles per hectare** births = deaths = 0.46 per vole per year, so **K = 6 voles/ha**. Below 6, births exceed deaths and the population grows; above 6, deaths exceed births and it shrinks.

(b) Net change at 4 voles/ha = births − deaths = 0.58 − 0.38 = **+0.20 per vole per year**. It is positive, so the population is **growing** at that density.

(c) New deaths: 0.35, 0.43, 0.51, 0.59, 0.67 at densities 2, 4, 6, 8, 10. New net change (births − deaths) at 4 = 0.58 − 0.43 = +0.15; at 6 = 0.46 − 0.51 = −0.05. The sign changes between 4 and 6, with a gap of +0.15 and −0.05 (total 0.20). The crossing is at 4 + 2 × (0.15 ÷ 0.20) = 4 + 1.5 = **5.5 voles/ha**. The new K (about 5.5) is **lower** than the old K of 6, because extra mortality at every density moves the crossing to a lower density.

SG-U3-E014

Answer: (a) rabbits peak year 2, foxes peak year 4, a lag of 2 years; (b) see below; (c) 60%; (d) negative feedback, density-dependent.

(a) The rabbit peak is at **year 2** (1,500) and the fox peak at **year 4** (90). The foxes lag the rabbits by **2 years**.

(b) When rabbits are abundant, foxes find plenty of food and raise more young, but fox numbers take time to rise because of the time needed for mating, gestation and young growing to adulthood. Foxes therefore keep increasing even after rabbits have started to decline. As foxes become numerous they eat more rabbits and the rabbit population falls; the food supply then falls and foxes starve or fail to raise young, so they decline after the rabbit decline.

(c) Percent decrease = (final − initial) ÷ initial × 100 = (600 − 1,500) ÷ 1,500 × 100 = **−60%**, a 60% decrease.

(d) It is a **negative feedback loop** (a rise in one population triggers changes that push it back down). Predation is **density-dependent**: the effect on prey depends on prey and predator density, and predators eat more when prey are plentiful.

SG-U3-E015

Answer: (a) 400 sheep; (b) 30%; (c) new K = 300, remove 220 sheep; (d) 13,140 metric tons demand, 1,140 metric ton shortfall.

(a) K = supply ÷ need per individual = 240,000 kg per year ÷ 600 kg per sheep per year = **400 sheep**.

(b) Percent over capacity = (N − K) ÷ K × 100 = (520 − 400) ÷ 400 × 100 = **30%**.

(c) Forage after drought = 240,000 × (1 − 0.25) = 180,000 kg per year. New K = 180,000 ÷ 600 = **300 sheep** (25% less forage means 25% fewer sheep). Sheep to remove = 520 − 300 = **220 sheep**.

(d) Demand per person per year = 0.45 kg per day × 365 days = 164.25 kg per year. Town demand = 80,000 people × 164.25 kg = 13,140,000 kg = 13,140,000 ÷ 1,000 = **13,140 metric tons per year**. Shortfall = demand − supply = 13,140 − 12,000 = **1,140 metric tons per year** (a deficit).

SG-U3-E016

Answer: a model experimental design follows.

(a) **Hypothesis:** If phosphorus is added to tanks of pond water, then the green alga population will grow larger (reach a higher population density) than in tanks with no added phosphorus.

(b) **Independent variable:** the amount of phosphorus added (phosphorus added or not). **Dependent variable:** the alga population (cell count per mL, or chlorophyll concentration, measured at regular intervals). **Controlled variables** (any two): light intensity and hours of light, temperature, volume and source of pond water, tank size and type, starting number of algae, same measuring times.

(c) **Treatment group:** 6 tanks receive the phosphorus solution (the same amount in each). **Control group:** 6 tanks receive an equal volume of water (or the solution without phosphorus) and are otherwise treated identically. Using six tanks per group gives replicates, so chance differences between tanks average out and the result is more reliable. Assign tanks to treatments at random so position near the light does not bias the result. Measure all tanks on the same schedule until the populations plateau.

(d) **Supporting results:** the phosphorus tanks reach a higher alga count (a higher growth rate or a higher plateau) than the control tanks, with the difference larger than the variation within a group. **Limitation:** small tanks in a lab are simpler than a whole pond, which has other nutrients (such as nitrogen), grazers, varying light and temperature, and water exchange, so the result might not scale to the pond. Other acceptable answers: a short study time, or other factors limiting growth in the pond.

SG-U3-E017

Answer: (a) P: 11.8, Q: 3.3; (b) Q; (c) P has a wide-based pyramid and greater momentum, Q has a narrow base.

(a) Support ratio = people aged 15–64 ÷ people aged 65 and older. Country P: 59 ÷ 5 = **11.8** (about 12 workers-age people per older person). Country Q: 66 ÷ 20 = **3.3**.

(b) **Country Q** faces greater pension and health-care strain. Its support ratio is much lower, so each person aged 65 or older is supported by only about 3 people of working age, compared with about 12 in P. A smaller group of workers and taxpayers must fund the costs of a large older group.

(c) **Country P** has a wide base that narrows quickly (a triangle-shaped pyramid), with 36% under 15, indicating rapid growth. **Country Q** has a narrow base (14% under 15) and a large top (20% aged 65 and older), indicating slow growth or decline. **P has the greater population momentum**, because its large young group is about to enter reproductive age.

SG-U3-E018

Answer: (a) parasite infection is density-dependent, flood mortality is density-independent; (b) totals depend on population size; (c) S 80 million tons, T 40 million tons, S has the larger impact.

(a) The **parasite** is **density-dependent**: the percent infected rises as density rises (4% at 5 hares/ha to 37% at 40 hares/ha), so the rate depends on crowding. The **flood** is **density-independent**: about the same percent is killed at every density (11–13%).

(b) The total number infected would rise with density even if the infection rate were the same, simply because there are more hares to be infected. A total mixes the effect of density with the effect of population size. The rate (percent infected) removes the effect of population size, so it is the correct test.

(c) Total use = population × use per person. Country S: 10,000,000 × 8 tons = **80,000,000 tons (80 million) per year**. Country T: 40,000,000 × 1 ton = **40,000,000 tons (40 million) per year**. **Country S has the larger impact** (twice that of T) despite having a quarter of the population, because its per-person consumption is eight times higher.

SG-U3-E019

Answer: (a) graph described below; (b) RNI 2.3%, doubling about 30 years, year 2000; (c) 1970 Stage 2, 1990 Stage 3, 2010 Stage 4.

(a) Put **year** (independent variable) on the x-axis, from 1950 to 2010 in equal steps of 10 years, and **rate per 1,000 people per year** on the y-axis, from 0 to 40 in steps of 5. Plot two data sets with a key:

- Birth rate: (1950, 36), (1970, 33), (1990, 21), (2010, 12).
- Death rate: (1950, 20), (1970, 10), (1990, 8), (2010, 9).

Join the points of each set with a line. The graph shows the death rate falling steeply first, then the birth rate falling later until the two lines nearly meet in 2010.

(b) RNI (1970) = (33 − 10) ÷ 10 = **2.3%**. Doubling time ≈ 70 ÷ 2.3 ≈ **30 years**. Year of doubling = 1970 + 30 = **2000** (about). This assumes the rate stays constant.

(c) **1970 = Stage 2**: the death rate (10) has fallen to a low level while the birth rate (33) is still high, giving a wide gap and rapid growth (2.3%). **1990 = Stage 3**: the birth rate has fallen to 21, clearly above the death rate (8), so growth is positive but slowing (1.3%). **2010 = Stage 4**: both rates are low (12 and 9) and close, so growth is very slow (0.3%).

Unit 4 — Earth Systems and Resources

SG-U4-E001

Answer: (a) Site 1 convergent (oceanic–continental subduction); Site 2 divergent; Site 3 transform. (b) The plates move apart and new crust forms by seafloor spreading. (c) Site 1.

(a) Site 1: a deep trench offshore with a line of volcanoes inland shows an oceanic plate sinking beneath a continental plate, so the boundary is convergent. Site 2: a mid-ocean ridge with new basalt and seafloor that grows older on both sides is a divergent boundary. Site 3: a road offset sideways with no volcanoes shows plates sliding horizontally past each other, a transform boundary (like the San Andreas Fault).

(b) At Site 2 the two plates move apart. Magma rises into the gap, cools and hardens into new basalt crust, and the older crust is carried away on both sides. This process of adding new crust is **seafloor spreading**.

(c) Site 1 is the most likely source of a tsunami, because a large subduction-zone earthquake can suddenly lift or drop the seafloor and displace the water above it. A transform fault slides sideways with little vertical displacement, and ridge earthquakes are usually small and shallow.

SG-U4-E002

Answer: (a) P = O, Q = A, R = E, S = B, T = C, U = R. (b) Leaching removes material from R and deposits it in S. (c) Q (the A horizon, topsoil).

(a) P is loose, partly decayed litter, so it is the **O** horizon. Q is dark, humus-rich and root-filled, so it is the **A** horizon. R is pale, sandy and nutrient-poor, so it is the **E** horizon. S is reddish and clay-rich, so it is the **B** horizon. T is broken, partly weathered parent material, so it is the **C** horizon. U is solid, unbroken bedrock, so it is the **R** horizon.

(b) Water percolating down through R carries away (leaches) fine clay, dissolved minerals, iron and humus, leaving behind the pale, coarse sand grains that hold few nutrients. The material removed from R is carried down and deposited in S, where it accumulates, which is why S is clay-rich and reddish.

(c) The farmer loses the topsoil, the dark, humus-rich A horizon (Q). On a bare field there is no litter layer to protect it, and if the litter (P) is present it is stripped first. Practices that reduce the loss include cover crops, no-till farming, contour plowing, terracing, strip cropping or windbreaks, which keep the surface covered, slow water or wind, and bind soil with roots.

SG-U4-E003

Answer: (a) sand 42%, silt 36%, clay 22%. (b) Loam. (c) Clay.

(a) Percent = component mass ÷ total mass × 100. Sand = 21.0 g ÷ 50.0 g × 100 = **42%**. Silt = 18.0 g ÷ 50.0 g × 100 = **36%**. Clay = 11.0 g ÷ 50.0 g × 100 = **22%**. These sum to 100%.

(b) Check the simplified key in order. Sand ≥ 85%? No (42%). Clay ≥ 40%? No (22%). Silt ≥ 80%? No (36%). Loam requires sand 23–52%, silt 28–50% and clay 7–27%: 42% sand, 36% silt and 22% clay all fall in range, so the texture class is **loam**.

(c) Clay holds the most water against gravity. Its particles are the smallest, so its pores are very tiny and have a huge total surface area; water is held tightly on the particle surfaces and in the narrow pores instead of draining out under gravity.

SG-U4-E004

Answer: (a) X 15%, Y 45%, Z 80% retained. (b) See below. (c) X sand, Y loam, Z clay. (d) Run replicates.

(a) Retained = poured − drained; percent retained = retained ÷ poured × 100. Soil X: 200 − 170 = 30 mL; 30 ÷ 200 × 100 = **15%**. Soil Y: 200 − 110 = 90 mL; 90 ÷ 200 × 100 = **45%**. Soil Z: 200 − 40 = 160 mL; 160 ÷ 200 × 100 = **80%**.

(b) Independent variable: the soil type (X, Y or Z). Dependent variable: the volume of water drained after 10 minutes. Controlled variables (any two): mass of dry soil (400 g), volume of water poured (200 mL), identical funnels and filter paper, the 10-minute collection time, water temperature, and pouring all the water at once.

(c) Soil X is the **sand**: it drained the most (170 mL) and retained only 15%, because large pores let water pass quickly. Soil Z is the **clay**: it drained the least (40 mL) and retained 80% because tiny pores hold water tightly. Soil Y is the **loam**, intermediate in both (110 mL drained, 45% retained).

(d) Repeat the test at least three times with each soil and average the results, which reduces the effect of random error and shows how consistent the results are. Acceptable alternatives: also record the time for drainage to stop, pack each soil to the same density, or test additional samples of each soil type.

SG-U4-E005

Answer: (a) 6.5°C per km of cooling. (b) Troposphere; stratosphere. (c) Ozone absorbs UV. (d) About 12 km.

(a) Average rate = (T lower − T higher) ÷ altitude difference = (18 − (−60)) ÷ (12 − 0) = 78°C ÷ 12 km = **6.5°C per km** (the temperature falls with height, as expected for the average lapse rate).

(b) The 0–12 km layer, where temperature falls with altitude, is the **troposphere**. The 16–50 km layer, where temperature rises from −60°C to −3°C, is the **stratosphere**.

(c) The trend reverses because the stratosphere contains the **ozone** layer. Ozone (O_3) absorbs ultraviolet radiation from the sun, which warms the air, so temperature rises with altitude instead of falling. The troposphere, in contrast, is heated from the ground below, so the air farthest from the surface is coldest.

(d) The tropopause is at about **12 km**, where the temperature stops falling and reaches its minimum (−60°C). The data show it staying at −60°C up to 16 km, so the turning point lies in the 12–16 km range, and 12 km is the first altitude at which the cooling trend ends.

SG-U4-E006

Answer: (a) The northeast trade winds, blowing from the northeast. (b) The Coriolis effect deflects the air to the right. (c) Toward the east, carried by the prevailing westerlies. (d) Sinking dry air at 30° N.

(a) At 15° N the ship is in the trade-wind belt (0°–30°). The prevailing wind there is the **northeast trade wind**, which blows *from* the northeast, toward the southwest, helping a ship cross from West Africa to the Caribbean.

(b) Surface air in the Hadley cell flows toward the equator, which by itself would be due south. Earth rotates eastward, and the air keeps the greater eastward speed of the higher latitude it left, so relative to the ground it appears to curve **to the right** in the Northern Hemisphere. A southward-moving flow turned to the right ends up moving toward the southwest, so the wind comes from the northeast instead of directly from the north.

(c) At 45° N the storm is in the **prevailing westerlies** belt (30°–60°), so it is most likely carried **west to east** (toward the east).

(d) Air that rose at the equator loses its moisture as rain, moves poleward aloft, cools, and sinks near 30°. Sinking air creates high pressure, warms as it descends and becomes drier, so clouds and rain are rare. This produces the Sahara and other deserts near 30° N.

SG-U4-E007

Answer: (a) 3.24×10^7 m³. (b) 4.23×10^7 m³. (c) About 31% increase. (d) Pavement reduces infiltration.

Formula: runoff volume = precipitation depth × area × C. Convert 1 km² = 10^6 m², so 120 km² = 1.2×10^8 m².

(a) 0.90 m × 1.2×10^8 m² × 0.30 = **3.24×10^7 m³**.

(b) Paved part: 0.90 m × (20×10^6 m²) × 0.85 = 1.53×10^7 m³. Remaining part: 0.90 m × (100×10^6 m²) × 0.30 = 2.70×10^7 m³. Total = $1.53 \times 10^7 + 2.70 \times 10^7$ = **4.23×10^7 m³**.

(c) Percent increase = (new − old) ÷ old × 100 = ($4.23 \times 10^7 - 3.24 \times 10^7$) ÷ $3.24 \times 10^7 \times 100$ = 0.99 ÷ 3.24 × 100 ≈ **31%**.

(d) Pavement and rooftops are impervious, so rain cannot soak into the ground (infiltration drops) and nearly all of it runs off the surface. The characteristic that changed is the **vegetation or soil cover** (land surface type): natural plant cover and permeable soil were replaced by impervious surfaces. (The runoff coefficient, and so the speed of flow to the stream, increased.)

SG-U4-E008

Answer: (a) 73.5° in June, 26.5° in December. (b) June. (c) Longer daylight in June. (d) The equator (0°).

(a) Noon sun angle = 90° − |latitude − latitude of the direct sun|. June solstice (direct sun at +23.5°): 90° − |40° − 23.5°| = 90° − 16.5° = **73.5°**. December solstice (direct sun at −23.5°): 90° − |40° − (−23.5°)| = 90° − 63.5° = **26.5°**.

(b) The **June solstice** gives more intense radiation per unit area. The higher sun angle (73.5°) concentrates the same beam of sunlight on a smaller area of ground, and the rays travel a shorter path through the atmosphere, so less is absorbed, scattered or reflected. In December the low sun spreads the energy over a larger area and passes through more air. (The sines make the point: sin 73.5° ≈ 0.96 and sin 26.5° ≈ 0.45, so June is about 2.1 times as intense.)

(c) Day length: in June the Northern Hemisphere is tilted toward the sun, so daylight lasts longer (about 15 hours at 40° N) than in December (about 9 hours), giving more hours of exposure.

(d) On the March equinox the sun is directly overhead at the **equator (0° latitude)**.

SG-U4-E009

Answer: (a) 4.4 times. (b) City N. (c) Rising, cooling air drops its moisture on the windward side. (d) Drier vegetation at N.

(a) 1,850 mm ÷ 420 mm = 4.405, so City M receives about **4.4 times** as much precipitation as City N.

(b) **City N** is in the rain shadow. The prevailing winds blow from the west, so the western foot (City M) is the windward side and the eastern foot (City N) is the leeward side.

(c) Moist westerly air meets the mountains and is forced to rise. As it rises it expands and cools, so water vapor condenses into clouds and falls as heavy rain or snow on the windward slope. The air, now much drier, crosses the crest, descends on the leeward side, is compressed and warms, so clouds evaporate and little rain falls. The result is a dry region downwind (the rain shadow).

(d) City N, with only 420 mm a year, would have drought-tolerant vegetation such as grassland, sagebrush or desert scrub, while wet City M would support lush forest. A coastal city has a smaller annual temperature range because water has a high specific heat: the ocean heats and cools slowly and moderates the air temperature, whereas inland air over land heats and cools more.

SG-U4-E010

Answer: (a) Year 1 El Niño, Year 2 neutral, Year 3 La Niña, Year 4 neutral. (b) −56%. (c) Upwelling collapses in El Niño. (d) Drier in Year 1, wetter in Year 3.

(a) Using the rule: Year 1 (+2.1°C ≥ +0.5) is **El Niño**; Year 2 (−0.3°C, between −0.5 and +0.5) is **neutral**; Year 3 (−1.6°C ≤ −0.5) is **La Niña**; Year 4 (+0.2°C) is **neutral**.

(b) Percent change = (new − old) ÷ old × 100 = (4 − 9) ÷ 9 × 100 = −55.6%, so the catch **fell by about 56%** from Year 2 to Year 1.

(c) In the El Niño year (Year 1) the trade winds weaken, so warm surface water spreads east toward South America and the thermocline deepens. Upwelling of cold, nutrient-rich water off Peru weakens or stops, so nutrients are not brought to the surface and phytoplankton decline. Anchovies, which feed on plankton, decline, and the catch is lowest (4 million tonnes).

(d) In Year 1 (El Niño) the rain shifts east, so Indonesia has **drought** (less rain, with greater wildfire risk). In Year 3 (La Niña) the strengthened trade winds pile warm water in the west and rain concentrates there, so Indonesia has **heavier than normal rain and flooding**.

SG-U4-E011

Answer: (a) 70 km per million years = 7.0 cm per year. (b) P gives the same. (c) 15 million years. (d) The plate moves away from the hot spot, along the chain toward the oldest island.

(a) Speed = distance ÷ age = 630 km ÷ 9.0 million years = **70 km per million years**. Since 1 km per million years = 0.1 cm per year, 70 × 0.1 = **7.0 cm per year**.

(b) Island P: 140 km ÷ 2.0 million years = 70 km per million years, the same speed (Island Q also gives 350 ÷ 5.0 = 70), so the data are consistent with a steady plate speed.

(c) Age = distance ÷ speed = 1,050 km ÷ 70 km per million years = **15 million years**.

(d) The older islands lie farther along the chain from the active volcano, so the plate is moving away from the hot spot in the direction from the active volcano toward the oldest island (the direction of the chain). The volcanoes are not at a plate boundary because a hot spot is a plume of hot rock rising from deep in the mantle beneath the interior of a plate. The plate drifts over the stationary plume, so each volcano forms over it and is then carried away, leaving a line of progressively older islands.

SG-U4-E012

Answer: (a) 11 t/ha/yr. (b) 8,250 t. (c) 0.96 mm per year. (d) Not sustainable.

(a) Net loss = erosion − formation = 12 − 1.0 = **11 t/ha/yr**.

(b) Total = net rate × area × time = 11 t/ha/yr × 25 ha × 30 yr = **8,250 t** (about 8.3×10^3 t).

(c) Depth lost = mass lost ÷ (bulk density × area). Per hectare per year: 12 t ÷ (1.25 t/m³ × 10^4 m²) = 12 ÷ 12,500 = 9.6×10^{-4} m = **0.96 mm per year**.

(d) **Not sustainable.** Soil is lost at 0.96 mm per year but forms at only about 0.08 mm per year, so loss is about 12 times faster than formation, and the topsoil is steadily depleted. (The formation rate agrees with 1.0 t/ha/yr: 1.0 ÷ 12,500 = 0.08 mm.) One practice that would move erosion toward the formation rate: no-till farming, cover crops, contour plowing, terracing, strip cropping or a vegetated buffer strip, each of which keeps the surface covered, slows runoff or wind and binds the soil with roots.

SG-U4-E014

Answer: (a) A 40 mm/h; B 5 mm/h. (b) A 0 mm; B 21 mm. (c) Field B.

(a) Infiltration rate = depth ÷ time. Field A: 30 mm ÷ (45 min = 0.75 h) = **40 mm/h**. Field B: 30 mm ÷ 6 h = **5 mm/h**.

(b) Runoff rate = rainfall rate − infiltration rate (only if positive). Field A: rain 12 mm/h is less than 40 mm/h, so all rain soaks in and runoff is **0 mm**. Field B: 12 − 5 = 7 mm/h, and over 3 hours 7 mm/h × 3 h = **21 mm** of runoff.

(c) **Field B** is more likely compacted clay because its infiltration rate is very low. The runoff from it carries sediment, fertilizer and other pollutants to the nearby stream, making the water cloudy (turbid), harming aquatic life, and increasing peak flow and the risk of flooding and bank erosion.

SG-U4-E015

Answer: (a) 3.78×10^7 m³. (b) 1.2 m³/s. (c) about 3,000 t (3,024 t). (d) about 0.19 mm per year.

(a) Runoff volume = depth × area × C. Depth = 1,200 mm = 1.2 m; area = 90 km² = 9.0×10^7 m². Volume = 1.2 m × 9.0×10^7 m² × 0.35 = **3.78×10^7 m³** per year.

(b) Discharge = volume ÷ time = 3.78×10^7 m³ ÷ 3.15×10^7 s = **1.2 m³/s**.

(c) Concentration: 80 mg/L × 1,000 L/m³ = 80,000 mg/m³. Mass = 80,000 mg/m³ × 3.78×10^7 m³ = 3.02×10^{12} mg. Divide by 10^9 mg/t: **≈ 3,024 t**, or about 3.0×10^3 metric tons of sediment per year.

(d) Cropland area = 0.15 × 90 km² = 13.5 km² = 1.35×10^7 m². Volume of soil = mass ÷ bulk density = 3,024 t ÷ 1.2 t/m³ = 2,520 m³. Depth = volume ÷ area = 2,520 m³ ÷ 1.35×10^7 m² = 1.87×10^{-4} m = **≈ 0.19 mm per year**.

SG-U4-E016

Answer: (a) 61.5° in June, 14.5° in December. (b) About 3.5 times. (c) 117 W/m². (d) Smaller changes in sun angle and day length near the equator.

(a) Noon sun angle = 90° − |latitude − latitude of the direct sun|. June: 90° − |52° − 23.5°| = 90° − 28.5° = **61.5°**. December: 90° − |52° − (−23.5°)| = 90° − 75.5° = **14.5°**.

(b) Intensity is proportional to the sine of the noon angle. sin 61.5° ≈ 0.879 and sin 14.5° ≈ 0.250. Ratio = 0.879 ÷ 0.250 ≈ **3.5**, so June noon sunlight is about 3.5 times as intense as December's.

(c) Energy absorbed = incoming × (1 − albedo). Snow: 180 × (1 − 0.80) = 36 W/m². Bare soil: 180 × (1 − 0.15) = 153 W/m². Increase = 153 − 36 = **117 W/m²** (check: 180 × (0.80 − 0.15) = 117 W/m²).

(d) A city at 5° N is near the equator, so its noon sun angle stays high all year (about 71.5° in June and 61.5° in December), and its day length stays near 12 hours. Intensity and hours of daylight change little, so temperatures

change little. At 52° N the noon sun angle drops from 61.5° to 14.5° and daylight changes greatly between summer and winter, causing large seasonal temperature swings.

Unit 5 — Land and Water Use

SG-U5-E001

Answer: (a) 112 cattle, 40% above the sustainable number. (b) Each rancher keeps the full gain and shares the loss. (c) A grazing quota, for example 10 cattle per rancher.

(a) Total cattle = 8 ranchers × 14 cattle = 112 cattle. Excess = 112 − 80 = 32 cattle. Percent over = 32 ÷ 80 × 100% = **40%**.

(b) A meadow open to all is a commons. An extra animal gives its owner the whole sale value, but the cost of lost grass is divided among all eight ranchers, so each rancher bears only about 1/8 of it. Adding an animal is therefore rational for each rancher even though everyone knows the meadow is overgrazed. If all eight add animals, the grass cover fails and every rancher loses. A common wrong answer blames greed or ignorance; the cause is self-interest combined with no enforceable limit.

(c) Rule: a permit system limiting the meadow to 80 cattle in total, for example 10 per rancher (8 × 10 = 80), enforced by inspection and fines. Drawbacks (any one): it costs money to monitor and enforce; an equal share may be unfair to ranchers with larger families or businesses; or it reduces each rancher's income in the short term. Privatizing the meadow in equal plots would be another acceptable rule, with the drawback that it could favor wealthier buyers.

SG-U5-E002

Answer: Cutting the trees removes shade, root-held soil and canopy interception, which raises stream temperature, sediment and peak flow. The clearcut also releases carbon as CO_2, and it is cheap and fast.

(a) *Temperature:* with no canopy, sunlight strikes the stream and the ground directly, so water warms (14 °C to 21 °C). Warmer water holds less dissolved oxygen, which stresses cold-water fish. *Sediment:* roots no longer hold the soil and rain hits bare ground, so erosion increases and eroded soil washes into the stream. *Peak flow:* with no canopy to intercept rain and no roots drawing water out of the soil, less water infiltrates or is taken up. More water runs off quickly over the surface, so the storm peak is higher and flooding downstream is more likely.

(b) The stored carbon in the cut trees is released as CO_2 when the wood is burned or decays, which adds to the greenhouse effect. The loss of the trees also removes a carbon sink, so less CO_2 is absorbed in future years. Either effect earns credit.

(c) Clearcutting is the cheapest and fastest harvest method. Machines can reach every tree, roads are simple, and the cleared land can be replanted with one fast-growing species. Freeing the land for crops, pasture or buildings is another acceptable advantage.

SG-U5-E003

Answer: (a) 100% increase; 21,600 t more grain. (b) 960,000 kg more nitrogen fertilizer. (c) Fertilizer: nutrient runoff and eutrophication; diesel: CO_2 emissions.

(a) Percent increase = (new − old) ÷ old × 100% = (3.6 − 1.8) ÷ 1.8 × 100% = **100%**. Extra yield per hectare = 3.6 − 1.8 = 1.8 t/ha. Extra grain = 1.8 t/ha × 12,000 ha = **21,600 t**.

(b) Extra nitrogen per hectare = 90 − 10 = 80 kg/ha. Extra for the district = 80 kg/ha × 12,000 ha = **960,000 kg** (960 t) per year.

(c) *Fertilizer:* plants take up only part of the nitrogen applied, so the rest runs off or leaches into streams and groundwater. The nutrients cause algae blooms. When the algae die, bacteria decompose them and use up dissolved oxygen, creating a low-oxygen dead zone (eutrophication). Nitrate in drinking water is also a risk for infants. *Diesel:* the extra diesel is 55 − 20 = 35 L/ha, or 35 × 12,000 = 420,000 L per year for the district. Burning it releases CO_2 (a greenhouse gas) and air pollutants, which adds to climate change and makes the farms depend on fossil-fuel prices.

SG-U5-E004

Answer: (a) 342,000 m³. (b) 240,000 m³. (c) 102,000 m³ saved per year; a farmer might keep furrow because it is cheap.

Water needed at the roots for the farm = 5,700 m³/ha × 40 ha = 228,000 m³.

(a) Furrow loses 1/3, so efficiency = 1 − 1/3 = 2/3. Water pumped = 228,000 ÷ (2/3) = **342,000 m³**.

(b) Drip loses 5%, so efficiency = 0.95. Water pumped = 228,000 ÷ 0.95 = **240,000 m³**.

(c) Savings = 342,000 − 240,000 = **102,000 m³ per year**. A farmer might keep furrow irrigation because it is inexpensive to build and run. Drip needs costly hoses, filters and maintenance, and it can clog. A small farm or a farm with low water prices may not recover the cost of switching. A common error is to multiply 228,000 by 1.33 and by 1.05, which gives a different answer; losses must be divided as shown.

SG-U5-E005

Answer: (a) 4%, 8% and 16%. (b) Humans apply the pesticide that decides which moths survive. (c) Use IPM or refuge areas.

(a) Generation 1 = 2%. Doubling each generation: generation 2 = 2 × 2% = **4%**; generation 3 = **8%**; generation 4 = **16%**. Resistance rose eightfold in three generations.

(b) Individual moths vary genetically, and some already carry a resistance allele. The insecticide, applied by the grower, kills most susceptible moths while the resistant ones survive and reproduce, so the resistant share rises each generation. Selection is driven by human action, hence *artificial* selection. The moths did not "learn" or change; the existing variation was sorted.

(c) Proposal (any one): use integrated pest management, spraying only when the moth count passes an economic threshold and using biocontrol such as parasitic wasps; leave an untreated refuge so susceptible moths keep breeding with resistant ones; or rotate insecticides with different modes of action. Drawbacks: IPM needs training, careful field scouting and sometimes more labor, and its results are slower than a quick spray. A refuge sacrifices some crop to pest damage.

SG-U5-E006

Answer: (a) 90,000 kcal of beef. (b) 450 person-days as grain, 45 as beef. (c) Methane and manure/nutrient pollution (also water use and antibiotics).

(a) Beef energy = grain energy × transfer = 900,000 kcal × 0.10 = **90,000 kcal**.

(b) Grain eaten directly: 900,000 kcal ÷ 2,000 kcal/day = **450 person-days**. As beef: 90,000 ÷ 2,000 = **45 person-days**. The same hectare feeds ten times as many people as grain, because 90% of the energy is lost as heat and body maintenance at the cattle step.

(c) Any two of the following. *Methane:* cattle are ruminants, and microbes in their guts produce CH_4, a greenhouse gas. *Water pollution:* concentrated manure and fertilizer for feed grain add nutrients and pathogens to waterways, causing eutrophication. *Water use:* growing feed and watering animals uses far more fresh water per gram of protein than plant foods. *Antibiotics:* routine use in feedlots can select for resistant bacteria.

SG-U5-E007

Answer: (a) 100 thousand t per year at a stock of 500 thousand t. (b) −26 thousand t. (c) The stock keeps falling and collapses.

(a) Maximum natural growth is the largest value in the data: **100 thousand tonnes per year at a stock of 500 thousand tonnes**. This is the maximum sustainable yield.

(b) Growth at 800 = 64; harvest = 90. Change = growth − harvest = 64 − 90 = **−26 thousand tonnes**. The stock falls from 800 to about 774 thousand tonnes after one year. Growth is below the harvest, so the stock declines even though the harvest is below the 100 maximum.

(c) At 200 the growth is only 64, so growth − harvest = 64 − 90 = −26 each year. The stock gets smaller, growth gets smaller still (the curve falls as the stock drops), and the gap widens, so the stock collapses. A harvest of 90 can be sustained only if the stock stays above the unstable point somewhere between 400 and 500 (where growth just equals 90). The fleet's harvest level is the same whatever the stock, so nothing reduces fishing as the stock falls. This is a tragedy of the commons: each boat gains the full catch, the cost of fewer fish is shared by everyone, and without enforceable quotas the stock is fished down until everyone loses. A fix is a quota set below the maximum sustainable yield and lowered when the stock is low.

SG-U5-E008

Answer: (a) 125,000 t of ore at 0.8%; 250,000 t at 0.4%. (b) 124,000 t and 249,000 t of tailings. (c) Lower grade means more rock per tonne of metal; reduce demand by recycling copper.

(a) Ore = metal ÷ grade. At 0.8% (0.008): 1,000 ÷ 0.008 = **125,000 t**. At 0.4% (0.004): 1,000 ÷ 0.004 = **250,000 t**.

(b) Tailings = ore − metal. At 0.8%: 125,000 − 1,000 = **124,000 t**. At 0.4%: 250,000 − 1,000 = **249,000 t**.

(c) Halving the grade doubles the rock that must be mined, hauled, crushed and processed for the same 1,000 t of copper. That raises energy, water and chemical costs and doubles the overburden and tailings, so there is more habitat loss, dust and risk of acid mine drainage or leaks from tailings ponds. Reducing demand for newly mined copper: recycle copper wiring and pipes, reuse products, or substitute another material. A drawback of recycling is that collection and sorting cost money and not all copper is recovered.

SG-U5-E009

Answer: (a) 1,500 m³. (b) 6,000 m³, four times as much. (c) Higher, earlier flood peaks and more pollutants in the stream; use permeable pavement or green space.

Site area = 40 ha × 10,000 m²/ha = 400,000 m². Rain depth = 25 mm = 0.025 m.

(a) Runoff = 400,000 m² × 0.025 m × 0.15 = **1,500 m³**.

(b) Impervious area = 0.60 × 400,000 = 240,000 m²; pervious area = 160,000 m². Impervious runoff = 240,000 × 0.025 × 0.90 = 5,400 m³. Pervious runoff = 160,000 × 0.025 × 0.15 = 600 m³. Total = **6,000 m³**. Ratio = 6,000 ÷ 1,500 = **4 times** the original.

(c) *Consequence (any one):* more water arrives faster, so the stream has a higher, earlier peak, causing flash flooding and bank erosion downstream; it also carries oil, metals, road salt and trash from the pavement; and less water infiltrates to recharge groundwater. *Design change (any one):* permeable pavement, a retention pond or rain garden, a green roof, trees and green space, or leaving part of the site undeveloped.

SG-U5-E010

Answer: (a) 120 and 90 million gha. (b) Deficit 30 million gha, 33% of biocapacity. (c) Reduce consumption and population growth; import resources, with consequences elsewhere.

(a) Total footprint = 6.0 gha/person × 20,000,000 = **120 million gha**. Total biocapacity = 4.5 × 20,000,000 = **90 million gha**.

(b) Deficit = 120 − 90 = **30 million gha** (equal to 1.5 gha per person). Percent of biocapacity = 30 ÷ 90 × 100% = **33%** (33.3%). The footprint is 120 ÷ 90 = 1.33 times biocapacity, so it would need to be cut by 30 ÷ 120 = 25% to reach balance.

(c) *Two ways to lower the footprint (any two):* cut fossil-fuel use through efficiency, renewable energy and public transportation; eat less meat; reduce waste and recycle; slow population growth. *Meeting demand without lowering the footprint:* import food, timber and other resources from countries with an ecological reserve (or overuse its own ecosystems more). The consequence is that the deficit is shifted to other places, which may deplete those countries' resources or raise their prices, and the importing country's own resources are not any more sustainable.

SG-U5-E011

Answer: (a) 77.8% reduction; 750 t saved. (b) Plan 4 (3 t/ha, 3 sprayings). (c) Plan 2 is cheaper and simpler.

(a) Percent reduction = (18 − 4) ÷ 18 × 100% = 14 ÷ 18 × 100% = **77.8%** (about 78%). Plan 4 saves 18 − 3 = 15 t/ha. Over 50 ha: 15 × 50 = **750 t of soil per year**.

(b) **Plan 4** has the lowest soil loss (3 t/ha/yr) and ties Plan 3 for the fewest sprayings (3, compared with 12 in Plans 1 and 2). Its likely tools are terracing (shortens the slope and slows runoff) with IPM: monitoring pest numbers and spraying only above an economic threshold, plus biocontrol such as natural predators, crop rotation or intercropping.

(c) Plan 2 cuts soil loss by nearly 78% with no change to the spraying routine. A farmer might choose it because it is cheap and simple. It needs no training, scouting or terrace construction, which is expensive and labor-intensive. IPM is complex, needs knowledge and monitoring, and results can be slower than a weekly spray. A farmer with a small budget, rented land or a short time horizon might accept more pesticide to avoid those costs.

SG-U5-E012

Answer: (a) 2,730 t of carbon. (b) 10,010 t of CO_2. (c) Selective cutting.

(a) Stored carbon = 140 t/ha × 30 ha = 4,200 t. Carbon released = 4,200 × 0.65 = **2,730 t C**.

(b) Mass of CO_2 = carbon × 44 ÷ 12 = 2,730 × 44 ÷ 12 = 2,730 × 3.667 = **10,010 t CO_2** (about 1.0×10^4 t).

(c) Harvest selectively, removing only some mature trees and leaving the canopy and most of the stored carbon in place. Strip cutting, harvesting no more than regrowth, reforestation after harvest, and reusing or recycling wood to reduce new harvest are also acceptable. A drawback of selective cutting is higher cost per tonne and more skill required.

SG-U5-E013

Answer: (a) −0.9 m/yr; 40 years. (b) −0.46 m/yr; still unsustainable. (c) Switch to drip irrigation, which is expensive.

(a) Net change = recharge − withdrawal = 0.2 − 1.1 = **−0.9 m/yr**. Years until dry = 36 m ÷ 0.9 m/yr = **40 years**.

(b) Pumping after a 40% cut = 1.1 × 0.60 = 0.66 m/yr. New net = 0.2 − 0.66 = **−0.46 m/yr**. The water table is still falling because withdrawal (0.66) is greater than recharge (0.2), so use is **not sustainable**. The aquifer would last about 36 ÷ 0.46 ≈ 78 years, which is longer but still ends in depletion. Sustainable use needs pumping ≤ 0.2 m/yr, a cut of about 82%.

(c) Replace flood or furrow irrigation with drip or spray irrigation, which loses far less water to evaporation and runoff. Drip is costly to install and maintain and can clog. Other acceptable changes include planting less water-demanding crops (the drawback being lower income) and scheduling irrigation by soil moisture.

SG-U5-E014

Answer: (a) 144 cows. (b) 178% above. (c) 175,000 kg of feed; manure pollution.

(a) Usable forage = 600 ha × 1,800 kg/ha × 0.40 = 432,000 kg/yr. Cows supported = 432,000 ÷ 3,000 = **144 cows**.

(b) Percent over = (400 − 144) ÷ 144 × 100% = 256 ÷ 144 × 100% = **178%** (177.8%). The ranch is overgrazed, so plant cover declines, soil is exposed and erodes, and the land may begin to desertify.

(c) Feed = FCR × mass gained = 7 × 25,000 kg = **175,000 kg of feed**. Environmental problem (any one): concentrated manure can contaminate water with nutrients and pathogens and releases methane and ammonia; routine antibiotics can promote resistant bacteria; the grain and soy grown for feed use land, water and fertilizer.

SG-U5-E015

Answer: (a) 2.0, 1.4 and 0.7 t per boat-day. (b) Catch rose then fell; CPUE fell every period; the stock is declining. (c) Total catch hid the decline; set a catch limit.

(a) CPUE = catch ÷ effort. 2005: 60,000 ÷ 30,000 = **2.0 t/boat-day**. 2015: 70,000 ÷ 50,000 = **1.4 t/boat-day**. 2025: 56,000 ÷ 80,000 = **0.7 t/boat-day**.

(b) Total catch rose from 60,000 to 70,000 t (2005 to 2015), then fell to 56,000 t (2025), while effort rose from 30,000 to 80,000 boat-days. CPUE fell continuously, to 35% of its 2005 value. A falling CPUE with rising effort means each boat finds fewer fish, so the stock is shrinking; the 2025 fall in total catch confirms it.

(c) In 2015 total catch was at its highest, which looks like a healthy fishery, but it was reached only by adding 20,000 boat-days; CPUE had already dropped 30%. A manager looking at catch alone would see no problem and allow more fishing. Management action (any one): catch quotas based on stock surveys; closed seasons or marine protected areas; limiting the number of boats; or larger mesh sizes. A drawback is lost income for fishers and the need for enforcement.

SG-U5-E016

Answer: (a) 20,000 m^3 rock; 0.006 m^3 gold. (b) 114 kg. (c) About $3.51 per gram.

(a) Rock volume = 50 × 40 × 10 = **20,000 m^3**. Gold volume = 20,000 × 3×10^{-7} = **0.006 m^3**.

(b) Convert: 0.006 m^3 × 10^6 cm^3/m^3 = 6,000 cm^3. Mass = 6,000 cm^3 × 19 g/cm^3 = 114,000 g = **114 kg**.

(c) Break-even price = cost ÷ mass = $400,000 ÷ 114,000 g = **$3.51 per gram** (3.5088). The cost of removing and processing the block is about the same however much gold it holds, so a lower grade gives less gold from

the same block (and the same cost) and the cost per gram rises. At half the grade the break-even price would double to about $7.02 per gram.

SG-U5-E017

Answer: (a) 2,190 m³. (b) 1,860 m³, a 15% reduction. (c) Permeable pavement can clog.

Depth = 3.0 cm = 0.030 m. Areas: roofs 30,000 m², asphalt 40,000 m², lawn 50,000 m².

(a) Roofs: 30,000 × 0.030 × 0.90 = 810 m³. Asphalt: 40,000 × 0.030 × 0.90 = 1,080 m³. Lawn: 50,000 × 0.030 × 0.20 = 300 m³. Total = **2,190 m³**.

(b) Half the asphalt (20,000 m²) stays at 0.90: 20,000 × 0.030 × 0.90 = 540 m³. Permeable section: 20,000 × 0.030 × 0.35 = 210 m³. New total = 810 + 540 + 210 + 300 = **1,860 m³**. Percent reduction = (2,190 − 1,860) ÷ 2,190 × 100% = 330 ÷ 2,190 × 100% = **15%** (15.1%).

(c) Permeable pavement can clog with sediment and lose its ability to drain unless it is cleaned. It also costs more than ordinary asphalt, and it may not be suitable for heavy truck traffic.

SG-U5-E018

Answer: a model experiment compares replicated mulched and bare slopes under the same conditions and measures the sediment in the runoff.

(a) *Hypothesis:* If bare construction slopes are covered with a layer of wood-chip mulch, then less sediment will wash into runoff than from bare slopes with no mulch.

(b) *Independent variable:* presence (or thickness) of wood-chip mulch on the slope. *Dependent variable:* mass of sediment in the runoff, in grams (or turbidity in NTU, or sediment per liter of runoff, measured after each storm). *Control group:* slopes with bare soil and no mulch. *Controlled variables (any two):* slope angle, soil type, plot area, rainfall amount or simulated rain, time of measurement, how the runoff is collected, and the starting soil condition.

(c) Use several replicate plots per treatment, for example five mulched and five bare plots of the same size, so that chance differences do not decide the result. Assign each treatment to plots at random so that a sunnier or steeper corner does not line up with one treatment. After each storm, collect all runoff from each plot in a bucket, then dry and weigh the sediment, and compare the treatment mean with the control mean. The hypothesis is supported if the mean sediment from the mulched plots is lower than the control mean by more than the scatter (range or error bars) among the replicates. Weak designs change two variables at once, use a single plot, or compare different sites or years.

Unit 6 — Energy Resources and Consumption

SG-U6-E001

Answer: (a) Uranium-235, natural gas and coal are nonrenewable; wind and sunlight are renewable. (b) Natural gas is nonrenewable because it exists in a fixed amount that formed over millions of years and is used far faster than it is replaced. (c) Uranium-235.

(a) Nonrenewable column: uranium-235, natural gas, coal. Renewable column: wind, sunlight.

(b) The test is whether a source is replenished at or near the rate we use it. Natural gas formed from buried organisms over millions of years, so on a human timescale it is not replaced at all. Burning more cleanly than coal describes its pollution, not its supply. A common wrong answer mixes up "clean" with "renewable."

(c) Uranium-235 (nuclear fission) releases no air pollutants or CO_2 during operation. It is still nonrenewable because the uranium ore is finite, and it still produces thermal pollution and hazardous solid waste.

SG-U6-E002

Answer: (a) Country A: 300 GJ per person; Country B: 30 GJ per person. (b) 10 times. (c) 27 EJ. (d) Industrialization and rising incomes will increase demand.

(a) Energy per capita = total energy ÷ population. Country A: 90 EJ = 90×10^{18} J; population 300 million = 3.0×10^{8}. 9.0×10^{19} J ÷ 3.0×10^{8} = 3.0×10^{11} J per person = **300 GJ per person**. Country B: 45×10^{18} J ÷ 1.5×10^{9} = 3.0×10^{10} J per person = **30 GJ per person**.

(b) 300 GJ ÷ 30 GJ = **10 times** greater.

(c) Coal energy = 0.60 × 45 EJ = **27 EJ**.

(d) Country B's per-person use is low, so as it industrializes (factories, steel, cement, transport) and incomes rise (more vehicles, appliances and electricity), demand per person will grow. With 1,500 million people, even a modest rise in per-person use raises total demand a great deal. Population growth would add to this. Any one well-explained reason earns credit.

SG-U6-E003

Answer: (a) 75%. (b) 50 MJ wasted. (c) Because separate production of the heat would need extra fuel, so the same outputs would take more than 200 MJ of fuel.

(a) Efficiency = (electrical output + useful heat) ÷ fuel input × 100 = (60 MJ + 90 MJ) ÷ 200 MJ × 100 = 150 ÷ 200 × 100 = **75%**.

(b) Energy is conserved: input = useful output + waste. Waste = 200 MJ − 150 MJ = **50 MJ**.

(c) Alone, the plant turns 60 of 200 MJ into electricity (30% efficiency), so the 60 MJ of electricity already uses all 200 MJ of fuel and the other 140 MJ is lost as heat. Supplying the 90 MJ of heating would then need a separate boiler burning additional fuel (at least 90 MJ even if the boiler were perfectly efficient, and more in practice). So the total fuel is greater than 200 MJ, with more pollution and CO_2. Cogeneration puts the plant's waste heat to use. (Reading used: the comparison plant keeps the 30% electrical efficiency and heat comes from a separate boiler.)

SG-U6-E004

Answer: (a) 60 years. (b) New production 5.0 bcm per year; R/P = 48 years. (c) Natural gas forms from buried marine organisms in sedimentary basins, so a country without them lacks the source rock.

(a) R/P = proven reserves ÷ annual production = 240 bcm ÷ 4.0 bcm/yr = **60 years**.

(b) New production = 4.0 × 1.25 = **5.0 bcm per year**. New R/P = 240 ÷ 5.0 = **48 years**. Faster use shortens the lifetime of a fixed resource (assuming no new reserves are found).

(c) Natural gas (like oil) formed from marine microorganisms that were buried in sediments in ancient sedimentary basins, then heated, and trapped under impermeable rock. A country with no such basins never had the right environment, burial or rock traps, so it has no deposit however much it needs one. It must import gas instead.

SG-U6-E005

Answer: (a) 44 kg CO_2. (b) 200 MWh of electricity and 300 MWh of waste heat. (c) Chemical energy of the fuel → thermal energy (heat) → kinetic energy of steam/turbine → electrical energy.

(a) Molar masses: CH_4 = 12 + 4(1) = 16 g/mol; CO_2 = 12 + 2(16) = 44 g/mol. 16 kg CH_4 = 16,000 g ÷ 16 g/mol = 1,000 mol. The equation is 1 : 1, so 1,000 mol CO_2 forms. Mass CO_2 = 1,000 mol × 44 g/mol = 44,000 g = **44 kg**. (Check: 44 ÷ 16 = 2.75 kg CO_2 per kg CH_4.)

(b) Electrical output = efficiency × input = 0.40 × 500 MWh = **200 MWh**. Waste heat = 500 − 200 = **300 MWh**.

(c) Burning the fuel releases chemical energy as heat; the heat boils water into high-pressure steam (thermal energy); the steam spins the turbine (kinetic energy); the turbine turns the generator, which produces electrical energy. Waste heat leaves at the condenser.

SG-U6-E006

Answer: (a) 40 units. (b) 48 days. (c) The isotope with the 24,000-year half-life stays hazardous for hundreds of thousands of years.

(a) n = 32 days ÷ 8 days = 4 half-lives. Remaining = 640 × (½)⁴ = 640 ÷ 16 = **40 units**.

(b) 640 → 320 → 160 → 80 → 40 → 20 → 10 is 6 halvings (640 ÷ 2^6 = 10). Time = 6 × 8 days = **48 days**.

(c) After about 10 half-lives of the first isotope (80 days) its activity is about 0.1% of the start, so it is essentially gone within a few months. The 24,000-year isotope loses only half of its activity every 24,000 years, so about 10 half-lives would be 240,000 years. The longest-lived isotope controls how long the waste must be isolated, which is why deep geologic repositories must work for such long periods.

SG-U6-E007

Answer: (a) EROEI = 2.0; net energy = 12 MJ per liter. (b) The crude-oil-like source (18 versus 2.0). (c) Habitat loss and loss of a carbon sink (more CO_2), plus soil erosion and fertilizer runoff.

(a) EROEI = energy delivered ÷ energy invested = 24 MJ ÷ 12 MJ = **2.0**. Net energy = 24 − 12 = **12 MJ per liter**.

(b) The oil source has an EROEI of 18 (18 MJ ÷ 1.0 MJ), far above the biofuel's 2.0, so it returns more energy per unit invested.

(c) Any two: replacing forest removes habitat and lowers biodiversity; it removes trees that act as a carbon sink and stores carbon that is released as CO_2 when cleared; bare farmland has more soil erosion; fertilizer and pesticide runoff pollutes streams and causes eutrophication; irrigation and land compete with food crops.

SG-U6-E008

Answer: (a) 19.0%. (b) $1,500 per year; payback 8.0 years. (c) 36,000 MJ. (d) Passive solar uses building design to absorb heat with no mechanical equipment; active solar uses pumps, fans and collectors to gather and store heat.

(a) Possible output = 6.0 kW × 8,760 h = 52,560 kWh. Capacity factor = 10,000 ÷ 52,560 × 100 = **19.0%**.

(b) Annual savings = 10,000 kWh × $0.15/kWh = **$1,500**. Payback = $12,000 ÷ $1,500 per year = **8.0 years**.

(c) 10,000 kWh × 3.6 MJ/kWh = **36,000 MJ** (3.6 × 10^4 MJ).

(d) Passive solar heating absorbs the sun's heat directly through building design (such as south-facing windows and thermal mass) without mechanical or electrical equipment, while active solar heating uses equipment such as pumps, fans and collectors to heat and move a fluid and store the energy.

SG-U6-E009

Answer: (a) 4,730,400 MWh. (b) About 473,000 homes. (c) Blocked fish migration and flooded habitat; advantage: no air pollution during operation (or on-demand power).

(a) Energy = capacity × hours × capacity factor = 1,200 MW × 8,760 h × 0.45 = **4,730,400 MWh** (≈ 4.7 × 10^6 MWh).

(b) 4,730,400 MWh = 4.7304 × 10^9 kWh. Homes = 4.7304 × 10^9 kWh ÷ 10,000 kWh per home = **473,040 homes** (≈ 4.7 × 10^5).

(c) Any two effects: the reservoir floods forests, farmland and river habitat; the dam blocks migrating fish such as salmon; it traps sediment so downstream floodplains and deltas lose nutrient-rich silt; it changes river temperature and flow; decaying vegetation in the reservoir releases methane and CO_2. Advantages for

justification: it emits no air pollutants during operation, is renewable, can supply power on demand, and provides flood control or irrigation. Name one.

SG-U6-E010

Answer: (a) 36% (20 ÷ 55 = 36.4%). (b) 49.5 kg CO_2. (c) The CO_2 is emitted upstream at the coal plant.

(a) Efficiency = electricity out ÷ electricity in × 100 = 20 kWh ÷ 55 kWh × 100 = **36.4%** (≈ 36%). About 64% of the electricity is lost as heat in the two steps.

(b) CO_2 = 55 kWh × 0.9 kg CO_2/kWh = **49.5 kg CO_2** per 1.0 kg of hydrogen.

(c) The fuel cell itself emits only water, but the hydrogen carries a large upstream carbon footprint: making it burned coal and released about 49.5 kg of CO_2. Hydrogen is an energy carrier, not an energy source, so its emissions depend on how the energy that made it was produced. It is zero-emission only if the electricity is clean (such as wind or solar).

SG-U6-E011

Answer: (a) 306,600 MWh. (b) About 276,000 tonnes CO_2 per year. (c) Birds and bats can be killed by the blades; siting away from migration routes (or slowing blades at certain times) reduces this.

(a) Capacity = 40 × 2.5 MW = 100 MW. Energy = 100 MW × 8,760 h × 0.35 = **306,600 MWh per year**.

(b) CO_2 avoided = 306,600 MWh × 0.9 t/MWh = 275,940 ≈ **276,000 tonnes CO_2 per year** (2.8×10^5).

(c) Drawback: birds and bats are killed when they fly into the spinning blades (other acceptable drawbacks: intermittency, noise, visual impact, land or seabed disturbance). Action: place turbines away from migration routes and flyways, or increase the cut-in speed (slow or stop the blades) at times of high bat activity. For intermittency, add storage or pair with other sources.

SG-U6-E012

Answer: (a) 480 gal (current) and 300 gal (hybrid). (b) $630 per year. (c) About 6.7 years. (d) For example, a change in gasoline price.

(a) Fuel used = distance ÷ fuel economy. Current: 12,000 mi ÷ 25 mi/gal = **480 gal**. Hybrid: 12,000 mi ÷ 40 mi/gal = **300 gal**.

(b) Gallons saved = 180 gal. Savings = 180 gal × $3.50/gal = **$630 per year**.

(c) Payback = extra cost ÷ annual savings = $4,200 ÷ $630 per year = **6.7 years**.

(d) Any one factor with its direction: if gasoline prices rise, savings grow and payback is shorter; if prices fall, it is longer. Rebound driving (driving more because it is cheaper) lengthens payback; a battery replacement or higher maintenance lengthens it; tax credits or rebates, or higher resale value, shorten it. Payback also ignores the environmental benefit of lower CO_2.

SG-U6-E013

Answer: (a) 82.5%. (b) 20 years. (c) The plant is overloaded after about 5.6 years because it is already at 82.5% and demand only needs to grow by about 21%.

(a) Demand = people × kW per person. City: 400,000 × 3.0 kW = 1,200,000 kW = 1,200 MW. Town: 90,000 × 5.0 kW = 450,000 kW = 450 MW. Total = 1,650 MW. Percent of capacity used = 1,650 ÷ 2,000 × 100 = **82.5%**.

(b) Doubling time = 70 ÷ 3.5 = **20 years**.

(c) Doubling is not needed to overload the plant. Capacity is 2,000 MW, and demand only has to rise by a factor of 2,000 ÷ 1,650 = 1.21 (a 21% increase). At 3.5% growth per year, $1.035^5 = 1.19$ and $1.035^6 = 1.23$, so demand passes capacity in the sixth year, about 5.6 years, far sooner than 20 years. Planners need to add capacity or conserve well before the doubling time.

SG-U6-E014

Answer: (a) 425 MW. (b) About 94 metric tons per hour. (c) About 880,000 metric tons CO_2 avoided.

(a) Power in = power out ÷ efficiency = 200 MW ÷ 0.32 = 625 MW. Waste heat = 625 − 200 = **425 MW**.

(b) Fuel energy per second = 625 MW = 625 MJ/s. Coal rate = 625 MJ/s ÷ 24 MJ/kg = 26.04 kg/s. Per hour = 26.04 × 3,600 = 93,750 kg = **≈ 94 metric tons per hour** (93.75 t).

(c) Electric energy = 200 MW × 8,000 h = 1.6×10^6 MWh = 1.6×10^9 kWh (assuming the gas plant delivers the same electricity). CO_2 avoided = energy × (old − new factor) = 1.6×10^9 kWh × (1.0 − 0.45) kg/kWh = 8.8×10^8 kg = **880,000 metric tons** (8.8×10^5 t).

SG-U6-E015

Answer: (a) 60 units. (b) 116 years. (c) 7 days.

(a) n = 87 ÷ 29 = 3 half-lives. Remaining = 480 × (½)³ = 480 ÷ 8 = **60 units**.

(b) 6.25% = 1/16 = (½)⁴, so 4 half-lives. Time = 4 × 29 = **116 years**.

(c) 0.80 → 0.40 → 0.20 → 0.10 mol is 3 halvings (0.80 ÷ 8 = 0.10). Half-life = elapsed time ÷ number of halvings = 21 days ÷ 3 = **7 days**.

SG-U6-E016

Answer: (a) 19.7% reduction. (b) 40 hectares and 200,000 m³ of irrigation water. (c) 200 MJ net gain.

(a) Baseline emissions = 200,000 L × 3.2 kg CO_2e/L = 640,000 kg. New fuel: biodiesel = 0.30 × 200,000 = 60,000 L; diesel = 140,000 L. Emissions = 140,000 × 3.2 + 60,000 × 1.1 = 448,000 + 66,000 = 514,000 kg. Percent reduction = (640,000 − 514,000) ÷ 640,000 × 100 = 126,000 ÷ 640,000 × 100 = **19.7%** (≈ 20%).

(b) Land = 60,000 L ÷ 1,500 L/ha = **40 ha**. Water = 40 ha × 5,000 m³/ha = **200,000 m³** per year.

(c) EROEI = 3.0 means 3.0 MJ delivered per 1.0 MJ invested. For 100 MJ invested, delivered = 300 MJ, so net gain = 300 − 100 = **200 MJ** (= (3.0 − 1) × 100).

SG-U6-E017

Answer: (a) About 266,000 kWh per year. (b) 66.6% of use; savings about $34,600 per year; about 120 metric tons CO_2 avoided. (c) Output is zero in the evening and at night, and low in summer when school is closed; use storage or net metering.

(a) Energy per day = area × sunlight × efficiency = 800 m² × 4.8 kWh/m²/day × 0.19 = 729.6 kWh/day. Annual = 729.6 × 365 = 266,304 kWh ≈ **2.66×10^5 kWh per year**.

(b) Percent supplied = 266,304 ÷ 400,000 × 100 = **66.6%**. Savings = 266,304 kWh × $0.13/kWh = **≈ $34,600 per year** ($34,620). CO_2 avoided = 266,304 kWh × 0.45 kg/kWh = 119,837 kg ≈ **120 metric tons** (this assumes all the PV output replaces grid electricity).

(c) Any one mismatch with a response: PV makes nothing at night or on cloudy days, while the school uses power for evening events and early-morning heating and lighting, so add batteries or stay grid-connected and buy power at night; or summer output is highest when the school is largely closed, so sell the surplus to the grid with net metering.

SG-U6-E018

Answer: (a) 1.2×10^8 m³. (b) City use 2.63×10^7 m³ per year; evaporation is about 457% of it (about 4.6 times as much).

(a) Area = 80 km² × 10^6 m²/km² = 8.0×10^7 m². Volume = 8.0×10^7 m² × 1.5 m = **1.2×10^8 m³**.

(b) Daily use = 600,000 people × 120 L = 7.2×10^7 L = 7.2×10^4 m³ (1 m³ = 1,000 L). Annual = $7.2 \times 10^4 \times 365 = 2.628 \times 10^7$ m³ ≈ **2.63×10^7 m³**. Percent = $1.2 \times 10^8 \div 2.628 \times 10^7 \times 100$ = **≈ 457%**. The reservoir loses about 4.6 times as much water to evaporation as the city uses, which is a hidden cost of reservoirs.

SG-U6-E019

Answer: (a) About 3.4 times. (b) Raise the cut-in speed to about 7 m/s (hold blades still below it) during bat activity. (c) The generator has a maximum power rating, and the blades are adjusted to shed extra wind.

(a) Power is proportional to the cube of wind speed: (9.0 ÷ 6.0)³ = 1.5³ = **3.375 ≈ 3.4 times** more power.

(b) Rule: keep the blades feathered (not generating) at wind speeds below about 7 m/s during the bat season or on migration nights, and let them run above 7 m/s. Justification: deaths per turbine-night are 0.9 at 3–5 m/s and 0.4 at 5–7 m/s but only 0.1 above 7 m/s, so most harm occurs at low speeds. Because wind power rises with the cube of speed, light winds hold little energy, so little electricity is lost.

(c) Rated power is the most the generator can deliver. Above rated speed the blades are pitched to spill wind, so output stays on a plateau. Beyond cut-out speed the turbine shuts down to protect the blades.

SG-U6-E020

Answer: model design below (any logical equivalent earns credit).

(a) Hypothesis: *If a wind farm is built, then the noise level (dB) at nearby homes will be higher than before it was built (and highest at the homes closest to the turbines).* Independent variable: the presence of the wind farm (before versus after), or distance from the turbines. Dependent variable: sound level measured in decibels (dB) at the homes.

(b) Design: Measure noise at several homes near the planned site before construction (baseline), at the same times of day and under similar weather and wind conditions. Choose a control location of similar land use and distance from roads that is far from the turbines and measure it on the same schedule. After the wind farm starts operating, measure again at the same homes and the control site using the same meter, height, time of day and season. Take replicate readings (for example, three at each location, on several days) and record wind speed as a controlled variable. Other acceptable variables: distance from the turbines, or turbine operating versus shut down.

(c) Compare the mean change after construction with the spread (range or standard deviation) of the baseline readings. If the average increase is larger than the natural variation in the baseline, and the control site shows no similar rise, the increase is attributed to the wind farm. Graph readings versus distance; noise typically falls with distance and levels off at the background value. Use averages from replicates, not single readings, and state the limits of the data.

Unit 7 — Atmospheric Pollution

SG-U7-E001

Answer: (1) primary, (2) secondary, (3) primary, (4) secondary, (5) primary; carbon monoxide is the asphyxiant.

(a) Carbon monoxide leaves the tailpipe as CO, so it is **primary**. Ground-level ozone forms in the air from NO_x and VOCs in sunlight, so it is **secondary**. Sulfur dioxide leaves the stack as SO_2, so it is **primary**. Nitric acid in rainwater forms when NO_x reacts with oxygen and water in the air, so it is **secondary**. Soot is emitted directly by the diesel engine, so it is **primary**.

(b) Ozone: nitrogen oxides (NO/NO_2) or VOCs (either is accepted; both are needed with sunlight). Nitric acid: nitrogen oxides (NO_x) from vehicles or power plants.

(c) The asphyxiant is **carbon monoxide (1)**: it binds to hemoglobin much more strongly than oxygen does, so the blood cannot carry enough oxygen to the tissues. A common wrong answer is soot, which is a particulate that damages the lungs but does not block oxygen transport.

SG-U7-E002

Answer: (a) NO_2 peaks at 8:00; ozone peaks at 14:00. (b) 6 hours. (c) Ozone is a secondary pollutant that needs time and sunlight to form. (d) Ozone would be lower and flatter.

(a) NO_2 is highest at 8:00 (40 ppb). Ozone is highest at 14:00 (95 ppb). The smaller NO_2 rise at 18:00 (30 ppb) is the evening rush and is lower than the morning peak.

(b) Time between peaks = 14:00 − 8:00 = **6 hours**.

(c) Morning traffic releases NO, which becomes NO_2, together with VOCs. Sunlight splits NO_2 into NO and an oxygen atom, and the oxygen atom joins O_2 to make O_3. VOCs convert NO back to NO_2 without using up ozone, so the cycle repeats and ozone accumulates. Because the reactions take hours and need rising sunlight and temperature, ozone builds through the morning and peaks in the afternoon, after NO_2 has already been consumed and has fallen.

(d) On a cool, cloudy day the ozone curve would be **lower and flatter**, with a smaller peak (and possibly less accumulation overall). The traffic and NO_x are the same, but less sunlight and lower temperature slow the reactions that make ozone, so fewer secondary pollutants form. NO_2 would stay high or fall more slowly because less of it is split by light.

SG-U7-E003

Answer: (a) Day Y, from 0 to 500 m. (b) +6 °C. (c) Day Y had 55 µg/m³; Day X had 15 µg/m³.

(a) On Day X the temperature falls steadily with altitude (14 → 12 → 10 → 8 °C), which is the normal pattern. On Day Y the temperature **rises** from 6 °C at 0 m to 9 °C at 250 m and 12 °C at 500 m, so the inversion layer extends from **0 m to 500 m**. Above 500 m it falls again (11 °C at 750 m), so normal mixing resumes above the warm layer.

(b) Change = 12 °C − 6 °C = **+6 °C** (a rise of 6 °C over 500 m).

(c) Day Y (inversion) had the higher value, **55 µg/m³**; Day X had **15 µg/m³**. On Day Y the cold, dense surface air is capped by warmer air, so it cannot rise and mix. Fine particles from vehicles, heating and industry are trapped in the lowest 500 m and build up. On Day X warm surface air rises, mixing and diluting the particles.

SG-U7-E004

Answer: (a) inversion days have much higher PM2.5; (b) 52 µg/m³ and 14 µg/m³; (c) 271%; (d) see below.

(a) Every inversion day (Days 3, 4, 6: 46, 58, 52 µg/m³) has PM2.5 far above every non-inversion day (Days 1, 2, 5: 14, 11, 17 µg/m³). The data are consistent with inversions trapping particulates near the ground, though six days is a small sample.

(b) Inversion days: (46 + 58 + 52) ÷ 3 = 156 ÷ 3 = **52 µg/m³**. Non-inversion days: (14 + 11 + 17) ÷ 3 = 42 ÷ 3 = **14 µg/m³**.

(c) Percent difference = (52 − 14) ÷ 14 × 100 = 38 ÷ 14 × 100 = 271.4% ≈ **271%**.

(d) Acceptable: restrict wood burning on forecast inversion days. Wood stoves and fireplaces are a large winter source of PM2.5, and during an inversion the smoke cannot disperse, so cutting the source lowers the trapped

concentration. Other acceptable answers: replace old wood stoves with cleaner-burning or electric heat, limit idling and driving on inversion days (with public transit), or add particulate controls to local industry. A good answer names the source, says how the action cuts PM2.5 and may add a drawback (cost, or reduced heating choices).

SG-U7-E005

Answer: (a) Homes 2 and 4 exceed 4.0 pCi/L; 40% of homes. (b) 4.6 pCi/L. (c) through foundation cracks and gaps (and in well water). (d) sub-slab pipe and fan.

(a) Compare each value with 4.0 pCi/L: Home 1 (2.1) no, Home 2 (5.6) **yes**, Home 3 (3.9) no (just below), Home 4 (8.4) **yes**, Home 5 (3.0) no. Percentage = 2 ÷ 5 × 100 = **40%**.

(b) Mean = (2.1 + 5.6 + 3.9 + 8.4 + 3.0) ÷ 5 = 23.0 ÷ 5 = **4.6 pCi/L**. The mean is above the action level even though only two homes are, because Home 4 is far higher than the rest.

(c) Uranium in rock and soil decays to radium and then to radon-222 gas. The gas moves up through soil and enters the house through cracks in the foundation and walls, gaps around pipes and floor drains. It can also be carried into the house dissolved in well water. In a closed house with poor ventilation it builds up.

(d) Install a **sub-slab depressurization system** (a vent pipe and fan under the foundation). It draws radon from the soil and vents it outdoors above the roof before it can enter the living space. Also acceptable: sealing foundation cracks (a supporting measure, less effective alone) or treating well water if water is the source. Retest afterward to confirm the level has fallen below 4.0 pCi/L.

SG-U7-E006

Answer: (a) 88,000 metric tons/yr; (b) 64.8% decrease; (c) 220 kg per person per year; (d) sludge; CO_2 (or NO_x).

(a) Plant A after scrubbing = 180,000 × (1 − 0.90) = 18,000 metric tons. Plant B = 70,000 metric tons. Total = 18,000 + 70,000 = **88,000 metric tons/yr**.

(b) Before: 180,000 + 70,000 = 250,000 metric tons/yr. Percent change = (88,000 − 250,000) ÷ 250,000 × 100 = −162,000 ÷ 250,000 × 100 = **−64.8%**, a decrease of about 65%.

(c) 88,000 metric tons × 1000 kg/metric ton = 8.8×10^7 kg. Per person = 8.8×10^7 kg ÷ 400,000 people = **220 kg per person per year**.

(d) By-product: **scrubber sludge** (solid calcium sulfite/sulfate) that needs disposal (wastewater is also acceptable). Pollutants a scrubber does not reduce: **CO_2** (also acceptable: NO_x, since the scrubber is designed for SO_2).

SG-U7-E007

Answer: (a) C > B > A; (b) 100 times; (c) SO_2 and NO_x; (d) dissolved CO_2 makes carbonic acid.

(a) A lower pH is more acidic, so from most to least acidic: **C (3.6), B (4.6), A (5.6)**.

(b) Difference = 5.6 − 3.6 = 2.0 pH units. Factor = 10^2 = **100 times** greater H^+ concentration in Sample C.

(c) **Sulfur dioxide (SO_2)**, human source: coal-burning power plants (also diesel fuel, smelters). **Nitrogen oxides (NO_x)**, human source: motor vehicle engines (also power plants). They form sulfuric and nitric acids.

(d) Natural rain absorbs CO_2 from the air, which forms weak carbonic acid, giving clean rain a pH of about 5.6, below the neutral value of 7.

SG-U7-E008

Answer: (a) 10 times; (b) limestone buffers acid; (c) Lake P; (d) scrubbers at the source (treats the cause), liming locally (treats symptoms).

(a) Difference = 6.0 − 5.0 = 1 pH unit. Factor = 10^1 = **10 times** more H^+.

(b) Limestone is calcium carbonate, which neutralizes acid. The bedrock and soil around Lake Q have high buffering capacity, so the incoming acid is neutralized before the pH can drop.

(c) **Lake P** is at greater risk. Granite bedrock and thin soil provide little buffering, so its pH fell to 5.0, a level at which many fish species decline and toxic aluminum is released. Lake Q stays near neutral and fish are not stressed by pH.

(d) At the source: install scrubbers on the power plants (or switch to low-sulfur fuel, natural gas or renewables) to cut SO_2 and NO_x emissions. This **treats the cause**. Locally: add crushed limestone to Lake P (liming) to raise its pH. This only treats the symptom, must be repeated, and does nothing to stop the acid from arriving.

SG-U7-E009

Answer: (a) 10,000 times; (b) 95 dB exceeds the ~85 dB damage level and exposure is long; (c) masking; (d) see below.

(a) Δ dB = 95 − 55 = 40 dB. Intensity ratio = 10^(40 ÷ 10) = 10^4 = **10,000 times** greater.

(b) Hearing damage depends on both level and time. Long exposure to about 85 dB or more causes permanent hearing loss, and this worker spends 8 hours a day at 95 dB, 10 dB (10 times the intensity) above that level, with no protection. The sensory hair cells in the inner ear are gradually destroyed and do not regrow.

(c) The frogs call at night to attract mates. Factory noise **masks** the calls, so females cannot hear or locate males, fewer pairs form, and breeding success falls. Noise can also raise stress hormones in the frogs.

(d) Source-level control: enclose or insulate noisy machines, install mufflers or quieter equipment, or add sound barriers (buffer zones for the wetland). Worker-level control: provide earplugs or earmuffs, or limit hours spent in the noisy area by rotating workers. Controlling at the source is preferred.

SG-U7-E010

Answer: (a) 65% decrease; (b) lead fell the most, atmospheric CO_2 rose; (c) other factors also changed.

(a) Percent change = (35 − 100) ÷ 100 × 100 = **−65%**, a 65% decrease in SO_2 emissions.

(b) Values compared with 100: SO_2 35 (down 65%), NO_x 52 (down 48%), lead 6 (down 94%), CO_2 118 (up 18%). **Lead** fell the most. **Atmospheric CO_2** is the one that rose; it is not a criteria pollutant targeted by the Clean Air Act limits.

(c) The graph shows a correlation between the time of regulation and falling SO_2, which supports effectiveness but does not prove cause. Other changes over the same 30 years, such as a switch from coal to natural gas or renewables, cleaner engines, low-sulfur fuel, economic changes or plant closures, could also have lowered emissions. To judge the law, you would need to show that the declines followed the regulation and rule out those other causes.

SG-U7-E011

Answer: model design below; a downwind rise alone does not prove the quarry is the cause.

(a) *If the quarry raises PM10, then PM10 will be higher at the monitor 1 km east (downwind) than at a monitor upwind of the quarry.*

(b) Independent variable: monitor location (downwind of the quarry versus upwind/background site). Dependent variable: PM10 concentration (µg/m^3). Controlled variables (any two): same type of instrument, same height above the ground, same hours and days of measurement, same averaging time, similar distance from roads.

(c) Place one monitor east (downwind) at about 1 km from the quarry and one **upwind** to the west of the quarry (and optionally more at several distances). Measure at the same times over many days, covering different weather and both working and non-working days of the quarry. The upwind site measures the background PM10 that is present anyway (dust, traffic, pollen). Quarry contribution ≈ downwind reading − upwind reading.

(d) A higher reading east of the quarry is only a correlation. Something else could differ between the sites: a road, a construction site, a farm field or bare soil near the east monitor, wind changing direction, or a different elevation. Without an upwind site, many days of replicated data and control of such confounding variables, the data are consistent with the quarry as the cause but do not prove it.

SG-U7-E012

Answer: (a) 4,200 kg/yr; (b) 3,360 kg/yr; (c) about $107 per kg removed.

(a) Miles = 400 buses × 30,000 miles/bus = 1.2 × 10^7 miles. Particulates = 1.2 × 10^7 miles × 0.35 g/mile = 4.2 × 10^6 g = 4.2 × 10^6 g × 1 kg/1000 g = **4,200 kg/yr**.

(b) Removed = 4,200 kg × 0.80 = **3,360 kg/yr**.

(c) Yearly cost = 400 buses × \$9,000 ÷ 10 years = \$3,600,000 ÷ 10 = \$360,000/yr. Cost per kg removed = \$360,000 ÷ 3,360 kg = \$107.14 ≈ **\$107 per kg**.

SG-U7-E013

Answer: (a) 4,000 metric tons S; (b) 8,000 metric tons SO_2; (c) 1,200 metric tons SO_2; (d) about \$2,200 per metric ton.

(a) Mass of S = 5.0×10^5 t × (0.8 ÷ 100) = **4,000 metric tons/yr**.

(b) Uncontrolled SO_2 = 4,000 t × (64 ÷ 32) = **8,000 metric tons/yr**.

(c) Removed = 8,000 × 0.85 = 6,800 t. Emitted = 8,000 − 6,800 = **1,200 metric tons/yr** (equivalently 8,000 × 0.15).

(d) Cost per ton removed = \$15,000,000 ÷ 6,800 t = \$2,205.9 ≈ **\$2,200 per metric ton of SO_2 removed** (2.2×10^3).

SG-U7-E014

Answer: (a) about 25 times; (b) pH 6, acidic; (c) about 126 times.

(a) Difference = 5.6 − 4.2 = 1.4 pH units. Factor = 10^1.4 = **≈ 25 times** greater H^+ in the forest rain.

(b) pH = $-\log_{10}(1 \times 10^{-6})$ = **6**. A pH below 7 is acidic, so the lake sample is (weakly) **acidic**, although it is less acidic than the 5.6 of clean rain.

(c) Difference = 6.3 − 4.2 = 2.1 pH units. Factor = 10^2.1 = **≈ 126 times** lower H^+ concentration in the lake than in the forest rain.

Unit 8 — Aquatic and Terrestrial Pollution

SG-U8-E001

Answer: (i) point; (ii) nonpoint; (iii) point; (iv) nonpoint.

(i) **Point source.** The exhaust leaves through one identifiable stack that can be measured.

(ii) **Nonpoint source.** The oil and tire residue is picked up from the whole surface of the city by rain and enters many storm drains, so there is no single discharge location.

(iii) **Point source.** All of the effluent leaves through one identifiable pipe.

(iv) **Nonpoint source.** Fertilizer runs off 200 separate fields at many places along the river, so the pollution is diffuse.

Regulation. A regulator can set a measurable discharge limit (for example, a maximum concentration or mass per day) only for the point sources, (i) and (iii), because the flow can be sampled and metered at one outlet and the owner can be held responsible. Nonpoint pollution has no single outlet to measure, so it is controlled by practices such as buffer strips, reduced fertilizer use and street sweeping rather than by a discharge limit.

SG-U8-E002

Answer: (a) lowest DO is 2.9 mg/L at km 30; (b) km 20, 30 and 40; (c) about 67%; (d) see below.

(a) Scanning the DO column, the minimum is **2.9 mg/L at km 30**.

(b) DO is below 5 mg/L at **km 20 (4.2), km 30 (2.9) and km 40 (3.6)**. Km 15 (6.5) and km 60 (6.1) are above 5.

(c) Percent decrease = (DO at km 10 − lowest DO) ÷ DO at km 10 × 100 = (8.8 − 2.9) ÷ 8.8 × 100 = 5.9 ÷ 8.8 × 100 = 67.0%, or **about 67%**.

(d) The effluent adds organic waste, which gives a high BOD. Aerobic bacteria decompose this waste and use dissolved oxygen as they do, so DO falls as long as decomposition removes oxygen faster than it enters from the air and from photosynthesis; this is true until the minimum at about km 30. After that the waste is largely used up, so BOD and the bacteria's oxygen demand decline. Re-aeration from the atmosphere and turbulence (and photosynthesis by plants and algae) then supply oxygen faster than it is consumed, so DO climbs from km 40 to km 80 and returns near its upstream value (8.0 against 8.8). A common wrong answer says the pollutant "disappears" or is "diluted" without mentioning oxygen consumption by decomposers.

SG-U8-E003

Answer: (a) Tank 1; (b) both trends are monotonic with phosphate (algae up, DO down); (c) see chain; (d) tank 1 oligotrophic-like, tank 4 eutrophic-like.

(a) The **control is tank 1**, which received 0 mg/L phosphate and shows the baseline conditions.

(b) As phosphate increases from 0 to 0.50 mg/L, algal density rises steadily from 2,000 to 210,000 cells/mL (about 105 times higher), while bottom DO falls steadily from 8.1 to 1.1 mg/L. The two variables move in opposite directions.

(c) Phosphate is a limiting nutrient, so extra phosphate lets algae reproduce rapidly (an algal bloom). The dense algae block light from reaching the deeper water and plants there. Algae and other organisms die, sink to the bottom, and are decomposed by aerobic bacteria. This decomposition consumes dissolved oxygen (high BOD), and because the bottom water is poorly mixed and receives little photosynthetic oxygen, bottom DO drops. Low DO stresses or kills fish and other aerobic organisms.

(d) Tank 1 (low algae, DO 8.1) is **oligotrophic-like**: nutrient-poor, clear and well oxygenated. Tank 4 (210,000 cells/mL, DO 1.1) is **eutrophic-like**: nutrient-rich with heavy algal growth and oxygen depletion.

SG-U8-E004

Answer: (a) about 34% decrease; (b) and (c) below.

(a) Percent decrease = (DO at 10°C − DO at 30°C) ÷ DO at 10°C × 100 = (11.3 − 7.5) mg/L ÷ 11.3 mg/L × 100 = 3.8 ÷ 11.3 × 100 = 33.6%, or **about 34%**.

(b) Any two of the following:

- Warm water speeds up metabolism, so fish need more oxygen at the same time that the water holds less.

- Warm water is above the temperature that cold-water species (such as trout) can tolerate, causing heat stress, reduced growth, poor reproduction or death.
- A sudden temperature change (thermal shock) can kill fish, for example when a plant shuts down in winter.
- Warm water favors tolerant species or invasive species and can promote algal growth, changing the community.

(c) Engineering solution (one): cool the water in cooling towers or cooling ponds before discharge, recycle the cooling water in a closed loop, or reuse the waste heat. Landscape solution (one): plant shade trees along the stream banks (riparian buffer), so the stream is shaded and the discharged water cools more quickly downstream.

SG-U8-E005

Answer: (a) 8.0, 8.1, 8.0; (b) 520; the others below.

(a) BMF = consumer concentration ÷ concentration in its food.

- Algae → minnows: 6.4 ÷ 0.8 = **8.0**
- Minnows → trout: 52 ÷ 6.4 = 8.125 → **8.1**
- Trout → heron: 416 ÷ 52 = **8.0**

(b) Overall factor = 416 ÷ 0.8 = **520** (the product of the step factors, 8.0 × 8.1 × 8.0 ≈ 520 within rounding).

(c) The pesticide is persistent (it is not broken down) and fat-soluble, so it is stored in the fatty tissue of each organism rather than excreted. Each consumer eats many individuals from the level below and keeps the pesticide from all of them, so the concentration increases by about eightfold at every trophic level. The top consumer, the heron, therefore holds the highest dose even if the water itself contains only a trace.

(d) Possible effects (one is enough): thinner eggshells that break, so fewer chicks hatch; reproductive failure; damage to the nervous or endocrine system; lower survival of adults, leading to a population decline.

SG-U8-E006

Answer: (a) 1.8 kg per person per day; (b) goal 1.44 kg per person per day, or 65,700 t per year; (c) below.

(a) 82,125 t × 1,000 kg/t = 82,125,000 kg per year. Per person per day = 82,125,000 kg ÷ (125,000 people × 365 days) = 82,125,000 ÷ 45,625,000 = **1.8 kg per person per day**.

(b) A 20% reduction leaves 80% of the present rate: 1.8 × 0.80 = **1.44 kg per person per day**. Annual mass = 1.44 kg × 125,000 people × 365 days = 65,700,000 kg = **65,700 t per year** (check: 82,125 t × 0.80 = 65,700 t).

(c) Plastics and metals: recycling (collection and reprocessing into new products). Food scraps: composting (or an anaerobic digester). Other acceptable answers include source reduction and reuse.

SG-U8-E007

Answer: (a) 40%; (b) 30 years now, 50 years under the plan; (c) below.

(a) Percent diversion = (25% + 15%) = **40%** of the 40,000 t, which is 16,000 t diverted per year.

(b) Current practice: 1,200,000 t ÷ 40,000 t/yr = **30 years**. Under the plan, buried waste = 40,000 × (1 − 0.40) = 24,000 t/yr, so 1,200,000 t ÷ 24,000 t/yr = **50 years**, an extension of 20 years.

(c) Benefit (one): recycling saves raw materials and energy, composting returns nutrients to soil, and less waste buried means less methane and leachate. Drawback (one): the program costs money for collection, sorting and processing; recycled or composted material may be contaminated or lack a market; or residents must change habits and may have additional collection costs.

SG-U8-E008

Answer: (a) 30%; (b) 87.5%; (c) about 91%; (d) below.

(a) Primary removal = (240 − 168) ÷ 240 × 100 = 72 ÷ 240 × 100 = **30%**.

(b) Secondary removal = (168 − 21) ÷ 168 × 100 = 147 ÷ 168 × 100 = **87.5%** of the BOD entering the secondary tank.

(c) Overall removal = (240 − 21) ÷ 240 × 100 = 219 ÷ 240 × 100 = 91.25%, or **about 91%**. The overall figure is not 30% + 87.5%, because the secondary percent is based on a smaller starting amount.

(d) Primary treatment is physical: sewage sits in a settling tank, so heavy solids sink as sludge and grease floats, which removes part of the BOD that is in particles. The dissolved and finely suspended organic matter remains. Secondary treatment is biological: aerobic microorganisms in the tank feed on the dissolved organic waste and convert it to CO_2, water and new cells that settle out. These microbes need oxygen to respire, so air is pumped into (aerated) the tank; without oxygen the process would slow or turn anaerobic and produce odors.

SG-U8-E009

Answer: (a) LD50 of X ≈ 23 mg/kg; LD50 of Y ≈ 191 mg/kg; (b) X is more toxic; (c) 0.58 mg; (d) below.

(a) Formula: LD50 ≈ low dose + [(50 − low response) ÷ (high response − low response)] × (high dose − low dose).

Insecticide X: 50% lies between 20 mg/kg (46%) and 40 mg/kg (72%). LD50 = 20 + [(50 − 46) ÷ (72 − 46)] × (40 − 20) = 20 + (4 ÷ 26) × 20 = 20 + 3.08 = 23.08, so **23 mg/kg**.

Insecticide Y: 50% lies between 200 mg/kg (52%) and 100 mg/kg (30%). LD50 = 100 + [(50 − 30) ÷ (52 − 30)] × (200 − 100) = 100 + (20 ÷ 22) × 100 = 100 + 90.9 = 190.9, so **191 mg/kg**.

(b) **Insecticide X is more toxic.** Its LD50 (23 mg/kg) is much lower than Y's (191 mg/kg), so a much smaller dose kills half the mice; X is about 8 times more toxic (191 ÷ 23 ≈ 8.3).

(c) Dose (mg) = LD50 × body mass = 23 mg/kg × 0.025 kg = 0.575 mg, or **about 0.58 mg**.

(d) LD50 measures only acute (24-hour) lethality in lab mice at one dose. It does not show chronic or sublethal effects (such as reproductive or hormonal harm), effects on fish or aquatic invertebrates, how the chemical behaves in water (persistence, bioaccumulation), or exposure at the low concentrations in a river. Any one of these is an acceptable limitation.

SG-U8-E010

Answer: (a) about 30 times; (b) drinking-water treatment; (c) and (d) below.

(a) 61 ÷ 2 = 30.5, so village C's rate is **about 30 times higher** than village A's.

(b) **Claim: untreated drinking water is most strongly associated with the difference between A and B.** Both villages have latrines connected to a treatment plant, so sewage handling is the same, yet B (untreated river water) has 14 cases per 10,000 versus 2 in A (chlorinated well water), which is 7 times higher (14 ÷ 2).

(c) In village C, sewage containing the cholera bacterium (*Vibrio cholerae*) from infected people is disposed of openly upstream of the intake. Rain carries it into the river, and the village draws that untreated water for drinking. People swallow the bacteria in the water, become infected and shed more bacteria in their waste, which repeats the cycle (fecal-oral transmission).

(d) This is an observational survey with three villages, so other differences (income, nutrition, hygiene, climate, local immunity or sanitation habits) could also explain the cases; association is not proof of cause. Low-cost interventions for C (one): chlorinate or boil drinking water, build latrines and move waste disposal downstream of the intake, or use simple household filters.

SG-U8-E011

Answer: (a) 200 kg per day; (b) 90 kg removed, 110 kg reaches the lake; (c) about 33,000 kg per year; (d) below.

(a) Convert volume: 25,000 m^3 × 1,000 L/m^3 = 2.5 × 10^7 L. Load = 2.5 × 10^7 L × 8 mg/L = 2.0 × 10^8 mg = 2.0 × 10^5 g = **200 kg per day**.

(b) Removed = 0.45 × 200 kg = **90 kg per day**. Reaching the lake = 200 − 90 = **110 kg per day**.

(c) Without the wetland the full 200 kg reaches the lake, so the additional amount is the 90 kg per day that was being removed: 90 kg/day × 365 days = 32,850 kg, or about **3.3 × 10^4 kg per year**.

(d) Any two: flood control (water storage), habitat for wildlife, groundwater recharge, filtering of sediment and other pollutants, carbon storage, recreation.

SG-U8-E012

Answer: model design below (other reasonable designs are accepted).

(a) **Hypothesis:** If phosphorus from field runoff causes the July algal bloom, then water with more phosphorus (the ditch and the pond inlet) will have higher phosphate concentrations and higher algal density than the pond with no farm nearby, and algal density will rise in the weeks after fertilizer is applied.

(b) The independent variable is the sampling site (distance from the fertilized field, hence the phosphorus level; or time since fertilizing). The dependent variables are the phosphate (phosphorus) concentration and the algal density (chlorophyll *a* or cell count; DO can also be measured). The **control is S4**, the similar pond with no farm nearby.

(c) Sampling plan: sample all four sites on the same day before fertilizing (baseline), then weekly through July and August (and also after a rainstorm, when runoff is greatest). At each visit collect at least three water samples (replicates) at each site at the same depth, and measure each sample with the same method (for example, a phosphate test kit and a count of algae in 1 mL). Average the replicates for each site.

(d) Two variables kept constant or recorded at every site: water temperature, sunlight/cloud cover, pond depth, pH, dissolved oxygen, turbidity, time of day of sampling, and rainfall. Any two are acceptable.

(e) Supporting results: phosphate is highest at S1, lower at S2 and S3, and lowest at S4; algal density follows the same pattern and rises after fertilizer application and storms, while S4 stays low, showing that the bloom follows the phosphorus supply. (Correlation alone does not prove cause, so a stronger test would add phosphate to a pond or tank and see if algae increase.)

SG-U8-E013

Answer: (a) about 32,700 t per year; (b) about 59,500 m^3 per year; (c) about 34 years; (d) about 1.4×10^6 kWh per year.

(a) Generated = 80,000 people × 1.6 kg/person/day = 128,000 kg/day = 128 t/day. The landfill receives 100% − 20% − 10% = 70%, so 128 t/day × 0.70 = 89.6 t/day. Annual mass = 89.6 t/day × 365 days = 32,704 t, or **about 3.27×10^4 t per year**.

(b) Volume = mass ÷ density = 32,704 t × 1,000 kg/t ÷ 550 kg/m^3 = 3.2704×10^7 kg ÷ 550 kg/m^3 = 59,462 m^3, or **about 5.9×10^4 m^3 per year**.

(c) Years = 2.0×10^6 m^3 ÷ 59,462 m^3/yr = 33.6, or **about 34 years**.

(d) Energy in the methane = 4.0×10^5 m^3 × 36 MJ/m^3 = 1.44×10^7 MJ. Electrical energy = 1.44×10^7 MJ × 0.35 = 5.04×10^6 MJ. Convert using 1 kWh = 3.6 MJ: $5.04 \times 10^6 \div 3.6 = 1.4 \times 10^6$ kWh, so **1.4×10^6 kWh per year**.

SG-U8-E014

Answer: (a) log axis; (b) about 0.5 mg/L; (c) shows background mortality.

(a) The concentrations span seven orders of magnitude (0.01 to 100 mg/L), with each step ten times the one before. On a linear axis, the low doses would be crowded at the origin and the high ones would stretch far out, so the S-shaped response could not be seen. A log axis spaces each tenfold step equally and spreads the data into a readable S-curve. Set-up: put $\log_{10}$ concentration on the x-axis (ticks at 0.01, 0.1, 1, 10, 100, equally spaced, i.e., exponents −2, −1, 0, 1, 2) and percent mortality (0–100%) on the y-axis; the control (0 mg/L) cannot be plotted on a log axis, so it is shown separately or noted beside the axis.

(b) The 50% mortality lies between 0.1 mg/L (24%, exponent −1) and 1 mg/L (61%, exponent 0). The fraction of the way = (50 − 24) ÷ (61 − 24) = 26 ÷ 37 = 0.70. The exponent = −1 + 0.70 = −0.30, so the concentration = $10^{-0.30} \approx 0.50$, or **about 0.5 mg/L**.

(c) The control shows how many shrimp die from causes other than the metal (handling, temperature, age, starvation). Here 3% died with no metal, so some deaths at each concentration are not due to the metal, and the control gives the baseline against which the effect of the metal is judged. It also confirms that the dishes and conditions themselves were not harmful. (Correcting for the 3% would change the estimate only slightly, since the interpolated value stays near 0.5 mg/L.)

Unit 9 — Global Change

SG-U9-E001

Answer: (a) the fall in stratospheric ozone. (b) Skin cancer or cataracts from UV; lung irritation from ground-level ozone.

(a) Stratospheric ozone absorbs most incoming UV-B and UV-C and converts the energy to heat. When its concentration falls, less UV is absorbed and more reaches the surface and the residents. Ground-level ozone sits in the troposphere, below the protective layer, so a rise in it does not change the UV arriving at the city. A common wrong answer says the rise in ground-level ozone "adds" UV; it does not, and it is a pollutant, not a shield.

(b) Increased UV: higher risk of skin cancer (UV damages DNA), cataracts, or immune suppression (any one). Increased ground-level ozone: irritated airways and lungs, triggering asthma, coughing and reduced lung function (any one).

SG-U9-E002

Answer: (a) Coolant X. (b) Coolant Y. (c) Coolant Z is the best choice, with leak and ignition controls.

(a) Coolant X is the only one containing chlorine. If it leaks, it can rise into the stratosphere, where UV frees chlorine atoms (Cl). $Cl + O_3 \rightarrow ClO + O_2$, then $ClO + O \rightarrow Cl + O_2$ regenerates the Cl, so each chlorine atom acts as a catalyst and destroys thousands of ozone molecules. Y and Z have no chlorine, so they cannot run this cycle.

(b) Coolant Y has the highest heat-trapping ability (GWP) per molecule, so per kilogram released it adds most to the greenhouse effect. X and Z have low GWP.

(c) Claim: choose Z. Reasoning: it has zero ozone harm (no chlorine) and low climate impact, whereas X harms ozone and Y is a potent greenhouse gas. Its flammability is a safety risk that can be managed. Precaution (any one): leak detectors and alarms, ventilation, keeping ignition sources away from the plant, sealed low-charge systems, staff training. Choosing Y with a leak-recovery plan could be argued but is weaker because climate harm is built into every leak.

SG-U9-E003

Answer: (a) 2,515 t CO_2e. (b) about 44.5% from methane. (c) CO_2: tractor/fuel use; CH_4: cattle digestion; N_2O: fertilizer.

(a) CO_2e = mass × GWP, then add. CO_2: 600 t × 1 = 600 t CO_2e CH_4: 40 t × 28 = 1,120 t CO_2e N_2O: 3 t × 265 = 795 t CO_2e Total = 600 + 1,120 + 795 = **2,515 t CO_2e**.

(b) Methane share = 1,120 ÷ 2,515 × 100 = 44.5% (about 45%), even though methane is only 40 t of the mass.

(c) CO_2: burning diesel in tractors and trucks (or heating fuel). CH_4: digestion by cattle, or manure lagoons and rice paddies. N_2O: synthetic fertilizer or manure applied to fields. (Other correct farm sources are acceptable.)

SG-U9-E004

Answer: (a) 50% increase. (b) The present concentration is far outside the natural range of the last 800,000 years. (c) Seasonal plant uptake causes the zigzag; the yearly average still rises.

(a) Percent change = (new − old) ÷ old × 100 = (420 − 280) ÷ 280 × 100 = 140 ÷ 280 × 100 = **50%**.

(b) 420 ppm is about 40% above the highest value in the ice-core record (300 ppm), so today's level is unprecedented in at least 800,000 years. Because the rise happened in about 250 years, it is far faster than natural glacial-interglacial changes and points to human activity.

(c) In the Northern Hemisphere growing season plants absorb CO_2 by photosynthesis and the concentration dips; in autumn and winter plants and soils release CO_2 (respiration and decay) and it climbs. This is a repeating seasonal cycle on top of the trend. Each year's average (and each peak and trough) is higher than the one before, so the long-term trend is upward.

SG-U9-E005

Answer: (a) 3.2 mm/yr (2000–2010) and 3.4 mm/yr (2010–2020). (b) Speeding up by 0.2 mm/yr. (c) Thermal expansion of seawater and melting of land ice.

(a) Rate = Δ sea level ÷ Δ time. 2000–2010: (32 − 0) mm ÷ 10 yr = **3.2 mm/yr**. 2010–2020: (66 − 32) mm ÷ 10 yr = 34 ÷ 10 = **3.4 mm/yr**.

(b) The rate rose from 3.2 to 3.4 mm/yr, so the rise is accelerating by 3.4 − 3.2 = **0.2 mm/yr** between the two decades.

(c) (1) Thermal expansion: ocean water expands as it warms. (2) Melting land ice (mountain glaciers and the Greenland and Antarctic ice sheets) adds new water to the ocean. Melting floating sea ice does not noticeably raise sea level.

SG-U9-E006

Answer: (a) positive (ice-albedo) feedback. (b) Bright clouds counteract the warming, so it is negative. (c) Loss of hunting or resting platform for an ice-dependent species.

(a) Warming lengthens the ice-free period, exposing dark water. Dark water has a low albedo, so it absorbs more sunlight than the bright ice it replaced. The extra absorbed energy warms the water and air, which lengthens the ice-free period further. The change (warming and ice loss) is amplified, so it is a positive feedback loop.

(b) Warmer air increases evaporation, forming bright low clouds that reflect more sunlight to space, which reduces the sunlight reaching the surface and so reduces the warming. The result opposes the original change, so it is a negative feedback loop. (Positive and negative describe direction, not whether the outcome is good or bad.)

(c) Any one: polar bears and seals lose sea-ice platforms for hunting, resting and breeding; under-ice algae at the base of the food web are lost; ice-dependent fish or waterfowl lose habitat as the ice season shortens.

SG-U9-E007

Answer: (a) 1.8°C. (b) The corals lost their symbiotic algae (zooxanthellae). (c) The shaded deeper area recovered; the shallow sunny flat suffered mortality.

(a) 30.8°C − 29.0°C = **1.8°C** above the normal summer maximum, held for six weeks.

(b) Heat stress made the algae produce harmful compounds, and the corals expelled them. Without algae the transparent coral tissue shows the white skeleton (bleaching). The algae supply most of the coral's food through photosynthesis, so the bleached coral is starving, more vulnerable to disease, and dies if the stress lasts too long. Bleached does not mean dead.

(c) Recovery: the shaded, deeper part of the reef, where corals "have regained their color," showing algae returned. Mortality: the shallow, sunny flat, where corals "have no living tissue" and are covered in algae growth. Likely reason: shallow sunlit water is warmer and has more light stress, so the stress there lasted or hit harder. This is an inference; the survey only shows the outcomes.

SG-U9-E008

Answer: (a) 1.26 times, a 26% increase in H^+. (b) $CO_2 + H_2O \rightleftharpoons H_2CO_3$. (c) Extra H^+ removes carbonate ions the oysters need.

(a) Factor = $10^{(pH_1 - pH_2)} = 10^{(8.20 - 8.10)} = 10^{0.10} \approx$ **1.26**. Percent increase = (1.26 − 1) × 100 = **26%** more H^+ (the pH scale is logarithmic, so a 0.1 drop is not small).

(b) $CO_2 + H_2O \rightleftharpoons H_2CO_3$ (carbonic acid). It then partly dissociates: $H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$.

(c) The released H^+ combines with carbonate ions ($H^+ + CO_3^{2-} \rightarrow HCO_3^-$), so less carbonate is available. Oysters build shells from calcium carbonate ($Ca^{2+} + CO_3^{2-} \rightarrow CaCO_3$), so building shells takes more energy, larvae grow weaker shells, and in more acidic water the shell can dissolve. The bay is still basic (pH above 7); it is becoming less basic.

SG-U9-E009

Answer: (a) −45%. (b) Fast growth/high reproduction and few natural enemies (also generalist tolerance). (c) Hand-pulling, with labor intensity as the drawback.

(a) Percent change = (new − old) ÷ old × 100 = (44 − 80) ÷ 80 × 100 = −36 ÷ 80 × 100 = **−45%** (sedge cover fell 45%, which is 36 percentage points).

(b) Any two: r-selected traits (rapid growth, early maturity, abundant seeds, good dispersal); generalist tolerance of wet and dry conditions; no natural herbivores or diseases in the new range (enemy release);

chemicals that suppress other plants (allelopathy). The shrub rose from 5% to 38% cover in six years, consistent with fast spread.

(c) Method options and drawbacks: hand-pulling or cutting (targeted, no chemicals, but labor-intensive and must be repeated); herbicide (fast over large areas, but can harm non-target native sedges and other wetland life, and pollutes water); biological control with a host-specific enemy (self-sustaining, but may attack natives or become invasive). Any one method with a matching disadvantage earns credit.

SG-U9-E010

Answer: (a) Species Q is more at risk. (b) A selective pressure is a factor that affects survival and reproduction. (c) Protect the bark-tree forest above 2,000 m; the limit is that it restricts logging.

(a) Species Q is more at risk because: (1) it is a specialist on the bark of one tree species, so losing that tree causes its loss; (2) it is K-selected, with slow maturity (5 years) and one young every two years, so its population recovers slowly; (3) it is confined above 2,000 m, so as warming pushes conditions upslope it has little higher ground and cannot disperse. Species P is a generalist, breeds fast and can move to nearby valleys, so it is more likely to adapt or relocate.

(b) A selective pressure is any environmental factor that changes the survival and reproduction (fitness) of organisms. Examples here: logging removing Q's food tree, warming shrinking Q's cold habitat, or competition for bark.

(c) Strategy: protect the habitat by creating a reserve or conservation easement covering Q's high-elevation forest and banning logging of its food tree (or replanting that tree with corridors upslope). Limitation: it restricts logging income and land use for local communities, requires enforcement, and cannot stop climate change from shifting the habitat. Captive breeding is another option but it is costly, and Q breeds slowly.

SG-U9-E011

Answer: (a) 64 km² originally; 36 km² in the four squares. (b) 64% and 36%. (c) The forest-interior bird loses much habitat; corridors or removing/reducing the barrier would help.

(a) Interior of a square = (side − 2w)². Original: $(10 - 2 \times 1)^2 = 8^2 =$ **64 km²**. Each small square: $(5 - 2 \times 1)^2 = 3^2 = 9$ km², so four squares = $4 \times 9 =$ **36 km²**.

(b) Total area is $10 \times 10 = 100$ km² in both cases ($4 \times 5 \times 5 = 100$). Interior share: original $64 \div 100 \times 100 =$ **64%**; after division $36 \div 100 \times 100 =$ **36%**. Interior fell by 28 percentage points, a 44% decrease ($28 \div 64$). This ignores the road's own width.

(c) The forest-interior bird loses almost half of its interior habitat, and the patches are smaller and isolated. Edge effects (more predators, nest parasites such as cowbirds, wind and temperature extremes, invasive species) reach most of each patch, populations in each patch are small and prone to local extinction, and the bird cannot easily cross the road to find mates. Measures (any one): a habitat corridor or wildlife overpass linking the patches, restoring forest along the road, making patches bigger, or routing the road along the edge instead of through the middle.

SG-U9-E012

Answer: (a) A: 15,000 t CO_2e; B: 100,000 t CO_2e. (b) A: \$1,200 per t; B: \$60 per t. (c) Fund Plan B first. (d) Both are mitigation.

(a) Plan A: 1,500 t CO_2e/yr × 10 yr = **15,000 t CO_2e**. Plan B: 400 t CH_4/yr × 25 = 10,000 t CO_2e/yr; × 10 yr = **100,000 t CO_2e**.

(b) Cost per tonne = total cost ÷ total CO_2e avoided. Plan A: \$18,000,000 ÷ 15,000 t = **\$1,200 per t CO_2e**. Plan B: \$6,000,000 ÷ 100,000 t = **\$60 per t CO_2e**.

(c) Claim: fund Plan B first, because it avoids almost 7 times more CO_2e for one-third of the cost, i.e. 20 times cheaper per tonne. Advantage: very low cost per tonne and a large reduction of a potent gas (GWP 25). Disadvantage (any one): it does not remove the underlying source or reduce fossil fuel use; the captured methane is usually burned, releasing some CO_2 (ignored in the problem); it needs ongoing maintenance and only addresses emissions at the landfill. (Plan A's coal-to-network switch could be funded second, as it also cuts air pollution.)

(d) Both are mitigation, because they reduce greenhouse gas emissions (the cause). Adaptation would reduce damage from changes already happening, such as sea walls.

SG-U9-E013

Answer: (a) −16.7%. (b) Ozone responds with a lag because chlorine was still above its 1985 level. (c) 45,000 kg CO_2e; about 4% of a CFC's ozone harm.

(a) Percent change = (new − old) ÷ old × 100 = (3.0 − 3.6) ÷ 3.6 × 100 = −0.6 ÷ 3.6 × 100 = **−16.7%** (about a 17% decrease).

(b) In 2015 chlorine was 3.0 ppb, still well above its 1985 level of 2.2 ppb, so more chlorine remains available to destroy ozone than in 1985 (ozone minimum 120 DU compared with 190 DU). CFCs persist for decades, and the chlorine cycle is catalytic, so existing chlorine keeps destroying ozone. Ozone can only recover after chlorine has fallen for years, and "recovered" means back to the earlier level. The minimum bottomed out (110 DU in 2005) after chlorine peaked (3.6 ppb in 1995), then started to improve to 120 DU.

(c) CO_2e = mass × GWP = 50 kg × 900 = **45,000 kg CO_2e** (45 t). ODP 0.04 means it destroys ozone at 4% the rate of a CFC (ODP 1), so it harms ozone **much less** (25 times less). But its GWP of 900 means it is a potent greenhouse gas, so ozone-safe is not climate-safe.

SG-U9-E014

Answer: (a) 80 units. (b) 62 units. (c) Energy out falls to 96 against 100 in, so Earth warms until a warmer surface emits more infrared (negative feedback) and out again equals 100.

(a) Surface budget: energy in = energy out. Surface absorbs 50 from the Sun + infrared from the atmosphere = 130 emitted. Returned infrared = 130 − 50 = **80 units**.

(b) The atmosphere absorbs the 22 units of sunlight plus the infrared that does not escape directly: 130 − 10 = 120. Total absorbed = 22 + 120 = 142 units. It emits 142 in total: 80 down, so to space = 142 − 80 = **62 units**. Check at the top of the atmosphere: reflected 28 + direct infrared 10 + atmosphere's 62 = 100, equal to the 100 arriving.

(c) Immediate effect: only 6 units now escape directly, so the atmosphere absorbs 4 more. Before the atmosphere and surface adjust, outgoing energy = 28 + 6 + 62 = 96 units, less than the 100 arriving. About 4 units per 100 are retained, so Earth is out of balance and gains energy. Long term: the surface and lower atmosphere warm. Feedback: a warmer surface emits more infrared (and the atmosphere more to space); this negative feedback continues until outgoing energy again equals 100 at a higher temperature. Other positive feedbacks (water vapor, ice-albedo) would amplify the warming along the way.

SG-U9-E015

Answer: (a) 825 t CO_2; net saving 7,575 t CO_2e/yr. (b) 37.9% (about 38%). (c) Target 12,000 t CO_2e; about 53 t CO_2e/yr more must be cut. (d) 12 years.

(a) CO_2 from burning ($CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$): 300 t × 44 ÷ 16 = **825 t CO_2**. CO_2e avoided from CH_4 = 300 × 28 = 8,400 t CO_2e. Net change = 825 − 8,400 = **−7,575 t CO_2e per year** (a reduction of 7,575 t CO_2e). The CO_2 from burning is counted as CO_2 with GWP 1.

(b) Percent reduction = 7,575 ÷ 20,000 × 100 = **37.9%** (37.875%).

(c) Target = 20,000 × (1 − 0.40) = **12,000 t CO_2e**. After the project the farm emits 20,000 − 7,575 = 12,425 t CO_2e. Remaining cut = 12,425 − 12,000 = 425 t CO_2e. Average yearly cut = 425 ÷ 8 = **about 53 t CO_2e per year**.

(d) Payback = cost ÷ annual income = $420,000 ÷ $35,000 per yr = **12 years**.

SG-U9-E016

Answer (model design): remove the plant from replicated, randomly chosen bays and compare DO and native fish species with untouched bays.

(a) Hypothesis: If the floating invasive plant is removed from a bay, then dissolved oxygen and the number of native fish species will be higher than in bays where the plant is left in place.

(b) Independent variable: presence or absence of the invasive plant (removal versus no removal). Dependent variables: dissolved oxygen (mg/L) and native fish species richness (number of species).

(c) Design: choose, for example, 12 similar bays and randomly assign 6 to plant removal (treatment) and 6 to be left alone (control group). Replication (6 per group) lets chance differences average out. Measure DO with the same meter at the same depth and time of day, and sample fish by the same method (nets or traps, same

effort) before removal and at set intervals (for instance monthly) for a full season or longer. Concurrent controls allow the treatment effect to be separated from year-to-year changes. Alternative acceptable designs: matched pairs of bays (one removed, one control), or a survey along a gradient of plant cover as an observational study (which shows association only).

(d) Variables kept similar (any two): bay size and depth, water temperature, water clarity/light, nutrient levels, season and sampling time, shoreline development, connection to the main lake, and sampling method.

(e) Supporting pattern: after removal, treatment bays show higher DO and more native fish species than control bays, with a gap that grows over time and is bigger than the variation among replicates; control bays show no change. Limitation (any one): small number of bays, one lake, short time, fish moving between bays, or unmeasured differences such as pollution or fish stocking; a survey without removal cannot show cause.

Part II — Book Two Classified Question Solutions

For each unit: MCQ answers and explanations first, then complete FRQ solutions.

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Unit 1 — The Living World: Ecosystems

Section A — MCQ Answers and Explanations

QB-U1-MCQ001

Answer: B. Both partners gain (the tree gets phosphorus and water, the fungus gets sugars), which defines mutualism. Commensalism (A) requires one species to be unaffected, and parasitism (C) requires harm to one partner; neither fits a relationship in which both grow better.

QB-U1-MCQ002

Answer: D. Permafrost, a very short growing season and mosses, lichens and low shrubs describe tundra. Taiga (A) is also cold, but it is a coniferous forest with trees and no continuous permafrost layer controlling vegetation.

QB-U1-MCQ003

Answer: A. The oceans are the primary reservoir of water at Earth's surface. Ice caps (B) and groundwater (C) are much smaller reservoirs, even though ice holds most of the fresh water.

QB-U1-MCQ004

Answer: C. Photosynthesis takes CO_2 from the air and builds it into sugars. Cellular respiration (A) and combustion (D) release carbon to the atmosphere, and decomposition (B) returns carbon from dead matter.

QB-U1-MCQ005

Answer: A. In nitrogen fixation, certain bacteria convert atmospheric N_2 into ammonia (which becomes ammonium in soil). B describes denitrification, C describes assimilation, and D describes ammonification.

QB-U1-MCQ006

Answer: C. The phosphorus cycle lacks a significant atmospheric component; its major reservoirs are rocks and ocean sediments. A is wrong because the nitrogen reservoir is atmospheric but phosphorus has none; D is wrong because phosphorus is scarce, since rocks weather slowly.

QB-U1-MCQ007

Answer: D. Counting up from the producer: grass is level 1, grasshopper (primary consumer) level 2, frog (secondary) level 3, snake (tertiary) level 4. Option B confuses the snake with the frog.

QB-U1-MCQ008

Answer: B. An estuary is where river freshwater meets the sea, so its salinity varies with tides and flow. A lake (D) stays fresh, and coral reefs and the open ocean (A, C) have stable, high salinity.

QB-U1-MCQ009

Answer: C. Producers → primary consumers → secondary consumers is two transfers: 50,000 × 0.10 × 0.10 = 500 kcal/m²/yr. D (5,000) counts only one transfer and gives the energy for primary consumers.

QB-U1-MCQ010

Answer: A. NPP = GPP − respiration = 6,000 − 4,500 = 1,500 kcal/m²/yr. D adds the two values instead of subtracting, and C ignores respiration.

QB-U1-MCQ011

Answer: D. Species that use the same resource (insects) in different places on the tree are partitioning resources, which reduces competition and allows coexistence. B is wrong because the competition described is between species (interspecific), not within one species.

QB-U1-MCQ012

Answer: B. Temperatures are warm all year (24–29 °C) with a pronounced wet season (about November to April) and a dry season (June to August), typical of a savanna. A tropical rainforest (A) would be wet in every month, and a temperate grassland (C) would have a cold winter.

QB-U1-MCQ013

Answer: C. Marine resources such as fish vary with nutrient availability, and nitrate is highest at Site X (18 µmol/L), so phytoplankton at the base of the food web should be most abundant there. Site W has little nitrate (A, D), and the very turbid water at Site Y limits light for photosynthesis, so B is wrong. Warm water does not hold more nutrients.

QB-U1-MCQ014

Answer: A. Inputs to the atmosphere: B + D + E = 118 + 90 + 10 = 218 Gt C/yr. Outputs: A + C = 120 + 92 = 212 Gt C/yr. Net change = 218 − 212 = +6 Gt C/yr. Option C (+10) counts only the fossil fuel flux, ignoring that the land and ocean fluxes nearly balance, and B (+8) is not the difference between total inputs and outputs.

QB-U1-MCQ015

Answer: D. Step T moves nitrate to N_2 gas in the atmosphere, which is denitrification; it is carried out by bacteria in oxygen-poor, waterlogged soil. Step Q (nitrification) needs oxygen and would slow down, and P (fixation) and S (ammonification) add nitrogen to the soil rather than remove it.

QB-U1-MCQ016

Answer: B. Adding phosphorus alone raised chlorophyll *a* from 4.1 to 17.8 µg/L, while nitrogen alone changed almost nothing (4.5), so phosphorus is the main limiting nutrient; once phosphorus was supplied, nitrogen became limiting and the combination gave the highest value (41.3). A fails because the 0.4 µg/L increase is negligible, and D fails because phosphorus alone caused a large increase.

QB-U1-MCQ017

Answer: A. Pavement and roofs prevent infiltration, so more precipitation becomes surface runoff and drains quickly to streams. B and D state the opposite, and C is wrong because replacing plants removes transpiration.

QB-U1-MCQ018

Answer: C. Total NPP = NPP per m^2 × area. Open ocean: $600 \times 3.6 \times 10^{14}$ $m^2 = 2.2 \times 10^{17}$ kcal/yr; rainforest: $9{,}000 \times 1.7 \times 10^{13} = 1.5 \times 10^{17}$; estuary: $8{,}000 \times 1.4 \times 10^{12} = 1.1 \times 10^{16}$. The ocean has the lowest NPP per area yet the largest total. A is the tempting error of ignoring area.

QB-U1-MCQ019

Answer: B. Matter is conserved and cycles between organisms and the environment through decomposers, whereas energy enters as sunlight, flows one way through trophic levels, and is lost as heat. A and C wrongly make energy cycle, and D wrongly makes matter leave the ecosystem.

QB-U1-MCQ020

Answer: D. Pike are tertiary consumers, three transfers from the algae: $2.5 \times 10^6 \div 0.10 \div 0.10 \div 0.10 = 2.5 \times 10^9$ kcal/yr. Option C counts only two transfers.

QB-U1-MCQ021

Answer: C. The snake eats voles (primary consumers, so the snake is a secondary consumer at level 3) and frogs (secondary consumers, so the snake is a tertiary consumer at level 4). The grasshopper and clover each feed at one level only, and the frog eats only grasshoppers.

QB-U1-MCQ022

Answer: A. In this web the frogs are the only predator of grasshoppers, so removing the frogs releases grasshoppers from predation and their numbers rise. B and D invent relationships that are not in the web (frogs are predators of grasshoppers and prey of snakes, not competitors of grasshoppers), and C contradicts the web, which shows grasshoppers eating only grass.

QB-U1-MCQ023

Answer: D. Biome distribution is dynamic and follows climate; warming lets the taiga move poleward into tundra. A, B and C all require movement opposite to the warming.

QB-U1-MCQ024

Answer: B. Algae in marine biomes supply a large share of Earth's oxygen and take CO_2 from the atmosphere. A, C and D describe processes that algae do not carry out; photosynthesis releases oxygen, it does not absorb it.

QB-U1-MCQ025

Answer: C. Residence time = reservoir volume ÷ flow rate = 8.0×10^3 $km^3 \div 16$ km^3/yr = 500 years. A inverts the ratio, and D multiplies the two numbers.

QB-U1-MCQ026

Answer: A. Blue-green light penetrates to depth while red light is absorbed near the surface, so deep-living photosynthesizers have pigments that capture the remaining wavelengths. Photosynthesis requires light (D), and red light is gone at 40 m (B).

QB-U1-MCQ027

Answer: B. Transfer efficiencies are 1,800 ÷ 20,000 = 9.0%, 160 ÷ 1,800 = 8.9% and 12 ÷ 160 = 7.5%, all near 10%. The remainder is lost as heat in metabolism, as the second law predicts. C is wrong because energy is conserved: it changes form rather than being destroyed.

QB-U1-MCQ028

Answer: D. Energy in harvested bass: 60 kg × 2,000 kcal/kg = 1.2×10^5 kcal/yr. Bass are three transfers above the algae: $1.2 \times 10^5 \div 0.10^3 = 1.2 \times 10^8$ kcal/yr of NPP. Area = $1.2 \times 10^8 \div 2{,}400$ kcal/m²/yr = 5.0×10^4 m² = 5.0 ha. Option C results from counting only two transfers.

QB-U1-MCQ029

Answer: C. During the Northern Hemisphere growing season, photosynthesis by land plants exceeds respiration, lowering CO_2; the steady rise (about 2.4 ppm per year) reflects fossil fuel burning. A is wrong because summer fossil fuel use does not drop, and D reverses the logic since net uptake exceeds decomposition in summer.

QB-U1-MCQ030

Answer: A. A valid test has replicated treatment and control plots that differ only in nitrogen. B has no replication and confounds nitrogen with location, C changes three factors at once, and D has no control group to separate nitrogen from year-to-year weather.

QB-U1-MCQ031

Answer: B. Bass fell, minnows rose, zooplankton fell and chlorophyll *a* rose: a trophic cascade in which fewer predators allow more minnows, which eat more zooplankton, which graze less on algae. A reverses the order shown by the data, and D fails because the stem gives no evidence that bass eat zooplankton; the zooplankton fell as minnows (their predators) rose.

QB-U1-MCQ032

Answer: D. Precipitation is the same, evapotranspiration fell by 250 mm (700 → 450), and runoff rose by 600 − 380 = 220 mm; with less water returned to the atmosphere, more leaves as streamflow (the rest, +30 mm, goes to storage). A is wrong because precipitation is equal; B is the tempting error, since clearing reduces infiltration, not increases it.

Section B — FRQ Solutions

QB-U1-FRQ001

(a) Eutrophication or excess nutrient pollution of the lake, producing summer algal blooms (algal scum, odor). *(1 point: identifies nutrient-driven algal growth as the problem.)*

(b) For example: "If phosphorus is the limiting nutrient, then bottles with added phosphorus will have more algal growth than bottles with only nitrogen added." Reasoning: phosphorus is scarce because rocks weather slowly and it has no atmospheric source, and the July water has a very high nitrogen to phosphorus ratio (0.90 mg/L N compared with 0.008 mg/L P). A nitrogen-limitation hypothesis with reasoning about fertilizer input and the nitrogen cycle is also acceptable if it is testable and justified. *(1 point: testable if-then or directional prediction with reasoning.)*

(c) (i) The nutrient treatment (nitrogen added, phosphorus added, both added, none). *(1 point)* (ii) Algal growth, measured as chlorophyll *a* concentration (or algal cell count or biomass). *(1 point)*

(d) Collect lake water and fill many identical clear bottles. Make four treatments: no nutrients (control), nitrogen only, phosphorus only, nitrogen and phosphorus. Use several replicate bottles (at least three) per treatment, and incubate them together in the lake or under the same lights. Measure algal growth at the start and after one to three weeks. *(1 point: treatments with a control; 1 point: replication/same conditions for reliable results.)*

(e) Light (or temperature, bottle size, volume of water, incubation time, water source). Different light or temperature would change algal growth independently of the nutrient, so the effect of nutrients could not be isolated. *(1 point: names variable with a reason linked to isolating the nutrient effect.)*

(f) The phosphorus-only bottles (and the nitrogen-and-phosphorus bottles) would show significantly higher algal growth than the control and nitrogen-only bottles. (If the hypothesis was nitrogen limitation, the corresponding result for nitrogen.) *(1 point)*

(g) Any one with justification, for example: establish vegetated buffer strips along the inlet stream so that phosphorus-rich soil and manure are trapped before reaching the lake; or reduce fertilizer and manure application and match it to crop needs. Justification: because phosphorus has no atmospheric source, the lake's supply depends on runoff and sediment, so cutting phosphorus inputs lowers the nutrient that limits algae. *(1 point: reasonable action; 1 point: justification tied to the phosphorus (or nitrogen) cycle.)*

QB-U1-FRQ002

(a) GPP is lowest in winter, rises rapidly from March to a peak in June–July (about 1,500–1,550 kcal/m²/month), then falls rapidly through autumn to a low in December. *(1 point: describes rise to a summer peak and decline, using data.)*

(b) Any two: longer days and higher solar radiation in July provide more light energy for photosynthesis; warmer temperatures in July speed enzyme activity and photosynthesis; the deciduous trees have full leaf area in July but no leaves in January; more liquid water is available in summer than when it is frozen in winter. *(1 point each, maximum 2; each must link the factor to photosynthesis.)*

(c) November (120 < 130), December (40 < 90), January (40 < 90) and February (80 < 90). NPP = GPP − respiration, so it is negative when plants respire more (using stored energy to maintain living tissue, such as trunks, roots and evergreen needles) than they capture in photosynthesis. *(1 point: identifies months; 1 point: explains that respiration exceeds GPP.)*

(d) Total GPP = 40 + 80 + 260 + 600 + 1,100 + 1,500 + 1,550 + 1,400 + 900 + 400 + 120 + 40 = 7,990 kcal/m²/yr. Total respiration = 90 + 90 + 150 + 280 + 450 + 620 + 700 + 650 + 480 + 260 + 130 + 90 = 3,990 kcal/m²/yr. Percentage = (3,990 ÷ 7,990) × 100 = 49.9% (about 50%). *(1 point: both totals; 1 point: correct percentage.)*

(e) The plants gain energy: annual NPP = 7,990 − 3,990 = 4,000 kcal/m²/yr, which is positive. *(1 point: claim with supporting number.)*

(f) GPP would decrease substantially in those months because less leaf area means less photosynthesis; annual NPP would decrease because plant respiration (maintenance of living tissue) does not fall in proportion to the leaf loss. *(1 point: predicts decrease with justification.)*

(g) For example: manage insect pests, protect forest from clearing, or thin and replant with diverse species so that leaves capture light every growing season; justified by the link between leaf area and GPP. *(1 point: reasonable action with justification.)*

QB-U1-FRQ003

(a) Area = 400 ha × 10^4 m²/ha = 4.0 × 10^6 m². Total NPP = 4.0 × 10^6 m² × 4,500 kcal/m²/yr = 1.8 × 10^{10} kcal/yr. *(1 point)*

(b) Cattle (primary consumers): 1.8 × 10^{10} × 0.10 = 1.8 × 10^9 kcal/yr. People (secondary consumers): 1.8 × 10^9 × 0.10 = 1.8 × 10^8 kcal/yr. *(1 point: cattle energy; 1 point: energy available to people.)*

(c) Annual need per person = 2,500 kcal/day × 365 days = 912,500 kcal/yr (9.1 × 10^5). People = 1.8 × 10^8 ÷ 9.125 × 10^5 = 197 people, about 2.0 × 10^2 people. *(1 point: annual need per person; 1 point: number of people.)*

(d) People = 1.8 × 10^{10} ÷ 9.125 × 10^5 = 19,700, about 2.0 × 10^4 people, which is 100 times the number in part (c). *(1 point)*

(e) According to the second law of thermodynamics, energy transfers are never 100% efficient; at each trophic level much of the energy is lost as heat in metabolism. Eating beef is two transfers beyond the plants, so only about 1% of the original energy reaches people, rather than all of it. *(1 point: heat loss at each transfer linked to fewer people.)*

(f) Much of plant NPP cannot be eaten or digested by people, for example roots, stems, leaves and cellulose; some energy is also needed to run the farm and is lost in processing. *(1 point: gives a valid reason.)*

(g) For example: shift part of the land from cattle to crops people can eat directly, or reduce meat in the diet; or use aquaculture or poultry, which convert feed energy more efficiently. Justification: feeding people at a lower trophic level avoids energy losses at the intermediate steps, so more people can be supported per hectare. *(1 point: proposal; 1 point: justification with trophic levels.)*

QB-U1-FRQ004

(a) Sea urchin density is higher in the fished area (22 per m², about 7 times the 3 per m² in the reserve), whereas kelp cover is much lower in the fished area (9% compared with 82%). *(1 point: describes both with direction.)*

(b) Sea urchins are primary consumers (second trophic level); large predatory fish that eat urchins are secondary consumers (third trophic level). *(1 point: both levels correct.)*

(c) Fishing removes the large predatory fish, so fewer urchins are eaten and urchin numbers increase. A larger number of urchins graze more kelp, so kelp cover declines. *(1 point: fewer predators lead to more urchins; 1 point: more urchins lead to less kelp.)*

(d) Decrease = (82 − 9) ÷ 82 × 100 = 89% (89.0%). *(1 point)*

(e) The data come from two sites compared once and are only correlational; other differences between the sites (wave exposure, depth, water temperature, nutrients, disease) could explain the difference, and the sites may have differed before the reserve was created. Strengthening: survey the same reefs before and after protection, or use many reserve and fished sites, or run an experiment that excludes urchins or predators from plots. *(1 point: identifies a confounding variable or lack of replication; 1 point: describes one improved design.)*

(f) Kelp cover will increase, because fewer urchins means less grazing, allowing kelp to recover (growth is rapid under favorable light and nutrients). *(1 point: predicts increase with explanation.)*

(g) For example: establish a no-fishing zone or catch limits for the predatory fish so their numbers rebuild and control urchins; or remove urchins manually while the fish recover. Justification: restoring the predator reduces grazing pressure and allows kelp to recover. *(1 point: reasonable action; 1 point: justification via the food web.)*

Unit 2 — The Living World: Biodiversity

Section A — MCQ Answers and Explanations

QB-U2-MCQ001

Answer: D. Genetic diversity is the variety of alleles (gene versions) among the individuals of one population or species. A counts species (species diversity), B counts habitat types (habitat diversity), and C is simply population size, which says nothing about how varied the genes are.

QB-U2-MCQ002

Answer: A. Absorbing storm surge and floodwater is a regulating service because the ecosystem moderates a natural hazard. Provisioning services supply goods such as food and timber, cultural services provide recreation and spiritual value, and supporting services (soil formation, nutrient cycling) underlie all the others.

QB-U2-MCQ003

Answer: C. Primary succession begins where there is no soil, such as newly exposed bare rock. The abandoned hay field, burned forest, and logged hillside (A, B, D) all keep their soil and seed bank, so they undergo secondary succession.

QB-U2-MCQ004

Answer: B. Ecological tolerance is the range of conditions, here temperature, that an organism can endure before injury or death. Carrying capacity (A) is the largest population an environment can sustain, biotic potential (C) is maximum reproductive rate, and species richness (D) is a count of species.

QB-U2-MCQ005

Answer: A. Near islands receive more immigrants, and large islands have lower extinction rates because they hold bigger populations and more habitat. The combination of high immigration and low extinction gives the highest equilibrium richness. Option D is the opposite combination and would have the fewest species.

QB-U2-MCQ006

Answer: D. A periodic process recurs at regular intervals, as a spring flood does. The meteorite impact, a single lightning-ignited fire, and an earthquake (A, B, C) are episodic or random events with no regular schedule.

QB-U2-MCQ007

Answer: B. Natural selection acts on heritable variation: individuals whose traits improve survival and reproduction leave more offspring, so the trait becomes more common over generations. A describes inheritance of acquired traits, C wrongly claims need-driven change within one generation, and D is wrong because mutations arise randomly and are not directed by the environment.

QB-U2-MCQ008

Answer: C. Pioneer species on bare rock are lichens and mosses, which can grow without soil and begin to break the rock down and build soil. Trees, shrubs, and grazing mammals (A, B, D) need soil or food plants that do not yet exist.

QB-U2-MCQ009

Answer: B. The table shows alleles fell from 14 to 5 during the bottleneck and stayed at 5 after the population recovered to 380. Lost alleles cannot be regained by population growth, so the population has less genetic variation to respond to stressors. A contradicts the table (alleles did not drop further), and C and D confuse genetic diversity with species or habitat diversity.

QB-U2-MCQ010

Answer: D. Wetlands regulate flooding by storing water and regulate water quality by filtering nitrates. More flooding and more nitrate are the loss of these regulating services. Provisioning, cultural, and supporting services (A, B, C) are not what the stem's symptoms point to.

QB-U2-MCQ011

Answer: A. Equilibrium is where the immigration and extinction curves cross. Large, near island: $24 - 0.4S = 0.2S$ gives $S = 40$. Small, far island: $12 - 0.2S = 0.6S$ gives $S = 15$. The difference is $40 - 15 = 25$ species. Choice B (10) is near-large versus far-large, C (16) is near-large versus near-small, and D (15) is the small, far island's richness alone.

QB-U2-MCQ012

Answer: C. Reading the ranges: P about 16–26°C, Q about 6–34°C, R about 24–38°C. At 30°C, P is outside its range but Q and R are inside. A is false because Q, not R, has the widest range (28 degrees versus 14). B is false because Q has the widest, not narrowest, range. D is false because R cannot grow at 20°C.

QB-U2-MCQ013

Answer: D. During glacial periods much water was locked in ice, sea level fell, and shallow shelves became land bridges that let mammals spread; rising seas later isolated the island populations. Mammals that cannot cross water would not have been carried by currents (B), and independent evolution or repeated mutation of the identical species (A, C) is extremely unlikely.

QB-U2-MCQ014

Answer: A. The distribution shifted toward thicker shells because snails with thicker shells survived crab attacks better and passed the heritable trait on. B describes inheritance of acquired characteristics, C wrongly says crabs cause mutations, and D says need drives genetic change, which is not how selection works.

QB-U2-MCQ015

Answer: C. Percent increase = (new − old) ÷ old × 100 = (35 − 22) ÷ 22 × 100 = 13 ÷ 22 × 100 ≈ 59%. Choice B (37%) divides by the new value (35) instead of the original, A (13%) is the raw difference, and D (160%) is the ratio 35 ÷ 22 expressed as a percent.

QB-U2-MCQ016

Answer: B. Habitat loss and fragmentation remove specialists with narrow habitat needs first and also species with large territory requirements. The 25-ha patches are too small for a 40-ha territory, so the specialist bird cannot persist. The generalist rodent, the generalist deer, and the weedy plant (A, C, D) use many habitats and tolerate edges and disturbance.

QB-U2-MCQ017

Answer: A. Nitrogen removal: 60 ha × $1,500/ha = $90,000. Oysters: 60 ha × 800 kg/ha = 48,000 kg × $6/kg = $288,000. Total = $90,000 + $288,000 = $378,000. B and C each count only one service, and D ($6,300) is the per-hectare total, not the total for 60 ha.

QB-U2-MCQ018

Answer: D. Island resources are limited, so many island species evolved as specialists. Introduced generalists such as rats can eat many foods and outcompete or prey on specialists. A has no basis (island populations are often less genetically diverse), and B and C are unsupported generalizations.

QB-U2-MCQ019

Answer: B. Tolerance is found by exposing replicate groups to a graded series of the factor while holding other variables constant and measuring survival. A tests only one level, C compares size rather than survival and confounds creek conditions, and D has no replication or controlled levels, so it cannot give a tolerance range.

QB-U2-MCQ020

Answer: C. Wildlife may move short distances for a season in response to a natural disruption such as a hurricane. The birds' relocation does not create a new species (A), does not begin primary succession (B), and says nothing about a loss of genetic diversity (D).

QB-U2-MCQ021

Answer: D. Adaptation by natural selection needs heritable variation, many generations, and gradual change. A short-lived insect with high variation meets all three. Options A, B, and C pair low variation or small populations with sudden change, so individuals are more likely to move away or die than adapt.

QB-U2-MCQ022

Answer: A. Mayfly density falls from 80–85 per m² upstream to 6 per m² at the outfall as oxygen falls from 8.8 to 4.1 mg/L, then recovers as oxygen recovers. That pattern fits an indicator species whose abundance signals water quality. B reverses cause and effect, C contradicts the data, and D ignores the clear association between low oxygen and low density.

QB-U2-MCQ023

Answer: C. Where the sea star was removed, richness fell from 15 to 6 and mussels took over, even though the sea star was only 2% of biomass. A species with a disproportionate effect on community structure is a keystone

species. B fails because the control pools changed only from 15 to 14. A confuses keystone with indicator species, and D mistakes the dominant competitor for the keystone.

QB-U2-MCQ024

Answer: B. Plant: $24 million + 20 × $1.2 million = $48 million. Wetland: 800 × $9,000 = $7.2 million, plus 800 × $150 = $120,000 per year × 20 = $2.4 million, total $9.6 million. Difference = $48 million − $9.6 million = $38.4 million. A ignores annual costs of both ($24 − $7.2 million), C omits the wetland's maintenance ($48 − $7.2 million), and D omits the plant's operating cost ($24 − $9.6 million).

QB-U2-MCQ025

Answer: A. Species lost = 31 − 24 = 7 (present in 2010 only); species gained = 31 − 24 = 7 (present in 2020 only). Richness is steady while the species list turns over, which matches an equilibrium of immigration and extinction. B and D deny any turnover, and C confuses the number shared with the number lost or gained.

QB-U2-MCQ026

Answer: D. At 31°C survival fell from 90% after one week to 55% after four weeks, so the effect of a condition depends on exposure time as well as intensity. A is wrong because survival at 33°C after a week was only 20% and 0% by week four. B is wrong because survival at 30°C declined to 85% at four weeks. C is wrong because the tolerable temperature range shrinks with longer exposure.

QB-U2-MCQ027

Answer: C. The resistant fraction rose from 0.02 to 0.90 because each spray selects for resistant survivors. Untreated refuges keep susceptible individuals in the population, and when they mate with resistant survivors they dilute resistance alleles. A, B, and D all increase or maintain selection pressure, and D delays the change until resistance is already widespread.

QB-U2-MCQ028

Answer: B. NPP ÷ biomass: 0.60 ÷ 0.8 = 0.75 yr^{-1} at 5 years, 1.20 ÷ 4.0 = 0.30 at 20 years, 1.00 ÷ 18.0 ≈ 0.056 at 80 years, and 0.90 ÷ 22.0 ≈ 0.041 at 200 years. Productivity per unit biomass falls while biomass rises. A is false because NPP peaks at 20 years (1.20) and then declines. C is false because biomass at 20 years (4.0 kg/m^2) is far lower than at 80 or 200 years. D is false because NPP at 200 years is lower than at 20 years.

Section B — FRQ Solutions

QB-U2-FRQ001

(a) (i) Chloride concentration decreases as distance from the highway increases (about 620 mg/L at 10 m down to 40 mg/L at 400 m). (ii) The number of egg masses increases with distance (4 at 10 m up to 31 at 400 m). *(1 point: describes both trends correctly, including direction.)*

(b) The data are correlational. Distance from the highway also changes other factors (canopy cover, pool depth or size, vehicle pollutants, predators, noise), and any of these could affect egg-laying. Only a controlled experiment that changes chloride alone can show cause. *(1 point: identifies a confounding variable or states that correlation does not show causation.)*

(c) Hypothesis: if chloride concentration increases, then wood frog egg hatching success will decrease, because the eggs have a limited tolerance for salinity and high chloride causes osmotic stress that kills embryos above a threshold. *(1 point: testable prediction with direction; 1 point: justification linked to ecological tolerance.)*

(d) Independent variable: chloride concentration of the water. Dependent variable: hatching success (percent of eggs that hatch) or embryo survival. *(1 point for each.)*

(e) Set up at least five containers with chloride concentrations spanning the field range, for example 0 (control), 50, 150, 300, and 600 mg/L, with at least three replicate containers per concentration and the same number of eggs (e.g., 30) in each. Hold constant any two of: water temperature, water volume, pond-water source, light, egg-mass age or source, pH, or time. *(1 point: concentrations with a control and replicates; 1 point: two valid controlled variables.)*

(f) Plot percent hatching against chloride concentration. The tolerance range is the interval over which hatching stays high; the concentration at which hatching drops sharply marks the tolerance limit. Compare this limit with the chloride values measured in the pools. *(1 point: uses a hatching-versus-concentration pattern to identify the limit.)*

(g) Reduce road-salt use near vernal pools by applying less salt or using alternatives (sand, brine pre-wetting, or less harmful de-icers), or install vegetated buffers or swales that filter runoff. Tradeoff: reduced salt can mean less ice control and more accident risk, and alternatives or buffers cost more. *(1 point: valid solution with a stated tradeoff or cost.)*

QB-U2-FRQ002

(a) Primary succession. The lava flows cooled to bare rock with no soil, and the earliest vegetation was lichens and mosses. *(1 point: primary; 1 point: evidence such as no soil or pioneer lichens and mosses.)*

(b) Richness rises rapidly from 3 to a peak of 26 at year 60, then declines slightly to 24 at year 120 and 22 at year 250. *(1 point: increase then slight decrease, with a peak near year 60.)*

(c) Percent increase = (9.5 − 2.1) ÷ 2.1 × 100 = 7.4 ÷ 2.1 × 100 ≈ 352%. *(1 point: correct value of about 352%.)*

(d) 15 years: 0.25 ÷ 0.4 = 0.625 yr^{-1} (about 0.63). 250 years: 1.00 ÷ 21.0 ≈ 0.048 yr^{-1}. *(1 point for each ratio.)*

(e) As trees form a closed canopy, light at ground level declines and light-demanding early species are shaded out; competition and the dominance of a few late-successional species also reduce the number of niches. Either reason is acceptable. *(1 point: reason; 1 point: links the reason to a loss of species.)*

(f) Recovery will be faster because the soil, nutrients, seed bank, and roots remain, so the community starts from an advanced stage as in secondary succession. Primary succession on bare rock must first build soil. *(1 point: faster; 1 point: justification based on soil already present.)*

QB-U2-FRQ003

(a) Area converted = 1,500 ha × 0.20 = 300 ha. Fish and crab value = 380 kg/ha/yr × $4.50/kg = $1,710/ha/yr. Carbon value = 6.0 t CO_2/ha/yr × $50/t = $300/ha/yr. All three services per hectare = $7,200 + $1,710 + $300 = $9,210/ha/yr. Total lost = $9,210/ha/yr × 300 ha = $2,763,000 per year. *(1 point each: 300 ha; $1,710/ha; $9,210/ha; $2,763,000/yr. Units are required.)*

(b) Developer's profit = $3,800/ha/yr × 300 ha = $1,140,000 per year. Net difference = $2,763,000 − $1,140,000 = $1,623,000 per year, with the loss exceeding the gain. *(1 point: $1,140,000; 1 point: $1,623,000 with units.)*

(c) CO_2 no longer sequestered = 6.0 t/ha/yr × 300 ha = 1,800 metric tons of CO_2 per year. *(1 point.)*

(d) (i) Fish and crab harvest: provisioning. (ii) Storm protection: regulating. *(1 point: both correct.)*

(e) Claim: the conversion does not benefit society as a whole. The services lost ($2,763,000/yr) exceed the developer's profit ($1,140,000/yr) by $1,623,000/yr, and those costs fall on the public, not on the developer. *(1 point: claim consistent with the calculation, supported with figures. A claim that it benefits the developer privately but not society is also acceptable.)*

(f) Acceptable: build ponds on land that is already degraded rather than mangrove; require protection or restoration of mangroves elsewhere; make the developer pay for lost services; or use payments for ecosystem services to conserve the forest. *(1 point: valid solution.)*

QB-U2-FRQ004

(a) Among near islands, species richness increases as area increases (12 species on 2 km^2 to 41 species on 200 km^2). *(1 point: positive relationship, with direction.)*

(b) Percent difference = (30 − 15) ÷ 30 × 100 = 50%. Island G has 50% fewer species than Island C. *(1 point: 50%.)*

(c) (i) Larger islands have larger populations and more habitat types, so extinction rates are lower, and they present a bigger target for colonists, so more species persist. (ii) Islands far from the mainland receive fewer colonists, so the immigration rate is lower, giving a lower equilibrium number of species. *(1 point for each of (i) and (ii).)*

(d) Percent change = (9 − 18) ÷ 18 × 100 = −50%, a 50% decrease. *(1 point: −50% or a 50% decrease.)*

(e) Island specialists evolved with limited resources and have narrow diets and nest sites, so they cannot switch when rats eat their eggs or compete for the food. Generalists can use many foods and habitats, so rats affect them less. *(1 point: specialists have narrow niches; 1 point: invasive generalists outcompete or prey on them while generalists can switch.)*

(f) Acceptable: eradicate the rats; control them near nesting areas; improve biosecurity at ports and boats; or move specialist birds to a rat-free island or captive-breeding program. *(1 point: valid solution.)*

(g) More than 30 species. The island is near the mainland and larger than Island C (50 km², 30 species) but smaller than Island D (200 km², 41 species), so its richness should fall between 30 and 41. *(1 point: more than 30; 1 point: justification using Islands C and D or the area and distance patterns.)*

Unit 3 — Populations

Section A — MCQ Answers and Explanations

QB-U3-MCQ001

Answer: C. A specialist depends on a narrow set of resources or conditions, so it does best in stable habitats and suffers when those conditions change. The option describing wide tolerance and success in changing habitats is the definition of a generalist, and the other two attribute generalist or r-selected traits to specialists.

QB-U3-MCQ002

Answer: D. K-selected species invest heavily in a few offspring, mature late, live long, and breed repeatedly. The first distractor lists the r-selected package, while the other two each mix one K-selected trait with traits that do not belong together (few offspring with no care; very many offspring with extensive care).

QB-U3-MCQ003

Answer: B. Very heavy mortality among the youngest individuals followed by relatively good survival of the few that persist is a Type III curve, which is typical of r-selected species that produce huge numbers of offspring. Pairing Type III with K-selected is the most tempting error, but K-selected species produce few offspring and follow Type I or II.

QB-U3-MCQ004

Answer: B. Carrying capacity (K) is set by resources and space and is the largest population the habitat can support over the long term. The first distractor describes biotic potential, and the second confuses K with the point of fastest absolute growth, which occurs near K/2.

QB-U3-MCQ005

Answer: D. When resources are abundant, per-capita reproduction stays high and a larger breeding base adds more rabbits each year, so growth accelerates (the early part of a J-shaped curve). Resources are finite, but they are not yet limiting, so growth does not slow until the population nears K.

QB-U3-MCQ006

Answer: A. A broad base means many children who will enter their reproductive years, which signals rapid growth. Small older groups do not signal decline here; they reflect high mortality and a young, fast-growing population. A stable population would give a more column-like shape.

QB-U3-MCQ007

Answer: A. A TFR of about 2.1 is the replacement level, meaning each generation just replaces itself, so the population is relatively stable. A TFR near 2 is not high enough to cause rapid growth or doubling in 35 years, and it is not low enough to cause a sharp decline.

QB-U3-MCQ008

Answer: C. Population size changes through births and immigration (gains) and deaths and emigration (losses). Each distractor pairs one gain with a loss.

QB-U3-MCQ009

Answer: A. Stage 1 is the pre-industrial stage: birth rates are high, but disease, famine, and poor sanitation keep death rates high as well, so the population grows very slowly. In Stage 2 death rates fall while birth rates stay high, which produces rapid growth.

QB-U3-MCQ010

Answer: D. Overshoot means the population is using resources faster than they are replaced; depletion leads to famine, disease, or conflict and a dieback. Carrying capacity does not increase because the population is large (it often falls after depletion), and competition rises, not falls, during overshoot.

QB-U3-MCQ011

Answer: C. A Type I curve stays high through most of the life span and then drops steeply in old age, as in humans in countries with good healthcare. Constant mortality describes Type II, and very high early mortality describes Type III.

QB-U3-MCQ012

Answer: B. Density-independent factors such as fires, storms, heat waves, and droughts affect the same proportion of a population regardless of its density. Competition for nest sites, disease transmission, and food depletion all become more severe as density rises, so they are density-dependent.

QB-U3-MCQ013

Answer: B. Biotic potential is the highest possible rate of reproduction when resources are unlimited and conditions are ideal. The first distractor is carrying capacity, and the second describes realized (not maximum) reproductive success after environmental resistance has acted.

QB-U3-MCQ014

Answer: C. Species 1 is a specialist and Species 2 is a generalist. When habitat changes, generalists are advantaged because they can use alternative resources, whereas specialists lose the single resource they depend on. The prediction that both decline equally ignores the difference in dietary breadth.

QB-U3-MCQ015

Answer: A. Biotic potential rises with the number of offspring per event and falls as age at first reproduction increases. Species P combines 400 eggs with maturity at 0.4 years, giving by far the highest maximum reproductive rate. Species S is the most tempting alternative, but its litter is 50 times smaller and it matures 2.5 times later than P.

QB-U3-MCQ016

Answer: C. K-selected species have few offspring, mature late, and have low reproductive rates, so losses from an invader are replaced slowly. Many r-selected species rebound quickly, and most invasive species are themselves r-selected. The distractors attribute r-selected traits (fast reproduction, short life span) to the K-selected native.

QB-U3-MCQ017

Answer: D. A K-selected mammal with few young and heavy parental investment has high survival through most of its life span, which gives the Type I shape shown by Curve 2. Curve 1 is the Type III pattern of heavy early mortality, produced by species with huge numbers of unprotected young, and Curve 3 is the straight line on a log axis that represents constant mortality (Type II).

QB-U3-MCQ018

Answer: A. A constant probability of death means a constant fraction of the remaining survivors dies each interval, which plots as a straight diagonal line on a log scale (Type II, Curve 3). Curve 2 shows very low early mortality and a collapse late in life, and Curve 1 shows extremely high early mortality followed by slower losses.

QB-U3-MCQ019

Answer: D. In logistic growth the number of individuals added per year, $rN(1 - N/K)$, is greatest when the population is about half the carrying capacity. Here $K = 600$, so the steepest part of the curve is at about 300 sheep, which is point S. Point P has more sheep, but resource limits have already slowed its growth, and point R is essentially at K, where growth is near zero.

QB-U3-MCQ020

Answer: B. Deer numbers climbed while browse fell, and the herd peaked when browse was nearly gone; the resulting starvation caused dieback that let browse recover slowly. Lower browse means a lower carrying capacity, not a higher one, and the weather records rule out a density-independent cause.

QB-U3-MCQ021

Answer: B. A valid test changes only the independent variable (surface area), keeps starting numbers, nutrients, and light the same, replicates treatments, and tracks the dependent variable (plant count) over time. The other designs vary starting numbers, light, or nutrients along with, or instead of, surface area, so any difference could not be attributed to surface area.

QB-U3-MCQ022

Answer: C. All three cultures grew at nearly the same rate early on, but each leveled off at a different population size that matched its glucose supply, showing that the limiting resource sets carrying capacity. The cultures did not reach the same plateau, and constant temperature rules out temperature, not a density-dependent resource limit.

QB-U3-MCQ023

Answer: C. Country C has by far the largest share of its population in the youngest age groups, so a large cohort will soon enter its reproductive years and births will greatly exceed deaths. Country D also has a broad base, but its younger groups are much smaller relative to its middle-aged groups, so its growth will be slower. Countries A and B have nearly constant or inverted shapes, indicating stable or declining populations.

QB-U3-MCQ024

Answer: B. Country B has a narrow base and its largest groups are middle-aged adults who will move into retirement while fewer young people enter the workforce, so its ratio of retirees to workers will rise the most. Country A is also fairly column-shaped, but it has a larger share of children and a smaller bulge in the middle-aged groups. Country C is very young, so its retiree ratio will stay low.

QB-U3-MCQ025

Answer: D. Education for girls and access to family planning are two of the main factors that lower TFR: they delay the age at first birth and give women more control over the number of children. Fertility is strongly influenced by social and policy factors, and the decrease does not require higher infant mortality; infant mortality typically falls as these changes occur.

QB-U3-MCQ026

Answer: B. Infant mortality is strongly affected by whether mothers have access to good healthcare and adequate nutrition. A higher TFR and more child labor are characteristics of regions with higher, not lower, infant mortality, and total population size does not determine a death rate per 1,000 births.

QB-U3-MCQ027

Answer: D. Doubling time ≈ 70 ÷ growth rate (%) = 70 ÷ 0.7 = 100 years. Multiplying instead of dividing gives 49, using 70 as the answer ignores the rate, and moving the decimal the wrong way gives 10.

QB-U3-MCQ028

Answer: A. RNI = CBR − CDR = 17 − 11 = 6 per 1,000 per year. Converting to a percentage: 6 ÷ 1,000 × 100 = 0.6% per year. Choosing 6% forgets that 6 per 1,000 is 0.6%, and 28 results from adding the two rates.

QB-U3-MCQ029

Answer: D. The percentage infected rose steadily with density (2% to 31%), which is the signature of a density-dependent factor; crowding increases transmission. A density-independent factor would affect the same proportion at every density. The data say nothing about carrying capacity, and the percentages are clearly not equal.

QB-U3-MCQ030

Answer: A. This pattern of falling deaths and still-high births is Stage 2 of the demographic transition model, shown here as Stage X, and it produces rapid growth. In Stage W both rates are high and balanced, in Stage Y the birth rate is falling toward the death rate, and in Stage Z both rates are low.

QB-U3-MCQ031

Answer: A. Developing countries typically have higher infant mortality and more children working, which are linked with higher birth rates. Each distractor pairs one correct characteristic with its opposite.

QB-U3-MCQ032

Answer: C. Malthusian theory holds that population grows geometrically while food production grows arithmetically, so growth is eventually limited by famine, disease, and conflict. The statement about families choosing smaller sizes describes the demographic transition, and the last two reject the central Malthusian premise.

QB-U3-MCQ033

Answer: C. r-selected and K-selected describe the ends of a continuum; many species have traits of both or change reproductive strategy depending on conditions. R-selected species generally have the higher biotic potential and are usually smaller than K-selected species, so the last two statements are reversed or overstated.

QB-U3-MCQ034

Answer: A. Carrying capacity is set by the availability of resources, so a sustained increase in forage raises it. Drought lowers it, cattle compete for the same forage and lower it, and adding bison changes the population, not the capacity of the habitat.

QB-U3-MCQ035
Answer: D. The two points at 90% and 96% enrollment have TFRs of 1.8 and 1.5, both below 2.1. The data show an association across ten countries, so they cannot guarantee a specific effect in any one country. The country with a TFR of 4.6 has only 41% enrollment, and the overall trend is clearly downward, not flat.

QB-U3-MCQ036
Answer: B. Past high fertility created a large cohort of young people who are now having children. Even though each woman now averages only enough children to replace herself and her partner, the number of births stays high because there are so many women of childbearing age, so growth continues for decades (population momentum). A TFR of 2.1 is the replacement level, and TFR is a measure of births per woman, not a measure that omits any group.

QB-U3-MCQ037
Answer: B. Carrying capacity = 150,000 kg ÷ 1,500 kg per elk = 100 elk. Excess = 130 − 100 = 30 elk, and 30 ÷ 100 × 100 = 30% above K. Dividing by the current herd (30 ÷ 130) gives 23%, the herd being 130% of K is not the percent by which it exceeds K, and 77% is K as a percent of the herd.

QB-U3-MCQ038
Answer: C. Doubling time = 70 ÷ 2.5 = 28 years, so 56 years is two doublings: 18 million × 2 × 2 = 72 million. Only one doubling gives 36 million, adding 18 million per doubling gives 54 million (linear thinking), and three doublings would give 144 million.

QB-U3-MCQ039
Answer: C. Total growth = (14 − 9 + 4) per 1,000 = 9 per 1,000 = 0.9% per year, so doubling time = 70 ÷ 0.9 ≈ 78 years. Using only RNI (0.5%) gives 140 years, using 1.8% (adding births and migrants) gives about 39 years, and 8 years results from forgetting to convert per 1,000 to percent.

QB-U3-MCQ040
Answer: A. The yearly increases are 70, 150, 210, 170, 90, 40, 10, 2, and −2, so the largest gain (210) occurs from year 2 to year 3, when the population is about half of K. The population levels off near 790, so K ≈ 790. An estimate of 480 pairs the right interval with the population at the end of that interval, not the plateau.

QB-U3-MCQ041
Answer: D. RNI = (41 − 11) ÷ 10 = 3.0% per year, so doubling time ≈ 70 ÷ 3.0 ≈ 23 years. A very high birth rate with a death rate that has already fallen is Stage 2. In Stage 4 both rates are low, and 230 years results from using 0.3% instead of 3.0%.

QB-U3-MCQ042
Answer: A. Deaths before age 1 = 5,000 − 400 = 4,600, and 4,600 ÷ 5,000 × 100 = 92%. Extremely high early mortality followed by much lower losses is a Type III curve. Choosing 8% reports the percent that survived, not the percent that died, and Type I would require high survival of the young.

QB-U3-MCQ043
Answer: D. Comparing both a generalist and a specialist across replicated cut and uncut plots that differ only in treatment, with equal effort, isolates the effect of cutting on each type. Counting only one species cannot show a difference between types, and comparing sites at different elevations or against another forest and decade confounds the result.

QB-U3-MCQ044
Answer: B. A density-dependent factor affects each individual more strongly as density rises, so eggs per female should fall with crowding as food and space become limited. Identical or random results indicate no density effect, and a steady increase with density would suggest the opposite trend.

Section B — FRQ Solutions

QB-U3-FRQ001
(a) Goat numbers rose from about 110 (2005) to a peak of about 430 (2015), then declined and leveled off at about 310 by 2024; forage fell from about 900 kg/ha to about 375 kg/ha, with the steepest forage loss occurring while the goat population was growing fastest. *(1 point: describes both trends with direction and approximate values)*

(b) If supplemental forage is provided to some winter ranges, then the goat populations in those ranges will be larger (or will have higher survival and more kids per female) than in ranges without supplemental forage, because winter forage limits the population. *(1 point: a prediction that links the independent and dependent variables and is testable)*

(c) *(1 point each)* (i) Independent variable: whether or not a winter range receives supplemental forage pellets (or the amount of pellets). (ii) Dependent variable, any one: number of goats per range; kid-to-adult ratio; winter survival of adults; body mass or condition of goats. (iii) Controlled variable, any one: size and elevation of the winter ranges; amount of natural forage per hectare at the start; absence of hunting; method and timing of the goat counts; predator density; human disturbance.

(d) Any two, *(1 point each)*: replicate (three ranges per treatment, as proposed, or more) so that chance differences average out; assign treatments to ranges that are similar in size, elevation, and starting forage (or assign at random) to avoid bias; run for several winters so that one unusual winter does not decide the outcome; use the same counting method, observers, and time of year in all ranges; include the no-supplement ranges as a control group.

(e) The fed ranges show higher goat numbers, higher winter survival, or more kids per female than the unfed ranges, with a difference larger than the variation among replicates. *(1 point)*

(f) Acceptable solutions, any one with justification *(1 point for a solution, 1 point for the justification)*: regulate the herd size through a limited harvest, relocation, or contraception program until the numbers are at or below the carrying capacity set by forage (about 300 goats in the long term, based on the data); restore or protect alpine meadows by limiting trampling and by reducing other grazing and human disturbance so forage biomass rises; monitor forage each spring and adjust herd targets. Justification: the data show that when the herd was near 430 forage fell steeply and the herd then declined, so keeping the herd at carrying capacity prevents overshoot, forage depletion, and dieback. *(Do not credit permanent supplemental feeding alone; it raises the herd above the natural carrying capacity and does not remove overshoot.)*

QB-U3-FRQ002

(a) The crude birth rate fell from 44 to 19 per 1,000 (a decrease of 25 per 1,000), with most of the decline after 1980. The crude death rate fell from 24 to 7 per 1,000, with most of the decline before 1980 and a nearly flat trend after 2000. *(1 point for each rate, with direction and a quantitative comparison or timing)*

(b) RNI = (CBR − CDR) ÷ 10 = (42 − 13) per 1,000 = 29 per 1,000 = 29 ÷ 10 = **2.9%** per year. *(1 point; units or the percent and correct subtraction)*

(c) Doubling time ≈ 70 ÷ 2.9 ≈ **24 years** (24.1). *(1 point)*

(d) Any two, *(1 point each)*: Rising girls' secondary school enrollment (9% to 84%) is associated with later age at first birth and greater opportunities for women, so women have fewer children. Falling IMR (140 to 17) means parents no longer need to have many births to ensure that some children survive, so desired family size falls. Other factors in the CED (access to family planning, government policy) may be mentioned only if tied to the data, but the explanation must link one of the listed table columns to lower TFR.

(e) Stage 2 *(1 point)*. Justification, any one *(1 point)*: between 1960 and 1980 the death rate had dropped sharply (24 to 13) while the birth rate stayed high (44 to 42), so the gap and the growth rate were large (RNI 2.9%). *(Also accept Stage 2 transitioning toward Stage 3 if the justification is data-based; Stage 1 or Stage 4 earns no credit.)*

(f) The 2040 diagram will have a narrower base (smaller share of children) and a larger share of older adults (a more column-shaped or top-heavy structure), because fertility has fallen to replacement level while the large cohorts born earlier age. *(1 point)*

(g) Any one with justification *(1 point for a reasonable action that addresses an aging population and a connected justification)*: strengthen pension and healthcare systems for the growing number of older adults; raise the retirement age or encourage later retirement to keep the worker-to-retiree ratio balanced; train more elder-care workers; encourage skilled immigration to supplement the workforce.

QB-U3-FRQ003

(a) Net change = births + immigrants − deaths − emigrants = 125,000 + 20,000 − 45,000 − 5,000 = 95,000 people per year. Growth rate = 95,000 ÷ 5,000,000 × 100 = **1.9% per year**. *(1 point for the net change including all four terms; 1 point for the correct percentage)*

(b) RNI = (births − deaths) ÷ population × 100 = (125,000 − 45,000) ÷ 5,000,000 × 100 = 80,000 ÷ 5,000,000 × 100 = **1.6% per year**. (Equivalent: CBR 25 − CDR 9 = 16 per 1,000.) *(1 point; do not credit if migration is included)*

(c) Doubling time ≈ 70 ÷ 1.9 = **37 years** (36.8). Population after one doubling time ≈ 2 × 5.0 million = **10 million** people. *(1 point for the doubling time; 1 point for the population)*

(d) Demand = 5,000,000 people × 0.55 kg/person/day × 365 days/year ÷ 1,000 kg/metric ton ≈ 1,003,750 t ≈ **1.0×10^6 metric tons per year**. *(1 point; units must be shown. Accept 1,003,750 or 1.0 million.)*

(e) Shortfall = demand − production ≈ 1,003,750 − 850,000 ≈ 153,750 t, or about **1.5×10^5 metric tons per year** (about 15% of demand). *(1 point; accept 150,000 to 154,000 t, or a value consistent with part (d))*

(f) Any one *(1 point)*: as incomes rise and girls' education and access to family planning expand, women marry and have children later and have fewer children, lowering birth rates; falling infant mortality lowers the number of births parents need; urbanization raises the cost of raising children; or the country moves from Stage 2 to Stage 3 of the demographic transition.

(g) *(1 point for a reasonable solution; 1 point for a justification that uses the quantitative results)* Examples: raise yields per hectare with improved seed, irrigation, and efficient fertilizer use and reduce post-harvest losses; slow population growth through education for girls and family planning; import grain or diversify the diet. Justification: the shortfall is already about 1.5×10^5 t per year, and if per-capita consumption stays the same, demand will double to about 2.0×10^6 t in about 37 years, which is about 2.4 times current production, so a one-time fix would not be enough; reducing growth lowers future demand while yield improvements raise supply.

QB-U3-FRQ004

(a) Any two *(1 point each)*: Species X loses few individuals early (950 of 1,000 survive to age 1) and most deaths occur late in life (after age 8), whereas Species Y loses most individuals before age 1 (60 of 1,000 survive); Species X still has survivors at age 14 while no Species Y individual is alive at age 10; at age 8, 700 of 1,000 X survive but only 5 of 1,000 Y survive; Species X's curve is nearly flat until about age 8 whereas Species Y's curve plunges in the first year. Each statement must cite or compare the data.

(b) Species Y: (1,000 − 60) ÷ 1,000 × 100 = **94%** died before age 1. Species X: 520 ÷ 1,000 × 100 = **52%** survived to age 10. *(1 point each)*

(c) Species X: Type I (high survival through most of life, deaths concentrated in old age) and K-selected. Species Y: Type III (very high early mortality) and r-selected. *(1 point for both curve types; 1 point for both strategies)*

(d) Because about 94% of Y's young die before age 1, the species must produce huge numbers of eggs so that enough individuals survive to reproduce and replace the parents; large numbers offset the very high juvenile mortality. *(1 point)*

(e) Claim: Species Y will recover faster. Justification, any one for Y and a contrast for X *(1 point for the claim with a Y characteristic; 1 point for the contrast with X)*: Y matures early, produces tens of thousands of eggs, and has a high biotic potential, so the population can rebuild within one or two generations; X produces one young at a time, matures late, and has a low biotic potential, so recovery takes many years. *(Accept a claim that X will recover faster only if the response is scientifically justified and consistent with the data; this is unlikely.)*

(f) Any one with justification *(1 point for a measure with a connected justification)*: set quotas or hunting seasons that protect breeding-age adults and limit total harvest; protect habitat and migration routes; monitor population size and adjust harvest rates. Justification: Species X is K-selected and its adults have the highest survival and produce the offspring, so removing them reduces the population faster than the slowly reproducing population can replace them.

Unit 4 — Earth Systems and Resources

Section A — MCQ Answers and Explanations

QB-U4-MCQ001

Answer: D. Subduction carries water-rich oceanic crust into the mantle, where melting produces magma that rises to build volcanoes on the overriding plate; when the overriding plate is oceanic, the volcanoes form an island arc. A rift valley and a mid-ocean ridge are features of divergent boundaries, where plates move apart, and a transform fault forms where plates slide past each other. The tempting rift-valley option confuses the collision with a divergent setting.

QB-U4-MCQ002

Answer: D. A locked fault resists motion while the plates continue to move, so elastic strain accumulates. When the stress overcomes the friction holding the fault, the rocks suddenly slip and release the stored energy as seismic waves. Plates do not stop moving, and smooth continuous slip releases energy gradually instead of producing a sudden earthquake. Volcanic activity can cause small tremors but is not the cause of fault earthquakes.

QB-U4-MCQ003

Answer: A. Parent material is the original rock or sediment (for example bedrock, glacial deposits, or river-deposited material) that is weathered, and sometimes transported and deposited, to supply the mineral part of a soil. The surface layer of humus is organic matter produced by decomposition, not parent material. Dissolved nutrients are products of weathering and decay, and unweathered bedrock alone is only one possible source, since transported sediment also serves as parent material.

QB-U4-MCQ004

Answer: C. Soil formation is faster where warm temperatures and abundant water speed chemical weathering of parent material and the growth and decomposition of organisms that add organic matter. Cold or dry conditions slow these processes. The cold, wet site has water but low temperatures and long freezing periods that slow chemical reactions and biological activity, so it forms soil more slowly than the warm, humid site.

QB-U4-MCQ005

Answer: B. As water percolates, soil particles trap suspended material, clay and organic matter adsorb many contaminants, and microbes break down some of them, which cleans the water. The ability is limited: soils can be overwhelmed by large pollutant loads, and they do not stop water from recharging groundwater. More runoff would carry pollutants to streams instead of filtering them.

QB-U4-MCQ006

Answer: A. Permeability is the rate at which water moves through a soil, and it depends on the size and connectedness of the pores. Sand has the largest particles and therefore the largest pores, so water drains quickly. Clay has very small pores that slow water movement even though they can hold a lot of water, silt is intermediate, and loam, a blend of all three, drains more slowly than sand.

QB-U4-MCQ007

Answer: B. Dry air is about 78% nitrogen, 21% oxygen, 0.93% argon, and 0.04% carbon dioxide, so the percentages in order identify W, X, Y, and Z. Reversing nitrogen and oxygen is the most tempting error because oxygen is the gas people associate with air, but it is the second most abundant. Carbon dioxide, although important for climate, is a trace gas far less abundant than argon.

QB-U4-MCQ008

Answer: D. The layers are defined by temperature gradients: temperature falls with altitude in the troposphere, rises in the stratosphere, falls again in the mesosphere, and rises in the thermosphere. The proportions of nitrogen and oxygen are nearly constant in the lower layers, so composition does not define them. Air pressure decreases gradually and never abruptly reaches zero, and water vapor is concentrated mainly in the troposphere but does not set the boundaries.

QB-U4-MCQ009

Answer: A. Because Earth rotates, the surface moves at different speeds at different latitudes, and moving air appears to be deflected to the right in the Northern Hemisphere and to the left in the Southern Hemisphere.

Differences in land and ocean heating cause local pressure differences, and the tilt of the axis causes the seasons, but neither deflects moving air in this way.

QB-U4-MCQ010

Answer: B. A divide is the line of higher ground along which precipitation drains to one watershed on one side and to a different watershed on the other. A floodplain is flat land next to a stream that floods, a delta is sediment deposited where a river enters a larger body of water, and a recharge zone is an area where water enters an aquifer.

QB-U4-MCQ011

Answer: C. The tilt of Earth's axis (about 23.5°) points the Northern Hemisphere toward the Sun for part of the orbit, which makes the Sun's rays more direct and lengthens daylight, so insolation increases. Earth is actually closest to the Sun in early January, so distance does not explain the seasons. The Sun's output is nearly constant over a year, and rotation speed does not change with season.

QB-U4-MCQ012

Answer: A. This is a rain shadow: moist air forced up the windward slope cools, condenses, and drops its water as rain or snow, and the drier air then descends on the leeward side, warms, and can hold more water vapor, so clouds and precipitation are rare. Mountains do not absorb water vapor, shade does not explain the pattern, and winds continue over the crest rather than reversing.

QB-U4-MCQ013

Answer: C. El Niño is defined by warmer-than-average surface water across the central and eastern tropical Pacific. Unusually cold water in that region marks La Niña, the opposite phase. El Niño is a Pacific phenomenon, and the warm water shifts toward the east rather than staying only in the western Pacific.

QB-U4-MCQ014

Answer: B. At Boundary II the arrows show Plates Q and R moving toward each other, which is a convergent boundary. Oceanic crust is denser than continental crust, so Plate Q subducts beneath Plate R, forming a trench offshore and a volcanic mountain range with earthquakes on the continent. Reversing the subduction direction ignores the density difference. Boundary I is divergent and Boundary III is transform, which are the settings for the other two descriptions.

QB-U4-MCQ015

Answer: C. Earthquakes occur in the brittle subducting slab as it descends, so foci are shallow near the trench and become deeper farther inland, tracing a plane that dips into the mantle. A vertical plume or a vertical transform fault would not produce a steady increase of focal depth with horizontal distance, and ridge push at a divergent boundary produces only shallow earthquakes.

QB-U4-MCQ016

Answer: A. The B horizon is the zone of accumulation: clay and minerals leached from the upper horizons collect there, so it is rich in clay and low in organic matter, which matches Y (36% clay, 1.2% organic matter). W is the organic-rich surface layer, X is the topsoil (A horizon) mixing humus and mineral particles, and Z, with rock fragments and lower clay, is the weathered parent material (C horizon).

QB-U4-MCQ017

Answer: D. Sample 3 is 60% clay. Clay's tiny pores give it a large total surface area and pore volume that hold water by adhesion, but the pores are so small that water moves through slowly, so water-holding capacity is high and permeability is low. Sample 1 is sandy: it drains quickly and holds little water. Sample 2 (40/40/20) is a loam, with intermediate properties, and silt in Sample 4 has larger particles than clay.

QB-U4-MCQ018

Answer: D. Porosity is the fraction of the soil volume that is pore space, while permeability depends on how large and how connected the pores are. Clay particles are the smallest of the soil particles, so clay has many very small pores that hold water tightly and offer high resistance to flow. The statement that clay particles are larger than sand is false, and clay actually has the greater water-holding capacity.

QB-U4-MCQ019

Answer: A. Water retained = water added − water drained. Soil X retained 200 − 150 = 50 mL, and Soil Y retained 200 − 60 = 140 mL, so Soil Y holds more water. The tempting errors are to read the drained volume as

the retained volume, or to conclude that adding equal volumes makes the soils equal; the same input shows different results because the soils differ in particle size and pore space.

QB-U4-MCQ020

Answer: C. Infiltration rate = depth ÷ time. Field P: 40 mm ÷ 16 min = 2.5 mm/min = 150 mm/h, which is much greater than the 30 mm/h rainfall, so P absorbs all of it. Field Q: 40 mm ÷ 200 min = 0.2 mm/min = 12 mm/h, so runoff = 30 − 12 = 18 mm/h. The claim that Field Q's infiltration is negligible is wrong, because 12 mm/h is not zero.

QB-U4-MCQ021

Answer: A. Sandy soil drains rapidly and holds little water, so a large weekly application mostly percolates below the roots and is wasted while the root zone dries out. Smaller, more frequent applications keep the water within reach of the roots. Larger applications would increase losses, and sand has a lower, not higher, water-holding capacity than loam or clay-rich soil.

QB-U4-MCQ022

Answer: B. In layer X temperature rises from about −56°C to about −2°C as altitude increases from 12 km to 50 km, because ozone concentrated in this layer absorbs ultraviolet radiation and warms the surrounding air. In W and Y temperature falls with altitude, and in Z temperature rises because molecules there absorb very high-energy solar radiation, not because of the ozone layer.

QB-U4-MCQ023

Answer: C. The arrows at X show air sinking. Descending air is compressed and warms, so its relative humidity falls and clouds dissipate, which creates the dry conditions of many deserts near 30° latitude. Rising air at the equator (W) and near 60° (Y) cools, condenses, and produces rain. Cold air at the surface and ocean winds do not explain the dryness at X.

QB-U4-MCQ024

Answer: B. Between the equator and about 30° N, surface air flows toward the equator (south). Earth's rotation deflects this moving air to the right in the Northern Hemisphere, so it moves toward the southwest, which means the wind blows from the northeast. These are the northeast trade winds. Winds from the southeast occur in the Southern Hemisphere, where deflection is to the left.

QB-U4-MCQ025

Answer: D. The most intense insolation arrives at the equator, heating the surface and the air above it. Warm, moist air is less dense and rises; as it rises it cools, water vapor condenses, and heavy rain falls. Sinking air is dry, the Coriolis effect is weakest (zero) at the equator, and mountains do not circle the equator.

QB-U4-MCQ026

Answer: A. The field drains into the eastern tributary between Sites 2 and 3. Water in streams flows downstream, so only sites downstream of that point can receive the nitrate: Site 3 (below the field on the same tributary) and Site 4 (below the confluence). Site 2 is upstream of where the runoff enters, and Site 1 is on a different tributary that joins the main stream at the confluence, upstream of Site 4.

QB-U4-MCQ027

Answer: B. Without vegetation and the litter layer, rainfall hits the soil directly, infiltration decreases, and more water runs quickly off the steep slopes, which raises peak flow and carries eroded soil into the stream. Bare soil does not absorb more water than forest soil, so recharge would decrease, and the loss of roots makes erosion worse, not better, so the water becomes muddier.

QB-U4-MCQ028

Answer: B. Area = 250 km^2 × (1,000 m)^2 per km^2 = 2.5 × 10^8 m^2. Precipitation depth = 1,000 mm = 1.0 m, so the precipitation volume is 2.5 × 10^8 m^2 × 1.0 m = 2.5 × 10^8 m^3. Runoff = 0.35 × 2.5 × 10^8 m^3 = 8.75 × 10^7 m^3 ≈ 8.8 × 10^7 m^3. The value 2.5 × 10^8 m^3 is the total precipitation, not the runoff, and 7.1 × 10^8 m^3 comes from dividing by 0.35.

QB-U4-MCQ029

Answer: A. The peak in U is 30 m^3/s at 4 h, compared with 9 m^3/s at 6 h in R, which is 30 ÷ 9 ≈ 3.3 times higher and 2 h earlier. Impervious surfaces prevent infiltration and send water rapidly to the stream, creating a sharper, earlier, higher hydrograph peak, while forest soil absorbs water and releases it slowly. At 10 h and 12 h, U's discharge is equal to or lower than R's, so U does not drain more slowly.

QB-U4-MCQ030

Answer: D. At the Northern Hemisphere summer solstice the North Pole is tilted toward the Sun, giving the hemisphere its most direct rays and longest days. At position Z the axis leans toward the Sun, and the Northern Hemisphere is mostly in the lit half. At X the axis leans away (Northern Hemisphere winter solstice), and W and Y are the equinoxes, when the axis is tilted neither toward nor away from the Sun.

QB-U4-MCQ031

Answer: C. Near the equator the Sun is nearly overhead, so a beam of sunlight covers a small area; at 60° N the beam arrives at a lower angle, so the same energy is spread over a larger area and travels through a longer path in the atmosphere. Differences in distance from the Sun are negligible, and at the equinox every latitude has about 12 hours of daylight. Air temperature does not determine how much radiation arrives.

QB-U4-MCQ032

Answer: B. The tilt of Earth's axis changes day length little near the equator and more at higher latitudes, so P (about 12 hours all year) is nearest the equator. City R has its longest days in December and shortest in June, which is the pattern of the Southern Hemisphere, where December is summer. City Q has long days in June and short days in December, which fits a Northern Hemisphere mid-latitude city.

QB-U4-MCQ033

Answer: D. Precipitation increases from the coast (W, 900 mm) to the windward slope (X, 2,300 mm) and then drops sharply behind the crest (Z, 350 mm). That pattern shows orographic lifting and cooling of moist air on the windward side, followed by descending, warming, drier air on the leeward side. A steady decline with distance from the ocean would not show X wetter than W. Elevation alone does not explain Z, and air rises, not sinks, on the windward slope.

QB-U4-MCQ034

Answer: D. Water has a high heat capacity, so the ocean warms slowly in summer and cools slowly in winter, and marine air blowing onshore keeps coastal temperatures from changing as much over the year. Inland cities respond quickly to changes in insolation and show large seasonal ranges. Ocean air is not always warmer than land air, and distance from the ocean does not change the intensity of insolation at a given latitude.

QB-U4-MCQ035

Answer: A. A +2.3°C anomaly indicates an El Niño. Weaker trade winds let warm surface water spread east and deepen, so upwelling brings little nutrient-rich cold water to the surface, phytoplankton decline, and the food web supporting anchovies collapses. Warm water holds less, not more, dissolved oxygen, El Niño weakens (not strengthens) the trade winds, and the water in Year 3 was warmer, not colder.

QB-U4-MCQ036

Answer: B. Speed = distance ÷ age: 300 km ÷ 6.0 million years = 50 km per million years = 5 cm/yr (300 km = 3×10^7 cm; 3×10^7 cm ÷ 6×10^6 yr = 5 cm/yr), and islands C and D give the same rate. Age = 1,600 km ÷ 50 km per million years = 32 million years. The other choices differ by factors of 10 or combine a correct rate with an incorrect age.

QB-U4-MCQ037

Answer: C. Continental crust is relatively low in density, so when two continental plates converge neither sinks easily. Instead the crust is folded, thrust upward, and thickened, building high mountains with strong earthquakes, but there is no subducting slab to produce magma and volcanoes. Collision zones are convergent, not transform or divergent, and an extended subduction process would eventually produce volcanoes.

QB-U4-MCQ038

Answer: D. In a controlled experiment only the independent variable (ground cover) should differ; here slope also differs, which is a confounding variable because steeper slopes usually erode more, making the result harder to interpret. Using the same slope for both trays isolates the effect of the grass. Different rain durations or different soil types would add confounding variables rather than remove one, and sediment mass, not runoff volume, is the measure of erosion.

QB-U4-MCQ039

Answer: A. In the stratosphere, warm air lies above cooler air because ozone absorbs ultraviolet radiation, which makes the layer stable and resistant to vertical mixing; the air also contains very little water, so rain and

snow do not wash particles out. The troposphere, in contrast, mixes strongly and has frequent precipitation. The stratosphere has little water vapor, little convection, and gravity there is almost the same as at Earth's surface.

QB-U4-MCQ040

Answer: D. Without rotation there would be no Coriolis effect, so moving air would not be deflected. Air heated at the equator would rise and move poleward aloft, cool, sink near the pole, and return along the surface toward the equator, so surface winds would blow directly from north to south in the Northern Hemisphere. The current three-cell pattern with east–west components depends on rotation, and heating would still be uneven.

QB-U4-MCQ041

Answer: C. June: z = 40° − 23.5° = 16.5°, so cos z = 0.959. December: z = 40° + 23.5° = 63.5°, so cos z = 0.446. The ratio is 0.959 ÷ 0.446 ≈ 2.1. The value 0.47 is the inverse of the ratio, 1.3 is the ratio of cos 16.5° to cos 40° (June compared with an equinox), and 3.8 is the ratio of the angles, not of their cosines.

QB-U4-MCQ042

Answer: C. At 45° N the prevailing winds are the westerlies, which blow from west to east. City P receives air that has passed over the ocean, which keeps winters mild and summers cool, while City Q receives air that has traveled over the land mass, which cools strongly in winter and heats in summer. Both cities are at the same latitude, trade winds blow at lower latitudes, and a rain shadow affects precipitation more than temperature range.

QB-U4-MCQ043

Answer: B. La Niña features unusually cold surface water in the eastern tropical Pacific, strong trade winds that pile warm water in the west, heavy rain over Indonesia, and dry conditions along the Peruvian coast. Year X matches all three (−1.6°C, 35% of normal Peruvian rainfall, 135% in Indonesia). Year W shows the opposite, a strong El Niño, Year Y is near normal, and Year Z is a weak warm event.

QB-U4-MCQ044

Answer: C. During El Niño the trade winds weaken, so the warm surface water that normally accumulates in the western Pacific moves east. Warm water deepens the warm surface layer off South America, so upwelling of cold, nutrient-rich water declines, and rising air and heavy rain move toward the central and eastern Pacific. The sequence that begins with strengthening trade winds describes La Niña, and the remaining sequences reverse or contradict the cause-and-effect order.

Section B — FRQ Solutions

QB-U4-FRQ001

(a) Soil erosion from the orchard (loss of topsoil) and the resulting sediment pollution (turbidity) of the stream. *(1 point: soil erosion or sediment runoff/stream sedimentation)*

(b) Example: *If* a grass cover is planted between the tree rows, *then* the mass of sediment lost from the block with grass will be less than from the block with bare soil during the same rain events. *(1 point: testable if–then statement that predicts the direction of the effect of cover on soil loss)*

(c) Independent variable: ground cover between the tree rows (grass versus bare soil). Dependent variable: soil loss, measured as the dry mass of sediment caught in the sediment trap (for example kg per rain event or per hectare); stream turbidity is also acceptable if measured the same way for each block. *(1 point for each variable; the dependent variable must be something measured and match the hypothesis)*

(d) Blocks 1 and 2: they have the same slope (6%) and the same soil (loam), so only the ground cover will differ between them. Slope and soil texture strongly affect erosion, so comparing blocks with different slopes or soils (for example Block 1 with Block 3 or 4) would confound the results. Blocks 3 and 4 have the same slope but different soil, so they are not a valid pair. *(1 point: Blocks 1 and 2; 1 point: reason that slope and soil type are the same, so cover is the only difference)*

(e) Sow grass between the rows in one of the two blocks (for example Block 2) and keep the other (Block 1) bare as the control. Install identical sediment traps at the downhill edge of each block and a rain gauge at the site. After each rain event, collect the sediment from each trap, dry it in an oven, and weigh it, and record the rainfall amount. Repeat for many storms (or one full rainy season, ideally for more than one year), and, if possible, use more than one trap per block, so that the results are reliable. *(1 point: grass in one block with the*

other as bare control and sediment traps with mass of dry sediment measured; 1 point: replication such as many storms or traps, with rainfall recorded)

(f) Support: the sediment mass from the grass block is consistently lower than from the bare block across storms, by an amount larger than the storm-to-storm variation (for example confirmed by a statistical test such as a t-test), and the difference holds when rainfall amount is similar. *(1 point: lower soil loss in the grass block than in the bare block, consistently across rain events)*

(g) Any one with a justification, for example: planting rows along the contour (contour farming), which slows runoff and increases infiltration; terracing steeper blocks (3 and 4), which shortens the slope length; adding mulch or leaving crop residue, which protects the soil from raindrop impact; planting a riparian buffer strip of vegetation between the orchard and the stream, which traps sediment before it reaches the water. *(1 point: valid practice with justification linked to reducing erosion or sediment reaching the stream)*

QB-U4-FRQ002

(a) El Niño years: 3 and 9. La Niña years: 5, 6, and 11. *(1 point: all five years correct; years 8 and 10 are not El Niño years because their anomalies are below +1.0°C)*

(b) El Niño mean = (3.2 + 1.9) ÷ 2 = 5.1 ÷ 2 = 2.55 ≈ 2.6 million metric tons. La Niña mean = (8.7 + 9.4 + 9.0) ÷ 3 = 27.1 ÷ 3 = 9.03 ≈ 9.0 million metric tons. *(1 point for each correct mean with units; work must be shown)*

(c) Percent decrease = (9.03 − 2.55) ÷ 9.03 × 100 = 6.48 ÷ 9.03 × 100 = 71.8 ≈ 72%. *(1 point: correct value; accept 71–72% when rounded means are used)*

(d) During El Niño the trade winds weaken and warm surface water spreads eastward toward South America, deepening the warm surface layer. Upwelling then brings up warm water instead of cold, nutrient-rich water, so the supply of nutrients to the surface falls, phytoplankton production declines, and the zooplankton and anchovies that depend on it decline or move away, so landings are lower. *(1 point: weaker trade winds, warm water or reduced upwelling; 1 point: connects reduced nutrients/phytoplankton to fewer anchovies)*

(e) Rainfall in northern Peru is much higher in El Niño years (640 mm and 910 mm, compared with 40–70 mm in La Niña years), which can cause flooding. Cause: the warm ocean water off the coast heats and moistens the air above it, so the air rises, cools, and condenses into heavy rain; the zone of rising air and rainfall that is normally over the western Pacific shifts east. *(1 point: higher rainfall in El Niño years supported by data; 1 point: a valid cause such as warm SST evaporating more water and rising air, or an eastward shift of the rainfall zone)*

(f) Any one, for example: diversify the catch or income (other fish species, aquaculture, tourism); set fishing quotas based on forecasts to avoid overfishing a stock that is already stressed; create savings or insurance funds for low-catch years; use El Niño forecasts to plan harvest, processing, and employment. *(1 point: a realistic solution that reduces economic risk or stress on the fishery)*

(g) Claim: Year 9 was the stronger El Niño. Justification (any two): its SST anomaly was larger (+2.4°C compared with +1.9°C), its anchovy landings were lower (1.9 compared with 3.2 million metric tons), and its rainfall was higher (910 compared with 640 mm). *(1 point: Year 9 with at least two correct pieces of data)*

QB-U4-FRQ003

(a) Area = 180 km^2 × 10^6 m^2/km^2 = 1.8 × 10^8 m^2. Precipitation depth = 1,100 mm = 1.1 m. Precipitation volume = 1.8 × 10^8 m^2 × 1.1 m = 1.98 × 10^8 m^3. Runoff = 0.40 × 1.98 × 10^8 m^3 = 7.92 × 10^7 m^3 per year (≈ 7.9 × 10^7 m^3). *(1 point: correct set-up with unit conversions of area and depth; 1 point: correct answer with units)*

(b) Discharge = volume ÷ time = 7.92 × 10^7 m^3 ÷ 3.15 × 10^7 s = 2.51 m^3/s (≈ 2.5 m^3/s). *(1 point: volume divided by seconds in a year; 1 point: correct answer with units)*

(c) Volume in liters = 7.92 × 10^7 m^3 × 1,000 L/m^3 = 7.92 × 10^{10} L. Mass = 7.92 × 10^{10} L × 150 mg/L = 1.188 × 10^{13} mg. In metric tons: 1.188 × 10^{13} mg ÷ 10^9 mg/t = 1.19 × 10^4 t (about 11,900 metric tons). *(1 point: multiplies runoff volume in liters by concentration; 1 point: correct unit conversion and answer)*

(d) Cropland area = 0.30 × 180 km^2 = 54 km^2 = 5.4 × 10^7 m^2. Soil volume lost = 1.188 × 10^4 t ÷ 1.3 t/m^3 = 9.14 × 10^3 m^3. Depth = 9.14 × 10^3 m^3 ÷ 5.4 × 10^7 m^2 = 1.69 × 10^{-4} m = 0.17 mm per year. *(1 point: converts mass to volume with the bulk density and finds the cropland area; 1 point: correct depth in mm)*

(e) Not sustainable. Soil is lost at 0.17 mm per year, about 3.4 times faster than the 0.05 mm per year at which it forms (0.169 ÷ 0.05 ≈ 3.4), so the soil is being depleted, which will reduce productivity over time. *(1 point:*

states not sustainable and compares the loss rate with the formation rate using the calculated value; accept 3.4 or 3.5 times)

(f) Any one with an explanation, for example: plant cover crops or leave crop residue (protects soil and slows runoff); no-till or conservation tillage (leaves the soil structure and residue in place); contour plowing or terracing (slows runoff along slopes); grass or tree buffer strips along the creek (trap sediment before it reaches the water). *(1 point: valid practice with an explanation of how it reduces sediment)*

QB-U4-FRQ004

(a) Any two supported differences: Harbor receives far more precipitation (1,730 mm per year compared with 310 mm at Plain) and has a pronounced wet season from October to March while Plain's precipitation is low and fairly even all year; Harbor has mild temperatures with a small annual range (7–18°C), while Plain has cold winters below freezing and hot summers up to 24°C (a range of 26°C); Plain has a freezing winter, Harbor does not. *(1 point for each difference that is described with data from the climatograms; a difference in temperature range or in total precipitation counts, but two statements about the same variable count only once)*

(b) Harbor total = 260 + 215 + 200 + 130 + 85 + 45 + 20 + 25 + 70 + 150 + 240 + 290 = 1,730 mm. Plain total = 30 + 25 + 30 + 28 + 35 + 30 + 15 + 12 + 18 + 22 + 32 + 33 = 310 mm. Percent less = (1,730 − 310) ÷ 1,730 × 100 = 1,420 ÷ 1,730 × 100 = 82.1 ≈ 82%. *(1 point: both annual totals found; 1 point: correct percent)*

(c) Moist air from the ocean is forced to rise up the windward slope of the mountains (orographic lift), where it cools, water vapor condenses, and most of the moisture falls as precipitation at Harbor and on the slopes. The air that crosses the crest is dry and descends the leeward side, where it is compressed and warms, so its relative humidity falls and clouds dissipate. Plain lies in this rain shadow. *(1 point: air rises, cools, and loses moisture as precipitation on the windward side; 1 point: dry air descends and warms on the leeward side)*

(d) Land heats and cools quickly, while water has a high heat capacity and warms and cools slowly. Harbor is moderated by air from the ocean, so its temperatures change little, while Plain is far from the ocean and its temperatures follow the changes in solar radiation, which gives hot summers and cold winters. Plain also lies in the lee of the mountains, so the ocean air that reaches it has less influence on its temperatures, and clear, dry skies increase night-time cooling. *(1 point: water's slow heating and cooling or the moderating effect of the ocean on Harbor; 1 point: Plain has a continental location with more extreme temperatures)*

(e) The prevailing westerlies (the westerly winds that blow from west to east at mid-latitudes). *(1 point: westerlies or prevailing westerly winds)*

(f) Any one, for example: grow drought-tolerant crops or use dry-land farming methods; use drip irrigation or other efficient irrigation; mulch or conserve soil moisture with residue; collect and store runoff in ponds or reservoirs; use no-till practices to reduce evaporation. *(1 point: a realistic water-conservation or supply solution suited to a dry climate)*

Unit 5 — Land and Water Use

Section A — MCQ Answers and Explanations

QB-U5-MCQ001

Answer: A. Because each family captures the whole benefit of an extra goat but bears only a fraction of the cost, adding goats is rational for each family and ruinous for the group. The prediction of spontaneous restraint is the opposite of the concept: the tragedy arises exactly when no rules or ownership limit use of the shared resource.

QB-U5-MCQ002

Answer: B. Removing the canopy raises stream temperature through direct sunlight, and removing roots and litter exposes soil to rain, increasing erosion, sediment, and runoff that can cause flooding. Forest floors infiltrate water better than bare soil, so the claim of increased infiltration is reversed.

QB-U5-MCQ003

Answer: A. Tilling breaks up soil structure and removes the plant cover that holds soil in place, so wind and rain can carry it away. Perennial strips, surface residue, and tree rows all protect the soil and are used as conservation practices.

QB-U5-MCQ004

Answer: D. Agriculture, mostly irrigation, accounts for roughly 70% of human freshwater consumption on a global scale. The remaining share is divided among industry and household use.

QB-U5-MCQ005

Answer: A. Pesticide use acts as artificial selection: individuals that already carry resistance alleles survive, reproduce, and raise the frequency of resistance. Acquired tolerance is not inherited, and the insecticide does not direct which mutations occur.

QB-U5-MCQ006

Answer: C. The description is a concentrated animal feeding operation (CAFO). CAFOs are economically efficient and cheap for consumers, but concentrated manure can reach waterways if poorly managed, and routine antibiotic use can promote antibiotic resistance. The other statements describe pasture-based systems or are false.

QB-U5-MCQ007

Answer: C. Overfishing can make a species extremely scarce, which can reduce biodiversity in the aquatic system and harm people who depend on fishing for food and income. As stocks shrink, catch per unit of effort falls, and genetic diversity generally declines rather than rises.

QB-U5-MCQ008

Answer: D. Overburden is the layer of soil and rock stripped away to expose ore. The material left after minerals are removed from the ore is called tailings, and slag is the by-product of smelting.

QB-U5-MCQ009

Answer: D. Pavement is an impervious surface: it prevents water from reaching the soil, so infiltration drops and runoff rises, increasing flooding. The same rainfall does not mean the same runoff, because the ground surface determines where the water goes.

QB-U5-MCQ010

Answer: A. Ecological footprints compare resource demands and waste production of a person or society, usually expressed as the area of productive land and water needed. Population size, renewable energy, and species loss may be related indicators, but none is the footprint itself.

QB-U5-MCQ011

Answer: A. Permeable pavement and trees allow water to soak into the ground, increasing infiltration and reducing runoff. Faster drains and concrete channels move water off the site more quickly, and paving grass removes infiltration area.

QB-U5-MCQ012

Answer: B. IPM combines methods such as biocontrol, intercropping, crop rotation, natural predators, and limited chemical use to control pests while reducing risks to wildlife, water, and people. Scheduled broad-

spectrum spraying, chemical-only control, and monoculture are conventional approaches that IPM seeks to reduce.

QB-U5-MCQ013

Answer: D. Terraces and contour plowing slow runoff and keep soil on the slope. Rows running downhill act as channels for water, bare soil is easily eroded, and perennial plants help hold soil in place.

QB-U5-MCQ014

Answer: D. The concept applies to a shared, open-access resource that individuals use for their own gain with no effective limits. The woodlot is privately owned, so the owner bears the full cost of her choice. The quota systems are management responses that prevent the tragedy rather than examples of it.

QB-U5-MCQ015

Answer: C. Carbon stored = 150 t C/ha × 50 ha = 7,500 t C. Carbon released = 0.60 × 7,500 = 4,500 t C. Converting to CO_2 multiplies by 44/12: 4,500 × 44/12 = 16,500 t ≈ 1.65×10^4 t CO_2. The value 4.5×10^3 stops before converting carbon to CO_2, and 2.75×10^4 assumes that all 7,500 t of carbon is released.

QB-U5-MCQ016

Answer: C. The Green Revolution aimed to raise food production using new crop varieties along with fertilization, irrigation, pesticides, and mechanization. These inputs depend on fossil fuels for machinery, fertilizer manufacture, and pumping, so energy use rose. The other statements describe lower-input farming, the opposite of the Green Revolution strategy.

QB-U5-MCQ017

Answer: A. Yield rose about 2.6 times (3.1 ÷ 1.2), while fertilizer rose 12 times, irrigated share more than quadrupled, and tractor density rose seven times. The data show a yield gain tied to large increases in inputs. The statement that yield rose faster than fertilizer is false, and the irrigation and tractor statements contradict the table.

QB-U5-MCQ018

Answer: D. Slash-and-burn farming oxidizes stored plant carbon to CO_2, and the ash gives only a temporary supply of nutrients that heavy tropical rain leaches away and crops quickly deplete. That is why farmers move on. A forest canopy and litter protect soil better than young crops, so erosion usually increases.

QB-U5-MCQ019

Answer: D. Feature W is a crust of salts left when irrigation water evaporates (salinization), and feature X shows the water table close enough to the surface that roots sit in saturated soil, so they cannot take in oxygen (waterlogging). The other options describe processes that do not match a pumped-groundwater, flood-irrigated inland field.

QB-U5-MCQ020

Answer: B. Furrow: applied = 6,000 ÷ (1 − 1/3) = 9,000 m^3/ha. Drip: applied = 6,000 ÷ 0.95 ≈ 6,316 m^3/ha. Savings = (9,000 − 6,316) × 25 ha ≈ 67,000 m^3 = 6.7×10^4 m^3. The value 4.3×10^4 m^3 wrongly applies the loss percentages to the crop requirement instead of the applied water, and 7.5×10^4 m^3 counts only the furrow losses.

QB-U5-MCQ021

Answer: D. Flood irrigation wastes about 20% of the water and can leave too much water sitting in the soil. Spray systems lose a quarter or less but need energy for pumping. Furrow irrigation loses about one-third, and drip irrigation, the most efficient at about 5% loss, is costly and therefore not often used.

QB-U5-MCQ022

Answer: D. Field A, sprayed 6 times per season, reached about 92% resistance by season 5, while Field B, sprayed twice, was still at 24%. This is stronger selection pressure acting on existing variation. Field B's resistant share is also rising (to 70%), so evolution is occurring there too, and the insecticide does not create the mutations.

QB-U5-MCQ023

Answer: A. Genetically engineered pest and disease resistance can raise yields, but widespread use of one variety lowers genetic diversity in the crop, so a disease or pest that overcomes the variety can damage most of

the region's crop. Pests can still evolve resistance to engineered traits, and nothing about the trait requires extra fertilizer.

QB-U5-MCQ024

Answer: A. Overgrazing removes plant cover, compacts soil, and drives erosion; matching livestock numbers to what the land can regenerate and rotating herds lets vegetation recover. More animals worsen the damage, and heavy irrigation in an arid area risks salinization and aquifer depletion. Removing fences concentrates animals unpredictably, defeating rotation.

QB-U5-MCQ025

Answer: A. Spawning stock biomass fell steadily from 900 to 330 thousand t between 1960 and 1985 even as landings rose, which can happen only if catching effort and efficiency rise. After 1985 the stock was too small to support large catches, and biomass kept falling, so no rebuilding occurred.

QB-U5-MCQ026

Answer: C. When easily accessible coal runs low, remaining coal must be reached through subsurface mining, which is very expensive. Subsurface mining does not eliminate methane release, and the shift reflects scarcer, costlier reserves, not a larger low-cost supply.

QB-U5-MCQ027

Answer: B. Impervious surfaces and storm drains deliver water to streams quickly, giving a short lag time and a high, sharp peak, as in Y. The more gradual response of X reflects infiltration and slower subsurface flow. The stem states that both watersheds received the same storm, so rainfall amount cannot explain the difference.

QB-U5-MCQ028

Answer: B. Country R's footprint (1.4) exceeds its biocapacity (0.9), so it runs a deficit even though its footprint is small. Country Q's footprint (4.2) is less than its biocapacity (4.9), so it has no deficit, and Country S has a large biocapacity but a modest footprint (2.9). Country P's footprint is 8.1 against a biocapacity of 3.2, so it is far above, not below, its biocapacity.

QB-U5-MCQ029

Answer: C. Earths needed = footprint ÷ biocapacity = 5.6 ÷ 1.6 = 3.5. Inverting the ratio gives 0.29, subtracting gives 4.0 (a difference in hectares, not a number of planets), and multiplying gives 9.0, which has no meaning here.

QB-U5-MCQ030

Answer: B. A valid comparison changes only one variable (surface type) and keeps area, slope, and rainfall the same, with runoff volume as the measured response. The other designs confound surface with storm, site, or slope, or lack a comparison group.

QB-U5-MCQ031

Answer: D. No-till had the lowest soil loss (4.5 t/ha/yr). Reduction = (18.0 − 4.5) ÷ 18.0 = 0.75, or 75%. Strip cropping reduced loss by (18.0 − 9.0) ÷ 18.0 = 50% and contour plowing by about 35%, both less than no-till. The value 25% is the no-till loss as a fraction of the conventional loss, not the reduction.

QB-U5-MCQ032

Answer: C. Aquaculture is efficient: it needs only small areas of water and little fuel compared with fishing fleets. High density actually increases disease, and farmed fish are commonly fed and release waste into the surrounding water.

QB-U5-MCQ033

Answer: C. Aquaculture can contaminate wastewater, and dense populations can develop diseases that are transmitted to wild fish; escaped fish may also compete or breed with wild fish. Each other option pairs at least one false statement with a possibly true one (for example, crowding raises, not lowers, disease risk).

QB-U5-MCQ034

Answer: B. Prescribed burns reduce the fuel that accumulates on the forest floor, lowering the chance that a natural fire becomes large and severe. They do not remove wildfire risk entirely, and they are not used to clear land or increase stored carbon.

QB-U5-MCQ035
Answer: C. Reforestation, sustainable harvesting, reuse of wood, and IPM with removal of affected trees reduce deforestation and protect against insects and pathogens. The alternatives increase forest loss, depend on heavy chemical use, or allow the outbreak to spread.

QB-U5-MCQ036
Answer: D. Only the fertilizer rate differs among the plots, and the replicated levels (including a zero control) let her attribute differences in runoff nitrate to fertilizer. The other designs vary rainfall, fertilizer type, slope, or soil at the same time, or lack a fertilizer comparison, so the effect of fertilizer cannot be isolated.

QB-U5-MCQ037
Answer: C. Net decline = 0.9 − 0.1 = 0.8 m/yr, so time = 24 m ÷ 0.8 m/yr = 30 years. Dividing by pumping alone gives 27 years (ignoring recharge), dividing by the sum of the rates gives 24 years, and dividing by recharge alone gives 240 years; none of these use the net rate.

QB-U5-MCQ038
Answer: B. Beef: 20 g × (50 kg ÷ 100 g) = 10 kg per serving; chicken: 20 g × (6 kg ÷ 100 g) = 1.2 kg per serving. Savings = 8.8 kg × 800 = 7,040 kg ≈ 7.0 t. The value 8.0 t ignores the emissions from the chicken, and 35 t forgets to scale from 100 g of protein to 20 g.

QB-U5-MCQ039
Answer: A. Pasture systems avoid concentrated manure and grain feeding and can improve soil with rotational grazing, but the CED notes they need more land (raising costs), carry runoff and erosion risks if animals are crowded, and may still use antibiotics. Cattle are ruminants and produce methane in any system, and neither extreme claim holds.

QB-U5-MCQ040
Answer: C. Ore needed = copper produced ÷ grade = 500 t ÷ 0.0025 = 200,000 t = 2.0×10^5 t (it was 500 ÷ 0.005 = 1.0×10^5 t at the higher grade). Halving the grade doubles the ore processed, and nearly all of it becomes tailings, which is why lower-grade ores increase resource use, waste, and pollution.

QB-U5-MCQ041
Answer: A. Urban depletion of coastal groundwater lowers the freshwater head so the saltwater boundary moves inland (saltwater intrusion). Impervious surfaces reduce, not increase, recharge, and the other two mechanisms are not supported by the described pumping increase.

QB-U5-MCQ042
Answer: B. Sustainable yield is the amount that can be taken without reducing the supply, and the curve shows the most the stock can regenerate is about 125 t/yr (at about 500 t). A fixed harvest of 150 t/yr is above the curve at every stock size, so the stock shrinks each year. The 725-ton equilibrium belongs to a harvest of 100 t/yr, not 150.

QB-U5-MCQ043
Answer: B. Random assignment of ladybug and no-ladybug treatments to similar benches, with all else held constant and repeated counts, isolates the effect of the predator. The other plans start with unequal aphid numbers, compare different years, or add a second treatment (insecticide) that confounds the result.

QB-U5-MCQ044
Answer: B. Limestone is basic and raises pH, and green manure from nitrogen-fixing legumes adds nitrogen and organic matter. Sulfur would lower pH further, repeated corn drains nitrogen without replacing it, and terracing and contouring conserve soil but do not change fertility.

Section B — FRQ Solutions

QB-U5-FRQ001
(a) Overgrazing (stocking beyond the land's ability to regenerate vegetation) leading to reduced plant cover, soil erosion, compaction, and risk of desertification. *(1 point: overgrazing, or erosion/desertification caused by it)*

(b) Example: "If cattle are grazed in rotation among paddocks, then plant cover (percent) at the end of the season will be higher than in continuously grazed paddocks." Equivalent acceptable: soil loss or bare ground will be lower under rotation. *(1 point: testable, compares the two systems, and states a direction)*

(c) *(1 point each)* (i) Independent variable: grazing system (continuous vs. rotational). (ii) Dependent variable, any one: percent plant cover, plant biomass (g/m²), amount of bare ground, soil compaction, sediment in the creek, species richness. (iii) Two controlled variables (both required for the point), any two: total cattle-days per hectare (the same overall grazing pressure, averaging 2 cattle/ha), paddock size, soil type, slope, starting vegetation, rainfall, cattle breed. *(Point for (iii) only if both are correct. Grazing schedule is the independent variable and is not an acceptable control.)*

(d) Place randomly located quadrats (for example, 1 m² frames) in each paddock, estimate or count percent cover in each (or clip and weigh biomass), and average the results per paddock. Repeat at the start and at set intervals (for example, monthly through the grazing season and again in the following year). *(1 point: a specific sampling method with replicated or random sampling; 1 point: repeated measurement over time, with a stated frequency)*

(e) The three rotational paddocks have higher average plant cover (or lower bare ground) than the three continuous paddocks, and the difference is larger than the variation among paddocks within each group. *(1 point: higher cover or lower bare ground/erosion under rotation, consistent with the hypothesis)*

(f) Measure turbidity or total suspended solids (or sediment trapped in a collector at the lower edge of each paddock) in runoff entering the creek after storms; expect lower turbidity or sediment from the rotationally grazed paddocks. *(1 point: names a creek measurement and the expected lower value for rotation)*

(g) Any one solution with an advantage: reseed with native grasses (rapid cover, holds soil); build contour trenches or small check dams in gullies (slow runoff and trap sediment); fence the creek bank with a vegetated buffer strip (filters runoff, reduces sediment); reduce herd size (fewer animals matches the land's carrying capacity). *(1 point: a suitable solution with a matching advantage)*

QB-U5-FRQ002

(a) Saturated thickness decreased steadily every decade, from 62 m in 1960 to 27 m in 2020 (a drop of 35 m, about 56%), with the decline slowing after 2000. *(1 point: describes a decrease with some supporting values; direction alone with no data is acceptable only if the decrease is clearly stated)*

(b) Rate = (58 m − 35 m) ÷ (2000 − 1970) = 23 m ÷ 30 yr ≈ 0.77 m/yr (0.8 m/yr). *(1 point: correct value with units; allow 0.77 or 0.8)*

(c) Pumping, although lower, still removed far more water (1,300 million m³/yr in 2020) than natural recharge returned, so the aquifer continued to lose water each year. Lower pumping only reduced the rate of decline (about 0.5 m/yr during 2000–2010 and 0.3 m/yr during 2010–2020, compared with 0.7 m/yr during 1990–2000). *(1 point: pumping still exceeds recharge, so there is a net loss; 1 point: reduced pumping slows the rate of decline but does not stop or reverse it, with supporting data or explicit reasoning)*

(d) Claim: not sustainable. Justification: saturated thickness fell in every period, including 2010–2020, when pumping was 1,300 million m³/yr, so water was withdrawn faster than it was replaced; pumping in 2020 was still more than four times the 1960 level (1,300 vs. 300). *(1 point: a claim consistent with the data; 1 point: justification using a specific value or trend from the table. Award neither point for the claim "sustainable" unless the response is supported by correct use of the data; a pumping rate equal to recharge is not shown in the data.)*

(e) Salinization: dissolved salts in the irrigation water stay in the soil when the water evaporates, and over time the soil becomes toxic to plants (white crust, reduced yields). Or waterlogging: too much water left on the field raises the water table so that roots cannot take up oxygen. *(1 point: a correct problem, salinization or waterlogging; 1 point: a correct cause-and-effect explanation of how it happens. Aquifer depletion is not accepted because it is already addressed in the other parts.)*

(f) Any one solution with one advantage and one drawback, for example: switch to drip irrigation (advantage: only about 5% of water lost, so less pumping; drawback: expensive to install); switch to spray irrigation (advantage: loses a quarter or less; drawback: costs more and needs energy); plant drought-tolerant crops (advantage: needs less water; drawback: may lower farm income); limit pumping with permits (advantage: protects remaining supply; drawback: farmers may lose income). *(1 point: a valid solution plus an advantage; 1 point: a valid drawback. Advantage and drawback must relate to the chosen solution.)*

QB-U5-FRQ003

(a) Area = 40 ha × 10,000 m^2/ha = 4.0 × 10^5 m^2. Rain volume = 4.0 × 10^5 m^2 × 0.050 m = 2.0 × 10^4 m^3 (20,000 m^3). *(1 point: correct value with unit; conversion of cm to m must be shown or evident)*

(b) Forest runoff = 0.10 × 20,000 m^3 = 2,000 m^3 (2.0 × 10^3 m^3). *(1 point)*

(c) (i) 0.90 × 20,000 m^3 = 18,000 m^3 (1.8 × 10^4 m^3). *(1 point)* (ii) Percent increase = (18,000 − 2,000) ÷ 2,000 × 100 = 800%. *(1 point: 800%; allow follow-through from an incorrect part (b) value used correctly)*

(d) Roofs and asphalt are impervious surfaces that do not allow water to soak into the soil, and removing trees reduces interception and the uptake of water, so far more of the rain runs over the surface. Effects on the stream (any one): increased flooding or a higher, sharper flow peak; more erosion of stream banks and sediment; warmer water or pollutants (oil, salt, metals) washed into the stream; less groundwater recharge and lower base flow. *(1 point: impervious surfaces reduce infiltration; 1 point: a valid effect on the stream)*

(e) Runoff = 20 ha: 0.90 × 10,000 m^3 = 9,000 m^3 plus 20 ha: 0.30 × 10,000 m^3 = 3,000 m^3, so total = 12,000 m^3 (1.2 × 10^4 m^3). Percent reduction = (18,000 − 12,000) ÷ 18,000 × 100 ≈ 33%. *(1 point: 12,000 m^3; 1 point: 33% reduction, allow follow-through)*

(f) Carbon stored = 40 ha × 120 t C/ha = 4,800 t C. Carbon released = 0.75 × 4,800 = 3,600 t C. CO_2 = 3,600 t C × (44 ÷ 12) = 13,200 t CO_2 (1.3 × 10^4 t). *(1 point: correct value, with the 44/12 conversion shown; 3,600 t C alone earns no point)*

(g) Any one action with an advantage, for example: preserve part of the forest or plant trees on the site (trees intercept rainfall, increase infiltration, and keep storing carbon); install green roofs or rain gardens (absorb runoff and plants take up CO_2); build a retention pond or bioswale (holds stormwater so peak flow is reduced); reduce parking area by building multi-story parking, or build up rather than out (less impervious area and less forest cleared). *(1 point: a valid action that addresses runoff or carbon, with an advantage; credit is given for linking it to both only if stated, but a single valid advantage earns the point)*

QB-U5-FRQ004

(a) Wild capture rose from 80 to 90 million t between 1990 and 2000 and then stayed constant at 90 million t. Aquaculture increased every period, from 15 to 90 million t (a six-fold increase), with its production rising by about 25 million t or more in each decade after 2000. *(1 point: wild capture rose then leveled off, with values; 1 point: aquaculture increased steadily and substantially, with values)*

(b) Total 2020 = 90 + 90 = 180 million t. Aquaculture share = 90 ÷ 180 × 100 = 50%. *(1 point: 50% with work shown)*

(c) A growing percentage of stocks was overfished (10% to 35%), meaning those stocks had been fished faster than they could reproduce, so their populations were too small to support larger catches regardless of fishing effort or technology. Added effort and technology only offset the declining catch per unit of effort, so total capture did not rise; the plateau also reflects regulation in some fisheries and fishing shifting to other species. *(1 point: overfished stocks are depleted/lower population so catches cannot increase; 1 point: more effort or technology only maintains catch or yields diminishing returns, or equivalent reasoning that links effort to a plateau)*

(d) Any one: it is highly efficient; it needs only small areas of water; it requires little fuel compared with fishing fleets; it supplies fish when wild stocks are depleted; it provides a reliable supply. *(1 point)*

(e) Any two: wastewater with nutrients, uneaten feed, or wastes contaminates surrounding water; escaped fish may compete or breed with wild fish; high density increases disease and parasites that can transmit to wild fish; use of antibiotics or chemicals; feed made from wild fish adds fishing pressure. *(1 point each for two different drawbacks)*

(f) Any one solution with justification, for example: set and enforce catch quotas based on sustainable yield (data: 35% of stocks overfished, so catches in those stocks must fall until the populations recover); create marine protected areas or seasonal closures (allow breeding and recovery); regulate gear to reduce bycatch and juvenile catch; expand sustainable aquaculture to reduce pressure on wild stocks (data: aquaculture supplied 50% of production in 2020 with wild capture constant); reduce demand through consumer education or sustainability labeling. *(1 point: a valid solution; 1 point: justification using data from the table or a correct advantage, such as allowing stocks to recover and protecting biodiversity and livelihoods)*

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Unit 6 — Energy Resources and Consumption

Section A — MCQ Answers and Explanations

QB-U6-MCQ001

Answer: A. Renewable sources are replenished naturally at or near the rate of consumption, which describes wind. Uranium and natural gas exist in fixed amounts that cannot be easily replaced, so discovering new deposits does not make gas renewable. The carbon dioxide released by burning biomass does not decide whether it is renewable; replenishment rate does.

QB-U6-MCQ002

Answer: A. Coal, oil, and natural gas together are the most widely used energy sources globally. Hydroelectric, nuclear, wind, and solar are all growing in some places but each supplies a much smaller share of global energy than fossil fuels.

QB-U6-MCQ003

Answer: B. Accessibility and low cost explain the heavy use of wood in developing countries. Wood has a lower energy content per kilogram than coal or oil, burning it indoors produces particulates and carbon monoxide, and overharvesting can cause deforestation, so it is not always replenished.

QB-U6-MCQ004

Answer: B. Natural gas is mostly methane (CH_4). Bitumen is the thick, tar-like component of tar sands, ethanol is a biofuel made by fermentation, and carbon dioxide is a product of combustion rather than a fuel.

QB-U6-MCQ005

Answer: D. Greater heat, pressure, and depth of burial produce higher-quality coal. Lignite is the lowest grade, bituminous is intermediate, and anthracite is the highest grade with the most carbon and energy per kilogram.

QB-U6-MCQ006

Answer: C. Fossil-fuel plants burn fuel to make steam; the steam turns a turbine, which spins a generator. Splitting nuclei describes fission in a nuclear plant, and combining hydrogen with oxygen to make electricity describes a fuel cell. A generator cannot be spun directly by heat.

QB-U6-MCQ007

Answer: D. Commercial reactors use fission: a neutron splits a U-235 nucleus, releasing a large amount of heat. Burning or chemically oxidizing uranium is not the source of the energy, and fusion of hydrogen nuclei is not used in commercial power plants.

QB-U6-MCQ008

Answer: D. Passive systems absorb heat directly from the sun without mechanical or electrical equipment. Active systems use pumps or fans to move a heated liquid or air, and photovoltaic cells convert light directly to electricity. Geothermal systems use heat from Earth’s interior, not sunlight.

QB-U6-MCQ009

Answer: B. Dams flood land upstream and change river flow, temperature, and sediment downstream, so habitats are lost or changed. Hydroelectric plants do not burn fuel, so they do not produce air pollution or waste; sulfur dioxide, radioactive waste, and hydrogen sulfide are associated with coal, nuclear, and geothermal plants respectively.

QB-U6-MCQ010

Answer: D. Wind turbines capture the kinetic energy of moving air, which spins the blades and the generator to produce electricity. The other sequences either invent chemical or thermal energy conversions that do not occur or run the process in reverse.

QB-U6-MCQ011

Answer: D. Lowering the thermostat reduces the energy used for heating. Raising the water-heater temperature, leaving lights on, and running partly filled dishwashers all increase energy use.

QB-U6-MCQ012

Answer: A. In a fuel cell, hydrogen combines with oxygen from the air to form water and release electrical energy, so water is the emission. Carbon dioxide, nitrogen oxides, and carbon monoxide are produced when carbon-containing fuels are burned.

QB-U6-MCQ013

Answer: C. Steam and fluids from geothermal reservoirs can carry hydrogen sulfide, which smells like rotten eggs. Carbon monoxide has no odor, ozone is formed in the atmosphere from other pollutants, and radon is an odorless radioactive gas from rock.

QB-U6-MCQ014

Answer: A. A renewable resource is replenished at or near the rate it is consumed. When use exceeds regrowth, the forest shrinks and deforestation results, so the fuel behaves as nonrenewable. Wood is not millions of years old like fossil fuels, and burning it does release CO_2.

QB-U6-MCQ015

Answer: A. Adding coal, oil, and natural gas gives about 26% for the low-income group, 74% for the middle-income group, and 75% for the high-income group. The jump occurs between low and middle income while biomass falls from 60% to 15%, which matches the trend that reliance on fossil fuels increases as countries develop. The middle-to-high change is only about one percentage point, hydroelectric exceeds wind and solar in the low- and middle-income groups, and biomass is largest in the low-income group.

QB-U6-MCQ016

Answer: B. The doubling time is 70 ÷ 3.5 = 20 years. Ten years results from dividing 35 by 3.5, 35 years comes from dividing by 2, and 245 years comes from multiplying 70 by 3.5.

QB-U6-MCQ017

Answer: C. Cogeneration generates both useful heat and electricity. Useful output is 35 units of electricity + 30 units of heat = 65 units per 100 units of fuel, so 65%. The 35% value counts electricity alone, 87% wrongly adds all 52 waste-heat units, and 30% counts only the recovered heat.

QB-U6-MCQ018

Answer: B. Tar sands contain bitumen mixed with clay, sand, and water, and crude oil can be recovered from them. The second description is coal, the first is peat, and the last describes gas-bearing shale.

QB-U6-MCQ019

Answer: C. Region A has 48% of reserves but 9% of consumption, about 5.3 times as much, so reserves and use are unevenly distributed. Region C uses 31% but holds only 8% of reserves, the regional shares differ widely, and Region E has 11% of reserves versus 21% of consumption.

QB-U6-MCQ020

Answer: A. The global distribution of coal, oil, and gas is not uniform and depends on regions' geologic history; coal comes from buried ancient vegetation that was changed by heat and pressure. Modern population, current rainfall, and government choices do not control where coal formed.

QB-U6-MCQ021

Answer: B. Fracking can contaminate groundwater and release volatile organic compounds. Hydrogen sulfide and flooding are tied to geothermal and hydroelectric projects, radioactive waste and thermal pollution to nuclear plants, and bird and bat deaths to wind turbines.

QB-U6-MCQ022

Answer: A. Coal close to the surface is removed by strip or surface mining. Crude oil is recovered by drilling wells, natural gas in tight shale is released by drilling and hydraulic fracturing, and tar sands are mined or heated in place to recover bitumen rather than fractured.

QB-U6-MCQ023

Answer: D. Thermal pollution happens when warmed cooling water is discharged into the river, which is location 4. Locations 1, 2, and 3 carry reactor water, steam, and condensed water inside closed loops that never mix with the river, so they do not directly warm it even though some are very hot.

QB-U6-MCQ024

Answer: C. The curve reaches 50% at 30 years, so the half-life is 30 years. Reaching 12.5% takes three half-lives (100% → 50% → 25% → 12.5%), or 3 × 30 = 90 years. Sixty years gives 25%, 120 years gives 6.25%, and 53 years comes from wrongly assuming the decline is linear.

QB-U6-MCQ025

Answer: A. Nuclear power does not produce air pollutants during fission and is considered cleaner for that reason, but it releases thermal pollution and radioactive solid waste. It does not burn fuel, and uranium exists in a fixed amount, so it is nonrenewable.

QB-U6-MCQ026

Answer: D. Carbon in ethanol came from the atmosphere through photosynthesis, so burning it does not add additional carbon the way fossil fuels do. However, growing, harvesting, fermenting, and distilling the crop takes a lot of energy, giving a low EROI. Ethanol is made from plants, not from carbon buried for millions of years.

QB-U6-MCQ027

Answer: C. Electricity = area × sunlight energy × efficiency = 20 m^2 × 5 kWh/m^2/day × 0.18 = 18 kWh/day. The value 100 kWh ignores efficiency, 3.6 kWh leaves out the sunlight energy, and 0.9 kWh leaves out the area.

QB-U6-MCQ028

Answer: B. Demand peaks near 19:00–20:00 at about 3.2 kW while PV output is close to zero, which shows that PV use is limited by the availability of sunlight. PV peaks around noon when demand is low, it supplies nothing at night, and photovoltaic cells convert light directly to electricity rather than storing it.

QB-U6-MCQ029

Answer: C. Turbines can be placed in small rivers where the flowing water spins them, avoiding the flooding and habitat loss of a reservoir. The trade-off is lower and less controllable output. Tidal energy is a different method that uses tidal flows.

QB-U6-MCQ030

Answer: B. Geothermal energy uses heat stored in Earth's interior, and the cost of accessing it is high where it is not easily accessible. It does not depend on sunlight, it is not used up like a fuel in the way described, and its main air emission concern is hydrogen sulfide rather than large amounts of CO_2.

QB-U6-MCQ031

Answer: A. A fuel cell combines hydrogen fuel with oxygen to form water and release electricity, with water as the only emission. It does not burn hydrogen to run a turbine and does not use nuclear splitting or nitrogen reactions.

QB-U6-MCQ032

Answer: C. Reading the graph, output is about 0.25 MW at 5 m/s and about 2.0 MW at 10 m/s, and 2.0 ÷ 0.25 = 8. Power rises roughly with the cube of wind speed ($2^3 = 8$) below the rated speed. A factor of 2 would assume output is proportional to wind speed, and 4 or 16 would assume squared or fourth-power relationships.

QB-U6-MCQ033

Answer: B. Power saved = 60 W − 10 W = 50 W. Energy saved = 50 W × 3 h/day × 365 days = 54,750 Wh ≈ 55 kWh. The 66 kWh value is the incandescent bulb's use alone, 11 kWh is the LED's use alone, and 548 kWh puts the decimal in the wrong place.

QB-U6-MCQ034

Answer: C. Old car: 12,000 ÷ 25 = 480 gallons. New car: 12,000 ÷ 40 = 300 gallons. Savings = 480 − 300 = 180 gallons. The 480 and 300 values are each car's use rather than the difference, and 15 simply subtracts the fuel economy values.

QB-U6-MCQ035

Answer: B. An EROI of 1.8 means 100 MJ invested delivers 180 MJ, so the net gain is 180 − 100 = 80 MJ. The 180 MJ value is the total energy delivered rather than the net gain, 56 MJ is 100 ÷ 1.8, and 44 MJ is 100 − 56.

QB-U6-MCQ036

Answer: A. Fuel power needed = 500 MW ÷ 0.38 = 1,316 MW = 1,316 MJ/s. Per hour: 1,316 MJ/s × 3,600 s = 4.74×10^6 MJ. Coal mass = 4.74×10^6 MJ ÷ 24 MJ/kg ≈ 1.97×10^5 kg ≈ 197 metric tons. The value 75 ignores efficiency, 28.5 multiplies by efficiency instead of dividing, and 54.8 leaves out the conversion from seconds to hours.

QB-U6-MCQ037

Answer: B. Maximum possible output = 900 MW × 8,760 h = 7.884×10^6 MWh. Capacity factor = $6.30 \times 10^6 \div 7.884 \times 10^6 \approx 0.80$, or 80%. The 70% value divides by a rounded 9.0×10^6, 125% inverts the ratio, and a capacity factor over 100% would be impossible.

QB-U6-MCQ038

Answer: A. A before-and-after comparison at multiple distances gives baseline data and a control, so changes can be linked to the activity and its distance. A single post-fracturing sample has no baseline, comparing different sites adds geology as an uncontrolled variable, and a survey is subjective and does not measure specific contaminants.

QB-U6-MCQ039

Answer: D. Fuel cell buses have no tailpipe CO_2, but energy is still needed to make hydrogen; using coal electricity moves the emissions upstream. Fuel cells do not burn hydrogen or emit CO_2, and the energy needed to split water is a real cost.

QB-U6-MCQ040

Answer: D. Most deaths occur at low wind speeds, when turbines produce very little power because output rises steeply with wind speed, so curtailing operation below 6 m/s protects bats at little cost. Shutting down every night wastes the most energy, stopping at high winds would remove the most productive hours while protecting bats little, and taller turbines do not address the pattern shown in the data.

QB-U6-MCQ041

Answer: C. Flooding created a new 38 km² reservoir, sediment fell by about 85%, and downstream summer water was 5 °C cooler, all consistent with loss of or change in habitat. The data show lower, not higher, sediment and temperature, and a reservoir does not eliminate all habitat. The data show an association over time, not proof of cause, so the correct choice limits itself to what was measured.

QB-U6-MCQ042

Answer: D. Availability, price, and government regulations influence which energy sources are used. The table shows cheaper gas and an emissions limit, and both gas and wind and solar grew as coal fell. Nothing in the table shows coal depletion, falling total demand, or hydroelectric expansion.

QB-U6-MCQ043

Answer: C. Net cost = \$2,400 − \$400 = \$2,000. Payback = \$2,000 ÷ \$250 per year = 8.0 years. The 9.6 years figure ignores the rebate, 1.6 years divides the rebate by the savings, and 11.2 years adds the rebate to the cost.

QB-U6-MCQ044

Answer: D. Pellet fossil CO_2 drops from 200 to 25 kg/MWh, while particulates rise from 0.01 to 0.30 and NO_x from 0.18 to 0.40 kg/MWh. Burning biomass produces CO_2, CO, NO_x, particulates, and VOCs, so renewable does not mean pollution-free.

Section B — FRQ Solutions

QB-U6-FRQ001

(a) Contamination of groundwater (the aquifer) by methane, fracking fluid chemicals, or brine from the gas well. Release of volatile organic compounds to air is a related fracking concern but does not address drinking-water wells. *(1 point: groundwater/drinking-water contamination)*

(b) Example: *If* the gas well is fractured, *then* dissolved methane concentration will increase most in the monitoring well nearest the pad (M1) and by progressively smaller amounts in M2 and M3, compared with their baseline values. *(1 point: directional prediction that links distance from the pad, or before/after fracking, to methane concentration)*

(c) Collect water samples from M1, M2, and M3 several times (for example monthly for six months) **before** drilling to establish a baseline, then at the same schedule during and after fracking. Collect at least 3 replicate samples from each well each time, use the same sampling method and a certified laboratory, and analyze for dissolved methane (mg/L). *(1 point: sampling at all three wells before and after fracking at stated intervals; 1 point: replication/repeated measurements or equivalent reliability measure)*

(d) Independent variable: distance of the monitoring well from the pad (M1, M2, M3), or alternatively time relative to fracking (before vs. after). Dependent variable: concentration of dissolved methane in the water (mg/L). *(1 point each; the independent variable must match the hypothesis written in (b))*

(e) Any one: sampling depth within the aquifer; same laboratory method or detection limit; same season or time of day; same type of well and purge procedure. Keeping it constant means differences in methane can be attributed to distance or fracking rather than to the sampling method. *(1 point: valid controlled variable with a reason)*

(f) Support: methane in M1 rises after fracking compared with its baseline, the rise in M2 is smaller, and M3 shows little or no change. To decide whether the increase is real, compare the post-fracking values with the range of the baseline samples (for example, outside the baseline mean ± 2 standard deviations) or apply a statistical test; consider that natural methane can occur in aquifers, so the baseline matters. *(1 point: pattern with distance/time; 1 point: comparison with baseline variability or statistical test)*

(g) Any one with justification: install and pressure-test steel casing cemented through the aquifer so fluids and gas cannot leak into it (isolates the aquifer from the well); recycle or safely treat wastewater instead of using unlined pits (prevents spills); set minimum distances between wells and drinking-water supplies (adds a safety margin); continue monitoring wells (detects problems early). *(1 point: reasonable action that addresses groundwater contamination, with a justification)*

QB-U6-FRQ002

(a) Coal generation decreased steadily, from 60 TWh in 2015 to 12 TWh in 2025 (a decline of 48 TWh, or 80%). Wind generation increased each period, from 6 TWh to 30 TWh (five times as much). *(1 point: coal trend with data; 1 point: wind trend with data)*

(b) Wind + solar in 2025 = 30 + 18 = 48 TWh. Percentage = 48 ÷ 108 × 100 = 44.4% (≈ 44%). *(1 point: correct sum of wind and solar and correct total of 108; 1 point: correct percentage)*

(c) Coal, the largest source in 2015, was replaced mostly by wind and solar (which release no CO_2 when generating) and some natural gas (which releases less CO_2 per TWh than coal), so each TWh generated releases less CO_2. Example data: coal fell by 48 TWh while wind and solar rose by 41 TWh. *(1 point: coal replaced by lower- or zero-emission sources, connected to the data)*

(d) Percent decrease = (73.5 − 29.1) ÷ 73.5 × 100 = 44.4 ÷ 73.5 × 100 = 60.4% (≈ 60%). *(1 point: difference of 44.4 million metric tons over the 2015 value; 1 point: correct percentage with a decrease or reduction stated)*

(e) Wind turbines kill birds and bats that fly into the blades. Data: deaths rose from 1,200 per year in 2015 to 12,000 per year in 2025 (ten times as many); because wind generation rose only five times, deaths per TWh doubled from 200 (1,200 ÷ 6) to 400 (12,000 ÷ 30). *(1 point: wildlife mortality identified; 1 point: supporting data comparison. Visual or noise impacts are not supported by the data tables.)*

(f) Any one with justification: shut down or slow blades at low wind speeds in migration seasons (deaths are highest when turbines produce little power); site new wind farms away from migration routes and nesting habitat (avoids collisions); use radar or detection systems to curtail turbines when flocks approach (reduces collisions). *(1 point: reasonable action with justification)*

QB-U6-FRQ003

(a) Energy = area × sunlight × efficiency × days = 1,200 m^2 × 5.0 kWh/m^2/day × 0.20 × 365 days/yr = 1,200 kWh/day × 365 = **438,000 kWh per year** (4.38 × 10^5 kWh/yr). *(1 point: correct setup using all four factors; 1 point: correct answer with units)*

(b) 438,000 ÷ 450,000 × 100 = **97.3%** (≈ 97%). *(1 point)*

(c) Savings = 438,000 kWh × \$0.12/kWh = **\$52,560 per year**. Payback = \$630,000 ÷ \$52,560/yr = **12.0 years** (≈ 12 years). *(1 point: savings; 1 point: payback with units; carry-through from an incorrect (a) is allowed)*

(d) CO_2 avoided = 438,000 kWh × 0.40 kg/kWh = 175,200 kg = **175 metric tons CO_2 per year** (1.75 × 10^2 t). *(1 point: multiplication by the emission factor; 1 point: conversion to metric tons with answer)*

(e) Advantage (any one): produces electricity with no air pollutants or CO_2 during operation; low environmental impact; sunlight is renewable. Limitation (any one): output depends on sunlight (none at night, less in winter or in cloudy weather); high cost; large solar farms can harm desert ecosystems. *(1 point: valid advantage; 1 point: valid limitation)*

(f) Any one that matches the limitation: add battery storage so daytime energy can be used at night; stay connected to the grid for evening and winter needs; pair PV with another source such as wind whose output peaks at different times; seek additional incentives or financing to reduce cost. The justification must state how the action solves the limitation. *(1 point: appropriate action with justification)*

QB-U6-FRQ004

(a) Energy = 1,000 MW × 0.90 × 8,760 h = **7.884×10^6 MWh** (7,884,000 MWh) per year. *(1 point: includes 0.90 capacity factor and 8,760 h; 1 point: correct value with units)*

(b) Thermal power = 1,000 MW ÷ 0.33 = 3,030 MW. Waste heat = 3,030 − 1,000 = **≈ 2,030 MW** (2.03×10^3 MW). *(1 point: thermal input; 1 point: subtraction giving waste heat.)*

(c) Mass flow = 150 m^3/s × 1,000 kg/m^3 = 1.5×10^5 kg/s. Waste heat = 2.03×10^6 kJ/s. $\Delta T = 2.03 \times 10^6$ kJ/s ÷ (1.5×10^5 kg/s × 4.18 kJ/kg·°C) = $2.03 \times 10^6 \div 6.27 \times 10^5$ = **3.2 °C**. *(1 point: mass flow and unit conversion of MW to kJ/s; 1 point: correct ΔT with units)*

(d) Any one: warmer water holds less dissolved oxygen, which can stress or kill fish and other aquatic organisms; temperature changes can alter species composition, spawning, or metabolic rates; warmer water can promote algal growth. *(1 point)*

(e) Any one: use cooling towers or cooling ponds so waste heat is released to the air rather than the river; recirculate cooling water; use cogeneration to put waste heat to a beneficial use such as heating buildings; reduce discharge by running at lower power when the river is warm or low. *(1 point: valid method that reduces heat discharged to the river)*

(f) CO_2 avoided = 7,884,000 MWh × 0.95 t/MWh = **7.49×10^6 metric tons** (≈ 7.5 million t) per year. *(1 point)*

(g) 100% → 50% → 25% → 12.5% → 6.25% is 4 half-lives; time = 4 × 29 years = **116 years**. *(1 point: 4 half-lives and correct time)*

Unit 7 — Atmospheric Pollution

Section A — MCQ Answers and Explanations

QB-U7-MCQ001

Answer: C. A secondary pollutant is not emitted directly; it forms in the atmosphere from reactions among other pollutants. Ground-level ozone forms when nitrogen oxides and VOCs react in sunlight. Carbon monoxide, lead, and sulfur dioxide (A, B, D) are all released directly from their sources, so they are primary pollutants.

QB-U7-MCQ002

Answer: A. Respiration and decomposition are natural sources of CO_2. Home furnaces, cement kilns, and diesel ship engines (B, C, D) are all human activities, so they are anthropogenic sources. Option C is tempting because limestone is a natural material, but the CO_2 in that case is released by an industrial process.

QB-U7-MCQ003

Answer: D. Noise can mask the sounds that animals use to find mates, warn of danger, and hunt, and it can cause stress and hearing damage. Noise does not change ultraviolet absorption, decomposition rates, or genetic diversity in a direct way (A, B, C).

QB-U7-MCQ004

Answer: B. Carbon monoxide binds to hemoglobin and prevents blood from carrying oxygen, so it is classified as an asphyxiant. Radon (A) causes lung cancer, formaldehyde (C) is a VOC irritant, and asbestos (D) is a particulate that damages lung tissue.

QB-U7-MCQ005

Answer: A. A vapor recovery nozzle captures gasoline fumes as the tank fills, so VOCs do not escape. A catalytic converter (B) treats vehicle exhaust, and the electrostatic precipitator and wet scrubber (C, D) treat industrial and power plant exhaust.

QB-U7-MCQ006

Answer: D. Sulfur dioxide and nitrogen oxides react with water in the atmosphere to form sulfuric and nitric acids. CO_2 and CH_4 (A) are greenhouse gases, CO and O_3 (B) are not the main acid precursors, and lead and mercury (C) are toxic metals rather than acid-forming gases.

QB-U7-MCQ007

Answer: C. In an inversion, air near the surface is cooler than the air above it, so the cooler, denser air stays put and pollutants are trapped. Option A states the opposite of the inversion; it describes a normal profile in which warm air rises and disperses pollutants. B and D describe conditions that disperse pollution.

QB-U7-MCQ008

Answer: B. Photochemical smog forms when sunlight drives reactions among nitrogen oxides and VOCs, producing ozone and other irritants. SO_2 and particulates (A) are linked to industrial smog and acid deposition, which does not require this sunlight-driven mix.

QB-U7-MCQ009

Answer: D. Under the Clean Air Act the EPA regulated lead in fuels, and eliminating lead additives in gasoline caused the sharp drop in airborne lead. Scrubbers (A) remove SO_2 and particulates, and vapor recovery nozzles (C) control VOCs; neither addresses lead. Limiting CO_2 (B) does not target lead.

QB-U7-MCQ010

Answer: A. Radon-222 is a radioactive gas produced by uranium decay in some rocks and soils. It moves up through the soil and enters through foundation cracks and basement floors; it can also enter dissolved in well water. It is not produced by carpet adhesives, gas combustion, or mold (B, C, D).

QB-U7-MCQ011

Answer: B. Figure 1 shows NO peaking near 7–8 a.m. with morning traffic, while ozone peaks near 2 p.m. Ozone is a secondary pollutant formed from nitrogen oxides and VOCs in sunlight, so it builds as light and temperature increase. Option A is wrong because vehicles do not emit ozone directly, and option C reverses the role of sunlight, which drives ozone production.

QB-U7-MCQ012

Answer: C. On Day 2 the temperature rises from 6 °C at the ground to 12 °C at 300 m, which is an inversion. The warmer layer aloft acts as a lid over cooler, denser air and limits vertical mixing, so pollutants accumulate near the ground. On Day 1 the temperature falls with height, which is the normal pattern that allows mixing (A is wrong for that reason).

QB-U7-MCQ013

Answer: A. A catalytic converter changes CO, NO_x, and unburned hydrocarbons in the exhaust into CO_2, N_2, and H_2O. Option B reverses the reaction. SO_2 and particulate removal (C) is done by scrubbers and precipitators, and lead and mercury (D) are not handled by converters.

QB-U7-MCQ014

Answer: D. Only formaldehyde rose substantially (11 to 78 ppb); CO, radon, and PM2.5 are essentially unchanged. Formaldehyde is a VOC released by composite wood, adhesives, and carpeting, which matches the renovation. Radon (A), furnace CO (B), and mold (C) would not raise formaldehyde alone, and the data show no change in radon or CO.

QB-U7-MCQ015

Answer: B. Limestone contains calcium carbonate, which neutralizes acid and buffers the lake, so Marl Lake stays near pH 6.8 even with acidic rain. Granite has little buffering capacity, so Flint Lake stays acidic. The rain pH is the same for both lakes, so option A is not needed to explain the difference, and limestone does not emit SO_2 (D).

QB-U7-MCQ016

Answer: C. Device 1 uses charged wires and collection plates, which is an electrostatic precipitator that removes fly ash particles. Device 2 sprays limestone slurry, which is a wet scrubber that absorbs SO_2 from the flue gas. Option A swaps the functions of the devices, and neither device removes CO_2 (D).

QB-U7-MCQ017

Answer: D. Volcanic ash, windblown dust, and sea spray are all natural particulate sources. Every other list includes a human-caused source: diesel trucks (A), coal fly ash (B), or tire wear (C).

QB-U7-MCQ018

Answer: A. Coal combustion releases CO_2, SO_2, particulates, and toxic metals such as mercury. Option B lists secondary pollutants and CFCs, which are not direct coal emissions. Option C lists indoor pollutants, and option D lists gases mostly associated with agriculture and landfills.

QB-U7-MCQ019

Answer: C. The leaf blower (95 dB), chainsaw (105 dB), and jackhammer (110 dB) are above the 85 dB line, so three of the six sources exceed it. Option A is wrong because the leaf blower and chainsaw also exceed the line, not only the jackhammer. Heavy traffic is 80 dB, below the line (B), and the chainsaw is above it (D).

QB-U7-MCQ020

Answer: B. Photochemical smog forms from nitrogen oxides and VOCs, so reducing vehicle emissions and fuel vapors attacks the cause. Scrubbers (A) target SO_2 from power plants, lead paint (C) is unrelated to smog, and taller stacks (D) only move pollution elsewhere without reducing its formation.

QB-U7-MCQ021

Answer: A. Acid deposition mainly affects communities downwind of coal-burning plants. Town E is 150 km downwind and its thin granite soils have little capacity to neutralize acid. Town G is also downwind, but limestone soils buffer the acid. Towns F and H are upwind, so they receive little deposition from this plant.

QB-U7-MCQ022

Answer: D. On clear, calm nights the ground loses heat by radiation and the air next to it cools; surrounding mountains keep the cool air from draining away or mixing. Strong winds (A) mix the air, clouds (B) slow the cooling, and a warm surface (C) promotes normal rising air.

QB-U7-MCQ023

Answer: C. The scrubber removes 92%, so 8% remains: 1,200 metric tons × 0.08 = 96 metric tons per year. Option D (1,104 metric tons) is the amount removed, not the amount emitted. Option A results from a decimal error (0.8%).

QB-U7-MCQ024

Answer: B. Mitigation is recommended at or above the 4.0 pCi/L action level. Home 3 (4.6) and Home 4 (7.2) are above it. Home 2 (3.9) is just below the action level, so it does not require mitigation by this standard, although retesting would be reasonable.

QB-U7-MCQ025

Answer: D. The urban peak is near 1 p.m., the suburban peak near 3 p.m., and the rural peak near 5 p.m. The delay is consistent with ozone forming as a polluted air mass travels downwind and reacts in sunlight. Option B treats ozone as a primary pollutant, and option C is contradicted by the curves at 2 p.m.

QB-U7-MCQ026

Answer: A. The upwind sensor gives the background concentration from other sources. Subtracting it from the downwind reading isolates the vent's contribution, like a control. It does not double the sample at the same location (B), calibrate the downwind sensor (C), or measure plant uptake (D).

QB-U7-MCQ027

Answer: B. Nest success rises as noise falls, from 38% at 72 dB to 74% at 45 dB, which is consistent with masking of communication. These are observational data, so other factors such as pollution, predators, and habitat were not ruled out; option A overstates the evidence. Options C and D contradict the data.

QB-U7-MCQ028

Answer: C. The pH difference is 5.6 − 4.3 = 1.3, so the hydrogen ion concentration ratio is 10^1.3 ≈ 20 times. Option A forgets the logarithmic scale, and option B treats it as a linear multiple of 10. Option D would correspond to a difference of 2.3 units.

QB-U7-MCQ029

Answer: A. Option 1 removes 0.90 × 8,000 = 7,200 metric tons, so \$36,000,000 ÷ 7,200 ≈ \$5,000/metric ton. Option 2 removes 0.60 × 8,000 = 4,800 metric tons, so \$14,400,000 ÷ 4,800 = \$3,000/metric ton. Option 2 is cheaper per ton. Choice B is the correct cost for scrubbers but the wrong option, and choices C and D result from dividing by the total emissions (8,000 metric tons) instead of by the amount removed.

QB-U7-MCQ030

Answer: D. Total NO_x = 250,000 vehicles × 12,000 mi × 0.4 g/mi = 1.2×10^9 g = 1,200 metric tons. A 75% cut prevents 0.75 × 1,200 = 900 metric tons per year. Option B is the starting emission, not the reduction, and option A is the amount that remains (25%).

QB-U7-MCQ031

Answer: B. Temperature rises with height from 4 °C at the surface to 11 °C at 300 m, which is an inversion layer. This stable layer limits upward mixing, so pollutants from surface sources stay concentrated near the ground. Above 300 m the temperature falls normally, but that is not where the pollutants are being trapped (A, C), and D ignores the stability of the layer.

QB-U7-MCQ032

Answer: C. The treatments differ in light and temperature as well as pH, so any difference in germination could be due to those variables. Controlling light and temperature for all dishes and adding a neutral-water control isolates pH as the only variable. Mixing varieties (A) or acid sources (B) adds more differences, and counting only the tallest seedling (D) biases the data.

QB-U7-MCQ033

Answer: D. The authors link doubling ship traffic, rising low-frequency noise, and fewer whale sightings, and they offer a mechanism (masking of calls). They also acknowledge that prey populations were not surveyed. No controlled experiment was done (A), the authors do not say that noise is irrelevant (B), and they do not reject the link because of different methods (C).

QB-U7-MCQ034

Answer: A. SO_2, NO_x, and lead all fell sharply after targeted regulation and controls (scrubbers, converters, unleaded fuel), while CO_2 rose from 100 to 116. Option B compares different sources and different pollutants, so it overreaches. Option C wrongly infers that one regulation caused another decline, and option D is not supported because CO_2 is not a direct health pollutant, and the graph shows the pollutants that harm health declining.

Section B — FRQ Solutions

QB-U7-FRQ001

(a) Volatile organic compounds (VOCs), such as formaldehyde, are the most likely cause. Sources: carpet and its adhesive, particleboard cabinets, or paint. *(1 point: a correct pollutant category with a matching building material)*

(b) Example: "If the renovation materials release VOCs, then the formaldehyde concentration in Wing B classrooms will be higher than in Wing A classrooms." *(1 point: testable, if/then, states the direction)*

(c) Independent variable: the wing (renovated Wing B versus non-renovated Wing A). Dependent variable: concentration of formaldehyde or total VOCs in the air. Controlled variables (any two): sampling time of day, height of the monitor, room size, temperature, humidity, ventilation rate, the same type of monitor. *(1 point for independent and dependent variables; 1 point for two valid controls)*

(d) Place identical calibrated formaldehyde or VOC monitors at the same height (for example 1 m, breathing height) in the middle of each of the four classrooms in both wings. Sample at the same times on several days (for example, mornings and afternoons for five school days) with the ventilation set the same way. Repeat the measurement to account for variability, and compare the wing averages. Alternative: include an outdoor sample as a baseline. *(1 point: monitors in all classrooms in both wings at the same height/conditions; 1 point: sampled over multiple days or times; 1 point: replication or averaging, or a baseline sample, that addresses variability)*

(e) The mean VOC or formaldehyde concentration in Wing B would be clearly higher than in Wing A, beyond the variation within each wing. *(1 point)*

(f) Examples: increase the flow of outdoor air through Wing B; run exhaust fans or open windows; allow the materials to off-gas before occupancy; replace the materials with low-VOC products; use air purifiers with activated carbon. *(1 point: a reasonable action)*

(g) For example, more outdoor air dilutes and removes the VOCs from the room, so concentration falls; low-VOC materials release smaller amounts of VOCs in the first place. *(1 point: links the action to lower pollutant concentration)*

QB-U7-FRQ002

(a) As SO_2 emissions decreased from 480 to 132 thousand metric tons per year, the pH of Lake Alder increased from 4.6 to 5.7, so lower emissions are associated with less acidic water (an inverse relationship). *(1 point: describes the inverse relationship with direction or data)*

(b) Percent decrease = (480 − 132) ÷ 480 × 100 = 348 ÷ 480 × 100 = 72.5%. *(1 point: the correct value, 72.5% or about 73%)*

(c) Limestone bedrock contains carbonate minerals (calcium carbonate) that react with and neutralize acid deposition. This buffering capacity keeps the lake water near neutral. Lake Alder is on granite and thin soils, which have little buffering capacity. *(1 point: limestone or carbonate neutralizes acid; 1 point: ties this to a stable pH despite the high emissions, or contrasts with granite)*

(d) SO_2 from the stack reacts with oxygen and water in the atmosphere to form sulfuric acid. Wind carries it downwind and it falls as acid rain (wet deposition) or as dry particles and gases (dry deposition). *(1 point: SO_2 reacts to form an acid (sulfuric acid) and is carried downwind and deposited)*

(e) Any two: acidification of soils and loss of nutrients such as calcium and magnesium; damage to forests; corrosion of buildings, statues, and bridges; release of toxic aluminum into water; harm to fish and other aquatic life; loss of income from fishing or forestry. *(1 point each for two valid effects; the effects must not be the acidification of lakes)*

(f) Claim: reducing SO_2 alone may not fully restore Lake Alder. Alder's pH rose from 4.6 to 5.7 but had not reached the near-neutral values (6.8–7.0) of Lake Birch by 2020, even though emissions fell 72.5%, and its gains slowed as emissions leveled off at 132. Alternative claim: Lake Alder is recovering and will be restored if emissions continue to fall, supported by the steady pH rise. *(1 point: a claim consistent with a specific data point or trend)*

(g) Examples: switch the plant to natural gas or renewables; reduce NO_x from vehicles by adding converters or promoting electric cars (NO_x also forms nitric acid); add lime to the lake (liming); reduce energy use through

conservation. Justification: these steps lower the amount of SO_2 or NO_x that is converted into acids. *(1 point: valid solution; 1 point: justification that links it to less acid formation)*

QB-U7-FRQ003

(a) Mass of sulfur = 2.0×10^6 metric tons coal × 0.015 = 3.0×10^4 metric tons S. *(1 point)*

(b) The molecular mass of SO_2 is twice the atomic mass of S (64 ÷ 32 = 2), so mass of SO_2 = 3.0×10^4 metric tons × 64/32 = 6.0×10^4 metric tons per year. *(1 point: uses the 64/32 ratio; 1 point: correct answer with units)*

(c) With 95% removal, 5% is emitted: $6.0 \times 10^4 \times 0.05 = 3.0 \times 10^3$ metric tons per year. *(1 point)*

(d) SO_2 removed = $6.0 \times 10^4 \times 0.95 = 5.7 \times 10^4$ metric tons. Cost per metric ton = \$48,000,000 ÷ 57,000 metric tons ≈ \$842 per metric ton, which is \$840 per metric ton to two significant figures. *(1 point: calculates the amount removed; 1 point: correct cost per ton with two significant figures)*

(e) Any two: acidification of lakes and streams, which harms fish; acidification of soils and loss of nutrients that damage forests; corrosion of buildings and statues; effects are greatest on communities downwind, especially where soils have little buffering capacity. *(1 point each for two valid effects)*

(f) Examples: switch to low-sulfur coal (trade-off: higher fuel cost or limited supply); switch to natural gas or renewable energy (trade-off: high capital cost or intermittent supply); conserve electricity so less coal is burned (trade-off: requires changes in behavior or investment). *(1 point: a valid alternative; 1 point: a valid trade-off for the chosen alternative)*

QB-U7-FRQ004

(a) Inversion days (Dec 4, 5, 6, 9) had much higher PM2.5 (61–92 µg/m³) than non-inversion days (12–23 µg/m³). The presence of an inversion is associated with higher PM2.5. *(1 point: describes the positive association using data)*

(b) Inversion days: (61 + 78 + 92 + 85) ÷ 4 = 316 ÷ 4 = 79 µg/m³. Non-inversion days: (14 + 18 + 23 + 12) ÷ 4 = 67 ÷ 4 = 16.75 ≈ 16.8 µg/m³. *(1 point: inversion mean; 1 point: non-inversion mean)*

(c) During an inversion a layer of warmer air lies above the cooler surface air. The warm layer acts as a lid, so the cool, dense surface air cannot rise and mix with the air above. Pollutants from vehicles, heating, and industry stay trapped in a shallow layer near the ground, and their concentrations build up over time. *(1 point: warm air layer above cooler air prevents vertical mixing; 1 point: pollutants trapped and accumulate near the ground)*

(d) Any one: wood-burning stoves and fireplaces; vehicle exhaust, including diesel; industrial combustion; construction dust; burning of coal or oil for heat. *(1 point)*

(e) Any one: respiratory irritation, asthma attacks, reduced lung function, bronchitis, heart disease, or premature death. *(1 point)*

(f) Examples: a temporary ban on wood burning on forecast inversion days; encourage public transit or free bus rides; restrict industrial emissions or require diesel idling limits on those days. Justification: these actions reduce the amount of PM2.5 emitted into the trapped air, so concentrations build up less. *(1 point: valid action tied to forecast inversion days; 1 point: justification linking it to lower emissions or concentrations)*

(g) Claim: eight days are not sufficient to conclude causation. The data show a strong association, but the sample is small and covers one short period; other factors such as weather, heating demand, and traffic could also differ on inversion days, and more days and a controlled comparison are needed. Alternative: argue that the data are consistent with the inversion mechanism, but note that correlation does not prove causation. *(1 point: a claim supported by a valid reason about sample size, confounding variables, or correlation versus causation)*

Unit 8 — Aquatic and Terrestrial Pollution

Section A — MCQ Answers and Explanations

QB-U8-MCQ001

Answer: C. Pollution that is diffused over a wide area and has no single identifiable source, such as agricultural runoff, is a nonpoint source. Options A, B and D each describe one identifiable pipe or tank, so they are point sources; D is the most tempting because it is "leaking," but a single tank is still a single, identifiable source.

QB-U8-MCQ002

Answer: B. Endocrine disruptors are chemicals that interfere with the endocrine (hormone) system of animals and can cause developmental problems and gender imbalances. A describes pathogens, C describes low-oxygen conditions, and D describes the effect of an oil spill.

QB-U8-MCQ003

Answer: A. Mangroves and wetlands provide flood protection, including buffering coasts against storm surge and wave energy, so clearing them removes that protection for nearby communities. B, C and D describe functions that mangroves do not perform; the other genuine services (habitat, water filtration) are not what the question asks about.

QB-U8-MCQ004

Answer: D. Warm water holds less dissolved oxygen than cold water, so thermal pollution lowers DO downstream of the discharge. A states the reverse relationship, which is the most common misconception about gas solubility.

QB-U8-MCQ005

Answer: B. POPs are synthetic, carbon-based molecules that resist breakdown, so they persist and can accumulate. A is wrong because POPs are synthetic, not naturally occurring elements; C and D describe fates that would remove the pollutant, the opposite of persistence.

QB-U8-MCQ006

Answer: A. The clay or plastic liner at the bottom prevents leachate (liquid that has percolated through the waste) from reaching groundwater. D is the job of the cap and the storm-water system, and C is the job of the methane collection system.

QB-U8-MCQ007

Answer: D. LD50 is the dose of a chemical that is lethal to 50% of the population of a particular species. A confuses lethality with sublethal symptoms, and B and C describe tissue concentration and half-life, which are different ideas.

QB-U8-MCQ008

Answer: C. Primary treatment is physical: screens and grates remove large objects and solids settle in a tank. A is secondary treatment, D is tertiary treatment, and B is disinfection before discharge.

QB-U8-MCQ009

Answer: B. Organisms have a range of tolerance for pollutants. Outside the optimum range, they experience physiological stress, limited growth and reduced reproduction before death occurs at extreme levels. A overstates the effect for a level below the lethal one.

QB-U8-MCQ010

Answer: A. Recycling rate = recycled ÷ generated × 100 = 120 t ÷ 480 t × 100 = 25%. B (33%) results from dividing by the 360 t that were not recycled.

QB-U8-MCQ011

Answer: D. When a bloom dies, microbes digest the dead algae and use up the dissolved oxygen in the process, producing hypoxia and fish die-offs. C is tempting, but phosphorus is a nutrient that harms fish indirectly through low oxygen rather than by poisoning gills.

QB-U8-MCQ012

Answer: C. Chlorophyll peaked in July (85 µg/L), then fell, while bottom DO kept falling to its minimum in September (1.2 mg/L), which fits decomposers consuming oxygen as dead algae sink. A fails because DO was

already falling in June and July while algae were increasing, so a shortage of algae cannot explain the decline. B fails because the highest chlorophyll month (July) did not have the highest DO.

QB-U8-MCQ013

Answer: A. Total load = 9,000 + 6,000 + 36,000 + 9,000 = 60,000 kg/yr. The two pipes are point sources (15,000 kg). Nonpoint load = 36,000 + 9,000 = 45,000 kg, and 45,000 ÷ 60,000 × 100 = 75%. B (25%) is the point-source share, and C (60%) counts only cropland.

QB-U8-MCQ014

Answer: B. Increased sediment reduces light infiltration, which affects the photosynthetic algae living in coral, and settling sediment smothers and disrupts habitat. A is not how sediment acts, and D is a different human impact (destructive fishing) not caused by sediment.

QB-U8-MCQ015

Answer: D. Some components of oil are denser than water and sink to the seafloor, killing bottom-dwelling organisms even where no oil is seen at the surface. A is tempting because surface oil is a real effect, but it cannot explain deaths on the bottom without surface oil.

QB-U8-MCQ016

Answer: C. Intersex fish downstream of effluent, with similar temperature and oxygen at both sites, point to endocrine disruptors in the wastewater. B and D are ruled out by the stated similarity of DO and temperature. A does not fit the observation of eggs in male fish.

QB-U8-MCQ017

Answer: A. Wetlands provide flood protection by soaking up storm water and releasing it slowly. Removing the marsh means rain runs off faster and more flows to the river at once. C reverses the effect of filling in a wetland.

QB-U8-MCQ018

Answer: B. Decrease = 10.1 − 8.3 = 1.8 mg/L, and 1.8 ÷ 10.1 × 100 ≈ 17.8%, which rounds to about 18%. C (22%) comes from dividing by the downstream value instead of the starting value, and D (82%) is the final value as a percentage of the starting value, not the decrease.

QB-U8-MCQ019

Answer: D. POPs do not break down easily, can travel long distances in air and water, are redeposited far from where they were used, and are stored in the fatty tissues of organisms. C fails because DDT is fat soluble, which is why it is retained rather than excreted.

QB-U8-MCQ020

Answer: C. Ratio = 60 ÷ 0.05 = 1,200. A (6) is the ratio between trout and osprey only, and B (60) is the osprey concentration itself.

QB-U8-MCQ021

Answer: A. Component 3 is the leachate collection layer lying above the liner, which carries liquid that percolated through the waste to be pumped out. Component 1 is the cap, 2 is a methane collection pipe, and 4 is the liner; the liner (4) is tempting because it also deals with leachate, but it blocks seepage rather than collecting the liquid.

QB-U8-MCQ022

Answer: D. E-waste contains hazardous heavy metals such as lead and mercury that can leach into groundwater, so certified recycling or reuse is the best option. A is tempting because landfills have liners, but leachate can still escape. B and C release or bury the metals directly.

QB-U8-MCQ023

Answer: B. Increase in composting = 25% − 10% = 15% of 150,000 t = 0.15 × 150,000 = 22,500 t less landfilled. D (67,500 t) is the total amount that would still be landfilled under the new plan (45%).

QB-U8-MCQ024

Answer: C. Tank Q is the secondary treatment stage: aeration speeds up the rate at which bacteria break down organic matter into carbon dioxide and sludge. A and B describe primary treatment, and D describes tertiary treatment (tank R).

QB-U8-MCQ025

Answer: B. Dose = 25 mg/kg × 0.040 kg = 1.0 mg per mouse. D (625 mg) divides instead of multiplying (25 ÷ 0.040), and C ignores the mouse's mass.

QB-U8-MCQ026

Answer: A. The curve for A reaches 50% mortality at about 20 mg/kg, while B needs about 80 mg/kg, so A is more toxic. B reverses the interpretation of a curve shifted to the right. D is wrong because the A curve is already near 10% mortality at 10 mg/kg.

QB-U8-MCQ027

Answer: D. As equatorial-type climate zones spread into subtropical and temperate regions, vectors and the infectious diseases they carry spread into areas where the disease did not previously occur. C is wrong because pathogens and vectors can occur regardless of the appearance of sanitary conditions.

QB-U8-MCQ028

Answer: C. A control plus separate nitrogen-only, phosphorus-only and combined treatments, with replication and constant conditions, isolates which nutrient limits growth. A cannot separate the two nutrients, B confounds the nutrient with the location in the lake, and D tests only nitrogen with no control or replication.

QB-U8-MCQ029

Answer: A. A testable hypothesis states a prediction that measurement can confirm or reject. Only A defines the measurement (tissue PCB concentration) and the comparison (otters versus their prey) needed to detect biomagnification. B is a policy opinion, C does not compare trophic levels, and D is vague and not specific to the question.

QB-U8-MCQ030

Answer: D. Below the outfall, decomposers consume oxygen faster than it is replaced by exchange with air, so DO falls. Further downstream, much of the waste has been broken down, consumption slows, and reoxygenation exceeds consumption, so DO recovers. A is incorrect because the waste is most concentrated near the outfall, not at the sag point.

QB-U8-MCQ031

Answer: B. Compound Y increases about tenfold at each trophic level, the signature of biomagnification, and this happens when a substance persists and is fat soluble, so it is retained in tissue. X stays constant, and Z decreases, so it is lost at each step. D gives the wrong property: water-soluble compounds are excreted and would not increase.

QB-U8-MCQ032

Answer: C. Because experiments exposing children are unethical, an observational study with a large sample across exposure levels, while recording confounders such as asthma and household smoking, gives the strongest evidence. A has no comparison group, B ignores other differences, and D relies on a poor, subjective measure. Cause and effect remains difficult to prove because people are exposed to many pollutants.

QB-U8-MCQ033

Answer: A. Energy released = 2.0×10^6 m³ × 36 MJ/m³ = 7.2×10^7 MJ. Electricity = $0.30 \times 7.2 \times 10^7$ MJ = 2.16×10^7 MJ. Convert: 2.16×10^7 MJ ÷ 3.6 MJ/kWh = 6.0×10^6 kWh. Households = $6.0 \times 10^6 \div 10{,}000$ = 600. B (2,000) forgets the 30% efficiency, and C (2,160) skips the MJ-to-kWh conversion.

QB-U8-MCQ034

Answer: B. The LD50 is the dose at which 50% die; the graph shows that at 40 mg/kg chemical B still kills about 11% of tadpoles, so it is not safe. A is the misconception that anything below the LD50 is harmless. C misreads the graph, and D compares a different chemical, which is irrelevant to whether B is safe.

Section B — FRQ Solutions

QB-U8-FRQ001

(a) The problem is eutrophication (nutrient enrichment leading to algal blooms and hypoxia/fish kills). *(1 point: identifies eutrophication, algal bloom or nutrient pollution; "fish kill" alone is not enough.)*

(b) Example: If runoff from the lawns of the housing development carries extra nitrogen and phosphorus into the pond, then nutrient concentrations and algal growth will be higher at the site receiving Inlet 1 than at the site receiving Inlet 2. *(1 point: if-then, testable and links fertilizer runoff to nutrients or algae.)*

(c) Independent variable: the source of the water (site/inlet, that is, residential runoff versus forested runoff) or the concentration of nutrients in the runoff. Dependent variable: algal abundance (e.g., chlorophyll or algal cell count) and/or nutrient concentration (nitrate, phosphate) in the water. *(1 point independent, 1 point dependent. Accept dissolved oxygen as a dependent variable.)*

(d) Collect water samples from S1, S2, S3 and S4 at the same time of day, several times through the season (for example weekly from May through August and after storms), with replicate samples at each site. Measure nitrate and/or phosphate, chlorophyll-a (or algal counts) and dissolved oxygen. Keep constant or record: depth of sampling, time of day, weather/rainfall, temperature, sampling method. *(3 points: 1 for sampling at all four sites with timing and repeated or replicated samples; 1 for a valid nutrient, algae or DO measurement; 1 for a variable kept constant or recorded.)*

(e) Nutrient and chlorophyll concentrations would be highest at S1 and also high at S3 and S4 downstream, while S2 would have low values. *(1 point: S1 higher than S2 with consistent downstream pattern.)*

(f) Excess nutrients cause an algal bloom. When the algae die, microbes decompose them and use up dissolved oxygen, causing hypoxia that kills fish. *(1 point: dead algae decomposed by microbes lowers DO. Must link decomposition to oxygen loss.)*

(g) Examples: plant a vegetated buffer or build a retention basin or rain garden to trap runoff; reduce or stop fertilizer use or use slow-release or phosphorus-free fertilizer. Justification: vegetation and basins absorb nutrients and slow runoff, so less nitrogen and phosphorus reach the stream. *(1 point: a valid action with a justification.)*

QB-U8-FRQ002

(a) Mercury concentration increases at each successive trophic level, from 0.02 mg/kg in plankton to 3.6 mg/kg in loons. *(1 point.)*

(b) 0.90 mg/kg ÷ 0.02 mg/kg = 45 times greater. *(1 point for 45 with work shown.)*

(c) Mercury (as methylmercury) is absorbed and stored in the tissues of organisms (bioaccumulation) and is not easily excreted. Each predator eats many prey containing mercury, so the concentration per unit of body tissue increases at successively higher trophic levels (biomagnification). *(1 point for stored in tissue/persists; 1 point for predators consuming many contaminated prey so concentration rises up the chain.)*

(d) Burning coal at the power plant releases mercury into the air, and it is deposited on the lake and watershed. Bacteria in the water convert elemental mercury to highly toxic methylmercury, which organisms absorb and which accumulates. *(1 point: coal burning or other mercury-releasing activity such as mining; 1 point: bacteria convert it to methylmercury.)*

(e) Claim: pike from Lake Alder are safe to eat but pike from Lake Birch are not. Justification: Alder pike contain 0.20 mg/kg, below the 0.5 mg/kg limit, while Birch pike contain 0.90 mg/kg, nearly double the limit. *(1 point for a correct claim; 1 point for supporting data, with both values compared to 0.5.)*

(f) Reduce emissions from the power plant (install scrubbers, switch to renewable energy or natural gas); drawback: scrubbers or switching sources cost money and results take years because mercury already in sediments continues to cycle. Accept: stop mercury mining/fish consumption advisories with a drawback such as not fixing the cause. *(1 point: valid solution; 1 point: a reasonable drawback.)*

QB-U8-FRQ003

(a) 250,000 people × 1.8 kg/person/day × 365 days/yr = 1.6425×10^8 kg/yr ÷ 1,000 kg/t = **164,250 metric tons per year** ($\approx 1.6 \times 10^5$ t). *(1 point for correct set-up with units; 1 point for the correct answer.)*

(b) Landfilled fraction = 100% − 22% − 12% = 66%. 164,250 t × 0.66 = **108,405 t/yr**. *(1 point.)*

(c) 108,405 t × 1,000 kg/t = 1.08405×10^8 kg; ÷ 600 kg/m³ = **180,675 m³/yr** ($\approx 1.8 \times 10^5$ m³). *(1 point for converting tons to kg and dividing by density; 1 point for answer.)*

(d) 4.0×10^6 m³ ÷ 180,675 m³/yr = **≈ 22 years** (22.1). *(1 point.)*

(e) Landfilled fraction = 50%. 164,250 t × 0.50 = 82,125 t = 8.2125×10^7 kg ÷ 600 kg/m³ = 136,875 m³/yr. Years = 4.0×10^6 ÷ 136,875 = **≈ 29 years** (7 years longer). *(1 point for volume per year; 1 point for years.)*

(f) Drawbacks of composting: odors and attraction of rodents, plus land and labor costs. Further proposal examples: incineration or waste-to-energy (reduces volume, but releases air pollutants); capturing landfill

methane for electricity (reduces landfill gas emissions and generates power); reduce waste at the source (reusable packaging, a pay-as-you-throw program). Justification must connect the proposal to less waste landfilled or a lower environmental impact. *(1 point: drawback; 1 point: valid proposal with justification.)*

QB-U8-FRQ004

(a) Mortality increases as the dose increases for both insecticides, following an S-shaped curve that approaches 100% at high doses. Insecticide N's curve is shifted to the left of M's (N kills a greater percentage at every dose). *(1 point: positive relationship/sigmoid; 1 point: N and M differ in position or in mortality at each dose.)*

(b) N: between 0.1 µg (30%) and 0.3 µg (72%), so LD50 ≈ 0.17 µg per bee (accept 0.15–0.20). M: between 0.3 µg (38%) and 1 µg (74%), so LD50 ≈ 0.45 µg per bee (accept 0.4–0.5). *(1 point each.)*

(c) Insecticide N is more toxic, because a smaller dose (about 0.17 µg per bee versus about 0.45 µg) kills half the bees; for instance, at 0.3 µg N kills 72% and M kills 38%. *(1 point: N with supporting data.)*

(d) 72% × 40,000 = 0.72 × 40,000 = **28,800 bees**. *(1 point.)*

(e) Any two: LD50 measures only death, not sublethal effects (impaired navigation, reduced reproduction, weakened immunity); it is measured over a short time (24 h) and a single dose, so chronic or repeated exposure may differ; it may not predict the effect on other species or on bees exposed to mixtures; real exposure depends on how much insecticide bees actually encounter. *(1 point for each valid reason, maximum 2.)*

(f) Examples: spray at night or when bees are not foraging; avoid spraying flowering plants; use integrated pest management, targeted application, or a less toxic insecticide; leave buffer strips. Justification: reduces the dose of insecticide that bees contact, so fewer bees are killed. *(1 point: valid practice; 1 point: justification linked to lower exposure.)*

Unit 9 — Global Change

Section A — MCQ Answers and Explanations

QB-U9-MCQ001

Answer: C. Ozone in the stratosphere absorbs UV-B and much UV-C, which damages DNA and would make life on land very difficult. The option about trapping infrared describes the greenhouse effect, a different process carried out by gases such as CO_2 and CH_4.

QB-U9-MCQ002

Answer: A. Greater UV exposure damages skin cells and the lens of the eye, raising rates of skin cancer and cataracts. Asthma and bronchitis are tied to air pollutants, including ozone at ground level, not to the loss of stratospheric ozone.

QB-U9-MCQ003

Answer: B. Chlorine atoms released from CFCs are what catalyze ozone destruction; HFCs lack chlorine. The claim of a zero global warming potential is false, because many HFCs are strong greenhouse gases.

QB-U9-MCQ004

Answer: A. The Montreal Protocol targeted ozone-depleting chemicals and is widely regarded as a successful international environmental treaty. The Kyoto Protocol (1997) addressed greenhouse gas emissions, Basel addresses hazardous-waste shipments, and Ramsar protects wetlands.

QB-U9-MCQ005

Answer: D. CO_2, CH_4 and N_2O (along with water vapor and CFCs) are the principal greenhouse gases. Nitrogen, oxygen and argon make up about 99% of the atmosphere but do not absorb infrared radiation significantly.

QB-U9-MCQ006

Answer: D. Greenhouse gases absorb and re-emit outgoing infrared radiation, raising the average surface temperature from roughly −18°C to about 15°C. Human activity adds to the effect (the enhanced greenhouse effect) but did not create it, and ozone depletion is a separate problem.

QB-U9-MCQ007

Answer: B. Meltwater from ice on land adds new volume to the ocean, and warmer water occupies more volume. Floating sea ice already displaces its own weight in water, so its melting adds almost nothing to sea level.

QB-U9-MCQ008

Answer: A. Warmer winters allow disease-carrying insects to survive farther from the equator, exposing new human populations to illnesses such as dengue. The other choices describe the opposite of the predicted trend.

QB-U9-MCQ009

Answer: C. Ice cores trap tiny samples of ancient air whose gas composition can be measured and matched to the age of each ice layer. Sales records and satellite images cover only recent history.

QB-U9-MCQ010

Answer: B. Heat stress causes corals to expel their zooxanthellae, which supply color and most of the coral's food, leaving the white skeleton visible. Some corals recover if temperatures fall; others die.

QB-U9-MCQ011

Answer: A. $CO_2 + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$, so more atmospheric CO_2 means more H^+ and a lower pH. Nutrient runoff causes eutrophication, a different problem.

QB-U9-MCQ012

Answer: D. A species is considered invasive when it lives, and often thrives, outside its normal range and harms native species. A non-native plant that stays where it was planted and harms nothing is not invasive.

QB-U9-MCQ013

Answer: B. Ships take on ballast water in one port and release it in another, carrying organisms across oceans. The other choices describe movement within native ranges, which is not an introduction of a non-native species.

QB-U9-MCQ014
Answer: D. Species with narrow habitat requirements and limited diets have few options when conditions change. Generalists and species that can move or adapt are less likely to face extinction.

QB-U9-MCQ015
Answer: C. Legal protection and enforcement reduce the hunting that is directly driving the decline. The other options would add predators, remove habitat or increase human access, all of which stress the population further.

QB-U9-MCQ016
Answer: A. HIPPCO stands for habitat destruction, invasive species, population growth, pollution, climate change and over exploitation. Poaching is part of over exploitation, and the other pairings are not in the acronym.

QB-U9-MCQ017
Answer: C. Roads, pipelines and clearings divide a large habitat into smaller isolated pieces. Regrowth, park expansion and reintroduction increase or restore habitat rather than splitting it.

QB-U9-MCQ018
Answer: B. In the dark polar winter chlorine compounds accumulate on ice crystals; when sunlight returns in spring they release chlorine atoms that destroy ozone catalytically. The 1980 line shows that UV alone does not produce a large October minimum.

QB-U9-MCQ019
Answer: A. Chlorine atoms act as a catalyst: each one can destroy thousands of ozone molecules before it is removed. The belief that chlorine is consumed in a one-to-one reaction underestimates how destructive a small amount of CFC can be.

QB-U9-MCQ020
Answer: A. The substitution removes the chlorine that destroys ozone, but HFCs can have global warming potentials hundreds to thousands of times that of CO_2. A good evaluation notes both the benefit and the drawback.

QB-U9-MCQ021
Answer: D. (200 t × 28) + (10 t × 265) = 5,600 + 2,650 = 8,250 t CO_2e. The value 5,600 counts only the methane, and 210 adds the masses without applying GWP.

QB-U9-MCQ022
Answer: A. Water vapor stays in the air only days, so its concentration is set by temperature (a feedback), not by long-lived emissions. It does absorb infrared radiation and is the most abundant greenhouse gas by volume.

QB-U9-MCQ023
Answer: A. Because most land area is in the Northern Hemisphere, seasonal plant growth causes a measurable annual cycle. The rising trend comes from fossil fuel combustion and deforestation, which continue all year.

QB-U9-MCQ024
Answer: C. Flooding, storm surge and saltwater contamination of freshwater make coastal areas uninhabitable, driving relocation. Saltwater intrusion degrades, rather than improves, drinking water and farmland.

QB-U9-MCQ025
Answer: B. Rate = (420 − 396) ppm ÷ 10 years = 2.4 ppm/yr. Remaining rise = 444 − 420 = 24 ppm, which takes 24 ÷ 2.4 = 10 years, so 2023 + 10 = 2033. Later years result from using a rate that is too small.

QB-U9-MCQ026
Answer: D. Each pass around the loop adds more warming than the last, so the change is amplified (positive feedback). Open ocean is dark and absorbs more sunlight than bright ice, which is why the process amplifies rather than limits warming.

QB-U9-MCQ027
Answer: A. Frozen soils store large quantities of organic carbon; when thawed, decomposition releases greenhouse gases that cause further warming, a positive feedback. Dark thawed ground actually absorbs more sunlight.

QB-U9-MCQ028
Answer: D. Dry, bare soil is easily carried away by wind and by heavy downpours, and repeated cycles damage soil structure and organic matter. Prolonged drought does not make soil resistant to erosion once intense rain arrives.

QB-U9-MCQ029
Answer: D. Flooded low-lying coastal land creates new shallow, sunlit habitat. Deeper water means less light reaches seagrass at its depth limit, a seawall blocks marsh migration, and seawater intrusion harms aquifers.

QB-U9-MCQ030
Answer: C. The points show a positive relationship, with one exception (Reef K). The four-week reef had 25% bleached, and the reefs with 0–2 weeks still had a few bleached colonies (3–9%), so the other statements contradict the graph or go beyond it.

QB-U9-MCQ031
Answer: B. Species track their preferred temperatures, moving poleward or deeper. Warmer water raises metabolic rates and holds less dissolved oxygen, which adds stress.

QB-U9-MCQ032
Answer: B. Carbon dioxide forms carbonic acid, which releases H^+ and lowers pH. The glucose-forming reaction is photosynthesis, and the sulfur reaction describes acid rain, not CO_2-driven ocean acidification.

QB-U9-MCQ033
Answer: C. Percent change = (91 − 60) ÷ 91 × 100 ≈ 34%. The value 31% is the drop in percentage points, not the percent change relative to the rate at pH 8.10.

QB-U9-MCQ034
Answer: B. $[H^+]$ ratio = 10^(8.2 − 8.0) = 10^0.2 ≈ 1.6, so H^+ is about 60% higher. A 20% increase comes from treating the 0.2 pH change as linear.

QB-U9-MCQ035
Answer: A. Acidification follows atmospheric CO_2, so cutting CO_2 emissions addresses the cause. Fertilizer adds nutrients, SO_2 controls address acid rain, and seawalls address flooding.

QB-U9-MCQ036
Answer: C. Invasive species are often generalist, r-selected species that can live in many conditions, reproduce quickly and spread widely. The other combinations describe K-selected specialists that struggle in new habitats.

QB-U9-MCQ037
Answer: C. 12 years ÷ 3 years = 4 doublings, and 40 ha × 2^4 = 640 ha. The value 160 ha comes from multiplying by 4 instead of doubling four times.

QB-U9-MCQ038
Answer: D. A new introduced species may itself become invasive or attack native species rather than the intended target. The other options are targeted and low-risk.

QB-U9-MCQ039
Answer: C. Eradication costs $2.0 million + (10 × $0.1 million) = $3.0 million; containment costs 10 × $0.35 million = $3.5 million; no action costs 10 × $0.6 million = $6.0 million. The $2.1 million figure counts only one year of monitoring.

QB-U9-MCQ040
Answer: D. Species with specific habitat requirements and a single food source cannot easily move or switch resources when conditions change. The other statements are not supported.

QB-U9-MCQ041
Answer: B. Percent decrease = (60 − 15) ÷ 60 × 100 = 75%. The value 25% is the percentage remaining, and 300% divides the change by the final value instead of the starting value.

QB-U9-MCQ042
Answer: B. Corridors reconnect fragmented populations so individuals can disperse and breed with other groups. A narrow strip adds little area and is mostly edge habitat, and it can even help invaders travel.

QB-U9-MCQ043

Answer: C. Managed populations are often bred from a few productive lines, narrowing the gene pool and making them more vulnerable to disease. The number of individuals is not the same as diversity.

QB-U9-MCQ044

Answer: A. Harvesting a species faster than it can replace itself is over exploitation, the O in HIPPCO. The scenario involves no habitat clearing, contaminant release or temperature change.

QB-U9-MCQ045

Answer: D. Cool-adapted species move up in elevation, but the summit limits how far they can go, so the available area shrinks. Lower slopes become warmer, not cooler.

QB-U9-MCQ046

Answer: A. The hypothesis links ozone amount directly to UV-B intensity, so both must be measured together over time. Skin cancer rates are influenced by behavior, and CFC or temperature data do not measure UV-B.

QB-U9-MCQ047

Answer: D. 4.0 kg × 1,430 ≈ 5,700 kg CO_2e. A zero ODP means no ozone damage but says nothing about climate warming, and each substance has its own GWP.

QB-U9-MCQ048

Answer: B. Avoided: 500 × 28 = 14,000 t CO_2e. Added: 500 × 2.75 = 1,375 t CO_2. Net = 14,000 − 1,375 = 12,625 t CO_2e decrease. Ignoring the CO_2 produced gives 14,000, and adding it gives 15,375.

QB-U9-MCQ049

Answer: A. Total warming depends on both GWP and the quantity emitted; CO_2 emissions are measured in tens of billions of tons, giving it the largest overall effect. CO_2 is the reference gas, with a GWP of 1, and methane is a potent greenhouse gas with large human sources.

QB-U9-MCQ050

Answer: B. Cold, salty water sinks in the North Atlantic and pulls warm surface water northward. Freshening reduces surface density and can slow this circulation, which would cool Europe regionally even as the global average warms.

QB-U9-MCQ051

Answer: B. Both curves rise and fall together and peak near 280 ppm, well below 420 ppm. A correlation alone cannot prove sole causation, and the peaks never reach 300 ppm.

QB-U9-MCQ052

Answer: C. A valid test changes only temperature, includes replicates and a control, and measures over time. Comparing different regions introduces many other differences, and a single unreplicated plot has no control.

QB-U9-MCQ053

Answer: C. Only pH, the independent variable, should differ; calcification rate is the dependent variable being measured. Anything else that affects coral growth must be controlled so differences can be attributed to pH.

QB-U9-MCQ054

Answer: D. The invasive curve is S-shaped and levels off near 1,500 plants per hectare, showing density-dependent limits. The native line levels off near 300 plants per hectare, so it declined but was not eliminated.

QB-U9-MCQ055

Answer: D. Halving time ≈ 70 ÷ 6 ≈ 11.7, or about 12 years. Using only births (70 ÷ 9 ≈ 8) or only poaching (70 ÷ 15 ≈ 5) ignores the net rate.

QB-U9-MCQ056

Answer: C. Before the road, interior area = (4 − 2) × (6 − 2) = 8 km². After the road, patch 1 (4 × 4 km) has (4 − 2) × (4 − 2) = 4 km² of interior and patch 2 (4 × 2 km) has none because every point lies within 1 km of an edge. Remaining = 4 ÷ 8 = 50%. The 67% option is the fraction of total forest area in patch 1.

Section B — FRQ Solutions

QB-U9-FRQ001

(a) When pH is lower (summer, lowest in July at about 7.85), larval survival is lower (about 46%); when pH is higher (winter, about 8.05), survival is highest (about 82%). *(1 point: states that survival rises and falls with pH, or that both are lowest in summer. Must connect the two variables.)*

(b) Ocean acidification: the ocean absorbs atmospheric CO_2, forming carbonic acid, which releases H^+ and lowers pH. Human activities: burning fossil fuels, vehicle emissions or deforestation. *(1 point: names ocean acidification, or CO_2 absorption by the ocean, and gives one valid CO_2-producing activity.)*

(c) Example: If oyster larvae are raised in seawater with lower pH, then the percentage that survive to day 14 will be lower than for larvae raised in seawater at a higher pH. *(1 point: if-then form, testable, links pH to survival.)*

(d) (i) Independent variable: seawater pH. Dependent variable: percentage of larvae surviving to day 14 (shell size or shell thickness are also acceptable). *(1 point: both correct.)* (ii) Any two of: water temperature, salinity, food (algae) type and amount, dissolved oxygen, light, tank size, larval density, age of larvae, parent stock. *(1 point: two valid controlled variables.)*

(e) Prepare tanks at several pH levels (for example 8.1, 7.9, 7.7) by bubbling CO_2 into seawater; the tank at the current baseline pH (8.1) is the control. Use at least three replicate tanks per level, each with the same number of larvae (for example 200), start them at the same time, and count survivors at day 14. *(2 points: 1 for at least two pH levels with a control at the baseline; 1 for replicate tanks or repeated trials.)*

(f) Survival would decrease progressively as pH decreases, with the lowest survival in the lowest-pH tanks and the highest survival in the control. *(1 point: describes a trend consistent with the hypothesis written in part (c).)*

(g) Example solutions: stop pumping water during low-pH periods and store higher-pH water in advance; add a base such as sodium carbonate to buffer the water; move the intake to a different location; use water monitoring to time spawning when pH is high; support policies that reduce CO_2 emissions. Disadvantages: added cost or energy, buffer chemicals might change salinity or water chemistry, storage tanks can allow disease to grow, shutting the pumps reduces production in summer, emission policies take decades. *(2 points: 1 for a reasonable solution; 1 for a valid disadvantage or unintended consequence of that solution.)*

QB-U9-FRQ002

(a) Examples: rapid reproduction (r-selected), generalist habitat needs, high dispersal ability, lack of natural predators, tolerance of many conditions, fast growth, or early leaf-out. *(1 point: one valid trait.)*

(b) Example: If invasive shrub density affects native plants, then plots closer to the trail (with high shrub density) will have fewer native wildflower species than plots farther from the trail. *(1 point: if-then, testable, links shrub density or distance to native richness.)*

(c) Independent variable: distance from the trail (or invasive shrub density). Dependent variable: number of native wildflower species (species richness) or native wildflower abundance. *(1 point: both correct; accept either independent variable as long as it is paired with a native-plant dependent variable.)*

(d) Establish several plots (for example 1 m × 1 m quadrats or 10 m × 10 m plots) at each of three or more distances from the trail (for example 5 m, 50 m, and 150 m). Use several replicate plots (at least 3–5) at each distance, located at random positions along the trail, and count the shrub stems and identify and count native wildflower species in each plot during the same week. *(2 points: 1 for plots at multiple distances with a defined plot size; 1 for replication and random placement.)*

(e) Any two of: soil type, slope, light (canopy cover), moisture, tree species composition, elevation, season or date of sampling, observer. *(1 point: two valid variables.)*

(f) Native wildflower richness would be lowest in the plots nearest the trail where shrub density is highest, and would increase with distance from the trail as shrub density decreases. *(1 point: consistent with the hypothesis in part (b).)*

(g) Examples: cutting or pulling the shrubs by hand, then replanting native species; applying herbicide; prescribed burning; introducing an approved insect that feeds on the shrub (biological control). Disadvantages: labor and cost, resprouting, herbicides harm non-target plants or contaminate water, burning can harm native

species, a biocontrol organism might attack native plants. *(2 points: 1 for a valid control method; 1 for a disadvantage or unintended consequence tied to that method.)*

(h) Hikers carry seeds on shoes and clothing, and pets or vehicles can spread seeds; trails create disturbed, open soil and light that seeds can colonize; the shrub was likely planted or dumped near entrances. *(1 point: gives a reason that links human use of the trail to seed dispersal or disturbed habitat.)*

QB-U9-FRQ003

(a) Chlorine increased from 1.7 ppb in 1980 to a peak of 3.4 ppb in 2000, then decreased to 2.5 ppb in 2020. *(1 point: increase then decrease, with the 2000 peak identified or implied.)*

(b) Percent decrease = (3.4 − 2.5) ÷ 3.4 × 100 = 0.9 ÷ 3.4 × 100 = 26.5% (accept 26% or 27%). *(1 point: correct answer with a correct setup.)*

(c) Rate = (100 − 225) DU ÷ (2000 − 1980) yr = −125 DU ÷ 20 yr = −6.25 DU per year (a decrease of 6.25 DU/yr). *(2 points: 1 for correct substitution; 1 for the correct answer with units, with negative sign or the word decrease.)*

(d) In the stratosphere UV light breaks CFC molecules apart, releasing chlorine atoms; each chlorine atom acts as a catalyst and destroys many ozone molecules ($O_3 \rightarrow O_2$) without being used up. Over Antarctica, ice crystals (polar stratospheric clouds) form in the winter darkness and chlorine compounds accumulate on them; when sunlight returns in spring the chlorine is released and rapidly destroys ozone. *(2 points: 1 for CFCs releasing chlorine under UV that destroys ozone catalytically; 1 for ice crystals or the return of sunlight in the Antarctic spring.)*

(e) The Montreal Protocol (international treaty that phased out CFCs and other ozone-depleting substances); accept an equivalent agreement or national bans on CFCs. CFCs have long residence times (decades to a century) in the atmosphere, so chlorine already in the stratosphere continues to destroy ozone long after production stops. *(2 points: 1 for a valid policy; 1 for explaining the delay by the long atmospheric lifetime of CFCs.)*

(f) Claim: the statement is not supported. Justification: in 2020 the minimum ozone column (125 DU) was still far below the 1980 value (225 DU) and the ozone hole was still 21 million km² compared with 3 million km² in 1980. *(1 point: claim that recovery is not complete, supported with a number or comparison from the table.)*

(g) Increased risk of skin cancer, cataracts (or other eye damage), or sunburn and weakened immune response. *(1 point: one valid effect.)*

QB-U9-FRQ004

(a) September sea ice extent decreased steadily from 6.3 million km² in 2000 to 3.9 million km² in 2020. *(1 point: decrease, with data or the word steady.)*

(b) Percent change = (3.9 − 6.3) ÷ 6.3 × 100 = −2.4 ÷ 6.3 × 100 = −38% (a 38% decrease). *(1 point: correct value with decrease indicated.)*

(c) Rate = (3.9 − 6.3) ÷ 2 decades = −1.2 million km² per decade. *(1 point: correct value with units.)*

(d) Remaining extent 3.9 million km² ÷ 1.2 million km² per decade = 3.25 decades = 32.5 years after 2020, so about 2052–2053. *(1 point: year 2052 or 2053, from a correct method.)*

(e) Examples: positive feedbacks (ice-albedo, methane from thawing permafrost) may accelerate the loss so the rate will not stay constant; natural variability in weather and ocean currents; emission changes will alter the rate; the five data points are too few to extrapolate with confidence. *(1 point: one valid reason.)*

(f) Snow and ice are bright and reflect most sunlight back to space. As warming melts the ice, dark ocean and land are exposed and absorb more solar energy, which causes more warming and more melting, a positive feedback loop that is strongest in the polar regions. *(2 points: 1 for ice reflecting sunlight and being replaced by darker surfaces that absorb more; 1 for the repeating positive feedback that amplifies warming.)*

(g) Cub survival decreases as sea ice extent decreases (a positive relationship): from 66% at 6.3 million km² to 35% at 3.9 million km². Explanation: bears hunt seals from the ice; less ice means shorter hunting seasons and longer swims or fasts on land, so mothers produce less milk and cubs starve or die. *(2 points: 1 for describing the relationship with direction or data; 1 for a valid ecological reason such as loss of hunting habitat or food.)*

(h) Examples: replace fossil fuels with renewable energy or nuclear power; carbon tax or cap-and-trade policies; improve energy efficiency; reduce methane emissions; reforest or protect carbon sinks. *(1 point: one valid action that reduces greenhouse gas emissions.)*

QB-U9-FRQ005

(a) 800 cows × 110 kg CH_4 per cow = 88,000 kg CH_4. *(1 point.)*

(b) CH_4: 88,000 kg × 28 = 2,464,000 kg CO_2e = 2,464 t CO_2e. N_2O: 4,000 kg × 265 = 1,060,000 kg CO_2e = 1,060 t CO_2e. Total = 2,464 + 1,060 = 3,524 t CO_2e. *(2 points: 1 for correct CO_2e of both gases with GWP applied; 1 for the sum with units, 3,524 t CO_2e.)*

(c) 2,464 ÷ 3,524 × 100 = 69.9%, or about 70%. *(1 point.)*

(d) Manure methane = 0.25 × 88,000 kg = 22,000 kg; captured = 0.80 × 22,000 kg = 17,600 kg CH_4; × 28 = 492,800 kg CO_2e = 492.8 t CO_2e per year (accept 493 t). *(2 points: 1 for 17,600 kg of methane captured; 1 for converting to 492.8 t CO_2e.)*

(e) 492.8 ÷ 3,524 × 100 = 14.0% (about 14%). *(1 point.)*

(f) Annual payment = 492.8 t × \$50/t = \$24,640 per year. Payback = \$350,000 ÷ \$24,640 per year = 14.2 years (about 14 years). *(2 points: 1 for \$24,640; 1 for about 14.2 years.)*

(g) Examples: add feed supplements or change feed (for example seaweed or fats) to reduce enteric methane; cover lagoons and capture gas; apply manure with less nitrogen loss (injecting rather than spreading) or use drier manure handling to reduce N_2O; improve cow productivity so fewer cows are needed. *(1 point: a valid action with a brief mechanism.)*

QB-U9-FRQ006

(a) Low: 4.0 mm/yr × 75 yr = 300 mm = 0.30 m. High: 10.0 mm/yr × 75 yr = 750 mm = 0.75 m. *(2 points: 1 for each scenario, with units.)*

(b) Low: 7 km² × 2,400 people/km² = 16,800 people. High: 22 km² × 2,400 people/km² = 52,800 people. *(2 points: 1 for each, using the area from the correct row.)*

(c) 52,800 − 16,800 = 36,000 more people. *(1 point.)*

(d) Thermal expansion: as ocean water warms, its volume increases. Melting of land-based ice (glaciers and ice sheets in Greenland and Antarctica) adds water to the ocean. *(2 points: 1 for each correct process with an explanation. Floating sea ice melt is not an acceptable answer, because it does not significantly change sea level.)*

(e) 14 km × \$18 million per km = \$252 million. *(1 point.)*

(f) \$252,000,000 ÷ 52,800 people ≈ \$4,770 per person (accept \$4,800). *(1 point: correct value with units.)*

(g) Examples: managed retreat that relocates residents and buildings to higher ground; restoring mangroves, marshes or dunes to absorb wave energy and trap sediment; raising buildings and improving drainage; zoning restrictions on development in flood zones; reducing greenhouse gas emissions to slow the rate of rise. A valid justification explains how the option lowers flood exposure, for example wetlands buffer storm surge at lower cost and provide habitat. *(1 point: valid solution with a justification.)*

Part III — Four Predicted Exam Solutions

For each exam: MCQ answers and explanations first, then complete FRQ solutions.

Predicted Practice Exam 1

Section I — MCQ Answers and Explanations

PE1-MCQ001

Answer: A. An estuary is a transition zone where river water and seawater mix, so salinity varies with tide and discharge. Nutrients carried by the river and trapped in the sediments make estuaries among the most productive biomes. The fresh-water and fully marine descriptions ignore the mixing, and the dark, nutrient-poor description fits the deep ocean, not a shallow estuary.

PE1-MCQ002

Answer: B. Legumes such as soybeans host Rhizobium bacteria that fix N_2 into ammonia, which is later converted to forms corn can use. Denitrification actually removes usable nitrogen from soil by returning it to the atmosphere, and leaching moves nitrate downward, not upward. Nitrification acts on ammonia, not on atmospheric N_2.

PE1-MCQ003

Answer: B. NPP = GPP − producer respiration = 3,400 − 1,500 = 1,900 kcal/m²/yr. The value 4,900 results from adding the two terms, and 3,400 is the gross value that has not yet been reduced by respiration.

PE1-MCQ004

Answer: A. The heron eats clams, which are primary consumers (making the heron a secondary consumer), and anchovies, which eat zooplankton and are therefore secondary consumers (making the heron a tertiary consumer). The anchovy eats zooplankton, so it is not a primary consumer, and zooplankton eat producers, making them primary consumers. The seal is a tertiary consumer, whereas the clam is a primary consumer.

PE1-MCQ005

Answer: B. Kelp is trophic level 1 and orcas are level 4, so three transfers occur: 480,000 × 0.10 × 0.10 × 0.10 = 480 kJ/m²/yr. The value 4,800 comes from counting only two transfers, and 48,000 from counting only one.

PE1-MCQ006

Answer: C. Carbon lost = 380 − 290 = 90 Mg C/ha; 90 × 3.67 ≈ 330 Mg CO_2/ha. The value 90 forgets to convert carbon to CO_2, and 1,390 converts the entire original stock (380 × 3.67) instead of just the loss. Draining converts a carbon sink into a carbon source because aerobic decomposers oxidize the stored organic carbon.

PE1-MCQ007

Answer: D. Regulating services control natural processes such as floods, erosion, climate, and storm impacts. Provisioning services deliver products such as food or timber, cultural services deliver recreational or spiritual benefits, and supporting services such as nutrient cycling underlie all the others.

PE1-MCQ008

Answer: B. This is primary succession on rock with no soil. Pioneer species such as lichens, mosses, and some microbes tolerate bare rock, and their weathering and decay begin to form soil. Trees, grasses, shrubs, and grazers all depend on soil or on plants that have not yet arrived.

PE1-MCQ009

Answer: A. Both sets of points rise with island area, and the near-mainland curve lies above the far curve at every area, matching island biogeography theory (larger and closer islands have higher immigration and lower extinction rates). The gap does not vanish above 10 km², richness on distant islands clearly rises with area, and no data show competition effects.

PE1-MCQ010

Answer: A. A species can persist only if the stress remains inside its tolerance range, and 8 ppt is within the clam's 5–30 ppt range only. The width of a range does not matter if the new condition falls outside it, and tolerance of high salinity does not imply tolerance of low salinity.

PE1-MCQ011

Answer: A. Richness peaks at intermediate times when pioneer and later species coexist, then declines as a closed canopy of shade-tolerant species outcompetes light-demanding species. Succession after a typhoon is secondary succession because soil remains, and the data show no repeated resets or total nutrient loss.

PE1-MCQ012

Answer: B. Specialists depend on a narrow range of resources, so they are vulnerable when that resource disappears. Generalists use a wide range of foods and can switch. The other options ignore this difference in niche breadth.

PE1-MCQ013

Answer: C. K-selected species invest heavily in a few young, live long, and have populations that fluctuate near the carrying capacity. The second description is the r-selected strategy, and the other two mix traits that do not occur together in either strategy.

PE1-MCQ014

Answer: D. A TFR well below replacement level means each generation is smaller than the previous one, and without immigration the population eventually shrinks while the median age rises. Low infant mortality is already built into the replacement value, and a TFR below 2.1 does not balance births and deaths.

PE1-MCQ015

Answer: C. Many offspring with little or no care and heavy early mortality gives a Type III curve, which on the log scale falls steeply at the start (Curve 1). Curve 2 is Type I (low early mortality, as in humans or whales), and Curve 3 is Type II (constant mortality). Oysters provide no parental care.

PE1-MCQ016

Answer: D. Density-dependent factors, such as competition, disease, and predation, intensify as population density rises, which pulls the population toward carrying capacity. Storms, eruptions, and freezes affect the population regardless of density and are density-independent.

PE1-MCQ017

Answer: A. Growth rate (%) = (births − deaths + immigration − emigration) per 1,000 ÷ 10 = (14 − 6 + 2 − 4) ÷ 10 = 0.6%. Ignoring migration gives 0.8%, ignoring emigration gives 1.0%, and forgetting to divide by 10 gives 6.0%.

PE1-MCQ018

Answer: D. In Stage 2, sanitation, medicine, and food supply reduce death rates while birth rates stay high, producing rapid growth. Stage 1 has no such improvements, Stage 3 is defined by falling birth rates, and Stage 4 has low, balanced rates.

PE1-MCQ019

Answer: B. Country 3's pyramid is narrow at the base and widest among older adults, so births will be fewer than deaths as the large older cohorts die. Country 1's broad base signals rapid growth, not high infant mortality, and Country 2's near-vertical sides indicate a roughly stable population.

PE1-MCQ020

Answer: C. Doubling time = 70 ÷ 2.8 = 25 years, so 50 years is two doublings: 600,000 × 2 × 2 = 2,400,000. One doubling gives 1,200,000, and 4,800,000 would require three doublings (75 years).

PE1-MCQ021

Answer: C. All three variables change together across provinces, which shows an association; the survey cannot separate the effects of education and contraceptive access or exclude other factors such as income, which is not even measured. Claiming that one factor has no effect, or that income is the sole driver, goes beyond the data.

PE1-MCQ022

Answer: C. The O horizon is the organic layer of litter and humus formed at the surface. The A horizon is a mixture of mineral particles and humus, the B horizon accumulates leached clay and minerals, and the C horizon is partly weathered parent rock.

PE1-MCQ023

Answer: B. Volcanic island arcs with an adjacent trench form where oceanic lithosphere subducts, melts, and feeds volcanoes on the overriding plate. Divergent boundaries make ridges, transform boundaries produce mainly earthquakes, and continental collisions produce folded mountains without trenches or island arcs.

PE1-MCQ024

Answer: C. The stratosphere contains most atmospheric ozone, and absorption of UV there warms it with height. Temperature falls with height in the troposphere and mesosphere, and the exosphere is the outermost layer where molecules escape to space.

PE1-MCQ025

Answer: B. Sand = 3.6 cm; silt = 7.2 − 3.6 = 3.6 cm; clay = 9.0 − 7.2 = 1.8 cm. Silt percent = 3.6 ÷ 9.0 × 100 = 40%. The value 80% treats the cumulative reading (7.2 cm) as the silt layer, and 20% is the clay fraction.

PE1-MCQ026

Answer: D. In the Hadley cell, warm air rises at the equator, cools, and drops rain, then descends near 30° as dry air that warms by compression, so clouds and rain are rare. The Coriolis effect deflects winds rather than stopping them, and insolation near 30° is high, not low.

PE1-MCQ027

Answer: B. This is a rain shadow: orographic lifting causes condensation and rain on the windward slope, and the air then descends, compresses, warms, and holds moisture instead of releasing it. Air descends, not rises, on the lee side, and elevation alone does not explain the dry valley because the coast is also low yet very wet.

PE1-MCQ028

Answer: B. Rain volume = 0.040 m × 5.0 × 10^6 m^2 = 200,000 m^3; runoff = 200,000 × 0.85 = 170,000 m^3. The 200,000 m^3 figure assumes all the rain runs off, and 30,000 m^3 is the volume that infiltrates or is retained (15%).

PE1-MCQ029

Answer: C. After urban development, the peak rises from about 18 to about 52 m^3/s and occurs at about 5 hours instead of 9.5 hours. Impervious surfaces reduce infiltration and storm drains speed delivery, shortening lag time and raising the peak, increasing flood risk. The other options describe the opposite effect or ignore the graph.

PE1-MCQ030

Answer: D. Negative anomalies mark La Niña, when strengthened trade winds drive warm surface water west and increase upwelling in the eastern Pacific; the deepest and longest negative interval on the graph is 2021–2022. Positive anomalies (2015–2016, 2023–2024) are El Niño events in which weakened trade winds suppress upwelling, and the 2017–2018 anomaly was weakly negative, not positive.

PE1-MCQ031

Answer: B. Absorbed = incoming × (1 − albedo). Ice absorbs 300 × 0.20 = 60 W/m^2; water absorbs 300 × 0.93 = 279 W/m^2. The increase is 279 − 60 = 219 W/m^2. The value 279 is the absorption by water alone, 240 is the reflected amount of ice, and 21 is the reflected amount of water.

PE1-MCQ032

Answer: A. When a shared resource is open to everyone, each user has an incentive to take more because the benefit is private and the cost of overuse is spread across the group, leading to depletion. Sustainable yield requires harvest limits, and the other two concepts do not describe resource sharing.

PE1-MCQ033

Answer: A. Removing all trees exposes steep soil to rain, increasing erosion and sediment in the stream, and eliminates shade so the water warms, both of which stress salmon. The other options reverse the effect of removing roots and canopy.

PE1-MCQ034

Answer: A. Sulfide minerals react with oxygen and water to form sulfuric acid, which lowers pH and mobilizes toxic metals. Eutrophication is caused by nutrients, thermal pollution by heated water, and saltwater intrusion by over-pumped coastal aquifers.

PE1-MCQ035

Answer: B. Water saved = 1,000 − 520 = 480 m^3/ha; 480 ÷ 1,000 × 100 = 48%. The 45% figure is the efficiency of flood irrigation (450 ÷ 1,000), 52% is the fraction of the original volume still used, and 86% is the efficiency of drip (450 ÷ 520).

PE1-MCQ036

Answer: D. Corn energy = 5,000 × 3,500 = 17,500,000 kcal. Beef produced = 5,000 ÷ 8 = 625 kg, which contains 625 × 2,500 = 1,562,500 kcal. The ratio is 17,500,000 ÷ 1,562,500 ≈ 11. The value 8 is only the feed conversion ratio, and 1.4 compares the energy density of corn and beef.

PE1-MCQ037

Answer: A. Effort rose every period (from 32 to 66 thousand boat-days between 1995 and 2020) while catch fell from 48 to 20 thousand tons, so catch per boat-day dropped from about 1.5 to 0.3 tons, a sign that fewer fish remain. Effort never decreased, so the catch decline was not caused by less fishing.

PE1-MCQ038

Answer: D. Permeable surfaces and bioswales let stormwater infiltrate and be filtered by soil and plants, reducing both the volume and the pollutant load that reaches the estuary. Drains and added pavement speed runoff and increase peak flows, and road salt adds a pollutant.

PE1-MCQ039

Answer: A. Yields rose alongside fertilizer use and irrigated area, so the likely environmental costs are nutrient runoff that triggers eutrophication and growing water withdrawals. The table shows yields increasing, not declining, and higher inputs of irrigation and fertilizer need more energy. The Green Revolution also tended to reduce crop genetic diversity.

PE1-MCQ040

Answer: A. Total footprint = 250,000 × 3.8 = 950,000 gha. Allowed footprint per person = 620,000 ÷ 250,000 = 2.48 gha. Required reduction = (3.8 − 2.48) ÷ 3.8 = 0.347, or about 35%. The value 53% is the overshoot (950,000 ÷ 620,000 = 1.53) minus one, and 153% reports the overshoot ratio itself.

PE1-MCQ041

Answer: A. Individual organisms do not adapt to a pesticide; survivors with heritable resistance reproduce, so resistant genotypes become common (natural selection). IPM slows this through monitoring, treatment thresholds, biological control, and varying control methods rather than repeated use of one chemical or higher doses.

PE1-MCQ042

Answer: B. Renewable sources are replenished on human time scales; waves are driven continuously by wind and solar heating. Natural gas, uranium, and coal are finite fuels.

PE1-MCQ043

Answer: A. Fission does not burn fuel and releases almost no CO_2 or SO_2 while operating, but spent fuel contains long-lived radioactive isotopes needing secure storage. Uranium is a finite resource, so nuclear power is not renewable.

PE1-MCQ044

Answer: C. Dams block migratory fish such as salmon and trap sediment in the reservoir, starving downstream habitats. Hydropower turbines use moving water, not combustion, and they do not release heat in meaningful amounts.

PE1-MCQ045

Answer: D. In 2024, wind generated 200 GWh and solar 300 GWh, for 500 GWh out of 1,000 GWh in total: 500 ÷ 1,000 = 50%. Wind alone gives 20%, solar alone 30%, and 38% is the diesel share.

PE1-MCQ046

Answer: D. Ratio = $(7.5 \div 5.0)^3 = 1.5^3 = 3.375 \approx 3.4$. A ratio of 1.5 ignores the cube, 2.25 (shown as 2.3) squares the speed ratio, and 3.0 has no mathematical basis in this relationship.

PE1-MCQ047

Answer: C. The two annual averages are equal, so Site 2's summer peak does not give more total energy, and its winter output is much lower (about 2.4 versus 4.0 kWh/m²/day in December). Site 2 exceeds Site 1 in April through September, so Site 1 is not higher in every month. A flatter seasonal profile reduces the storage or backup needed.

PE1-MCQ048

Answer: B. Power saved per bulb = 40 − 8 = 32 W; for 200 bulbs = 6,400 W = 6.4 kW. Operating time = 6 × 180 = 1,080 h. Energy saved = 6.4 kW × 1,080 h = 6,912 kWh. The value 8,640 kWh is the energy used by the old bulbs alone, and 1,728 kWh is the energy used by the LEDs.

PE1-MCQ049

Answer: A. Fracking injects water, sand, and chemicals into shale, and disposal of the wastewater can contaminate water supplies or induce small earthquakes. The other options describe nuclear waste, large dams, and biomass combustion.

PE1-MCQ050

Answer: B. Annual energy = 40 MW × 8,760 h × 0.32 = 112,128 MWh = 112,128,000 kWh; households = 112,128,000 ÷ 10,000 = 11,213, or about 11,200. The value 35,040 ignores the capacity factor, 112,128 is the MWh figure mistaken for households, and 1,120 is off by a factor of ten.

PE1-MCQ051

Answer: C. Hydrogen is an energy carrier, not an energy source. If it is made by electrolysis using coal-based electricity or by reforming natural gas, the upstream emissions can be large, so the benefit depends on production. Fuel cells produce electricity electrochemically and emit mainly water.

PE1-MCQ052

Answer: C. Secondary pollutants form in the atmosphere from reactions among primary pollutants; tropospheric ozone is the classic example. The other three are released directly from sources and are primary pollutants.

PE1-MCQ053

Answer: C. Radon is a radioactive gas produced by the decay of uranium and radium in rock and soil, and it can enter basements through cracks. Natural gas combustion produces CO and NO_2, and furniture and glues release formaldehyde.

PE1-MCQ054

Answer: D. Photochemical smog forms when sunlight drives reactions of nitrogen oxides and VOCs, producing ozone with a delay of some hours after the morning NO_x emissions. Ozone is a secondary pollutant, not a tailpipe emission, and the graph shows that its peak does not coincide with the NO_2 peak.

PE1-MCQ055

Answer: D. A thermal inversion occurs where temperature rises with height (here 200–500 m); the warm air above is less dense than the cooler air below, so rising air from the surface cannot pass through and pollutants accumulate. A normal decline in temperature with height promotes mixing, and the profile clearly varies by several degrees.

PE1-MCQ056

Answer: A. Remaining fraction = 1 − 0.92 = 0.08; 4,800 × 0.08 = 384 tons per year. The value 4,416 is the amount removed (4,800 × 0.92), and 440 comes from multiplying by 0.092 rather than 0.08.

PE1-MCQ057

Answer: C. Calcium carbonate in limestone reacts with and neutralizes hydrogen ions, buffering the water against acid deposition; granite has little buffering capacity. Rock type does not change the amount of rainfall, and limestone raises, not lowers, pH.

PE1-MCQ058

Answer: C. Only the valley residential site changes strongly with season, which suggests a seasonal local source (wood heating) concentrated by inversions that are common in cold, calm valleys. Traffic-dominated sites barely change, an industrial source would affect all sites, and photochemical pollution peaks in summer.

PE1-MCQ059

Answer: D. Excess nutrients cause an algal bloom; when the algae die, bacterial decomposition consumes oxygen, producing hypoxia and fish kills. Blooms block light and eventually reduce oxygen, nutrients do not cool water, and added sediment would reduce, not increase, light.

PE1-MCQ060

Answer: D. The solubility of oxygen in water falls as temperature rises, and warmer water also increases the metabolic oxygen demand of fish, so cold-water species such as trout are stressed. Warm water does not change salinity.

PE1-MCQ061

Answer: D. Ratio = 20 ÷ 0.002 = 10,000. The value 700 is the seal-to-algae ratio, 286 is the bear-to-cod ratio (20 ÷ 0.07), and 20 is the bear's concentration in ppm. Biomagnification of fat-soluble, persistent compounds produces the increase up the food chain.

PE1-MCQ062

Answer: B. Oxygen demand from decomposition of organic waste peaks some distance downstream as the waste is carried and broken down, creating the oxygen sag; farther on, the waste is depleted and reaeration restores oxygen. Photosynthesis adds oxygen, and no data suggest temperature or depth changes at that point.

PE1-MCQ063

Answer: C. Diverted = 21,600 + 14,400 = 36,000 tons; 36,000 ÷ 120,000 × 100 = 30%. The values 18% and 12% count only recycling or only composting, and 70% is the share that goes to the landfill.

PE1-MCQ064

Answer: C. The LD50 is the dose that kills half of the test animals: about 10 mg/kg for Chemical A and about 100 mg/kg for Chemical B, so A is about 10 times more toxic. A curve farther to the right shows a less toxic chemical, and the maximum effect alone does not describe toxicity.

PE1-MCQ065

Answer: C. Concentration 5 mg/L = 5 g/m³, so the daily load = 5 g/m³ × 40,000 m³ = 200,000 g = 200 kg. The marsh removed 60%, or 120 kg/day, and that amount now reaches the bay. The value 80 kg is the load that already reached the bay with the marsh in place, and 200 kg is the total load.

PE1-MCQ066

Answer: A. CFCs are stable in the troposphere, drift to the stratosphere, and are broken apart by UV light; each chlorine atom can destroy many ozone molecules as a catalyst. The other options describe processes that do not occur.

PE1-MCQ067

Answer: D. Sunlight warms the surface, which emits infrared radiation; gases such as CO_2, CH_4, and water vapor absorb it and re-radiate some of it downward, raising surface temperature. Without this effect Earth would be much colder.

PE1-MCQ068

Answer: B. Invasive species often reproduce quickly, tolerate or use a wide range of resources, and are released from the predators and pathogens that controlled them at home. The other sets describe species that are vulnerable or well controlled.

PE1-MCQ069

Answer: D. CITES restricts international trade in listed species and their products. The Montreal Protocol addresses ozone-depleting substances, the Kyoto Protocol addresses greenhouse gases, and the Basel Convention addresses hazardous waste.

PE1-MCQ070

Answer: A. CFCs are very stable with atmospheric lifetimes of 50 to 100 years or more, so the chlorine reservoir declines slowly even after emissions stop. The replacements (such as HFCs) do not contain chlorine, and ozone continues to be formed by UV light acting on O_2.

PE1-MCQ071

Answer: D. CO_2 equivalent = 50 t CH_4 × 28 = 1,400 t CO_2e. The value 1.8 divides instead of multiplying, 50 ignores the warming potential, and 78 incorrectly adds the two numbers.

PE1-MCQ072

Answer: B. Percent decrease = (7.8 − 3.9) ÷ 7.8 × 100 = 50%. The value 3.9% mistakes the final extent for the percent, 100% divides by the final value instead of the starting value, and 200% reports the ratio 7.8 ÷ 3.9 as a percentage.

PE1-MCQ073
Answer: C. A positive feedback amplifies the original change; permafrost thaw adds greenhouse gases that drive further warming and thaw. Plant uptake of CO_2 and cooling from sea ice would be negative feedbacks, and warming reduces sea ice rather than increasing it.

PE1-MCQ074
Answer: A. Seawater expands as it warms, and melting land ice adds water to the ocean. Floating sea ice and ice shelves already displace their own weight in water, so their melting adds almost nothing to sea level, and warming water expands rather than contracts.

PE1-MCQ075
Answer: D. Dissolved CO_2 forms carbonic acid, which lowers pH and reduces the carbonate ions needed to build calcium carbonate shells. Acidification reduces, not increases, carbonate availability, and it does not change salinity or nitrate directly.

PE1-MCQ076
Answer: B. Change = 421 − 316 = 105 ppm over 2023 − 1959 = 64 years; 105 ÷ 64 ≈ 1.6 ppm/yr. The value 105 is the total change, 0.61 inverts the ratio, and 6.6 divides the final concentration by the number of years.

PE1-MCQ077
Answer: A. Tropospheric ozone forms from NO_x and VOCs in sunlight and is harmful to respiratory systems and crops; stratospheric ozone forms naturally and shields life from UV-B. The other options swap the roles or ignore the harm caused near the surface.

PE1-MCQ078
Answer: C. The pH decrease is 8.12 − 8.04 = 0.08, so $[H^+]$ increased by a factor of 10^0.08 ≈ 1.20, or about 20%. The value 8% treats the pH change as a percent, 1% is the relative change in the pH number (0.08 ÷ 8.12), and 80% multiplies the pH change by 10.

PE1-MCQ079
Answer: D. Forest cover is similar in the middle and high bands (80% and 85%) yet bird density is more than four times higher at the top, pointing to disease pressure: the cold limits mosquitoes. Warming would raise the elevation at which mosquitoes can live, reducing refuge habitat. The data do not mention human presence, and forest cover cannot explain the large difference between the upper two bands.

PE1-MCQ080
Answer: B. Replicated plots with a manipulated treatment (crabs excluded or present) and matched conditions isolate the effect of crabs. Comparing two estuaries confounds crab effects with other differences, a single plot has no replication or control, and a one-time survey shows only a correlation.

Section II — FRQ Solutions

PE1-FRQ001
(a) Cultural eutrophication (nutrient pollution from fertilizer runoff) leading to algal blooms and hypoxia (low dissolved oxygen) in the bay. Accept "hypoxia/dead zone caused by excess nitrogen." *(1 point: identifies nutrient enrichment/eutrophication or hypoxia caused by fertilizer runoff)*

(b) Example: If the restored marsh intercepts nitrogen, then the nitrate concentration at H2 (near the bay) will be lower than at H1 (near the farmland) in Harbor Creek, whereas in Mill Creek (no marsh) the nitrate concentration at M2 will be about the same as at M1. Accept a directional prediction such as "nitrate will decrease by a larger percentage between upstream and downstream in Harbor Creek than in Mill Creek." *(1 point: a testable if/then or prediction statement that compares nitrate upstream and downstream and relates it to the marsh)*

(c) (i) Independent variable: the location relative to the marsh (upstream point versus downstream point, i.e., H1 versus H2, with M1 versus M2 as the comparison creek); accept "presence or absence of the marsh." *(1 point)* (ii) Dependent variable: nitrate concentration (mg/L of nitrate-nitrogen) in the water; accept dissolved oxygen only if the hypothesis in (b) is about oxygen. *(1 point)*

(d) Any one valid controlled variable with a reason, for example: time of day and tide stage (tides change dilution and salinity), recent rainfall or irrigation events (they flush nitrate into the creeks), depth and method of sampling, water temperature, the test method and instrument. Example: sample at the same tide stage at every

point because incoming seawater dilutes nitrate and would hide differences caused by the marsh. *(1 point: names a relevant variable and explains that it could otherwise affect nitrate concentration/confound the comparison)*

(e) Example: Once a week for 12 weeks during the growing season, collect three replicate water samples at each of H1, H2, M1, and M2 on the same morning at the same tide stage, 10 cm below the surface, in clean bottles; measure nitrate immediately with the same test kit or spectrophotometer and record water temperature, salinity, and DO. *(2 points: 1 point for a workable sampling design that specifies locations and timing/frequency; 1 point for replication or repeated measurements and a stated measurement method)*

(f) Example: Calculate the mean nitrate concentration at each point and the percent reduction from upstream to downstream in each creek; graph the means with error bars and compare Harbor and Mill Creek, using a statistical test such as a t-test on the differences. The hypothesis is supported if the reduction in Harbor Creek is significantly larger than in Mill Creek. *(1 point: describes a comparison of averages (or percent change) between the creeks and a way to decide whether the difference is meaningful)*

(g) Acceptable solutions: restore additional wetland or a riparian buffer on Mill Creek; use controlled-release or precision application of fertilizer, plant winter cover crops, or install constructed treatment wetlands at the ditch outlets; upgrade or expand buffer strips. Justification examples: wetland plants and microbes take up nitrate and convert it to N_2 by denitrification, so less nitrogen reaches the bay; lower nitrogen inputs mean less algal growth and decomposition, so dissolved oxygen stays high enough for oysters. *(2 points: 1 point for a realistic solution that reduces nutrient input or interception; 1 point for linking it to reduced algal blooms/oxygen demand)*

PE1-FRQ002

(a) Active-layer depth increased steadily from 42 cm in 2000 to 77 cm in 2025, and the rate of increase was greater in later years (3 cm in 2000–2005 versus 9 cm in 2020–2025). *(1 point: states an increase and supports it with data from the table)*

(b) Percent increase = (77 − 42) ÷ 42 × 100 = 35 ÷ 42 × 100 = 83.3%, or about 83%. *(1 point: correct setup and answer of about 83%)*

(c) Permafrost holds large amounts of frozen organic matter. When it thaws, microbes decompose the organic matter and release CO_2 (and CH_4 in waterlogged soils), which are greenhouse gases. The added greenhouse gases enhance warming, which thaws more permafrost and releases more gas, so the change amplifies itself. *(2 points: 1 point for thaw releasing CO_2/CH_4 from decomposing stored organic matter; 1 point for the loop in which the added warming causes more thawing)*

(d) Accept one of: dark shrub leaves and stems absorb more sunlight than snow, so the lower albedo increases heat absorption in spring; shrubs trap blowing snow, and the thick snow insulates the ground in winter, keeping the soil warmer and thawing it deeper. *(1 point: a mechanism that explains additional warming, such as reduced albedo or insulation by trapped snow)*

(e) (i) Claim: the data are consistent with the claim that rising temperature contributes to the increase in shrub cover. *(1 point: a claim that the data support or are consistent with the student's statement)* (ii) Evidence: from 2000 to 2025 July temperature rose from 3.1 °C to 5.7 °C while shrub cover rose from 12% to 38%, and every five-year rise in temperature was accompanied by a rise in shrub cover. Limitation: the data are only correlational and come from a single site, so they cannot rule out other factors such as longer growing seasons, deeper snow, nutrient release from thawing soil, or grazing changes. *(1 point: cites specific data from the table and states a valid limitation or the need for controlled experiments)*

(f) The depth increased by 9 cm between 2020 and 2025, so a continued trend gives 77 + 9 = about 86 cm in 2030. Accept 84–90 cm if the explanation refers to the recent rate of increase (a linear estimate from the whole record, 1.4 cm per year, gives about 84 cm). *(1 point: a value in the accepted range with a statement of how it was obtained)*

(g) Examples: build on pilings or elevated foundations anchored in permafrost; install thermosyphons or insulating gravel pads to keep the ground frozen; relocate buildings to bedrock or stable sites; monitor the ground temperature and adapt maintenance. Justification: pilings and insulation reduce heat transfer into the ground so thaw settling does not damage the structures; relocation avoids the risk entirely. *(2 points: 1 point for a feasible solution; 1 point for a justification linked to preventing ground thaw or settlement)*

PE1-FRQ003

(a) Energy = nameplate capacity × hours × capacity factor = 250 kW × 8,760 h × 0.18 = 394,200 kWh per year. *(2 points: 1 point for a correct setup using capacity, hours, and capacity factor; 1 point for the correct answer with units)*

(b) The farm uses 310,000 kWh, so the surplus is 394,200 − 310,000 = 84,200 kWh. Benefit = 310,000 kWh × \$0.13/kWh + 84,200 kWh × \$0.06/kWh = \$40,300 + \$5,052 = \$45,352 per year. *(2 points: 1 point for correctly splitting the output into used and surplus electricity; 1 point for the correct total of about \$45,350)*

(c) Payback = cost ÷ annual benefit = \$325,000 ÷ \$45,352 per year = 7.2 years. *(1 point: correct answer, 7.2 years, or consistent with the value in (b))*

(d) Annual CO_2 avoided = 394,200 kWh × 0.40 kg/kWh = 157,680 kg = 157.7 metric tons; over 25 years = 157.68 t × 25 = 3,942 metric tons CO_2 (about 3,900 t). *(1 point: correct answer with units, including the conversion from kg to metric tons)*

(e) N applied = 160 kg/ha × 120 ha = 19,200 kg; savings = 19,200 kg × 0.25 = 4,800 kg of nitrogen per year. *(1 point: correct answer with units)*

(f) Acceptable limitations: solar output is intermittent (no output at night and low in winter, so the farm still needs grid or backup power); the temperate climate has cloudy, rainy winters and a low capacity factor (18%); a ground-mounted array would convert farmland or habitat (rooftops avoid this); manufacturing panels and batteries requires mined materials and energy. *(1 point: a valid limitation linked to solar or to this site)*

(g) Examples: add battery storage that charges from the surplus solar output; install a small wind turbine or a biogas digester that burns dairy manure to produce electricity in the evening and winter; stay connected to the grid with net metering; shift loads (such as milking equipment or water heating) into daylight hours. Justification: batteries store daytime surplus for use at night; a manure digester provides dispatchable power regardless of weather and reduces methane emissions from manure; wind often blows in winter when solar output is low. *(2 points: 1 point for a feasible solution that addresses night or winter supply; 1 point for a justification that explains why it fills the gap)*

Predicted Practice Exam 2

Section I — MCQ Answers and Explanations

PE2-MCQ001

Answer: B. Warm, wet conditions make decomposition fast, so nutrients move from litter back into roots and biomass almost immediately and the soil holds little. The cold-ground option describes tundra, not the tropics. Fires are not a regular feature of intact rainforest, and shade does not stop decomposers; the tropical forest floor decomposes litter within weeks.

PE2-MCQ002

Answer: B. Estuaries mix fresh river water with seawater, producing brackish conditions, and they trap nutrients and sediment, which supports high productivity and nursery habitat. The other options describe a fast stream, an open-ocean surface layer in the tropics, and the deep ocean floor.

PE2-MCQ003

Answer: A. NPP = GPP − plant respiration = 3,200 − 2,000 = 1,200 g C/m²/yr. 5,200 results from adding the two values; 1,600 is half of GPP, and 2,000 is simply the respiration figure.

PE2-MCQ004

Answer: B. Three transfers occur: 80,000 × 0.10 × 0.10 × 0.10 = 80 kcal/m²/yr. 800 counts only two transfers, 8,000 only one, and 8 counts four.

PE2-MCQ005

Answer: B. Peat is a large carbon reservoir. Draining exposes it to oxygen, and fire oxidizes it, transferring carbon to the atmosphere as CO_2. Fossil fuels form over millions of years, and burning does not make carbonate rock or reverse the flow of carbon into the soil.

PE2-MCQ006

Answer: C. Grouper is the only predator of parrotfish in this web, so parrotfish increase and graze the algae down. Sardines would also increase, which would reduce, not increase, zooplankton. Groupers do not eat algae, and sharks feed only on groupers, so they would decline.

PE2-MCQ007

Answer: C. Hotspots combine high endemism with severe habitat loss, which is why they are conservation priorities. Intact wilderness, generalist-dominated areas and fully protected areas do not define hotspots.

PE2-MCQ008

Answer: D. Regulating services moderate environmental conditions such as storms and erosion, and provisioning services supply goods such as seafood. The other pairings swap the categories.

PE2-MCQ009

Answer: C. On a logarithmic axis, 300 ha lies about halfway (in log terms) between 100 ha (38 species) and 1,000 ha (52 species): $10 + 14 \times \log_{10}(300) \approx 45$. Linear interpolation between 100 and 1,000 ha gives 41, which ignores the log scale; 38 and 52 are the values at 100 and 1,000 ha.

PE2-MCQ010

Answer: B. Because soil and seeds survive, the process is secondary succession, which starts with pioneers such as vines and light-demanding trees. Lichens and soil formation belong to primary succession; canopy trees and a climax community appear only after decades.

PE2-MCQ011

Answer: C. At 30 °C the curve for species A has dropped to zero (beyond its upper limit of tolerance), species B is still growing well, and species C is at its optimum. A wide range makes B the most tolerant, not the first to be lost, and C benefits from the warming.

PE2-MCQ012

Answer: A. Generalists tolerate changing conditions and can switch food and nest sites; specialists depend on a narrow set of resources that the conversion removes. Specialization reduces competition but increases vulnerability, and plantation trees do not form old-growth hollows quickly.

PE2-MCQ013

Answer: B. r-selected species invest in rapid reproduction of many offspring rather than in individual survival. The second option describes K-selected species, and the other two mix traits that do not occur together in either strategy.

PE2-MCQ014

Answer: A. Many eggs, no parental care and heavy hatchling mortality, with the few survivors living for decades, give a Type III curve: a very steep early decline on the log axis (curve Z). Curve X (Type I) shows high survival until old age, and curve Y (Type II) shows a constant death rate; minimal parental care alone does not make a curve Type II.

PE2-MCQ015

Answer: A. Carrying capacity is the maximum population the environment can support, set by resources such as food. Removing fruit trees lowers it. A corridor adds resources and habitat; a typhoon changes population size, not the resource limit; and a hunting ban changes mortality rather than the carrying capacity.

PE2-MCQ016

Answer: B. Growth rate = (births + immigration − deaths − emigration) ÷ 1,000 × 100 = (14 + 5 − 6 − 1) ÷ 10 = 1.2%. 0.8% ignores migration, 1.3% ignores emigration, and 1.4% adds emigration instead of subtracting it.

PE2-MCQ017

Answer: A. Country Q has a narrow base and a large share of older adults, so it is aging and likely to shrink; Country P's wide base means many future parents, so it will keep growing through population momentum. The other statements reverse or ignore what the shapes show.

PE2-MCQ018

Answer: A. In dense cities, costly housing, education and job opportunities for women lower fertility. High infant mortality, limited contraception and agricultural labor needs all tend to raise fertility.

PE2-MCQ019

Answer: C. Population momentum occurs when large cohorts of young people move into their childbearing years: even at replacement fertility, births exceed deaths for decades. Rising deaths would reduce growth, and the other statements are not generally true.

PE2-MCQ020

Answer: C. Doubling time = 70 ÷ 2.8 = 25 years, so 50 years is two doublings: 8 → 16 → 32 million. 16 million is only one doubling, 19 million is a linear estimate (8 × 2.4), and 64 million counts three doublings.

PE2-MCQ021

Answer: D. The dashed line falls in Stage 3, where birth rates are falling toward the already low death rates; the shaded gap is still positive, so the population is growing more slowly. Stage 2 has high birth rates and falling death rates, and Stage 4 has rates that are nearly equal.

PE2-MCQ022

Answer: A. Subduction produces strong earthquakes and tsunamis and, as the sinking plate melts, explosive volcanoes in an arc. Sliding plates describe transform boundaries, and ridge volcanism occurs at divergent boundaries.

PE2-MCQ023

Answer: D. Topsoil lies at the surface, and without roots and canopy to protect it, runoff and raindrop impact remove it first. Bedrock and parent material lie deep below, and the subsoil holds less organic matter than the topsoil.

PE2-MCQ024

Answer: A. The troposphere is the layer people live in, where weather and smog occur. The other layers are higher and are described by their correct features, none of which includes surface weather.

PE2-MCQ025

Answer: D. Percolation rate falls from 40 to 2 mm/h as clay rises from 10% to 45%, consistent with small pores that slow water movement. Clay particles are the smallest, not the largest, and Soil Z drains slowest, so it is poorly suited to rapid-drainage crops.

PE2-MCQ026

Answer: A. Strong solar heating at the equator causes air to rise (the intertropical convergence zone); as it rises it cools, and water vapor condenses into heavy rain. Sinking air produces deserts near 30° latitude, and the other options do not describe equatorial circulation.

PE2-MCQ027

Answer: D. A watershed drains to a common outlet, so land uses upstream affect the water downstream. Removing forest from steep slopes raises erosion and runoff into the reservoir; the other options are practices that protect water quality.

PE2-MCQ028

Answer: C. Earth's 23.5° axial tilt changes sun angle and day length strongly at mid-latitudes but only slightly near the equator. Earth's distance from the Sun does not cause the seasons, and the Moon does not warm the tropics.

PE2-MCQ029

Answer: A. This is the rain shadow effect: orographic lifting causes condensation and rainfall on the windward side, and the air that crosses the crest is dry and warms as it sinks. The other options are not explanations of mountain rain shadows.

PE2-MCQ030

Answer: A. With a single column per treatment, chance differences cannot be separated from real effects; replication and averaging improves reliability. Changing soil types or water volumes adds confounding variables, and the 0% column is the control needed for comparison.

PE2-MCQ031

Answer: A. Large positive anomalies (El Niño conditions) coincide with weakened trade winds and suppressed upwelling; with fewer nutrients, plankton and anchovy populations fall. Stronger upwelling would raise productivity, warmer water holds less (not more) oxygen, and the data do not show effort.

PE2-MCQ032

Answer: D. This is the tragedy of the commons: individual gain from using a shared resource outweighs the individual share of the cost, so the resource is overused. Private ownership and quotas would limit this behavior.

PE2-MCQ033

Answer: B. Removing all trees exposes soil and increases runoff, erosion and sedimentation. Clearcutting also reduces shade and increases stream temperature, and decaying roots release, not retain, nutrients.

PE2-MCQ034

Answer: D. Mining exposes sulfide minerals; oxidation produces acid mine drainage, which lowers pH and dissolves metals. Eutrophication, CFCs and thermal pollution do not produce low-pH, metal-rich streams.

PE2-MCQ035

Answer: A. Drip irrigation puts water directly at the roots, which minimizes evaporation and runoff and can cut water use substantially, though it costs more to install. Flood and furrow irrigation lose much water to evaporation and seepage, and midday spraying increases evaporation.

PE2-MCQ036

Answer: B. Extra feed = 90,000 kg × (7.0 − 2.2) = 90,000 × 4.8 = 432,000 kg per day. 198,000 kg is the feed for chicken, 630,000 kg is the total feed for beef, and 828,000 kg adds the two.

PE2-MCQ037

Answer: D. Catch per unit effort has fallen from about 0.6 t per boat in 2000 to about 0.1 in 2020 even as boats increased, indicating a declining stock. Boat numbers rose, not fell, and 2020's catch per boat is far below the 1980 value.

PE2-MCQ038

Answer: D. Pond effluent with excess nutrients, antibiotics and waste is a known problem, along with mangrove loss and escapes. Trawling fuel use and bycatch are problems of wild-capture fishing, and aquaculture's yield per hectare is high.

PE2-MCQ039

Answer: C. A high, early peak is typical of urban runoff: pavement prevents infiltration and drains carry water rapidly to streams. Permeable pavement, green roofs and retention basins increase infiltration or storage and lower the peak. Forest soils absorb water and produce lower, later peaks.

PE2-MCQ040

Answer: B. Deficit per person = 3.0 − 1.2 = 1.8 gha; × 50 million = 90 million gha. 60 million is total biocapacity, 150 million is total footprint, and 210 million is the sum of the two.

PE2-MCQ041

Answer: D. IPM uses monitoring and thresholds, and chemical control only when pest numbers justify the cost. Because the count is below the threshold and predators are present, the best action is to keep monitoring. Broad-spectrum spraying kills predators, calendar spraying ignores thresholds, and introduced predators can become invasive.

PE2-MCQ042

Answer: B. Coal and uranium exist in finite deposits and are not replenished on a human time scale. Biomass, geothermal energy and wind are renewable.

PE2-MCQ043

Answer: A. Coal has a higher carbon content and contains sulfur and ash, so it emits more CO_2, SO_2 and particulates than natural gas, which is the cleanest-burning fossil fuel.

PE2-MCQ044

Answer: C. Anthracite, the highest rank, has the highest carbon content and energy density. Peat is not yet coal, and lignite and bituminous coal are lower ranks.

PE2-MCQ045

Answer: D. Country X gets about 80% of its electricity from hydroelectric power, which depends directly on water flow. The other countries rely mostly on thermal plants that may use cooling water but do not depend on river flow to generate power the way a dam does.

PE2-MCQ046

Answer: A. Fission produces electricity without direct CO_2 emissions but creates long-lived radioactive waste. Nuclear plants do not release SO_2, uranium is finite, and cooling water discharge can cause thermal pollution.

PE2-MCQ047

Answer: C. Reservoirs trap sediment and the nutrients it carries, so floodplains and deltas receive less, lowering soil fertility and causing coastal erosion. Dams also moderate floods and block fish migration.

PE2-MCQ048

Answer: C. Biomass carbon is part of the short-term carbon cycle, so it can be near carbon-neutral only if harvest is sustainable. Smoke from incomplete combustion contains PM and CO, and wood contains little sulfur compared with coal.

PE2-MCQ049

Answer: B. Maximum possible output = 250 MW × 8,760 h = 2,190,000 MWh. Capacity factor = 438,000 ÷ 2,190,000 = 0.20 = 20%. The other values come from incorrect time periods or arithmetic.

PE2-MCQ050

Answer: B. Power saved = 36 W × 1,000 = 36 kW; energy saved = 36 kW × 4,000 h = 144,000 kWh/yr; savings = 144,000 × \$0.12 = \$17,280/yr. Payback = \$27,000 ÷ \$17,280/yr ≈ 1.6 yr. Using 60 W instead of the savings gives 0.9, and using 24 W gives 2.3.

PE2-MCQ051

Answer: B. Energy = 180 MW × 0.35 × 8,760 h = 551,880 MWh = 5.52×10^8 kWh; ÷ 4,000 kWh = about 138,000 households. 394,000 ignores the capacity factor, and the others are unit-conversion errors.

PE2-MCQ052

Answer: D. Secondary pollutants form in the atmosphere from reactions among primary pollutants; ozone is the classic example. The others are emitted directly.

PE2-MCQ053

Answer: C. Incomplete combustion of wood, dung or charcoal indoors produces PM and CO, causing respiratory and cardiovascular disease. The other options refer to the stratospheric ozone problem or to outdoor industrial pollutants.

PE2-MCQ054

Answer: A. In a thermal inversion, the warm air above stops vertical mixing, so pollutants accumulate near the ground. If cool air were above warm air, normal convection would carry pollutants upward.

PE2-MCQ055

Answer: D. Scrubbers spray a basic slurry (such as limestone) that reacts with SO_2. Electrostatic precipitators remove particles, catalytic converters reduce NO_x, CO and hydrocarbons from vehicles, and PCV valves recirculate engine gases.

PE2-MCQ056

Answer: B. Ozone is a secondary pollutant; it forms through photochemical reactions that need strong sunlight and time, so it lags behind the morning peaks of NO and NO_2. Vehicles emit NO and NO_2 rather than ozone.

PE2-MCQ057

Answer: A. Wind determines where plumes travel, so differences along the line could reflect wind rather than distance; recording it allows researchers to account for it. Random heights, single readings and different instruments would add variability or error.

PE2-MCQ058

Answer: C. Each pH unit is a tenfold change in H^+ concentration. The difference is 5.6 − 4.2 = 1.4 pH units, so the factor is 10^1.4 ≈ 25. A factor of 1.4 uses the pH difference directly, and 14 or 140 are miscalculations.

PE2-MCQ059

Answer: D. A point source has a single, identifiable discharge location. Runoff from fields and streets and atmospheric deposition come from many diffuse sources.

PE2-MCQ060

Answer: A. Liners and leachate collection keep contaminated liquid from seeping into the soil and groundwater. Open burning pollutes the air, e-waste adds toxic metals, and rain on uncovered waste creates more leachate.

PE2-MCQ061

Answer: C. Microbial breakdown of organic matter uses dissolved oxygen (high BOD); the lowest DO occurs where oxygen use exceeds reaeration, and then DO recovers as BOD falls. The figure shows BOD decreasing with distance, not increasing.

PE2-MCQ062

Answer: B. 0.400 ÷ 0.002 = 200. Increase in concentration at successively higher trophic levels is biomagnification; bioaccumulation refers to build-up within one organism over its lifetime. A 20-fold figure is a miscalculation.

PE2-MCQ063

Answer: B. Total waste = $12 \times 10^6 \times 1.1$ kg/day × 365 = 4.82×10^9 kg = 4.82 million t. Not recycled = 78% × 4.82 = 3.76 million t. 1.06 million t is the recycled portion, and 4.82 million t is all of the waste.

PE2-MCQ064

Answer: C. The LD50 (the dose that kills 50%) is about 8 mg/kg for Q and about 40 mg/kg for P; a lower LD50 means higher toxicity, so Q is about 5 times more toxic. A larger LD50 indicates lower toxicity, and reaching 100% at high doses does not make toxicity equal.

PE2-MCQ065

Answer: B. The strong negative association supports the claim that waterborne fecal contamination drives cholera, and sanitation is a logical intervention. The data show a correlation, not that sewers are the only cause; cholera is waterborne, not airborne; and a low (but nonzero) rate in district 1 still fits the trend.

PE2-MCQ066

Answer: D. In the stratosphere, UV releases chlorine atoms that act as catalysts, destroying many thousands of ozone molecules each. The other options misdescribe the mechanism.

PE2-MCQ067
Answer: D. The Montreal Protocol phased out ODS, allowing stratospheric chlorine to decline. The other actions address climate, SO_2 and lead pollution.

PE2-MCQ068
Answer: B. Greenhouse gases absorb and re-radiate outgoing infrared radiation, keeping Earth warmer than it would otherwise be. They do not trap UV or reflect sunlight, and the ozone layer is a different issue.

PE2-MCQ069
Answer: C. Invasive species often reproduce quickly, eat a wide range of foods, tolerate a range of conditions, and lack the predators and parasites of their native range. The other traits would limit spread.

PE2-MCQ070
Answer: C. CO_2e = 150 kg × 28 = 4,200 kg CO_2e per hectare per year. Dividing by 28 gives 5.4, and leaving out the multiplier gives 150.

PE2-MCQ071
Answer: C. Land plants, mostly in the Northern Hemisphere, take up CO_2 in spring and summer and release it by respiration and decay in winter, creating the oscillation; the steady upward trend reflects human emissions. The other options misattribute both patterns.

PE2-MCQ072
Answer: A. Warming expands the water and meltwater from glaciers and ice sheets on land adds volume to the oceans. Floating sea ice displaces its own weight in water, so its melting adds almost no volume.

PE2-MCQ073
Answer: A. The years with bleaching above 35% (2015, 2016, 2019, 2020, 2023, 2024) all had maximum SST of 30.9 °C or higher, consistent with heat stress causing corals to expel their symbiotic algae. The data show no steady decline and a clear relationship to temperature.

PE2-MCQ074
Answer: C. A 0.1 decrease in pH multiplies $[H^+]$ by 10^0.1 ≈ 1.26, an increase of about 26%. The change is not 10%, even though the pH change appears small, because the scale is logarithmic.

PE2-MCQ075
Answer: B. CITES lists species and restricts or prohibits international trade. The Montreal Protocol addresses ozone-depleting substances, Ramsar protects wetlands, and Basel regulates the movement of hazardous waste, including e-waste.

PE2-MCQ076
Answer: A. Percent loss = (loss ÷ original) × 100: P = 1.8 ÷ 6.0 = 30%; Q = 2.8 ÷ 14.0 = 20%; R = 0.3 ÷ 3.0 = 10%. Region Q lost the largest area (2.8 million ha), but not the largest percentage.

PE2-MCQ077
Answer: D. Ice reflects most sunlight (high albedo); open water absorbs most of it, which increases warming and further melt, a reinforcing (positive) feedback. Lower albedo means less reflection.

PE2-MCQ078
Answer: C. Carbon stored = 5,000 × 6 = 30,000 t C per year. Each tonne of C corresponds to 44/12 = 3.67 t of CO_2, so CO_2 removed = 30,000 × 3.67 ≈ 110,000 t. 30,000 t omits the conversion, and 8,200 t uses 12/44 instead.

PE2-MCQ079
Answer: D. Because removal patches were also nearer to the reserve, a second variable could account for the increase; random assignment would remove this confounding. Before-and-after counts and a control group are strengths, and more replicates improve, not weaken, a study.

PE2-MCQ080
Answer: D. Without chlorine, HFCs do not destroy ozone, but they are potent greenhouse gases that leak from equipment, which is why the Kigali Amendment phases them down. The other options misstate either the ozone or climate effects.

Section II — FRQ Solutions

PE2-FRQ001

(a) Habitat fragmentation (and habitat loss) caused by converting forest to plantations, which creates edge effects and threatens biodiversity. Accept deforestation or biodiversity loss if linked to the fragments. *(1 point: names fragmentation, habitat loss, deforestation or edge effects as the problem)*

(b) Example: "Species richness of forest-interior understory birds will increase as distance from the forest edge increases." Accept the reverse only if consistent with a reasoned prediction, but the stated hypothesis must be testable and name both variables and a direction. *(1 point: testable hypothesis relating distance to richness with a direction)*

(c) Independent variable: distance from the forest edge (m). Dependent variable: number of forest-interior understory bird species (species richness) recorded. *(1 point for the independent variable; 1 point for the dependent variable)*

(d) Sample points at several set distances from the edge (for example 0, 100, 200, 300 and 400 m) along transects perpendicular to the edge. Replicate by using several transects in the same fragment and, ideally, several edges or fragments (at least 3 per distance). Count birds in a standardized way, for example 10-minute point counts or mist-netting for the same number of hours at each point, at the same time of day (dawn), in the same season, by trained observers with the same method. *(3 points: 1 for sampling at multiple distances along transects; 1 for replication; 1 for a standardized counting method and sampling effort)*

(e) Examples: time of day/season of sampling, observer or method, sampling effort per point, weather conditions, vegetation type or elevation. The explanation must state why: so differences in richness can be attributed to distance and not to another factor (for example, birds are more active and easier to detect at dawn). *(1 point: variable plus a reason it must be held constant)*

(f) Species richness is lowest near the edge and increases with distance into the forest (a significant positive relationship, which might level off in the interior), with similar results across replicates. *(1 point: describes the pattern consistent with the stated hypothesis)*

(g) Examples: establish forest corridors or buffer zones of native forest around fragments; stop further clearing next to the fragment; restore forest along the edge; enlarge protected areas. Justification: larger or connected habitat patches have a smaller proportion of edge habitat and support higher species richness and larger populations (island biogeography and edge effects), and corridors allow gene flow and recolonization. *(1 point: reasonable solution and justification that links to an ecological principle)*

PE2-FRQ002

(a) Annual mean PM2.5 generally increased from 118 µg/m³ in 2017 to 134 µg/m³ in 2021 (with a small dip in 2020). *(1 point: describes an increasing trend with values or direction)*

(b) Percent change = (final − initial) ÷ initial × 100 = (92 − 134) ÷ 134 × 100 = −42 ÷ 134 × 100 = −31.3%, a decrease of about 31%. *(1 point: correct setup using (92 − 134) ÷ 134; 1 point: answer of about −31% with the decrease indicated)*

(c) There is a positive relationship: as PM2.5 rises, respiratory admissions rise, and when PM2.5 falls (2022–2024), admissions fall. *(1 point)*

(d) Examples: cigarette smoking, indoor air pollution, seasonal infections (influenza, COVID-19), temperature or humidity, pollen, population age, or hospital access. Explanation: the other factor could change over the same years and cause some of the change in admissions, so the correlation does not prove that PM2.5 alone is the cause. *(1 point: identifies a factor and explains why it limits the conclusion)*

(e) Figure 1 shows that PM2.5 and the number of inversion days are both highest in the winter months (Dec, Jan) and lowest in summer (Jun–Aug), when no inversions occur. In an inversion, a layer of warm air lies above cooler surface air and acts as a lid that blocks vertical mixing, so emissions from vehicles, industry and heating accumulate near the ground and concentrations rise. *(1 point: describes that high PM2.5 coincides with high inversion days; 1 point: explains that the warm layer traps pollutants close to the ground)*

(f) Claim: the policies reduced PM2.5. Evidence: before the policies, PM2.5 rose from 118 to 134 µg/m³ (2017–2021); after the policies began in 2022, it fell each year, from 134 in 2021 to 92 in 2024, a 31% decrease, and admissions fell as well (462 to 372). Reasoning: coal boilers and old diesel vehicles are major

sources of primary particulates, so removing them lowers emissions. *(1 point: claim supported with data from the table; 1 point: reasoning linking the policy to reduced emissions)*

(g) Examples: expand public transit or electrify buses; require particulate filters or scrubbers on industry; ban open burning of crop residue and waste; use cleaner cooking fuels; plant urban trees or green belts. Advantage or disadvantage example: electrified buses reduce tailpipe particulates and NO_x, but require expensive infrastructure and electricity generation that might be supplied by coal. *(1 point: valid action with one advantage or disadvantage)*

PE2-FRQ003

(a) Energy = capacity × capacity factor × hours = 800 MW × 0.50 × 8,760 h = 3,504,000 MWh (about 3.5×10^6 MWh per year). *(1 point: correct setup; 1 point: correct answer with units)*

(b) CO_2 = 3,504,000 MWh × 0.95 t/MWh = 3,328,800 t, about 3.3×10^6 metric tons of CO_2 per year. *(1 point: correct setup using the answer to (a); 1 point: correct answer with units)*

(c) Volume = 200 $km^2 \times 10^6$ $m^2/km^2 \times 1.2$ m = 2.0×10^8 $m^2 \times 1.2$ m = 2.4×10^8 m^3 per year. *(1 point: correct conversion and answer with units)*

(d) City use = 5,000,000 people × 150 L/person/day × 365 days = 2.7375×10^{11} L/year ÷ 1,000 L/m^3 = 2.74×10^8 m^3 per year. Evaporation ÷ city use = $2.4 \times 10^8 \div 2.74 \times 10^8 \times 100$ = 87.7%, about 88%. *(1 point: city's annual water use in m^3; 1 point: percentage of about 88% [accept 87–88%])*

(e) Examples: the dam traps sediment and nutrients, causing erosion of the river channel and delta and reducing floodplain fertility; it blocks fish migration; it changes flow and temperature regimes, harming downstream species and fisheries. *(1 point: one correct disadvantage with a downstream effect)*

(f) Examples: build fish ladders or passages; release environmental flows that mimic seasonal floods; install sediment bypass or flushing; cover or shade the reservoir (or use floating solar panels) to reduce evaporation; relocate and compensate displaced residents. Justification must be linked to the problem: for example, fish ladders allow migratory species to reach spawning grounds, which maintains biodiversity and fisheries; floating solar on the reservoir reduces evaporation and adds renewable electricity. *(1 point: a valid action; 1 point: justification explaining how it reduces the stated effect)*

Predicted Practice Exam 3

Section I — MCQ Answers and Explanations

PE3-MCQ001

Answer: D. Grass is the producer, the zebra is the primary consumer, and the lion, which eats the zebra, is the secondary consumer. Only about 10% of the energy in one trophic level is passed on to the next; the rest is lost as heat or used in respiration. Receiving 90% reverses the 10% rule, and a lion in a three-link chain is not a tertiary consumer.

PE3-MCQ002

Answer: C. Symbiotic bacteria in root nodules convert N_2 from the air into ammonia (NH_3/NH_4^+) that plants can use; this is nitrogen fixation. Nitrification converts ammonium to nitrate, denitrification returns nitrogen to the atmosphere as N_2, and ammonification converts organic nitrogen in dead matter to ammonium.

PE3-MCQ003

Answer: D. Site Q is warm all year (about 22–26 °C) and receives about 900 mm of rain, nearly all of it in a wet season of roughly six months, with a distinct dry season from June to September; this is the savanna pattern of grasses with scattered trees. Site P is the desert (under 50 mm per year), Site R has cold winters (temperate grassland), and a rainforest would have rain in every month.

PE3-MCQ004

Answer: B. Three transfers separate the producers from the tertiary consumer: 250,000 × 0.10 = 25,000 kJ (grasshopper), × 0.10 = 2,500 kJ (lizard), × 0.10 = 250 kJ (hawk). Stopping after two transfers gives 2,500 kJ, the amount reaching the lizard, a secondary consumer.

PE3-MCQ005

Answer: A. NPP = GPP − plant respiration = 3,200 − 1,950 = 1,250 g C/m²/yr, the energy (as carbon) available to consumers. 5,150 results from adding rather than subtracting, and 3,200 ignores respiration.

PE3-MCQ006

Answer: A. Tillage aerates the soil and breaks open aggregates that protect organic matter. Aerobic decomposers respire faster, converting soil organic carbon to CO_2, and the annual crop returns less root biomass than perennial prairie grasses. Plowing does not eliminate decomposers; it favors them.

PE3-MCQ007

Answer: A. Changing when or how an organism acts, such as becoming nocturnal, is behavioral. Concentrated urine is physiological, and fleshy water-storing stems and large heat-releasing ears are structural adaptations.

PE3-MCQ008

Answer: D. Moderating floods is a regulating service, along with climate regulation and water purification. Provisioning services supply goods such as food and water, cultural services provide recreation or spiritual value, and supporting services include nutrient cycling and soil formation.

PE3-MCQ009

Answer: A. Secondary succession begins on soil that already exists, so surviving roots and the seed bank allow rapid regrowth. Primary succession starts on bare rock or new surfaces without soil. Lightning fires are natural, and recovery still passes through intermediate stages.

PE3-MCQ010

Answer: B. Comparing W with X (same area, different distance) shows distance matters, and comparing W with Y (same distance, different area) shows area matters. Larger, closer patches hold more species, which is the prediction of island biogeography for habitat islands.

PE3-MCQ011

Answer: C. The range of tolerance is the span along the axis where a species can live. Species 2 has a broad range that includes 300 mm, so it declines but survives; Species 1 has a narrow range that does not reach 300 mm. A high peak shows optimum conditions, not tolerance, and Species 1 is the specialist.

PE3-MCQ012

Answer: C. A specialist depends on a narrow range of resources, so losing its single food source causes steep decline. A generalist could switch foods. Specialists are generally more vulnerable than generalists to habitat change.

PE3-MCQ013

Answer: D. Long life, few young, and heavy parental care are K-selected traits, typical of populations that remain near carrying capacity. r-selected species mature early and produce many offspring, and K-selected species do not reproduce rapidly.

PE3-MCQ014

Answer: B. Female education, access to family planning, urbanization, and lower infant mortality are all associated with lower TFR. The other options describe conditions (more labor value of children, less access to contraception, higher infant mortality) that raise fertility.

PE3-MCQ015

Answer: D. Curve A falls steeply early in life and then flattens, the Type III pattern of high juvenile mortality. Curve B (Type I) shows high survival until old age, as in humans, and Curve C (Type II) shows a constant death rate across ages.

PE3-MCQ016

Answer: C. The peak exceeds the long-term level of roughly 1,400 sheep, so the peak is an overshoot. Overgrazed vegetation raised death rates until the population fell toward the sustainable carrying capacity. The island is isolated, ruling out immigration, and the curve levels off rather than growing exponentially.

PE3-MCQ017

Answer: B. Doubling time ≈ 70 ÷ growth rate (%) = 70 ÷ 2.8 = 25 years. 196 results from multiplying 70 × 2.8, and 7 years from dividing 70 by 10 instead of by the growth rate.

PE3-MCQ018

Answer: B. Rate = (births − deaths + immigration − emigration) ÷ 10 per 1,000 → (9 − 11 + 6 − 2) ÷ 10 = +0.2%. Leaving out migration gives −0.2%, leaving out emigration gives +0.4%, and not converting per 1,000 to percent gives 2.0.

PE3-MCQ019

Answer: A. This is population momentum. Country A's wide base means many young people will soon have children, so total births remain high even at replacement fertility. Country B's narrow base means few future parents, and replacement-level fertility yields eventual stability, not growth.

PE3-MCQ020

Answer: B. Ratio = 64 ÷ 24 = 2.7 workers per older person. 0.4 inverts the ratio (24 ÷ 64), 5.3 uses the under-15 group (64 ÷ 12), and 3.2 counts everyone under 65 ((12 + 64) ÷ 24).

PE3-MCQ021

Answer: A. Natural increase is the gap between the birth and death curves. It is about 2 per 1,000 in 1900, about 25 in 1960, 23 in 1980, and 8 in 2020, so 1960 is the maximum (late Stage 2 of the demographic transition) when sanitation and medicine had lowered death rates before birth rates fell. The 1980 gap is close but smaller.

PE3-MCQ022

Answer: B. The troposphere, the lowest layer, holds most of the atmosphere's mass and nearly all weather. The stratosphere contains the ozone layer, and the mesosphere and thermosphere lie higher.

PE3-MCQ023

Answer: C. Plates moving apart at a divergent boundary produce rift valleys, volcanism, and geothermal heat. Convergent boundaries make trenches, volcanic arcs, or folded mountains, and transform boundaries cause earthquakes without creating new crust.

PE3-MCQ024

Answer: B. The A horizon is the topsoil, a mix of mineral particles and humus with the highest biological activity. The B horizon is subsoil where leached clay and minerals accumulate, and the C horizon is partly weathered parent material above bedrock.

PE3-MCQ025

Answer: A. Air rises at the equator, cools, and loses its moisture as heavy rain; the dry air then sinks near 30° in the Hadley cell, warming as it descends and suppressing cloud formation. Rising air, as in the other options, would produce clouds and rain.

PE3-MCQ026

Answer: B. Slow infiltration (6 mm/h) combined with the highest water retention (42%) is characteristic of clay, whose small pores resist water movement but hold water by adhesion. Soil P, with rapid infiltration and low retention, is sandy, and soil R, intermediate in both, resembles loam.

PE3-MCQ027

Answer: A. Without plant cover and roots, soil is compacted and erodes, so rain runs off quickly, carrying sediment and manure nutrients downstream, which cloud the water and raise flood peaks. Bare, compacted soil absorbs less water and reduces recharge.

PE3-MCQ028

Answer: C. Net loss = (9.0 − 1.5) t/ha/yr × 20 yr × 40 ha = 7.5 × 20 × 40 = 6,000 t. 7,200 t ignores soil formation, 150 t is the loss per hectare, and 1,200 t is the formation alone.

PE3-MCQ029

Answer: B. Seasons result from Earth's 23.5° axial tilt, which changes the angle and duration of sunlight on each hemisphere as Earth orbits. Distance from the Sun is not the cause, and Earth is actually nearest the Sun in early January, during Northern Hemisphere winter.

PE3-MCQ030

Answer: C. Moist air forced up the windward side cools and drops rain (Sites 1 and 2); after crossing the crest the dried air descends, warms, and rarely produces rain, creating a rain-shadow desert at Site 3.

PE3-MCQ031

Answer: D. In El Niño years (positive anomalies such as 2002–03 and 2010) the catch drops to about 6–7 million t, while in cold years it is about 10 million t or more. During El Niño the trade winds weaken, the thermocline deepens, and upwelling of cold, nutrient-rich water declines, reducing phytoplankton that anchovies depend on.

PE3-MCQ032

Answer: D. When individuals use a shared resource for personal gain and the costs are spread over the group, the resource is overused, the tragedy of the commons. The other concepts concern toxins in food chains, community change, and caution under uncertainty.

PE3-MCQ033

Answer: D. Rotational grazing lets plants regrow between grazing periods, maintaining cover, root systems, and soil. Continuous or repeated grazing of the same area and removing cover cause loss of vegetation and erosion.

PE3-MCQ034

Answer: A. In dry climates, applied water evaporates rapidly, leaving salts from the river water at the surface. Over time, salinization stunts crops, and a white crust is typical. Drip irrigation reduces the risk.

PE3-MCQ035

Answer: D. High-yield varieties raised food production but needed large inputs of fertilizer, pesticides, and water, and planting a few uniform varieties reduced genetic diversity. They produce higher yields, and their larger water and fertilizer use creates, rather than prevents, pollution.

PE3-MCQ036

Answer: A. IPM combines monitoring, biological and cultural controls, and targeted, minimal pesticide use when pests exceed a threshold. Routine spraying, monocropping, and removing natural enemies go against IPM principles.

PE3-MCQ037

Answer: A. Catch per unit effort estimates stock abundance. It fell from about 200 to 70 kg per boat-day as effort tripled, so each boat finds fewer fish; total catch also dropped after 2012 despite more effort, a sign of overfishing.

PE3-MCQ038

Answer: A. Surface mining removes vegetation and soil, creating pits and waste piles; exposed sulfide minerals can generate acid drainage. Removing cover increases sediment, not reduces it, and mined spoil is typically low in fertility.

PE3-MCQ039

Answer: B. Usable forage = 800 ha × 2,400 kg/ha × 0.50 = 960,000 kg/yr; 960,000 ÷ 4,000 = 240 cows. 480 cows results from ignoring the 50% limit.

PE3-MCQ040

Answer: B. Deficit per person = 4.5 − 2.7 = 1.8 gha; for 25 million people, 1.8 × 25 million = 45 million gha. 67.5 million is total biocapacity, 112.5 million is total footprint, and 1.8 million treats the per-person value as the total.

PE3-MCQ041

Answer: A. Before: C = 0.5(0.90) + 0.5(0.20) = 0.55 → 0.025 m × 400,000 m² × 0.55 = 5,500 m³. After: C = 0.2(0.90) + 0.8(0.20) = 0.34 → 3,400 m³. The decrease is 5,500 − 3,400 = 2,100 m³; 3,400 is only the new volume and 8,900 adds the two volumes.

PE3-MCQ042

Answer: D. Uranium ore is a finite resource that is not replaced on human timescales. Geothermal heat, wind, and flowing water are replenished continuously.

PE3-MCQ043

Answer: A. Natural gas (mainly methane) is the cleanest-burning fossil fuel, emitting roughly half the CO_2 of coal per unit of energy and little SO_2. It still releases CO_2, is nonrenewable, and leaked methane is a potent greenhouse gas.

PE3-MCQ044

Answer: D. Radioactive waste requires isolation for very long times, a major challenge. Nuclear plants emit almost no CO_2, SO_2, or particulates during operation because fuel is not burned.

PE3-MCQ045

Answer: C. Geothermal power uses heat from Earth's interior, accessible at reasonable depths where volcanic or tectonic activity brings heat near the surface. It does not depend on sunlight, wind, or fossil fuels.

PE3-MCQ046

Answer: C. Fracking injects large volumes of water, sand, and chemicals into shale, and returned wastewater or leaks can contaminate groundwater; injected wastewater can also trigger small earthquakes. Spent fuel is a nuclear issue and acid mine drainage is a mining issue.

PE3-MCQ047

Answer: D. Biomass is carbon-neutral when regrowth removes from the atmosphere the carbon released in burning. Burning always releases CO_2, and permanent clearing prevents the re-absorption.

PE3-MCQ048

Answer: B. 90 years is 90 ÷ 30 = 3 half-lives, so activity = 800 × (½)³ = 100 units. 400 units is after one half-life, 267 units divides by 3 instead of halving three times, and 25 units is after five.

PE3-MCQ049

Answer: B. Power saved = 200 × (80 − 30) W = 10,000 W = 10 kW; energy saved = 10 kW × 3,000 h = 30,000 kWh; savings = 30,000 kWh × $0.12/kWh = $3,600. $2,160 is the cost of running the LED fixtures only and $5,760 the cost of the old fixtures.

PE3-MCQ050

Answer: C. Country Z: 20% coal and 40% natural gas, so emissions = 0.20 × 1.0 + 0.40 × 0.45 = 0.20 + 0.18 = 0.38 kg CO_2/kWh. 0.18 and 0.20 each leave out one fuel, and 0.60 treats gas as emitting as much as coal.

PE3-MCQ051

Answer: B. Electrical output = 0.33 × 3,000 = 990 MW; waste heat = 3,000 − 990 = 2,010 MW. 990 MW is the electricity produced, and 9,091 MW comes from dividing by 0.33.

PE3-MCQ052

Answer: C. In an inversion the normal temperature decrease with height reverses, and the warm layer acts like a lid, preventing vertical mixing so pollutants accumulate at ground level. Normal conditions, with warm air rising, disperse pollutants.

PE3-MCQ053

Answer: B. Radon forms from the decay of uranium in rocks and soil and seeps through foundations; it is the leading cause of lung cancer in nonsmokers. The other descriptions fit carbon monoxide, VOCs, and mold.

PE3-MCQ054

Answer: B. The pH scale is logarithmic; a 1.0 unit difference represents a 10-fold change in hydrogen-ion concentration (10^(5.6−4.6) = 10). 1.0 times would mean the rain is no more acidic, and 100 times would require a 2-unit pH difference.

PE3-MCQ055

Answer: C. The 20 dB difference is two 10-dB steps, so intensity increases 10 × 10 = 100 times. 20 times assumes the scale is linear, and 1.3 times divides 80 by 60.

PE3-MCQ056

Answer: C. Reduction = (480 − 30) ÷ 480 × 100 ≈ 94%. 6% is the LPG value as a share of the open-fire value (30 ÷ 480), 60% is the reduction achieved by the improved biomass stove, and 1,500% reverses the comparison.

PE3-MCQ057

Answer: D. Reducing the source and venting emissions outdoors is the most effective control. Air fresheners do not remove particles or CO, timing does not alter emissions, and a single window only slightly improves ventilation.

PE3-MCQ058

Answer: C. NO and NO_2 peak with rush-hour traffic, but ozone is a secondary pollutant formed when sunlight photolyzes NO_2 and the resulting oxygen atoms and VOC-derived radicals react; the intense afternoon sunlight explains the delayed peak. Cars do not emit ozone, and inversions occur at night or early morning.

PE3-MCQ059

Answer: D. Secondary treatment uses bacteria in aerated tanks to decompose organic waste, reducing biochemical oxygen demand. Primary treatment is physical, tertiary targets nutrients, and chlorination kills pathogens.

PE3-MCQ060

Answer: C. Fecal coliform bacteria live in intestines, so their presence is an indicator of fecal contamination and possible pathogens such as cholera or typhoid. Nitrate, metals, and oxygen are measured with different tests.

PE3-MCQ061

Answer: A. The lowest DO on the graph is about 2.5 mg/L, around 12 km below the outfall, where decomposition of organic waste by aerobic bacteria consumes oxygen faster than the river replaces it. Only the tubifex worm tolerates less than 3 mg/L; the others need more.

PE3-MCQ062

Answer: D. The LD50 is the dose that kills 50% of test animals: about 20 mg/kg for Toxicant 1 and 200 mg/kg for Toxicant 2. A lower LD50 means higher toxicity, and a curve farther to the left means a smaller dose is needed.

PE3-MCQ063

Answer: C. Biomagnification factor = 25 ppm ÷ 0.04 ppm = 625. 5 compares the osprey with the large fish (25 ÷ 5), and the other values result from decimal errors.

PE3-MCQ064

Answer: C. Total waste = 200,000 × 2.0 kg = 400,000 kg/day = 400 t/day. Landfilled = 400 × 0.70 = 280 t/day × 365 = 102,200 t/yr. 43,800 t is the diverted amount, 146,000 t is all waste, and 280 t/day was not annualized.

PE3-MCQ065

Answer: A. The percentage is highest at the outfall and drops with distance as the effluent is diluted or broken down, which points to the effluent as the source. Because the study is observational and measured no specific chemical, it cannot prove which compound is responsible. The 4% at the reference site indicates a low natural background rather than no effect.

PE3-MCQ066

Answer: C. UV light in the stratosphere breaks CFC molecules and releases chlorine atoms; these atoms act as catalysts, so a single atom can destroy thousands of ozone molecules. They are not consumed in one-to-one reactions.

PE3-MCQ067

Answer: A. Greenhouse gases such as CO_2, CH_4, and N_2O absorb outgoing infrared radiation and re-radiate it in all directions, including downward, warming the lower atmosphere. Reflecting sunlight would cool Earth, and greenhouse gases do not change the Sun's output.

PE3-MCQ068

Answer: B. Invasive species usually reproduce quickly, tolerate a wide range of conditions, and escape the predators, herbivores, and diseases that limited them at home. Slow growth, narrow tolerance, and specialized requirements restrict spread.

PE3-MCQ069

Answer: D. The Montreal Protocol (1987) phased out CFCs and other ozone-depleting substances, and the ozone layer is slowly recovering. Climate treaties target greenhouse gases, CITES regulates trade in endangered species, and the Clean Water Act regulates water pollution.

PE3-MCQ070

Answer: C. Methane emitted = 5,000 × 60 kg = 300,000 kg = 300 t CH_4. CO_2-equivalent = 300 t × 28 = 8,400 t. 300 t ignores the warming potential, 11 divides by 28, and 84,000 is a decimal error.

PE3-MCQ071

Answer: D. In the Northern Hemisphere, photosynthesis removes CO_2 during the growing season and decomposition and respiration return it in fall and winter, producing the annual cycle. The steady upward trend of about 2.4 ppm per year matches human emissions from burning fossil fuels and deforestation.

PE3-MCQ072

Answer: A. A positive feedback amplifies the original change; thawing permafrost releases greenhouse gases that cause more warming. The other options describe negative feedbacks that counter the change.

PE3-MCQ073

Answer: B. Warmer water expands, raising sea level, and heat stress makes corals expel their symbiotic algae. Floating ice melt does not raise sea level, warm water holds less dissolved oxygen and gas, and ocean pH falls because of CO_2 uptake.

PE3-MCQ074

Answer: D. The increase per 3-year interval grows from 70 ha to 440 ha, so growth is accelerating, as in exponential growth. A larger infestation is harder and costlier to remove, so early detection and rapid response pay off.

PE3-MCQ075

Answer: A. Introducing a non-native control agent can harm non-target native species, and an agent that spreads uncontrolled can become a new invasive species. Insects do not add pesticide residues to groundwater or salts to soil.

PE3-MCQ076

Answer: A. Killing animals for horns is overexploitation, addressed by enforcement, protected areas, and reducing demand for horn products. Wildlife corridors help when habitat loss is the threat, but they do not stop poaching.

PE3-MCQ077

Answer: C. Because pH is logarithmic, $[H^+]$ increases by a factor of 10^(8.20−8.10) = 10^0.1 ≈ 1.26, an increase of about 26%. A 0.1 pH change is not 10% or 1%, and a 100% rise would be a factor of 2.

PE3-MCQ078

Answer: B. After fragmentation each 5 × 5 km fragment has an interior of (5 − 2) × (5 − 2) = 9 km²; four fragments give 36 km². Before fragmentation the interior was (10 − 2)² = 64 km², so 28 km² is the interior lost, not the amount left.

PE3-MCQ079

Answer: D. A controlled, replicated experiment manipulates grazing (the independent variable) while holding soil and seed input constant, so differences in survival can be attributed to antelope. A single pair lacks replication, a correlation cannot isolate cause, and removing shrubs only from fenced plots changes a second variable.

PE3-MCQ080

Answer: D. In the extremely cold polar winter, polar stratospheric clouds provide surfaces on which chlorine compounds are converted to reactive forms. When sunlight returns in spring, it energizes the catalytic cycle and ozone falls sharply. Ice does not hold CFCs, and the ozone hole forms because of reactions, not transport to the equator.

Section II — FRQ Solutions

PE3-FRQ001

(a) (i) An invasive (non-native) species is spreading, degrading the native grassland and reducing rangeland productivity (habitat degradation or loss of biodiversity is also acceptable). *(1 point: identifies invasive species or a closely related problem)* (ii) Soil moisture decreases as invasion increases: about 9.1% in uninvaded plots, 7.4% in lightly invaded plots, and 4.8% in heavily invaded plots. *(1 point: states a negative relationship and cites at least one value)* (b) If the shrub is removed from heavily invaded plots, then perennial grass cover (or biomass) and soil moisture will be higher after one growing season than in similar plots where the shrub is left in place. *(1 point: an if/then statement that names the manipulated variable, the measured variable, and a direction)* (c) Independent variable: shrub treatment (shrub removed versus shrub left in place). Dependent variable: perennial grass cover or biomass (soil moisture is also acceptable as a dependent variable). *(1 point for each variable correctly identified)* (d) Any one of: livestock grazing (fence all plots), plot size, soil type and slope, rainfall received (use plots in the same area), or starting grass cover. Controlling it ensures that differences in grass cover are caused by shrub removal and not by the other factor. *(1 point: a valid variable with a linked explanation)* (e) Example: Select 20 heavily invaded plots of equal size on similar soil and slope, fence them against livestock, and randomly assign 10 to shrub removal (cutting the shrubs and treating the stumps) and 10 to be left as controls. Measure grass cover in five 1 m² quadrats per plot, and soil moisture with a probe at 0–20 cm at several points per plot, before treatment and again at the end of the next growing season and the dry season, on the same dates for all plots. *(1 point: treatment and control groups with replication and randomization or matching; 1 point: a specific measurement method with timing)* (f) Calculate mean grass cover and soil moisture for each group and compare them (for example with a statistical test). If the removal plots show significantly higher grass cover and soil moisture than the control plots, the hypothesis is supported; if there is no difference, or grass cover does not improve, it is refuted. *(1 point: compares treatment with control and states what result supports or refutes)* (g) Examples: reseed or replant native grasses after removal, use rotational grazing and deferred grazing so that grass can recover, introduce a carefully tested biological control agent, or provide alternative fuel sources (such as improved stoves or biogas) so that villagers no longer depend on planting the shrub. Justification: reseeding restores competition from native plants and soil cover so shrubs cannot re-establish; rotational grazing lets grass regain vigor; alternative fuels remove the reason for planting. *(1 point: a reasonable solution with a justification that explains how it addresses the cause)*

PE3-FRQ002

(a) The TFR of Country C decreased steadily from about 6.0 children per woman in 1960 to about 1.9 in 2020, falling below the replacement level of 2.1 between 2010 and 2020. *(1 point: describes a decline and gives values or the crossing below replacement)* (b) Growth rate (%) = (birth rate − death rate + net migration) ÷ 10 = (41 − 9 + (−1)) ÷ 10 = 31 ÷ 10 = 3.1% per year. *(1 point: correct value of 3.1% with work shown, including migration; 3.2% without migration earns no credit)* (c) 46% of Country A's population is under age 15. These children will reach reproductive age over the next 15–30 years, so the number of parents will increase for decades. Even if each woman has fewer children, the total number of births will remain high and the population will keep growing; this is population momentum. *(1 point: cites the large share under 15; 1 point: explains that*

a large pre-reproductive group will enter reproductive years, producing more births than deaths) (d) Stage 2. The birth rate is still very high (41 per 1,000) and TFR is above 5, while the death rate is low (9 per 1,000) and falling, with infant mortality cut in half from 88 to 44 per 1,000 between 2000 and 2020, so natural increase is rapid. *(1 point: Stage 2; 1 point: two data values that support it, such as high birth rate and a low or falling death rate or infant mortality)* (e) Claim: Country B. It already has the largest share of people aged 65 or older (22%), a very small share under 15 (13%), a TFR of about 1.3 (far below replacement), and a death rate (11) above its birth rate (8), so the number of working-age people paying for pensions and care will shrink relative to older people. *(1 point: claim naming Country B; 1 point: justification using at least two data values such as the 22% over 65, 13% under 15, TFR 1.3, or death rate greater than birth rate)* (f) Examples: expand education for girls and women, provide access to family-planning services and contraception, or reduce infant mortality through basic health care so that parents do not rely on having many children. Justification: Country C's TFR fell from 6.0 to 1.9 over 60 years as infant mortality fell (28 to 12 between 2000 and 2020) and education and family planning spread; Country A's TFR remains high (5.1) even though infant mortality is falling, so these measures should accelerate its decline. *(1 point: a valid solution; 1 point: a justification supported by the data, such as the experience of Country C or the infant mortality trend)*

PE3-FRQ003

(a) Energy per hectare = 3,000 kg × 16 MJ/kg = 48,000 MJ; for the ranch, 48,000 MJ/ha × 2,500 ha = 1.2×10^8 MJ per year. *(1 point: correct value with units)* (b) Energy consumed by cattle = 1.2×10^8 MJ × 0.40 = 4.8×10^7 MJ; energy stored in body tissue = 4.8×10^7 MJ × 0.10 = 4.8×10^6 MJ per year. *(1 point: applies 40% to find energy eaten; 1 point: applies 10% conversion and gets 4.8×10^6 MJ)* (c) Edible energy = 4.8×10^6 MJ × 0.40 = 1.92×10^6 MJ. Annual need per person = 10 MJ/day × 365 days = 3,650 MJ. People fed = 1.92×10^6 MJ ÷ 3,650 MJ/person ≈ 526 people. *(1 point: edible energy 1.92×10^6 MJ; 1 point: divides by 3,650 MJ per person per year to get about 526 people)* (d) Sorghum energy = 1,000 kg/ha × 15 MJ/kg × 2,500 ha = 3.75×10^7 MJ; people fed = 3.75×10^7 MJ ÷ 3,650 MJ/person ≈ 10,274 people (about 10,300). Ratio = 10,274 ÷ 526 ≈ 19.5, so about 20 times as many people. *(1 point: about 10,300 people; 1 point: ratio of about 19.5 or 20)* (e) Allowed forage = 3,000 kg/ha × 0.40 × 2,500 ha = 3.0×10^6 kg per year. Herd demand = 900 × 4,000 kg = 3.6×10^6 kg. The herd exceeds the plan by 0.6×10^6 kg, which is 0.6 ÷ 3.0 × 100 = 20% over. (Equivalent: the allowed herd is 750 cattle and 900 is 20% more.) *(1 point: compares 3.6×10^6 kg with 3.0×10^6 kg and states a 20% excess)* (f) Only about 10% of the energy at one trophic level is passed to the next because most is lost as heat in respiration and in material not eaten or digested. People eating sorghum feed at the producer level, whereas people eating beef feed one level higher, so much of the plant energy is lost before it reaches people. *(1 point: refers to energy loss (about 90%) between trophic levels and connects it to eating at a lower trophic level)* (g) Examples: reduce the herd to about 750 cattle; rotate cattle among paddocks so the grass can recover; destock during droughts. Justification: matching the herd to the amount of forage the land can sustainably supply keeps ground cover, roots, and soil in place, preventing erosion and loss of future forage. *(1 point: a valid solution with justification that links it to reduced overgrazing)*

Predicted Practice Exam 4

Section I — MCQ Answers and Explanations

Answer key. 1 A · 2 D · 3 D · 4 A · 5 A · 6 C · 7 C · 8 B · 9 D · 10 A · 11 A · 12 B · 13 C · 14 C · 15 D · 16 C · 17 B · 18 A · 19 D · 20 A · 21 B · 22 A · 23 A · 24 D · 25 D · 26 D · 27 A · 28 C · 29 B · 30 D · 31 B · 32 B · 33 A · 34 D · 35 D · 36 B · 37 B · 38 C · 39 A · 40 C · 41 C · 42 C · 43 A · 44 C · 45 B · 46 D · 47 B · 48 C · 49 B · 50 C · 51 D · 52 A · 53 D · 54 C · 55 D · 56 D · 57 A · 58 B · 59 A · 60 D · 61 D · 62 C · 63 B · 64 A · 65 B · 66 A · 67 C · 68 A · 69 B · 70 A · 71 B · 72 B · 73 C · 74 C · 75 D · 76 D · 77 C · 78 B · 79 B · 80 C

PE4-MCQ001

Answer: A. Permafrost beneath a shallow active layer and the absence of trees are the defining features of tundra. Taiga also has cold winters but is dominated by conifer trees, which the survey did not report. Grasslands and cold deserts lack continuous permafrost.

PE4-MCQ002

Answer: D. Brackish water produced by mixing of river and ocean water, high nutrient input, and use as a nursery are characteristics of estuaries. A coral reef requires clear, warm, fully saline water, a littoral zone is freshwater, and the pelagic zone is open, nutrient-poor water far from shore.

PE4-MCQ003

Answer: D. Denitrification converts nitrate to N_2 (and some N_2O) under anaerobic conditions, removing usable nitrogen from the soil. Nitrification does the opposite kind of conversion, turning ammonium into nitrite and nitrate. Nitrogen fixation converts N_2 into ammonia, and ammonification releases ammonium from decaying organic matter.

PE4-MCQ004

Answer: A. Producers to secondary consumers is two transfers, and 600 ÷ 60,000 = 0.01 = (efficiency)², so each transfer is 10%. The primary consumers hold 6,000 kJ/m²/yr, the secondary 600, and the tertiary 600 × 0.10 = 60 kJ/m²/yr. The 1% answers mistake the two-step overall transfer for the per-level efficiency.

PE4-MCQ005

Answer: A. NPP = GPP − plant respiration = 1,300 − 780 = 520 g C/m²/yr, the energy stored in new plant biomass and available to consumers. 2,080 results from adding the two values, and 1,300 and 780 are the GPP and respiration terms themselves.

PE4-MCQ006

Answer: C. Annual GPP = 0 + 40 + 230 + 60 = 330 and annual respiration = 45 + 35 + 160 + 80 = 320, so net uptake is +10 g C/m². The 330 value ignores respiration, and summer net uptake (+70) is not equal to the combined losses in the other seasons (−60), so the site is not exactly neutral. Because the margin is so small, warmer fall and winter soils could easily turn the site into a source.

PE4-MCQ007

Answer: C. Moderating the effects of storms, floods, or climate is a regulating service. Provisioning services supply goods such as fish, timber, or fresh water, cultural services provide recreation or spiritual value, and supporting services such as nutrient cycling underpin all the others.

PE4-MCQ008

Answer: B. All of the cultivars are the same species, so the variation in resistance genes within the species (genetic diversity) is what gives some plants a chance to survive. Species, ecosystem, and habitat diversity describe variety among species or among environments, which does not differ between the two farms.

PE4-MCQ009

Answer: D. Succession that begins on soil that survived a disturbance is secondary succession, and it proceeds faster than primary succession because the seed bank and nutrients remain. Primary succession starts on bare rock or new substrate with no soil. Pioneer species occur in both types, and a climax community takes decades to centuries to return.

PE4-MCQ010

Answer: A. The figure shows about 45 species for near islands but only about 30 for far islands of the same area, so distance matters. In island biogeography theory, greater isolation lowers the immigration rate, which lowers the equilibrium number of species. Area is identical for P and Q, so extinction rate due to area cannot explain the gap.

PE4-MCQ011

Answer: A. At 28°C species X is already on the declining side of its curve, and its upper limit is 30°C, so growth falls to zero at 31°C. Species Y has a broader range to 36°C and retains about 60% relative growth at 31°C. Neither species benefits from the warming, and 31°C is below Y's upper limit.

PE4-MCQ012

Answer: B. Specialists such as lichen-dependent caribou are vulnerable when their key resource disappears. Generalists such as coyotes switch among foods and habitats, so they tolerate disturbance better. A natural cause does not protect a specialist from the loss of its food supply.

PE4-MCQ013

Answer: C. Short generation time, many offspring, and no parental care describe an r-selected species, which can rebuild rapidly after a disturbance. K-selected species have few offspring, parental care, and slow recovery. r-selected species typically live far below carrying capacity, not near it.

PE4-MCQ014

Answer: C. Very high early mortality followed by a low death rate for survivors gives a Type III curve, shown by Curve C. Curve A is a Type I pattern with high survival until old age, and Curve B has a constant death rate at every age (Type II). Curve D has low mortality early and a steep decline in midlife, which does not match.

PE4-MCQ015

Answer: D. Net change = births − deaths + immigrants − emigrants = 110,000 − 40,000 + 20,000 − 10,000 = 80,000. Growth rate = 80,000 ÷ 5,000,000 × 100 = 1.6%. The 1.4% answer comes from ignoring migration (70,000 ÷ 5,000,000).

PE4-MCQ016

Answer: C. The J-shaped rise to about 5,600 followed by a crash is a classic overshoot: the population grew beyond what the food supply could sustain, grazing damaged the lichen, and starvation followed. The island has no predators, and an increase in carrying capacity would not cause a crash.

PE4-MCQ017

Answer: B. Country X has a wide base, so many young people will become parents over the next decades; even with fewer children per woman the number of births stays high, an effect called population momentum. Country Z has a small young population and would decline, and Country Y would be nearly stable. Growth does not stop immediately when fertility reaches replacement.

PE4-MCQ018

Answer: A. The data show a consistent inverse association between education and TFR. Correlation alone cannot prove causation because other factors such as income, health care, and contraception access also changed. The TFR fell below 4 by 2010, and 2.2 in 2020 is still slightly above 2.1, so growth continues.

PE4-MCQ019

Answer: D. A TFR far below the replacement level of about 2.1 produces small younger cohorts and an aging population, so the old-age dependency ratio rises. A doubling or a rise in the child share would require high fertility. Stage 2 is marked by falling death rates and still-high birth rates, which does not describe this country.

PE4-MCQ020

Answer: A. In 1970, birth rate = 38 and death rate = 14 per 1,000, so natural increase = 24 per 1,000 = 2.4% per year. Doubling time = 70 ÷ 2.4 ≈ 29 years. Adding the two rates gives 5.2%, and the other options contain decimal-place or division errors.

PE4-MCQ021

Answer: B. Before the drought births equal deaths at 8 per km². Births follow $b = 0.60 - 0.04N$ and deaths $d = 0.04 + 0.03N$; adding 0.07 gives $d = 0.11 + 0.03N$. Setting $0.60 - 0.04N = 0.11 + 0.03N$ gives $0.49 = 0.07N$, so $N = 7$ per km². The answer of 8 is the original carrying capacity.

PE4-MCQ022

Answer: A. Subduction of an oceanic plate creates a trench, melts mantle rock that feeds a volcanic arc, and produces powerful earthquakes. Ridges and rift valleys form at divergent boundaries, and transform faults involve sideways sliding.

PE4-MCQ023

Answer: A. Plant roots and cover protect soil from wind and water. Removing the cover exposes topsoil, which is detached and carried away. Soil formation is very slow, bare soil does not gain organic matter, and crusted bare ground generally reduces infiltration.

PE4-MCQ024

Answer: D. Large sand particles leave large pore spaces, so water moves through quickly but is not held. Clay has tiny pores that hold water tightly but drain slowly. A soil cannot have both the highest permeability and the highest water retention.

PE4-MCQ025

Answer: D. Layer X is the stratosphere, where the ozone layer is found; absorption of UV radiation by ozone heats the air so temperature rises with altitude. In layer W (troposphere) temperature falls with altitude because air is warmed from the surface, and that surface heating cannot explain a temperature rise higher up. Layer Y is the mesosphere, where temperature falls again.

PE4-MCQ026

Answer: D. In the Hadley cell, warm moist air rises at the equator and drops its rain there. The dry air spreads poleward, descends near 30°, warms by compression, and absorbs moisture, creating high pressure and deserts. Sunlight is most direct at the equator, not at 30°.

PE4-MCQ027

Answer: A. The soil mass and funnel size differ along with soil type, so the results cannot be attributed to soil texture alone. Controlling those variables isolates the effect of soil type. Changing water volume or time introduces more differences, and testing only soil A cannot support a comparison.

PE4-MCQ028

Answer: C. Absorbed = incoming × (1 − albedo). Ice absorbs 200 × 0.15 = 30 W/m², and open water absorbs 200 × 0.93 = 186 W/m². The difference is 156 W/m². The values 14 and 170 are the energy reflected by ocean and ice, and 30 is the energy absorbed by ice alone.

PE4-MCQ029

Answer: B. Orographic lifting cools air on the windward slope, where water vapor condenses and falls as rain or snow. The now-dry air descends the leeward side, warms by compression, and evaporates moisture from the land, forming a rain shadow desert.

PE4-MCQ030

Answer: D. Point P is a strongly positive anomaly (about +2.3°C), typical of an El Niño. Trade winds weaken, warm water spreads east, upwelling of cold nutrient-rich water declines, anchovy catches drop, and rain increases along the Peruvian coast. Point Q, a strongly negative anomaly, would instead fit La Niña conditions: strong trade winds, enhanced upwelling, and drought along the coast.

PE4-MCQ031

Answer: B. Precipitation depth = 0.600 m; volume = 25×10^6 m² × 0.600 m = 1.5×10^7 m³. Runoff = 0.30 × 1.5×10^7 = 4.5×10^6 m³. The 1.5×10^7 answer is the total precipitation volume, and the others contain unit-conversion errors.

PE4-MCQ032

Answer: B. When a shared resource is open to all and individuals gain personally from extra use while costs are spread among the group, the resource tends to be overexploited, which is the tragedy of the commons. Maximum sustainable yield is a management goal, not an explanation of overuse.

PE4-MCQ033

Answer: A. High-yield varieties require large inputs of irrigation water, synthetic fertilizer, pesticides, and machinery, which raise water use and pollution. Irrigation tends to increase salinization, monocultures reduce genetic diversity, and mechanization and fertilizer increase fossil fuel use.

PE4-MCQ034

Answer: D. In dry climates irrigation water evaporates quickly and leaves dissolved salts in the soil, forming a crust that stunts plants. Acid deposition and eutrophication do not produce a salt crust, and heavy rain is not a feature of an arid valley.

PE4-MCQ035

Answer: D. IPM combines monitoring, biological and cultural controls, and targeted chemical use only when pest populations reach a threshold, which reduces cost and resistance. The other options rely on preventive or routine spraying, which promotes resistance and harms nontarget organisms.

PE4-MCQ036

Answer: B. Deficit = footprint − biocapacity. For Q it is 5.6 − 1.9 = 3.7 gha per person, the largest. P has a reserve of 2.7, R is balanced, and S has a deficit of 0.4 despite its small footprint, because its biocapacity is even smaller.

PE4-MCQ037

Answer: B. Water withdrawn = water needed ÷ efficiency. Flood: 4,500 ÷ 0.50 = 9,000 m^3. Drip: 4,500 ÷ 0.90 = 5,000 m^3. The saving is 9,000 − 5,000 = 4,000 m^3. The value 5,000 is the drip withdrawal, not the saving.

PE4-MCQ038

Answer: C. Catch per unit effort = catch ÷ boats: 90,000 ÷ 150 = 600; 95,000 ÷ 300 ≈ 317; 70,000 ÷ 450 ≈ 156. Falling catch per boat with rapidly rising effort indicates declining stocks. Total catch rose slightly and then fell, so it did not increase every decade, and more boats lowered rather than raised efficiency.

PE4-MCQ039

Answer: A. Permeable surfaces and basins let water infiltrate or be stored, lowering the volume and peak of runoff. Faster conveyance and added pavement increase peak flows downstream, and road salt does nothing to reduce runoff volume.

PE4-MCQ040

Answer: C. Feed = 200,000 × 1.3 = 260,000 kg. Fishmeal = 260,000 × 0.30 = 78,000 kg. Wild fish = 78,000 × 4.5 = 351,000 kg. The 78,000 value stops at fishmeal, and 1,170,000 forgets that only 30% of the feed is fishmeal.

PE4-MCQ041

Answer: C. Resistant insects = 0.001 × 2,000,000 = 2,000. Susceptible = 1,998,000, and 1% survive, giving 19,980. Survivors total 21,980, so resistant = 2,000 ÷ 21,980 = 9.1%. Repeating the spray each generation drives the resistant fraction higher, the basis of the pesticide treadmill.

PE4-MCQ042

Answer: C. Coal, natural gas, and uranium formed over geologic time or are mined from finite deposits, so they are nonrenewable. Each of the other lists includes at least one renewable resource: geothermal heat, managed wood, or hydropower.

PE4-MCQ043

Answer: A. Growing biofuel crops competes with food crops for land and water, and in a desert the irrigation demand is especially significant. Ethanol is not radioactive, corn does take up CO_2 through photosynthesis, and ethanol has a useful energy content, although less per liter than gasoline.

PE4-MCQ044

Answer: C. In a fuel cell, hydrogen combines with oxygen to produce electricity, heat, and water. There is no carbon in the fuel, so no CO_2 or soot forms, and the low-temperature reaction does not produce nitrogen oxides at the tailpipe. Emissions can occur upstream, depending on how hydrogen is produced.

PE4-MCQ045

Answer: B. Commercial reactors use fission of uranium-235 to boil water, so operation emits almost no CO_2, but high-level waste must be isolated for very long periods. Fusion is not used commercially, and uranium is not burned. Reactors use steam turbines and need large amounts of cooling water.

PE4-MCQ046

Answer: D. Efficiency = output ÷ input = 3.6 ÷ 9.0 = 0.40, or 40%. Waste heat = input − useful output = 9.0 × 10^9 − 3.6 × 10^9 = 5.4 × 10^9 kJ. The options with 60% use the fraction wasted as if it were the efficiency.

PE4-MCQ047

Answer: B. Energy = capacity × capacity factor × hours = 50 MW × 0.30 × 8,760 h = 131,400 MWh. Without the capacity factor, 50 × 8,760 = 438,000 MWh, which assumes full output day and night. The other values come from using the wrong percentage.

PE4-MCQ048

Answer: C. Power saved per bulb = 60 − 9 = 51 W, or 10.2 kW for 200 bulbs. Hours = 8 × 180 = 1,440 h. Energy saved = 10.2 kW × 1,440 h = 14,688 kWh. The value 17,280 uses 60 W instead of the saving, and 29,784 uses 365 days.

PE4-MCQ049

Answer: B. Splitting water with renewable electricity produces hydrogen with almost no combustion emissions. Reforming natural gas and gasifying coal release CO_2 as the carbon in the fuel is converted, and electrolysis powered by coal simply moves the emissions to the power plant.

PE4-MCQ050

Answer: C. 2,000 GWh = 2.0×10^9 kWh. The saving per kWh is 1,000 − 450 = 550 g, so $2.0 \times 10^9 \times 550$ g = 1.1×10^{12} g = 1.1×10^6 metric tons. Using only the natural gas value gives 9.0×10^5, and using only coal gives 2.0×10^6.

PE4-MCQ051

Answer: D. Corn ethanol: input 7.0 + 1.0 + 10.0 = 18.0 MJ, so net energy = 21 − 18 = 3.0 MJ/L. Switchgrass: input 3.0 + 1.5 + 5.5 = 10.0 MJ, so net = 11.0 MJ/L. The ratio is 11.0 ÷ 3.0 ≈ 3.7. The ratio 2.1 is the switchgrass energy return per unit of input (21 ÷ 10), and 1.8 is the ratio of corn's total input to switchgrass's total input.

PE4-MCQ052

Answer: A. A pollutant formed in the atmosphere from chemical reactions of other pollutants is a secondary pollutant. Primary pollutants are emitted directly from a source, such as NO_x and VOCs themselves. Ozone is a gas, not a particle, and is not an organic compound.

PE4-MCQ053

Answer: D. Burning fossil fuels releases SO_2 and NO_x, which react with water, oxygen, and other chemicals in the atmosphere to form sulfuric and nitric acids. CO_2 and CH_4 are mainly greenhouse gases, and CFCs and halons deplete stratospheric ozone.

PE4-MCQ054

Answer: C. Ozone is a secondary pollutant. After the morning peak in NO and NO_2, photochemical reactions powered by increasing sunlight and temperature build ozone through the day, and early in the morning NO reacts with O_3 and lowers it. Ozone is not emitted by vehicles, NO_2 does not fall to zero in the graph, and afternoons are warmer, not cooler.

PE4-MCQ055

Answer: D. This is a thermal inversion: dense cold air under warmer air is stable and does not rise, so emissions stay near the surface. Normally, warm surface air rises and disperses pollutants. Winter sunlight is weak, and cold air does not rise.

PE4-MCQ056

Answer: D. Particles smaller than 2.5 micrometers bypass the body's filtering defenses, reach the alveoli, and are linked to asthma, heart disease, and lung disease. Larger particles are mostly caught in the nose and throat. Smoke particles are not radioactive and do not deplete stratospheric ozone.

PE4-MCQ057

Answer: A. Radon enters from the soil through cracks and openings, so sealing and sub-slab depressurization stops it from entering and vents it outside. A CO detector does not detect radon, ozone generators add a lung irritant, and the stove is not the source.

PE4-MCQ058

Answer: B. Sulfur = $2.0 \times 10^6 \times 0.015$ = 30,000 t. SO_2 formed = 30,000 × 2.0 = 60,000 t. The scrubber removes 92%, so 8% is emitted: 60,000 × 0.08 = 4,800 t. The 55,200 value is the amount removed, and 60,000 is the amount with no scrubber.

PE4-MCQ059

Answer: A. Nitrogen and phosphorus fuel algal blooms; when the algae die, aerobic decomposers consume dissolved oxygen, causing hypoxia and fish kills. Blooms shade submerged plants rather than helping them, and the other sequences reverse cause and effect.

PE4-MCQ060

Answer: D. Source reduction prevents waste from being created in the first place, so it ranks above reuse, recycling, composting, and disposal. Recycling bins, composting, and waste-to-energy all manage waste that has already been generated.

PE4-MCQ061

Answer: D. Endocrine disruptors mimic, block, or alter natural hormones, so they can change development and reproduction at very low concentrations, far below lethal levels. Immune damage, acidification, and lethal doses describe different kinds of harm.

PE4-MCQ062

Answer: C. A smaller LD50 means a smaller dose is lethal to half the population, so A is more toxic. The ratio is 480 ÷ 12 = 40. Every LD50 corresponds to 50% mortality by definition, so that does not make the chemicals equal, and toxicity is compared by ratio, not by difference.

PE4-MCQ063

Answer: B. Factor = 4,000 ÷ 5 = 800. This is biomagnification: concentrations of persistent, fat-soluble pollutants increase at each higher trophic level because they are stored in tissue and not excreted. The values 8 and 12.5 are single-step factors, and 3,995 is a difference, not a factor.

PE4-MCQ064

Answer: A. The dose at 50% mortality is about 8 mg/kg for X and 120 mg/kg for Y, so X needs one-fifteenth of the dose (120 ÷ 8 = 15). A steeper curve means mortality changes more sharply over a small range of doses, not a higher toxicity. All LD50s correspond to 50% mortality, and a larger LD50 indicates lower toxicity.

PE4-MCQ065

Answer: B. Landfilled share = 100 − (28 + 12 + 10) = 50%, or 300,000 t/yr. Volume = 300,000 ÷ 0.75 = 400,000 m^3/yr. Years = 12,000,000 ÷ 400,000 = 30. Sending all the waste to the landfill would give 15 years, and omitting the density conversion gives 40 years.

PE4-MCQ066

Answer: A. CFCs release chlorine atoms when broken down by ultraviolet radiation in the stratosphere, and each chlorine atom can destroy many ozone molecules. SO_2 and NO_x cause acid deposition, CO_2 and CH_4 are greenhouse gases, and VOCs contribute to ground-level ozone.

PE4-MCQ067

Answer: C. The natural greenhouse effect keeps Earth habitable because heat-trapping gases absorb outgoing infrared and radiate some back toward the surface. Reflection of sunlight is the role of clouds and ice, and UV absorption by ozone is a separate process from the greenhouse effect.

PE4-MCQ068

Answer: A. Water expands as it warms, and meltwater from land-based ice such as glaciers and the Greenland and Antarctic ice sheets adds new water to the ocean. Floating ice already displaces its own weight of water, so melting it changes sea level very little.

PE4-MCQ069

Answer: B. Invasive species typically reproduce quickly, are generalists, and escape the predators, parasites, and diseases that limited them in their native range. The other sets describe specialist, slow-growing species that rarely spread widely.

PE4-MCQ070

Answer: A. CFCs are chemically stable and have atmospheric lifetimes of many decades, so the chlorine burden declines slowly even after emissions stop. Production of CFCs has been almost completely phased out, and HFC replacements do not contain chlorine. Ozone formation does not depend on CO_2 declining.

PE4-MCQ071

Answer: B. Most of Earth's land area is in the Northern Hemisphere, so photosynthesis during the growing season lowers atmospheric CO_2 and decomposition afterward raises it, producing an annual cycle of a few ppm. Fossil fuels are burned year-round, eruptions are irregular, and the ocean pattern is not what creates the sharp seasonal signal.

PE4-MCQ072

Answer: B. The two curves rise and fall together, showing a positive association, and the present value of 425 ppm is well above the roughly 180–290 ppm range of the record. The figure does not show which changed first, so the claim that CO_2 always leads temperature is not supported. It also does not show that CO_2 was the only driver of temperature.

PE4-MCQ073

Answer: C. The pH fell from 8.11 to 8.05, a change of 0.06 units. The ratio of $[H^+]$ is 10^0.06 = 1.15, an increase of about 15%. Dividing 0.06 by 8.11 gives 0.7%, which ignores the logarithmic scale, and multiplying 0.06 by 100 gives 6%.

PE4-MCQ074

Answer: C. Corals depend on photosynthetic algae for food and color. When water temperatures exceed the coral's tolerance, the algae are expelled and the white skeleton shows through; corals can starve if the heat persists. Ocean pH is decreasing, not rising, and the other options describe processes that do not cause mass bleaching.

PE4-MCQ075

Answer: D. CITES (Convention on International Trade in Endangered Species) controls international trade in listed wildlife to prevent overexploitation. The Endangered Species Act is a U.S. law, the Montreal Protocol addresses ozone-depleting substances, and the Kyoto Protocol addresses greenhouse gas emissions.

PE4-MCQ076

Answer: D. Fragmentation shrinks interior habitat, exposes animals to predators and weather at the edges, and isolates small populations, which reduces genetic exchange and raises extinction risk. Generalist species typically benefit from edges, increasing competition and predation.

PE4-MCQ077

Answer: C. CO_2e = mass × GWP = 5,000 t × 28 = 140,000 t CO_2e. Dividing by 28 gives 179 and adding 28 gives 5,028, both of which misuse the GWP. The value 1,400,000 contains a decimal-place error.

PE4-MCQ078

Answer: B. Target = 480 × 0.70 = 336 Mt. Reduction still needed = 440 − 336 = 104 Mt over 10 years = 10.4 Mt per year. The value 14.4 is 30% of 480 divided by 10, which ignores the reductions already achieved, and 5.2 divides by 20 years instead of 10.

PE4-MCQ079

Answer: B. A negative feedback counteracts a change. A warmer surface emits more outgoing infrared radiation, which acts to cool it. Permafrost carbon release, reduced albedo from melting ice, and increased water vapor all amplify warming, so they are positive feedbacks.

PE4-MCQ080

Answer: C. Values fell from about 225 DU in 1979 to about 100 DU by 1999, then turned upward to roughly 125 DU, still well below 1979. The decline continued for a decade after 1987 because ozone-depleting substances persist in the atmosphere, so an immediate rise is not shown. The data do not show full recovery or a continuing decline.

Section II — FRQ Solutions

PE4-FRQ001

(a) The endocrine-disrupting effects (feminization, abnormal gonads, and reduced tadpole survival) in frogs possibly caused by herbicide runoff from irrigated cotton fields. *(1 point: identifies herbicide/pesticide pollution or endocrine disruption of the frog population. Acceptable: "agricultural runoff is harming amphibians.")*

(b) If frogs are exposed to higher concentrations of the herbicide, then a larger percentage of male frogs will have ovarian tissue (intersex gonads). *(1 point: a testable, if–then statement that names both herbicide concentration and frog abnormality and gives a direction.)*

(c) Independent variable: herbicide concentration in the water (µg/L), or the sampling site as a measure of exposure. Dependent variable: the percentage of male frogs (or tadpoles) with abnormal gonads, or tadpole survival rate. *(1 point for each. The independent and dependent variables must match the hypothesis; do not credit "frogs" alone.)*

(d) Any two of: species and age or developmental stage of the frogs; number of animals examined per site; water temperature and pH; time of year of collection; method of gonad examination; exposure duration; food availability. *(1 point for two valid controlled variables.)*

(e) Procedure: At each of Sites 1–4, collect water samples and capture a standard number of frogs or tadpoles (for example, 30 per pond), then measure herbicide concentration in the lab and examine the gonads under a microscope using the same method for all animals. (i) Site 4 is a reference (control) pond without upstream farms. It shows the background rate of abnormalities when herbicide exposure is near zero, so any extra abnormalities at Site 3 can be attributed to herbicide. *(1 point)* (ii) Replication: collect multiple water samples per site (for example, three), repeat on several dates or seasons, and examine many frogs; averaging reduces the effect of random variation and outliers. *(1 point. Accept replicate ponds or repeating the experiment.)*

(f) Collect herbicide concentration (µg/L) at each site and the percentage of male frogs with abnormal gonads at each site. The hypothesis is supported if abnormality rates are highest where concentrations are highest (Sites 2 and 3), and are low at Site 4, showing a positive relationship. *(1 point for the data and the result that supports the hypothesis. Both parts are required.)*

(g) Examples: plant a vegetated buffer strip or build a constructed wetland between the fields and the drain; switch to drip irrigation to reduce return flow; apply integrated pest management or reduce herbicide use; use a less persistent herbicide with a no-spray zone near drains. *(1 point for a reasonable solution.)* Justification: a buffer strip or wetland filters sediment and absorbs or breaks down the herbicide (advantage), but takes land out of production and costs money (disadvantage). Drip irrigation lowers runoff volume and saves water but is costly to install. *(1 point for an advantage or disadvantage that fits the solution.)*

PE4-FRQ002

(a) Atmospheric CO_2 increased steadily, from 369 ppm in 2000 to 425 ppm in 2025, with no year of decrease. *(1 point: states increasing trend with data; acceptable to note a slight acceleration after 2010.)*

(b) Rate = (413 − 369) ÷ 20 years = 44 ÷ 20 = **2.2 ppm per year**. *(1 point: correct value with units; accept 2.2 ppm/yr only.)*

(c) Percent increase = (61 − 42) ÷ 42 × 100 = 19 ÷ 42 × 100 = **45%** (45.2%). *(1 point: correct setup and answer.)*

(d) As the active layer deepens, more previously frozen organic matter thaws, and soil microbes decompose it (aerobically releasing CO_2 or anaerobically releasing CH_4 in waterlogged soils). *(1 point for the gas and decomposition process.)* Warming releases GHGs, which cause more warming, which thaws even more permafrost, so the change reinforces itself: a positive feedback. *(1 point for explaining the reinforcing cycle.)*

(e) Ice is bright, with high albedo, and reflects most incoming sunlight. As the ice shrinks, dark ocean water is exposed. *(1 point: albedo change.)* The ocean absorbs more solar energy, increasing temperatures, which melts more ice (an ice–albedo positive feedback). *(1 point: more absorption leads to more warming/melting.)*

(f) 2020–2025 rate = (425 − 413) ÷ 5 = 2.4 ppm/yr. From 2025 to 2040 = 15 years: 425 + 15 × 2.4 = 425 + 36 = **461 ppm**. *(1 point: correct answer with work; accept 460–461 ppm.)*

(g) Any one of: a carbon tax or cap-and-trade system; replacing coal with wind, solar, or nuclear power; improving energy efficiency; protecting forests and wetlands; capturing methane from landfills or farms. *(1 point for a reasonable action that reduces CO_2 emissions or increases carbon storage.)* Justification using data: CO_2 has risen about 2.2–2.4 ppm per year and the active layer and sea-ice trends move in the same direction as CO_2, so cutting emissions would slow the warming-linked changes (or reduce the risk of permafrost thaw feedback). *(1 point: refers specifically to data such as the rate of CO_2 increase or the declining sea ice or the deeper active layer.)*

PE4-FRQ003

(a) 50 buses × 48,000 km × (50 L ÷ 100 km) = 50 × 48,000 × 0.50 = **1,200,000 L per year (1.2×10^6 L)**. *(1 point: correct value with units.)*

(b) 1,200,000 L × 3.2 kg CO_2e/L = 3,840,000 kg = **3,840 metric tons CO_2e per year**. *(1 point: correct value with units.)*

(c) B20 volumes: biodiesel = 0.20 × 1,200,000 = 240,000 L; diesel = 0.80 × 1,200,000 = 960,000 L. Emissions = (960,000 × 3.2) + (240,000 × 1.1) = 3,072,000 + 264,000 = 3,336,000 kg = **3,336 t CO_2e**. *(1 point.)* Percent reduction = (3,840 − 3,336) ÷ 3,840 × 100 = 504 ÷ 3,840 × 100 = **13%** (13.1%). *(1 point. Consequential credit if (b) was wrong but the method is correct.)*

(d) 240,000 L ÷ 1,200 L/ha = **200 hectares per year**. *(1 point: correct value with units.)*

(e) 200 ha × 5,500 m^3/ha = **1,100,000 m^3 (1.1×10^6 m^3) of water per year**. *(1 point: correct value with units; accept consequential credit from (d).)*

(f) Any one of: irrigation uses large amounts of scarce desert water, which can lower aquifers and river flows; irrigated land can become saline; fertilizer and pesticide runoff pollutes waterways; cropland conversion destroys desert habitat; growing fuel instead of food raises food prices. *(1 point.)*

(g) Claim (any reasonable one): e.g., "The county should not adopt B20 as proposed; it should use waste cooking oil biodiesel or electric buses instead." *(1 point for a clear claim.)* Evidence examples (2 points, 1 each, must be specific): B20 cuts life-cycle emissions by only about 13% (504 t CO_2e per year); it requires 200 ha of land and 1.1 million m^3 of irrigation water per year, a major demand in a desert; the water used per liter of biodiesel is 1,100,000 ÷ 240,000 ≈ 4.6 m^3 per liter; waste cooking oil requires no added cropland or irrigation, and electric buses can eliminate tailpipe emissions (if the electricity comes from low-carbon sources). *(Alternative claim: B20 should be adopted because it reduces emissions by 13% without changing buses, with evidence from the calculations and an acknowledgement of the water cost. Credit any claim that is supported by two valid data-based pieces of evidence.)*

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